

System Multimeter PM2530

Operation Manual

4822 872 30448
920116



PHILIPS

Thank you for purchasing this PHILIPS instrument. It has been designed and manufactured to the highest quality standards to give you many years of troublefree and accurate measurements.

The powerful measuring functions listed in Section 2 have been combined with an easy, logical operation to let you use the full power of this instrument every day.

If you have any suggestions about how this product could be improved, please contact your local FLUKE/PHILIPS representative. Addresses are published on the last pages of this manual.

IMPORTANT

In correspondence concerning this instrument, please quote the type number and serial number as given on the type plate at the rear:

Model number : PM2530/...
 9447 325 30...
Serial number: DM

NOTE: *The design of this instrument is subject to continuous development and improvement. Consequently, this instrument may incorporate minor changes in detail from the information contained in this manual.*

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INITIAL INSPECTION

Check the contents of the shipment for completeness and note whether any damage has occurred during transport. If the contents are incomplete, or if there is damage to the instrument or its accessories, a claim should be filed with the carrier immediately, and the FLUKE/PHILIPS sales or service office nearest you should be notified in order to facilitate the repair or replacement of the instrument or accessories. Addresses are published at the back of this manual.

The following items should be included in the shipment:

Qty	Item	Model number or ordering number
1	System Multimeter	PM2530
1	Operation Manual	4822 872 30448
1	Command Reference Manual	4822 872 80173
1	SCPI Manual	4822 872 80149
1	Line cord:	
	Line cord (European type) or	4822 321 21616
	Line cord (North American type) or	4822 321 10446
	Line cord (British type) or	4822 321 21617
	Line cord (Swiss type) or	4822 321 21618
	Line cord (Australian type)	4822 321 21781
1	Set of 2 leads (including probes)	PM9260
1	Spare Fuse 2.5 AF / 250 V (current input protection)	4822 070 12502
2 (1)	Spare Line Fuses 160 mA, 250 V (1 if line voltage setting is 230 V)	4822 070 31601
2 (1)	Spare Line Fuses 315 mA, 250 V (1 if line voltage setting is 115 V)	4822 070 36309
2	Rack mounting kit (rack ears)	5322 401 11428

Special PM2530 accessories which are not included in the shipment:

4 Wire Ohms cable with Kelvin clips and 4 mm banana connectors	PM9268
Low thermal copper EMF plugin cable	FLUKE 5440A-7002
Pt100 Probe type 385 Pt (DIN 43760) with 4x4 mm banana connector	PM9252

Note: for a complete list see FLUKE/PHILIPS catalogue.

A performance check of the instrument may be carried out by following the procedures outlined in section 6.5 "PERFORMANCE TESTS".

ABOUT THIS MANUAL

Section 1 OPERATOR SAFETY

This section provides information and warnings which must be followed to ensure safe operation and maintenance of the meter. Read this section carefully before installing or using the instrument.

Section 2 THE PM2530 SYSTEM MULTIMETER

This section provides a summary of the PM2530's features.

Section 3 INSTALLATION INSTRUCTIONS

This section details the preparations required before switching the instrument on. It also deals with the mechanical installation of the instrument and its accessories. You are strongly advised to read this section before installing your instrument.

Section 4 OPERATION INSTRUCTIONS, LOCAL CONTROL

This section describes how to start, set up measurements and operate your PM2530 from the front panel.

Section 5 REMOTE PROGRAMMING

This section contains information about controlling your PM2530 via the GPIB (IEEE-488.2) bus interface.

Section 6 MAINTENANCE

This section describes how to perform basic maintenance and repairs (e.g., replacing fuses). It also includes a performance test, and other general service information.

Section 7 OPERATING FEATURES REFERENCE

This section describes the push buttons, connectors, and display messages.

Appendix A SPECIFICATIONS

This appendix contains the complete technical specification of your PM2530.

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1 OPERATOR SAFETY

This section provides information and warnings which must be followed to ensure safe operation and retain the meter in safe condition. Read this page carefully before beginning to install and use the instrument.

1.1 INTRODUCTION

The instrument described in this manual is designed to be used by qualified personnel only. Adjustment, maintenance and repair of the exposed equipment shall be carried out only by qualified personnel, who are aware of the hazards involved.

1.2 SAFETY PRECAUTIONS

For the correct and safe use of this instrument it is essential that both operating and servicing personnel follow generally-accepted safety procedures in addition to the safety precautions specified in this manual. Specific warning and caution statements, where they apply, will be found throughout the manual.

Where necessary, the warning and caution statements and/or symbols are marked on the apparatus.

1.3 CAUTION AND WARNING STATEMENTS

CAUTION: is used to indicate correct operating or maintenance procedures in order to prevent damage to or destruction of the equipment or other property.

WARNING: calls attention to a potential danger that requires correct procedures or practices in order to prevent personal injury.

1.4 SYMBOLS

Symbol Notion of symbol.



To preserve the instrument from damage the operator must refer to an explanation in the operation manual.



Protective earth (grounding) terminal.



High voltage terminal (≥ 1000 V).

1.5 IMPAIRED SAFETY

Whenever it is likely that safety has been impaired, the instrument must be turned off and disconnected from the line power. The matter should then be referred to qualified technicians. Safety is likely to be impaired if, for example, the instrument fails to perform the intended measurements or shows visible damage.

1 SICHERHEIT

Lesen sie diese Seite bitte vor dem Anschliessen und der Inbetriebnahme des Geräts. Das Gerät erfüllt die Sicherheitsanforderungen von IEC 348, Klasse 1.

1.1 EINLEITUNG

Das in dieser Bedienungsanleitung beschriebene Gerät ist geeignet zur Bedienung durch eingewiesenes Personal. Justierung, Reparatur und Wartung des geöffnetes Geräts sollte nur von geschultem Personal ausgeführt werden, das über alle mögliche Gefährdungen unterrichtet ist.

1.2 SICHERHEITSHINWEISE

Wie bei allen technischen Geräten sind auch bei diesem Gerät einwandfreie Funktion und Betriebssicherheit nur dann gewährleistet, wenn bei der Bedienung und beim Service sowohl die allgemein üblichen Sicherheitsvorkehrungen als auch die speziellen Sicherheitshinweise in dieser Bedienungsanleitung beachtet werden. Soweit erforderlich, sind entsprechende Stellen des Geräts mit warnenden Hinweisen und Symbolen gekennzeichnet.

Beim öffnen von Abdeckungen oder Entfernen von Teilen, ausser wenn dies von Hand möglich ist, können spannungsführende Teile freigelegt werden. Auch können Anschlussstellen spannungsführend sein.

Vor dem Abgleich, einer Wartung, einer Instandsetzung oder einem Austausch von Teilen muss das Gerät von allen Spannungsquellen getrennt sein, wie ein Öffnen des Geräts erforderlich ist.

Wenn danach ein Abgleich, eine Wartung oder eine Reparatur am geöffneten Gerät unter Spannung unvermeidlich ist, so darf das nur durch eine Fachkraft geschehen, die mit den damit verbundenen Gefahren vertraut ist.

1.3 VORSICHTS- UND WARNUNGSHINWEISE

VORSICHTSHINWEISE: Sollen auf eine korrekte Bedienung oder Wartung hinweisen, damit weder dieses Gerät noch andere daran angeschlossenen beschädigt werden.

WARNHINWEISE: Geben eine potentielle Gefahrenquelle an durch die bei unsachgemässer Behandlung für Bedienungspersonal oder Dritte gefährliche Situationen entstehen können.

1.4 SYMBOLE



Um eine Beschädigung des Geräts zu vermeiden, lesen Sie bitte die Bedienungsanleitung.



Schutzerdeanschluss.



Hochspannungs Buchse (≥ 1000 V).

1.5 BEEINTRÄCHTIGUNG DER SICHERHEIT

Wenn aus irgendeinem Grunde angenommen werden kann, dass die Sicherheit beeinträchtigt ist, muss das Gerät ausser Betrieb gesetzt und so gekennzeichnet werden, dass es nicht versehentlich von Dritten wieder in Betrieb genommen wird. Ausserden ist der Kundendienst zu benachrichtigen. Die Sicherheit kann z.B. beeinträchtigt sein, wenn das Gerät nicht wie vorgeschrieben arbeitet oder sichtbar beschädigt ist.

1 SECURITE

Lisez attentivement cette page avant d'installer et d'utiliser cet appareil. Celui-ci satisfait les impératifs de sécurité CEI 348, classe I.

1.1 INTRODUCTION

L'appareil décrit dans la présente notice est conçu pour être utilisé par un personnel ayant bénéficié d'une formation appropriée. Les travaux de réglage, d'entretien et de réparation sur un appareil ouvert ne pourront être effectués que par un personnel qualifié, pleinement conscient des risques encourus.

1.2 SECURITE

Pour assurer un emploi correct et sûr de cet appareil, il faut absolument que tout le personnel d'exploitation et de maintenance suive les procédures de sécurité courantes, en plus des précautions exposées ici même. Tout au long de la notice, l'utilisateur trouvera, aux endroits appropriés, des textes ayant particulièrement trait à des avertissements ou de mises en garde.

Si nécessaire, des textes et/ou des symboles d'avertissement ou de mise en garde sont inscrits sur l'appareil.

1.3 REMARQUES DE MISE EN GARDE ET D'AVERTISSEMENT

MISE EN GARDE: cette notion est utilisée pour indiquer les procédures correctes d'exploitation ou de maintenance, afin d'éviter tout dégât ou destruction de l'équipement, ou de tout autre matériel.

AVERTISSEMENT: cette notion est utilisée pour attirer l'attention sur un danger potentiel requérant des procédures ou des pratiques correctes afin d'éviter toute blessure de personnes.

1.4 SYMOLES



pour éviter que l'instrument ne subisse des dégâts, l'opérateur doit se référer à l'explication exposée dans la notice d'utilisation.



Borne de terre de protection.



Borne à haute tension (≥ 1000 V).

1.5 COMPROMISSION EVENTUELLE DE LA SECURITE

Si l'on a des raisons de croire que le fonctionnement sûr est menacé, l'instrument devra être mis hors circuit et protégé contre toute exploitation fortuite. L'organisation de service après vente appropriée devra alors être mise au courant.

Il y a compromission de la sécurité si, par exemple, l'instrument n'effectue plus les mesures voulues ou s'il présente des dégâts visibles.

2 THE PM2530 SYSTEM MULTIMETER

The PM2530 System Multimeter is designed for System and Bench applications, and has a built-in GPIB interface. It provides high accuracy voltage, -current, -resistance, -temperature and -frequency/period measurements at up to 7 1/2 digit resolution.

The standard multimeter functions can directly be selected. Operating menus enable easy access to a lot of auxiliary functions.

Beside control via the keyboard, the PM2530 can be remotely controlled via the GPIB interface, using SCPI (Standard Commands for Programmable Instruments).

The main PM2530 features are listed below.

STANDARD MULTIMETER FUNCTIONS

Measurement functions

- Volts DC, 100 nV sensitivity, 20 ppm basic accuracy (2 V range).
- Volts AC; AC coupled or DC coupled RMS, up to 1 MHz.
- Volts low frequency; DC coupled, RMS from 5 mHz to 500 Hz.
- Current DC.
- Current AC; DC coupled, RMS.
- Resistance 2-wire/4-wire; resistance in 2-wire and 4-wire configuration.
- Frequency and Period.
- Temperature; temperature with Pt100 Resistance Temperature Detector (RTD).

Measurement control functions

- fast automatic or manual ranging.
- internal or external triggering.
- selectable reading rate: low if resolution is critical, high if high speed is required.

Filters

- selectable analog low pass filter for Volts DC and Resistance functions.
- selectable RMS converter response time in AC functions.
- weighting filter (exponential smoothing).
- block averaging filter.

AUXILIARY FUNCTIONS

Data acquisition functions

- NULLING eliminates the effect of input offsets in the Volts DC, Ohms, and Current DC measurement functions.
- MIN/MAX constantly monitors and stores the minimum and maximum readings.
- LIMITS counts and signals passings of previously defined upper and lower limits.
- BURST stores the most recent measurement results (up to 15000) at any reading rate (up to 50000 measurements per second) in a shift register. The storage operation may be independently triggered to capture a particular group of successive results, that may occur before, after or both before and after the trigger instant.

Calculation functions

- (AX+B) / Z scales measurement result X (offset/scale/ratio)
- DELTA% compares measurement results with a stored nominal value. The reading is the percentual deviation.
- dBm calculates dBm of a measured value.
- REL REF relative reference, compares subsequent results with a stored reference value. The reading is the algebraic difference.

Output functions

- READ MIN/MAX to read the results stored in the acquisition function MIN/MAX
- LIMITS COUNT to read the results stored in the acquisition function LIMITS
- READ BURST to read the results stored in the acquisition function BURST
- BUS ADDR sets the GPIB interface address.
- CHECK performs built-in self tests, and enables closed case calibration (no internal calibration adjustments).

Miscellaneous functions

- BEEPER sets the beeper on/off and selects the conditions on which the beeper sounds.
- DISPLAY sets the display backlight on/off and selects the display length and contrast.
- SAVE/RECALL saves and recalls instrument configurations.
- DATE/TIME sets the PM2530 (battery maintained) date and time.

3 INSTALLATION INSTRUCTIONS

This section details the preparations required before switching the instrument on. It also gives details of the mechanical installation of the instrument. You are strongly advised to read this section before installing your instrument.

3.1 SAFETY INSTRUCTIONS

3.1.1 GROUNDING

Before any connection is made to the inputs, the instrument must be connected to protective ground using the 3-conductor line cord provided, or an equivalent.

WARNING: To avoid shock hazard or instrument damage, connect the instrument power line ground to earth ground. Do not break this protective connection by using a 2-conductor extension cord, or using a line voltage outlet without earth connection. Intentional interruption is dangerous and will void the instrument warranty.

When the instrument is brought from a cold into a warm environment, condensation may form causing hazardous conditions. Therefore, the equipment grounding requirements must be strictly complied with.

3.1.2 POWER CORD, LINE VOLTAGE SETTING

Different power cords are available for the various local line voltage outlets. Refer to "INITIAL INSPECTION" for the part numbers. The type of power cord delivered is determined by the particular instrument version ordered.

WARNING: The instrument must be disconnected from all voltage sources before the instrument line voltage setting is changed.

Before connecting the PM2530 to the line power, make sure that the instrument is set to the local mains voltage (115 V or 230 V nominal). Refer to the power line cord connector at the rear of the instrument (Fig. 3.1).

NOTE: If the line plug has to be changed for the local outlets, the work must be carried out by a qualified technician.

To change the line voltage setting (located at the rear of the instrument), proceed as follows (refer to Fig. 3.1):

- disconnect the instrument from all input voltage sources and unplug the line cord.
- carefully pry at the top centre of the cover to open the fuse holder compartment.
- remove the line voltage selection drum.
- remove the fuse holder with the fuse.
- replace the mains fuse with the required one (see Table 3.1 on the next page).
- insert the line voltage selection drum in such a way that the required line voltage marking is visible when the fuse holder is closed.
- close the fuse holder compartment (press firmly).

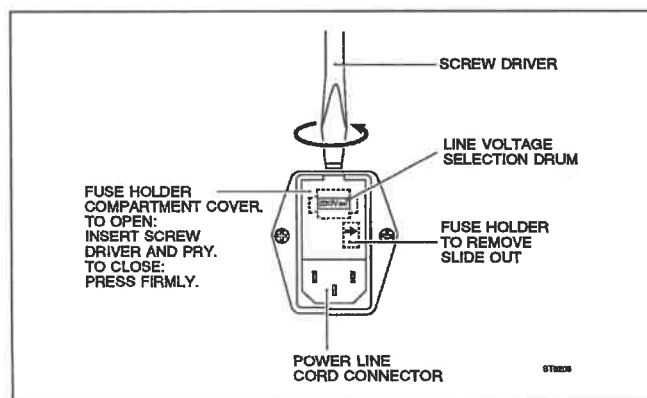


Fig. 3.1 Line voltage setting and line fuse replacement.

The instrument accepts a mains voltage frequency of 50, 60, and 400 Hz without the need for any modification.

3.1.3 LINE FUSE REPLACEMENT

The instrument uses a 250 V Slow Blow line fuse in series with the power supply. For line fuse ratings see Table 3.1.

WARNING: The instrument must be disconnected from all voltage sources before replacing the line fuse.

Table 3.1 Line fuse ratings

line voltage	required fuse	Parts Number
230 V nom.	160 mA, 250 V, Slow Blow	4822 070 31601
115 V nom.	315 mA, 250 V, Slow Blow	4822 070 36309

The line fuse holder is located in the line input socket, on the rear panel. To replace the fuse, proceed as follows (see Fig. 3.1):

- disconnect the instrument from all input voltage sources and unplug the line cord
- carefully pry at the top centre of the cover to open the fuse holder compartment.
- remove the fuse holder with the fuse as shown in Fig. 3.1.
- fit a new fuse of the correct rating and refit the fuse holder.
- close the fuse holder compartment (press firmly).

The instrument is shipped with a replacement fuse.

WARNING: Make sure that only fuses of the required current and voltage rating, and of the specified type, are used for replacement. The use of repaired fuses, and/or the short-circuiting of the fuse holder, is prohibited.

3.2 OPERATING POSITION OF THE INSTRUMENT

The instrument may be operated in the horizontal position. It may also be operated in a sloping position, with the stand-up brackets folded down. The specifications detailed in Appendix A are guaranteed for these operating positions.

3.3 COOLING CONSIDERATIONS

Do not position the instrument on any surface which produces or radiates heat, or in direct sunlight.

3.4 19 INCH RACK MOUNTING (optional)

You can mount the PM2530 in a standard 19-inch rack panel using the rack mounting kit provided. To install the rack mount kit, proceed as follows:

- remove the 2 foremost screws at each side of the instrument (Fig. 3.2).
- attach the rack ears with the screws provided.

You can remove the bottom feet if necessary.

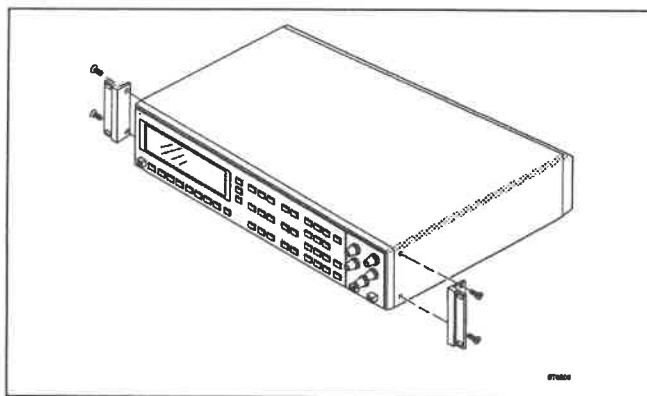


Fig. 3.2 Installing the rack mount kit.

4 OPERATING INSTRUCTIONS, LOCAL OPERATION

4.1 GETTING STARTED

4.1.1 TURNING POWER ON

Connect the instrument to the line voltage in accordance with the instructions in Sections 3.1.1 and 3.1.2. It is turned on and off with the POWER ON (I) /OFF (0) switch on the front panel. After being switched on the meter performs a self test and is operable within 10 seconds. Any errors found are shown on the right-hand side of the display, refer to Section 6.5.1 for details. After the required warm-up period (1 hour for 90 days specifications), the specifications listed in Appendix A are valid.

On completion of the power up sequence the meter assumes the power up measurement configuration as detailed in Appendix A.2.15.

4.1.2 START-UP EXAMPLE

The example below will have you operating the PM2530 in a matter of minutes. It covers the basic multimeter functions. Subsequent sections of the manual explain the operation of the instrument in more detail.

Example: To measure a DC voltage.

1. Select the front panel input terminals ("INPUTS REAR/FRONT" switch to FRONT).
2. Select internal guarding ("GUARD INT/EXT" switch to INT).
3. Press , to select the Volts DC measurement function. The display shows V=.
4. Press . The PM2530 assumes its default configuration: autoranging, internal triggering, reading rate SPD2 and all auxiliary functions off.
5. Connect the DC Voltage to be measured, to the front "VOLTS INPUT" terminals VΩ- COM (see Fig. 4.1).

The PM2530 will display the value of the input signal, e.g. +123.4567 mV (see Fig. 4.2).

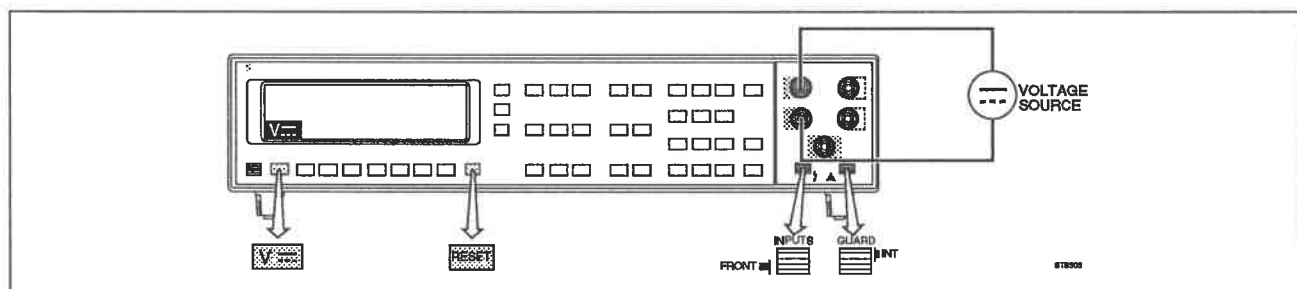


Fig. 4.1 Start-up example, basic instrument functions.

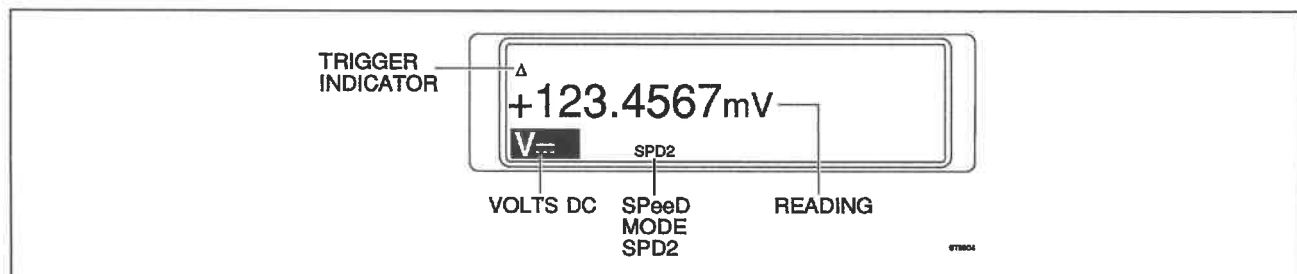


Fig. 4.2 Start-up display example.

Steps 6-9 show how to alter the ranging, the reading rate, triggering, and filter settings (see Fig. 4.3). Step 10 shows the use of the relative reference function.

6. Press **RANGE** **MAN**, **UP** or **DOWN** to select manual ranging. The display will show MAN. Press **UP** or **DOWN** for another range.
7. Press **READING RATE** **UP** or **DOWN**. The display shows the selected speed (SPD). Note, a higher reading rate results in a lower resolution (reduced scale length).
8. Press **FILTER** to invoke the FILTER menu. The right-hand side of the display shows the filter menu. Press **1** to select/deselect (toggle) the analog filter. Press **0** to exit the filter menu. The display shows A-FILT to indicate that the analog filter is active.
9. Press **TRIGGER** **SINGLE** to select manual triggering and start a measurement. Press **TRIGGER** **SINGLE** again to start a new measurement. Press **INT** to select internal triggering again.
10. Press **REL**. The present reading is stored as reference value. Subsequent readings are the difference between the input value and the reference value. The input value is displayed in the upper left display part. Press **REL** again to cancel the relative reference function.

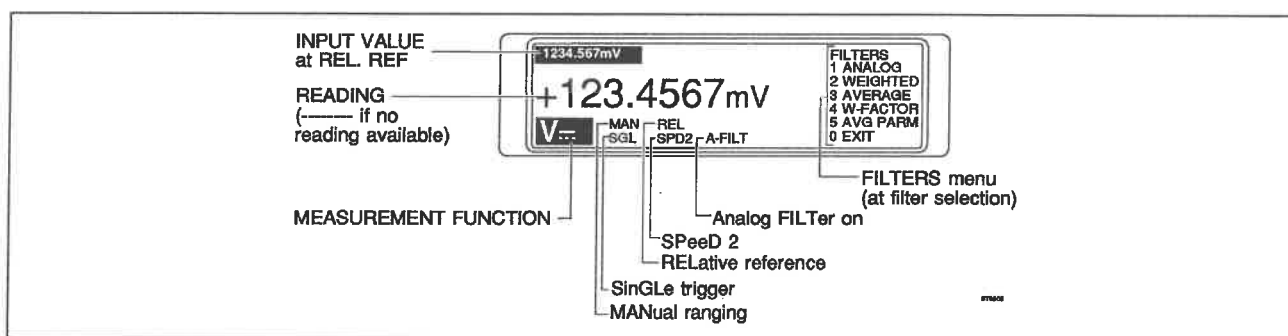


Fig. 4.3 Start-up example display.

OPERATING MENUS

The basic multimeter functions can be directly selected with the front panel push buttons. The other functions (e.g. auxiliary functions), are selected from operating menus. A brief description of these menus is given below. Refer to Section 7.4 for further details.

Press **TEMP** or **FILTER**, and follow the rules given below.

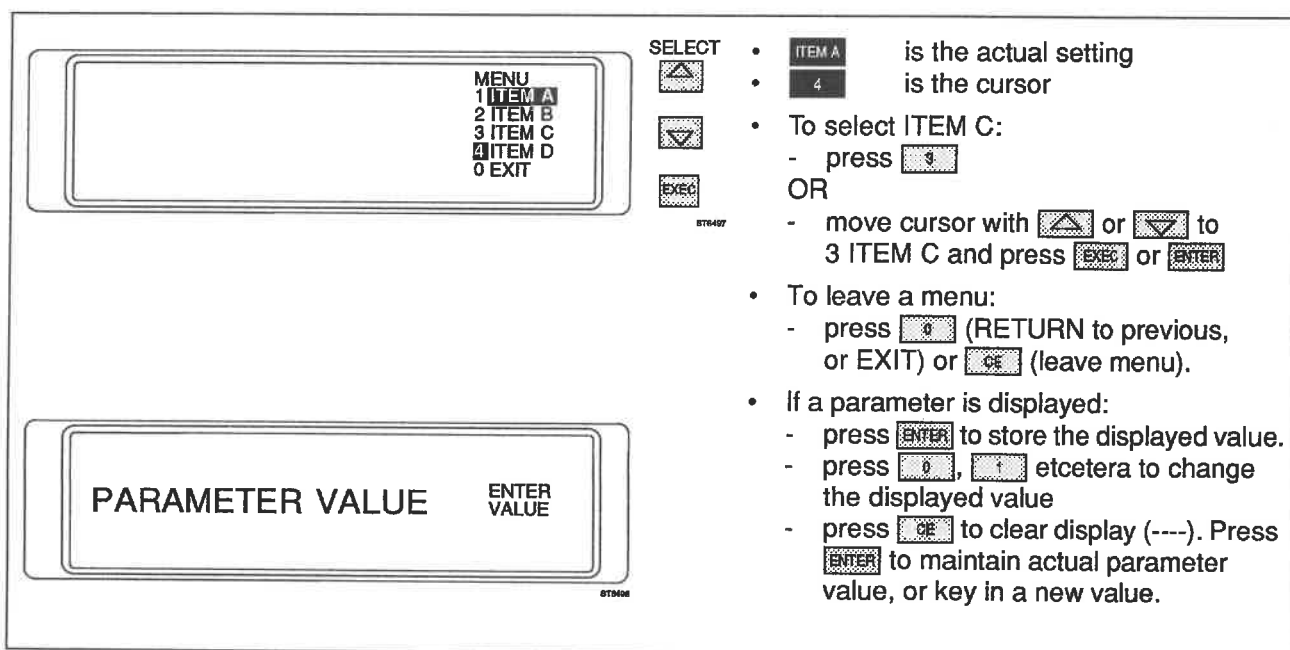


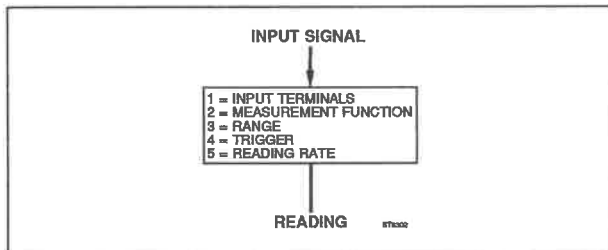
Fig. 4.4 The PM2530 operating menu.

4.2 SETTING UP MEASUREMENTS

This section gives a summary of the steps required to set up the PM2530 for making measurements. Details of each step are given in Sections 4.3 up to 4.21. The steps are described in the order in which they influence the signal being processed.

BASIC MEASUREMENT SET UP

For making simple measurements, you can use the basic instrument functions, as shown in Fig. 4.5. Refer to 1-5 (below).



DEFAULT INSTRUMENT CONFIGURATION

After pressing **RESET** only the basic instrument functions are active, with default settings (autoranging, internal triggering, reading rate SPD2). Refer to section 4.17 for details.

Fig. 4.5 Basic instrument functions.

1. INPUT TERMINALS

Connect the input signal to the appropriate input terminals, depending on the setting of the INPUTS REAR/FRONT switch and the selected measurement function (refer to Section 4.4).

Connect the GUARD, either internally (GUARD INT/EXT switch in the INT position), or externally by a third wire (GUARD INT/EXT switch in the EXT position). Refer to Section 4.3 for detailed information about guarding.

2. MEASUREMENT FUNCTION

Select the required measurement function:



- Volts DC (Section 4.4.2).
- Volts AC or Volts AC+DC (Section 4.4.3).
- Volts low frequency (Section 4.4.4).
- Resistance, 2-wire or 4-wire (Section 4.4.5).



- Current DC (Section 4.4.6).
- Current AC+DC (Section 4.4.7).
- Frequency or period (Section 4.4.8).
- Temperature with Pt100 probe (Section 4.4.9).

3. RANGE

Select the correct range. Refer to Section 4.5.1 for more details.

RANGE



- automatic or manual ranging.
- select a higher range in manual ranging (selects also manual ranging).
- select a lower range in manual ranging (selects also manual ranging).

4. TRIGGER

Select the required trigger mode, then trigger a measurement. Refer to Section 4.5.2 for more details.

TRIGGER



- triggers are internally supplied.
- you must supply a trigger manually by pressing **SINGLE**.
- other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

5. READING RATE

Select the measurement speed mode (reading rate SPD1...6). The measurement resolution depends on the selected speed mode. Refer to Section 4.5.3 for more details.

READING RATE

- | | |
|---|---|
|  | a lower reading rate is selected, resulting in a higher resolution. |
|  | a higher reading rate is selected, resulting in a lower resolution. |

ENHANCED MEASUREMENT SET UP

The PM2530 provides several functions in addition to the basic multimeter functions. Fig. 4.6 gives an overview of these additional functions and shows the order in which they influence the input signal (if selected). You can use this order to configure the instrument. Refer to items 6...12.

6. FILTER (ANALOG and AVERAGING)

Various types of filters are available, depending on the selected measurement function. The ANALOG, and AVERAGING are active at this level (before BURST). Refer to Section 4.6 for further details.

7. BURST MEASUREMENT MODE

The BURST measurement mode performs measurements and stores the most recent results (up to 15000), continuously updated, in the BURST memory. The storage operation may be independently triggered to capture a particular group of successive results. These can occur before, after or 'before and after' the trigger instant. The BURST mode enables high speed measurements (up to 50000 measurements per second in Volts DC). The stored results are only influenced by the input signal conditioning actions in 1-6 above. During a BURST measurement results are displayed when system time is available. If no system time is left (e.g. at high measurement speeds) to display a result, the display shows --- (dashes).

The BURST memory can be read using the OUTPUT RDBURST function. During reading, the results will be processed by all selected functions further on in the flow (items 8 to 12). Refer to Section 4.9 for further details.

8. FILTER (WEIGHTING)

The digital WEIGHTING filter, if selected, is active in the signal flow at this level. BURST measurement results are not processed by the digital weighting filter until they are read. Refer to section 4.6 for further details.

9. CALCULATION FUNCTIONS

You can select one of the calculation functions, dBm (calculation of the dBm value), $(AX+B)/Z$ (scaling) or DELTA% (computing the deviation from a preset value in terms of percentage). Only one of these CALC functions can be active at a time. Refer to Section 4.10 for further details.

10. RELATIVE REFERENCE

When the relative reference function is on, the readings are the difference between the measured value (including previous processing) and the relative reference. Refer to Section 4.7 for further details.

11. STORING MIN/MAX READINGS / COUNTING LIMITS CHECKS

The MIN/MAX and/or LIMITS function can be active at the same time.

When the MIN/MAX function is active, the lowest and highest result is stored in the MIN MEMORY and MAX MEMORY, respectively. When a new MIN or MAX result occurs, it overwrites the current value in the memory and is signalled, if selected, on the display or the beeper. The MIN MEMORY and MAX MEMORY can be read using the output function RD MIN/MAX. Refer to Section 4.11 for further details.

When the LIMITS function is active, an upper and lower limit can be set. If a result is outside these limits, the limits counter is incremented and it can be signalled on the display, the beeper and at the rear limits connector. The count is stored in the limit counter memory, which can be read using the output function LIM COUNT. Refer to Section 4.12 for more details.

12. SET DISPLAY AND BEEPER FEATURES

This function gives you some control of the display features (e.g. number of digits, annunciators, backlighting and contrast) and some of the beeper features (e.g. on/off, beeper active conditions). Refer to Sections 4.13 (DISPLAY) and 4.14 (BEEPER) for more detailed information.

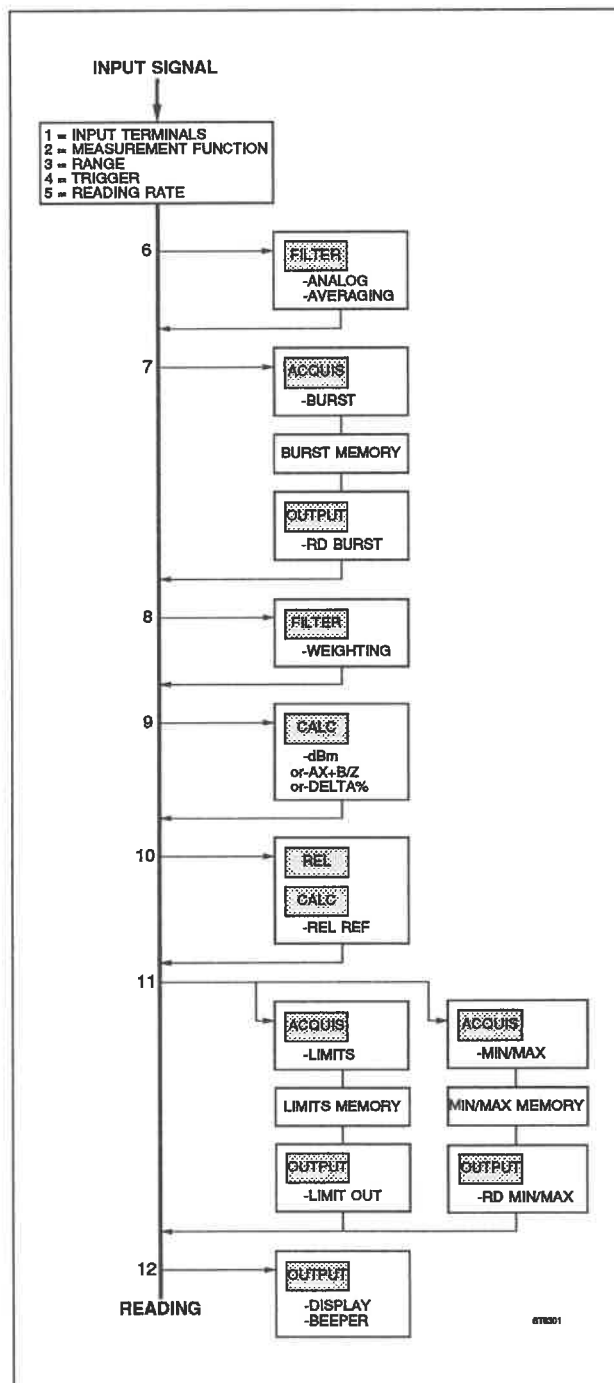


Fig. 4.6 Measurement set up flow diagram.

4.3 GUARDING

The guard is a shield around the electrically floating (not connected to earth) circuitry of the multimeter. The purpose of the guard is to prevent measurement errors due to common mode currents in the multimeter input measurement circuit. The guard is always connected to the front or rear GUARD terminal (whichever is selected).

If the GUARD INT/EXT switch is set to the INT position, the guard is connected the Lo (supply zero) of the PM2530 input circuit. This Lo is directly connected to the COM terminal for Volts, Current and Frequency/Period measurements. For Resistance and Temperature measurements, this Lo is not directly connected to any of the input terminals!

If the GUARD INT/EXT switch is set to the EXT position, the guard is floating, and can be externally connected. This can be done via an additional wire, or via the cable shield.

LEAVING THE GUARD TERMINAL DISCONNECTED WHEN THE GUARD SWITCH IS IN THE EXT POSITION CAN CAUSE A READING ERROR.

WARNING: If the GUARD switch is in the INT position, the voltage at the COM input will also arise at the GUARD terminal, and wiring connected to this terminal!

GUARD USAGE IN VOLTAGE, CURRENT AND FREQUENCY/PERIOD MEASUREMENTS

In the following situations you must select EXTERNAL GUARD access, and connect the GUARD terminal to the Lo of the signal source (Fig. 4.7), to minimize measurement errors:

- when Common Mode voltage exists.
- when making accurate measurements.
- when making sensitive measurements.
- when using long signal leads.

In other situations, the guard can be connected internally (GUARD switch in the INT position, Fig. 4.8).

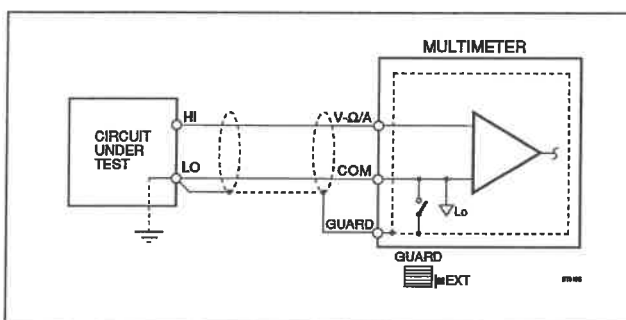


Fig. 4.7 External guarding at Volts, Current, and Freq/Per measurements.

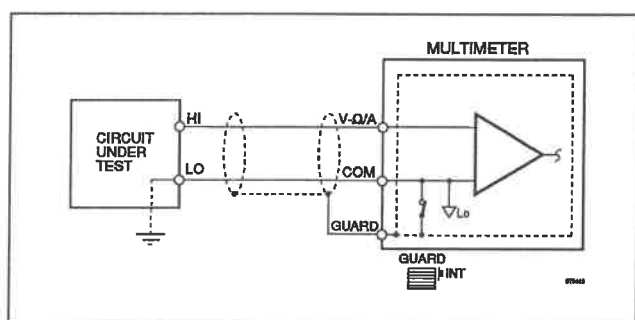
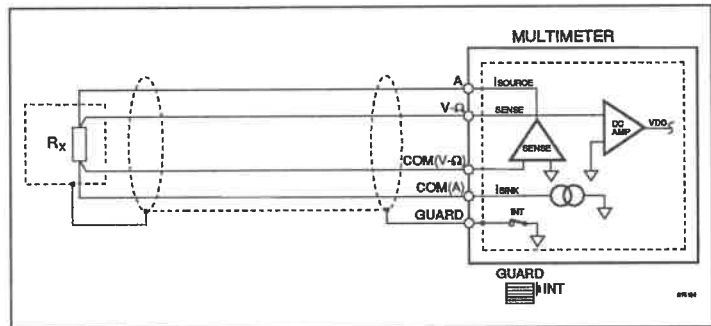


Fig 4.8 Internal guarding at Volts, Current, and Freq/Per measurements.

GUARD USAGE IN RESISTANCE AND TEMPERATURE MEASUREMENTS

Select INTERNAL guarding. Connect the cable shield to the GUARD terminal. See Fig. 4.9. Do not connect the GUARD to the COM terminals.

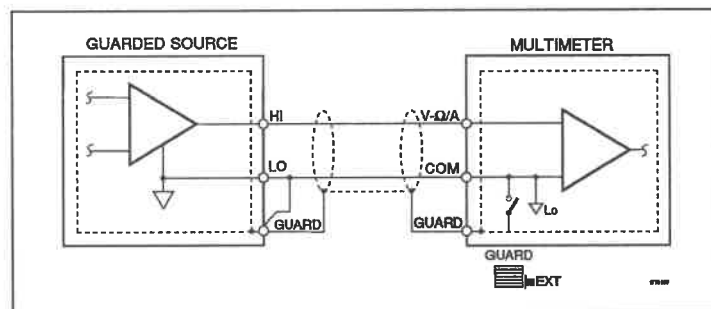
Fig. 4.9 GUARD at resistance measurements.



GUARD USAGE WHEN USING A GUARDED SOURCE

Fig. 4.10 shows guarding when using a guarded source. Connect only one LO to the GUARD terminal. For most applications, it should be placed at the SOURCE end (PM2530 guard switch in the EXT position).

Fig. 4.10 Guarding when using a guarded source.



THEORY (Fig. 4.11)

Unguarded multimeter

The interference voltage V_{cm} (e.g. a ground potential difference) causes a current to flow via the test leads, and impedances Z_1 and Z_2 . As there is a difference between the resistance of both test leads, and a difference between Z_1 and Z_2 (Z_1 is much lower than Z_2), the current will cause a voltage between the HI and LO of the multimeter input amplifier.

Guarded multimeter

The guard is connected to the interference voltage V_{cm} . No current will flow via Z_1 or Z_2 as the guard is at V_{cm} potential and also the HI and LO of the multimeter are at V_{cm} potential. A common mode current I_{cm} will flow via the guard wire. Notice that connecting the guard to the multimeter COM input, will cause the current to flow via the test lead! This will cause a measurement error, specially at sensitive measurements.

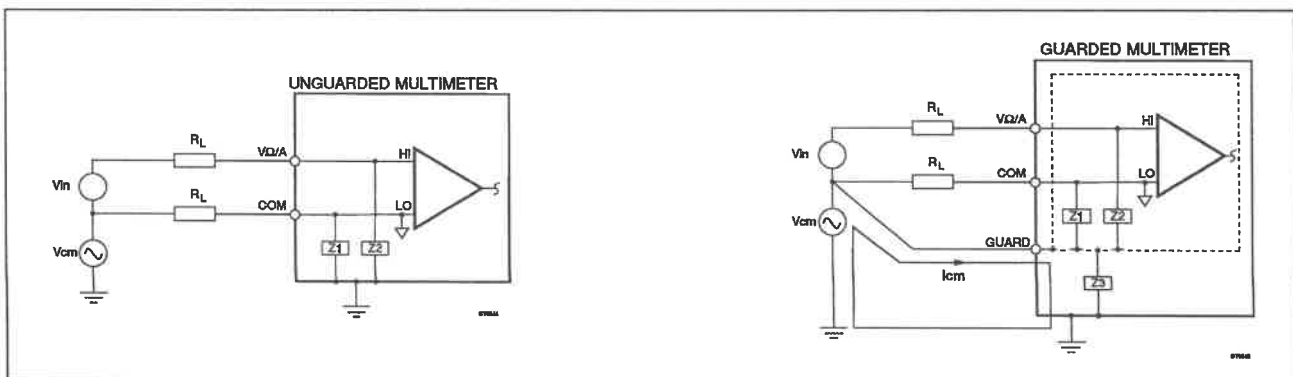


Fig 4.11 Guarding theory.

4.4 MEASUREMENT FUNCTIONS

4.4.1 SELECTING A MEASUREMENT FUNCTION

Press , , , , , , , or to select a measurement function.

When selecting a new measurement function, the settings of the old measurement function are stored as long as the power is on. If the old function is selected again, its stored settings are recalled. Settings stored per measurement function are:

- AUTO or MAN ranging; the range in MAN ranging.
- Reading rate (SPD..).
- Filters ON/OFF, filter parameters.
- Calculation functions ON/OFF (dBm, SCALE, DELTA%), including parameters.
- Relative reference function ON/OFF, including reference value.
- MIN/MAX function ON/OFF state.
- LIMITS function ON/OFF state, including set limits.

For details refer to Appendix A, Section A.2.15. For settings after power up refer to Section 4.17.

Sections 4.4.2 to 4.4.9, inclusive, provide detailed information on the various measurement functions. Refer to Appendix A for the specifications.

4.4.2 VOLTS DC MEASUREMENTS.

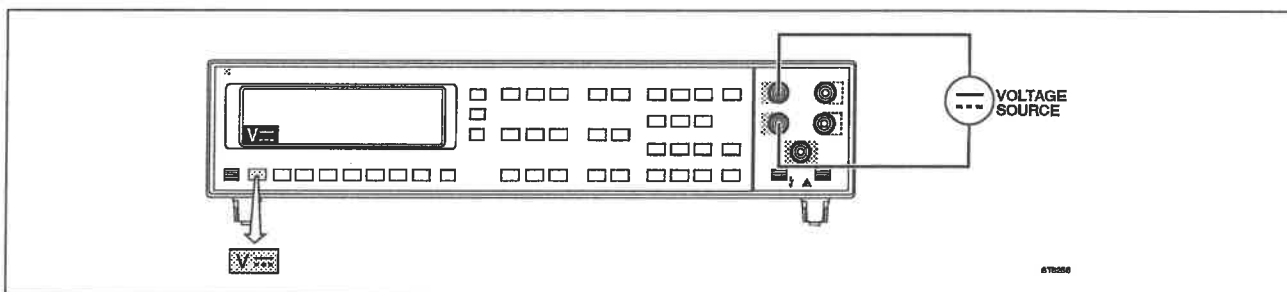


Fig. 4.12 Volts DC measurements.

MEASUREMENT FUNCTION

Press to select the Volts DC measurement function.

INPUT CONNECTIONS

Set the INPUTS REAR/FRONT switch for the required input terminals and connect the test leads to these terminals, as shown in Fig. 4.12. Configure the PM2530, as indicated below, before connecting the input signal. The maximum input voltage is 1000 V.

Shielded test leads are advised for accuracy. These should not run parallel to any heavy current cables. For low level DC Voltage measurements, thermal voltages can be an additional source of measurement errors. Thermal voltages, which can easily exceed 1µV, typically arise where wires are attached to binding posts. The following techniques help to reduce the effects of thermal voltages:

- use similar metals for connections wherever possible (copper-to-copper; input terminals are made of low thermal EMF copper).
- use tight and clean connections (especially free of grease, dirt and oxidation; avoid hand contact).
- use the input offset compensation function (NULLING), see below.
- avoid temperature differences between the input terminals.

Advised test leads: FLUKE 5440A-7002 low thermal copper EMF plug-in cable.

GUARDING

Select EXTERNAL guarding (GUARD INT/EXT switch in EXT position) for sensitive and accurate measurements. Connect the GUARD terminal to the LO of the input signal source. If you select INTERNAL guarding (GUARD INT/EXT switch in the INT position), the GUARD terminal is connected internally to the COM (V Ω) input terminal. Refer to Section 4.3 for the optimal GUARD connections.
NEVER LEAVE THE GUARD TERMINAL DISCONNECTED WHEN SWITCHED TO EXTERNAL GUARDING!

DEFAULT SET-UP

Press **RESET** to set the PM2530 to a simple configuration (autoranging, INTERNAL triggering, reading rate SPD2, all FILTERS and auxiliary functions off). Refer to section 4.17 for further details.

RANGING

Select AUTO or MAN ranging. For ranging details see Section 4.5.1. Provided ranges are 20 mV, 200 mV, 2 V, 20 V, 200 V and 1000 V.

NOTE 1: No automatic down ranging is provided to the 20 mV range, or to the 200 mV range at SPD1.

NOTE 2: the 20 mV range is not provided in speed modes SPD5 and SPD6.

TRIGGERING

INT triggers are internally supplied.
SINGLE you must supply a trigger manually by pressing **SINGLE**.
ACQUIS other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

For further details refer to Section 4.5.2.

INPUT OFFSET COMPENSATION (NULLING)

You can compensate for input offset voltages (e.g. thermal e.m.f.'s) during low level measurements with the NULLING function. This can be done as follows:

short the input with the test leads, press **ACQUIS**, **2** (NULLING menu), **1** (NULLING ON), **3** (to NULL the current range) or **4** (to NULL all ranges), **SINGLE** (to establish null), **0** (to RETURN), **0** (to EXIT).

Refer to Section 4.8. for a more detailed description.

READING RATE - RESOLUTION

Press READING RATE **UP** **DOWN**, to select the required READING RATE, indicated by SPD.. on the display. SPD1 indicates the lowest and SPD6 the highest reading rate. A higher reading rate results in a lower resolution (reduced scale length), see Table 4.1. Select a high reading rate if measurement speed is critical and a low reading rate for high resolution. SPD1, SPD2 and SPD3 provide increased line related noise suppression. Refer to Section 4.5.3 for further details.

NOTE: The measurement time for SPD1 is approximately 5 seconds. However, a provisional result is displayed after 0.5 seconds, when SPD1 is selected or on a step input change.

Table 4.1. Scale length, resolution, and reading rate for Volts DC

SPEED MODE	RANGE	FULL SCALE	RESOLUTION	READINGS PER SECOND ¹⁾	
				50 Hz	60 Hz
SPD1	20 mV 200 mV 2 V - 1000 V	200000 2000000 20000000	0.1 μ V 0.1 μ V 0.1 μ V - 0.1 mV	0.2	0.24
SPD2	20 mV 200 mV - 1000 V	200000 2000000	0.1 μ V 0.1 μ V - 0.1 mV	2	2.4
SPD3	20 mV 200 mV - 1000 V	20000 200000	1 μ V 1 μ V - 1 mV	20	24
SPD4	20 mV 200 mV - 1000 V	2000 20000	10 μ V 10 μ V - 10 mV	200	
SPD5	200 mV - 1000 V	20000	10 μ V - 10 mV	1000 5000 ²⁾	
SPD6	200 mV - 1000 V	2000	100 μ V - 100 mV	1000 50000 ²⁾	

¹⁾ In internal trigger mode with default trigger interval.

²⁾ In burst mode only.

RELATIVE REFERENCE MEASUREMENTS

Press **REL** to select or cancel the relative reference function. If it is not already in use (no REL in the display), the present reading is taken as new the reference value and the function becomes active. If the function is active, it is cancelled, when you press **REL**. For further details refer to Section 4.7.

FILTERS

You can select the following filters:

- ANALOG FILTER (low pass, 5 Hz, 20 dB/decade), to reduce the influence of external noise.
- WEIGHTING FILTER (exponential smoothing filter), to reduce large variations in the reading, due to unstable input signals.
- AVERAGING FILTER (e.g. takes 10 samples and shows the average value).

For selection and further details refer to Section 4.6.

INPUT OVERLOAD

If the input signal is out of range, the display shows OL. If the input circuit clips (e.g. due to input spikes), the display shows $\uparrow$ to indicate that the reading is invalid.

4.4.3 VOLTS AC MEASUREMENTS

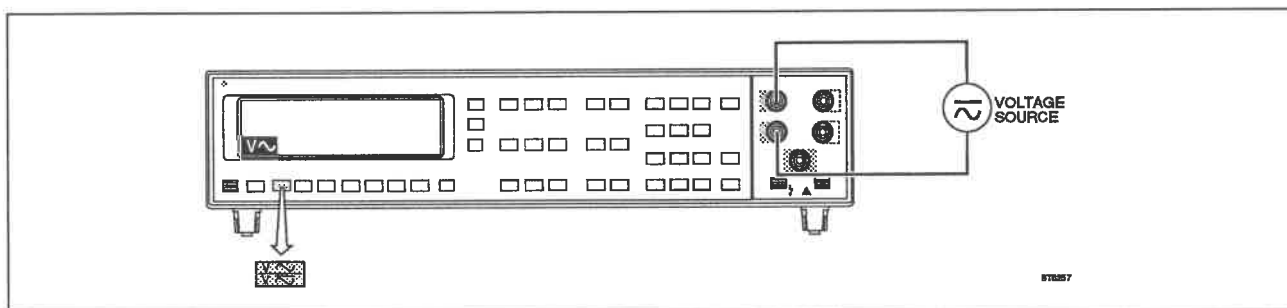


Fig. 4.13 Volts AC measurements.

MEASUREMENT FUNCTION

Press  to toggle between the Volts AC or Volts AC + DC measurement function. In measurement function $V\sim$, the RMS value of the AC component is measured (AC coupled input). In $V\sim$, the RMS value of the AC and DC component is measured (DC coupled input).

INPUT CONNECTIONS

Connect the test leads to the REAR or FRONT input terminals, as selected with the INPUTS REAR/FRONT switch, as shown in Fig. 4.13. It is recommended that shielded test leads are used. These should not run parallel to any heavy current cables. Configure the PM2530 as indicated below, before connecting the input signal. The maximum input voltage is 700 V RMS, 1000 V peak.

GUARDING

Select EXTERNAL guarding (GUARD INT/EXT switch in EXT position) for sensitive and accurate measurements. Connect the GUARD terminal to the LO of the input signal source. If you select INTERNAL guarding (GUARD INT/EXT switch in the INT position), the GUARD terminal is connected internally to the COM ($V\Omega$) input terminal. Refer to Section 4.3 for optimal GUARD connections.

NEVER LEAVE THE GUARD TERMINAL DISCONNECTED WHEN USING EXTERNAL GUARDING!

DEFAULT SET-UP

Press  to set the PM2530 to a simple configuration: autoranging, INTERNAL triggering, reading rate SPD2, RMS-filter >40 Hz (normal response), WEIGHTING and AVERAGING FILTER off, auxiliary functions off. Refer to Section 4.17 for further details.

RANGING

Select AUTO or MAN ranging. For ranging details see Section 4.5.1. Available ranges are 20 mV, 200 mV, 2 V, 20 V, 200 V, and 700 V.

NOTE 1: No automatic down ranging is provided to the 20 mV range at SPD1.

NOTE 2: The 20 mV range is not provided in speed modes SPD5 and SPD6.

TRIGGERING

- INT** triggers are internally supplied.
SINGLE you must supply a trigger manually by pressing **SINGLE**.
ACQUIS other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

For further details refer to Section 4.5.2.

FILTERS

You can select the:

- RMS FILTER frequency, for the optimal response depending on the input signal frequency. The filter setting (>10Hz, >40 Hz, >400Hz) indicates the lowest frequency that can be measured with accurately and stable readings. E.g. for a 60 Hz input signal you should select >40Hz; if you select >400Hz, you will get a faster response, but a larger measurement error (unstable reading).
- WEIGHTING FILTER (exponential smoothing filter), to reduce variations in the reading, due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.

READING RATE - RESOLUTION

Press READING RATE **UP** **DOWN**, to select the required READING RATE, indicated by SPD.. on the display. SPD1 indicates the lowest and SPD3 the highest available reading rates. A higher reading rate results in a lower resolution (reduced scale length), see Table 4.2. Select a high reading rate if measurement speed is critical and a low reading rate if a high resolution is required. Refer to Section 4.5.3 for further details.

NOTE: The measurement time for SPD1 is approximately 5 seconds. However, a provisional result is displayed after 0.5 seconds, when SPD1 is selected or on a step input change.

Table 4.2. Scale length, resolution, and reading rate for Volts AC.

SPEED MODE	READINGS PER SECOND ¹⁾	RANGE	FULL SCALE	RESOLUTION
SPD1	0.2	20 mV 200 mV - 700 V	200000 2000000	0.1 μ V 0.1 μ V - 1 mV
SPD2	2	20 mV - 700 V	200000	0.1 μ V - 10 mV
SPD3	20	20 mV - 700 V	20000	1 μ V - 10 mV

¹⁾ In internal trigger mode with default trigger interval.

RELATIVE REFERENCE MEASUREMENTS

Press **REL** to select or cancel the relative reference function. If this function is not currently active (no REL in the display), the present reading is taken as new the reference value and the function becomes active. If it is active, it is cancelled when you press **REL**. For further details refer to Section 4.7.

CREST FACTOR

In either the Volts AC or Volts AC+DC measurement functions, the PM2530 can handle an input peak voltage (positive or negative) of **4 x RANGE** maximum. If an input voltage is larger than this value, the input circuit clamps the signal. This causes an erroneous measurement result. In this situation the crest factor limit of the instrument is exceeded. This is indicated on the display, by "↑" next to the (erroneous) reading, and in the remote control status subsystem questionable condition register bit 10 refer to section 5.3.14.

The crest factor limit of an instrument is the maximum ratio between the positive or negative peak value of the input signal and its rms value for a correct measurement result. The crest factor limit increases as the measurement result goes down scale, according to

crest factor limit = $4 * \frac{\text{full scale}}{\text{reading}}$, with a maximum of 10.

At full scale the PM2530 crest factor limit is 4, at mid-scale it is 8, etc.

4.4.4 LOW FREQUENCY VOLTAGE MEASUREMENTS

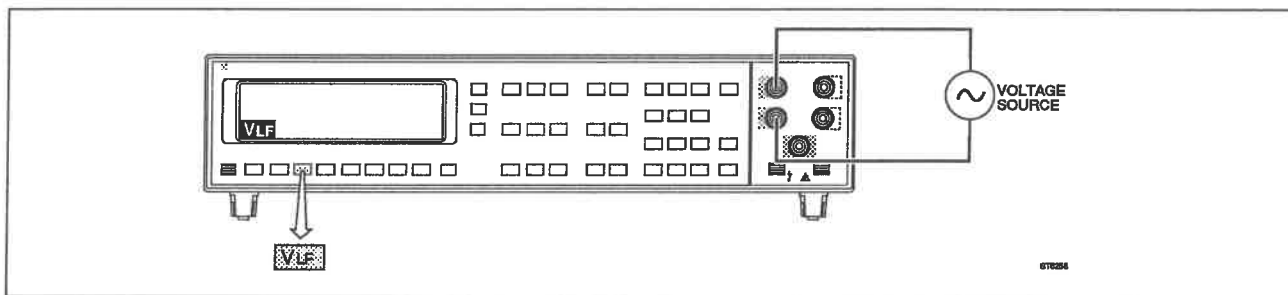


Fig. 4.14 Low frequency voltage measurements.

MEASUREMENT FUNCTION

Press **VLF** to select the Low Frequency Voltage measurement function. This measures the RMS value of the (DC coupled) input signal. Signals over a frequency range 2 mHz to 500 Hz (5 Hz with the analog filter on) can be measured. The input signal frequency is shown in the upper left display part. de pag attention to the relatively long measurement time. See below.

INPUT CONNECTIONS

Connect the test leads to the REAR or FRONT input terminals, as selected with the INPUTS REAR/FRONT switch, as shown in Fig. 4.14. It is recommended that shielded test leads are used. These should not run parallel to any heavy current cables. Configure the PM2530 as indicated below, before connecting the input signal. The maximum input voltage is 1000 V continuously and 1400 V peak (<10 ms).

GUARDING

Select EXTERNAL guarding (GUARD INT/EXT switch in EXT position) for sensitive and accurate measurements. Connect the GUARD terminal to the LO of the input signal source. If you select INTERNAL guarding (GUARD INT/EXT switch in the INT position), the GUARD terminal is connected internally to the COM (V Ω) input terminal. Refer to Section 4.3 for optimal GUARD connections. NEVER LEAVE THE GUARD TERMINAL DISCONNECTED WHEN USING EXTERNAL GUARDING!

DEFAULT SET-UP

Press **RESET** if you want to set the PM2530 to a simple configuration (autoranging, INTERNAL triggering, READING RATE SPD2, all FILTERS and auxiliary functions off). Refer to Section 4.17 for further details.

RANGING

Select AUTO or MAN ranging. For ranging details see Section 4.5.1. Available ranges are 200 mV, 2 V, 20 V, 200 V, and 1000 V.

NOTE: The range applies to the peak value of the input signal and not to the RMS value displayed.

TRIGGERING

INT triggers are internally supplied.

SINGLE you must supply a trigger manually by pressing **SINGLE**.

ACQUIS other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

For further details refer to Section 4.5.2.

READING RATE - RESOLUTION

The reading rate depends on the frequency of the input signal, see table 4.3. The scale length is 20000 counts for all speeds, which results in a minimum resolution of 10 μ V (200 mV range).

Table 4.3 Maximum measurement time for the various speed modes.

SPEED MODE	MEASUREMENT TIME (s)	FREQUENCY RANGE
SPD1	2000	1 mHz - 500 Hz
SPD2	200	10 mHz - 500 Hz
SPD3	20	100 mHz - 500 Hz
SPD4	2	1 Hz - 500 Hz
SPD5	0.2	10 Hz - 500 Hz

A time of approx. 4s, used for calculations, has to be added to the measurement time for all speeds.

The selected speed mode indicates the lowest frequency that can be measured. For example: when SPD3 is selected, frequencies lower than 100 mHz can not be measured.

Speed selection algorithm

The measurement algorithm selects a speed mode that depends on the signal frequency. When the V_{LF} function is selected, the first measurement is done in SPD5. When no valid measurement can be done, a new measurement will be done in SPD4, and so on, until a valid measurement is done, with the restriction that the algorithm does not downrange to a speed mode lower than the selected one.

Example 1:

The signal frequency is 300 mHz, and SPD2 is selected. After selecting V_{LF} the first valid measurement is done in SPD3, so the instrument stays in SPD3, although SPD2 is selected (and 300 mHz also can be measurement in this speed mode).

Example 2:

The signal frequency is 300 mHz, the instrument is measuring in SPD3. When the signal frequency is changed to 7 Hz, the instrument ranges to SPD4, because in this speed mode the first valid measurement is done.

FILTERS

You can select the following filters:

- ANALOG FILTER (low pass, 5 Hz, 20 dB/decade), to reduce the influence of external noise.
- WEIGHTING FILTER (exponential smoothing filter), to reduce variations in the reading, due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.

NOTE: *BURST measurements are not possible in the V_{LF} measurement function.*

4.4.5 RESISTANCE MEASUREMENTS (DIODE MEASUREMENTS)

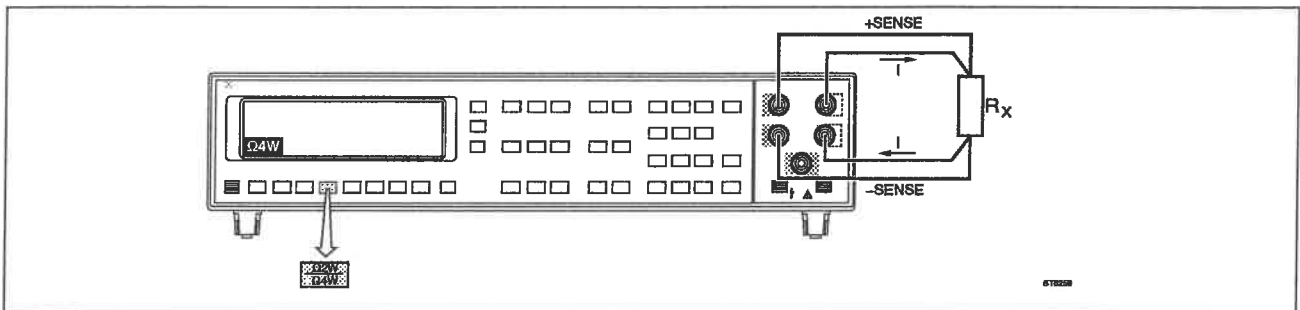


Fig. 4.15 4-wire resistance measurements.

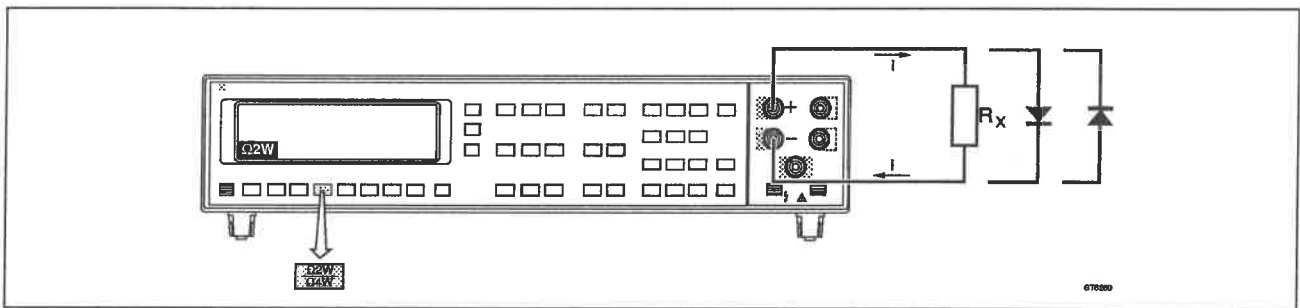


Fig. 4.16 2-wire resistance measurements.

MEASUREMENT FUNCTION

Press **Ω4W/Ω2W** to toggle between the 2-wire and 4-wire resistance measurement functions. Use the 4-wire function to eliminate lead resistance errors, especially for low resistance measurements.

INPUT CONNECTIONS

Connect the test leads to the REAR or FRONT input terminals, as selected with the INPUTS REAR/FRONT switch, as shown in Figs. 4.15 and 4.16. It is recommended that shielded test leads are used at measuring high resistance values. These should not run parallel to any heavy current cables. The FLUKE/PHILIPS PM9286 4-Wire Ohms cable is advised for accurate measurements. If the resistance under test is non-floating (at ground or at some potential), you must connect the non-floating side to the upper (VΩ, A) input terminals.

GUARDING

Select INTERNAL guarding (GUARD INT/EXT switch in INT position), and connect the GUARD terminal to the cable shield (if present). Refer to Section 4.3 for optimal GUARD connections.

DEFAULT SET-UP

Press **RESET** if you want to set the PM2530 to a simple configuration (autoranging, INTERNAL triggering, READING RATE SPD2, all FILTERS and auxiliary functions off). Refer to Section 4.17 for further details.

RANGING

Select AUTO or MAN ranging. For ranging further details see Section 4.5.1. Available ranges and currents are listed below.

RANGE	20Ω	200Ω	2 kΩ	20 kΩ	200 kΩ	2 MΩ	20 MΩ	200 MΩ
CURRENTS	10 mA	1 mA	1 mA	100 μA	10 μA	1 μA	100nA	10 nA

NOTE 1: In the 200 MΩ range the scale length is reduced by two digits.

NOTE 2: The 20Ω range is not available in SPD5, the 20 MΩ and 200 MΩ range not in SPD4 and SPD5.

NOTE 3: No automatic down ranging to the 20Ω range and the 200Ω range at SPD1.

TRIGGERING.

INT triggers are internally supplied.

SINGLE you must supply a trigger manually by pressing **SINGLE**.

ACQUIS other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

For further details refer to Section 4.5.2.

READING RATE - RESOLUTION

Press READING RATE **UP** **DOWN**, to select the required READING RATE, indicated by SPD.. in the display. SPD1 indicates the lowest and SPD5 the highest reading rate. A higher reading rate results in a lower resolution (reduced scale length), see Table 4.4. Select a high reading rate if measurement speed is critical, and a low reading rate if a high resolution if required. Refer to Section 4.5.3 for further details.

NOTE: The measurement time for SPD1 is approximately 5 seconds. However, a provisional result is displayed after 0.5 seconds, when SPD1 is selected or on a step input change.

Table 4.4. Scale length, resolution, and reading rate for Resistance.

SPEED MODE	READINGS PER SECOND ¹⁾		RANGE	FULL SCALE	RESOLUTION
	50 Hz	60 Hz			
SPD1	0.2	0.24	20 Ω 200 Ω - 2 MΩ 20 MΩ 200 MΩ	2000000 20000000 2000000 200000	10 μΩ 10 μΩ - 0.1 Ω 10 Ω 1 kΩ
SPD2	2	2.4	20 Ω - 2 MΩ 20 MΩ 200 MΩ	2000000 200000 20000	10 μΩ - 1 Ω 100 Ω 10 kΩ
SPD3	20	24	20 Ω - 2 MΩ 20 MΩ - 200 MΩ	200000 20000	100 μΩ - 100 Ω 1 kΩ - 10 kΩ
SPD4	200	200	20 Ω - 200 kΩ 2 MΩ	20000 2000	1 mΩ - 10 Ω 1 kΩ
SPD5	1000	1000	200 Ω - 200 kΩ 2 MΩ	20000 2000	10 mΩ - 10 Ω 1 kΩ

¹⁾ In internal trigger mode with default trigger interval.

INPUT OFFSET COMPENSATION (TRUE OHM)

In 4-wire resistance measurements you can compensate for input offset voltages (e.g. thermal e.m.f.) during low level measurements using the NULLING function, as follows:

- Connect the test leads to the resistor under test.
- press **ACQUIS**, **2** (NULLING menu), **1** (NULLING ON), **3** (to NULL the current range) or **4** (to NULL all ranges), **SINGLE** (to establish null), **0** (to RETURN), **0** (to EXIT).

At nulling the PM2530 measures the input voltage with no current applied. Refer to Section 4.8. for a more detailed description.

RELATIVE REFERENCE MEASUREMENTS

Press **REL** to select or cancel the relative reference function. If it is not already in use (no REL in the display), the present reading is taken as new the reference value and the function becomes active. If it was in use, it is cancelled when you press **REL**. For further details refer to Section 4.7.

FILTERS

You can select the following filters:

- ANALOG FILTER (low pass, 5 Hz, 20 dB/decade), to reduce the influence of external noise.
- WEIGHTING FILTER (exponential smoothing filter), e.g. to reduce variations in the reading, due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.

DIODE MEASUREMENTS

You can use the 2-wire resistance measurement function to test diodes. Select the 2 k Ω range, or a higher range. The 2 k Ω range is advised because its 1 mA test current provides a typical operating point, and its 2V full-scale voltage is sufficient to turn on most diodes. If a diode is good, when forward biased it will measure about 0.6 k Ω to 0.7 k Ω for silicon (0.25 k Ω to 0.3 k Ω for germanium), and when reverse biased it will cause the PM2530 to indicate overload (OL).

4.4.6 CURRENT DC MEASUREMENTS

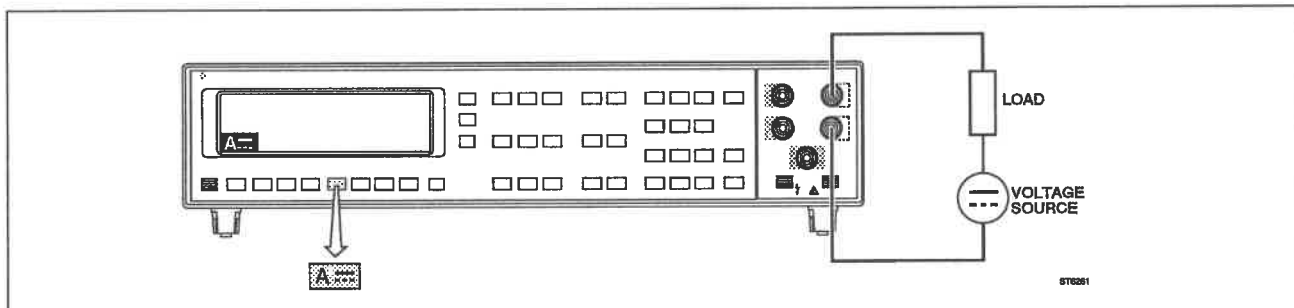


Fig. 4.17 Current DC measurements.

MEASUREMENT FUNCTION

Press  to select the Current DC measurement function.

INPUT CONNECTIONS

Connect the test leads to the REAR or FRONT input terminals, as selected with the INPUTS REAR/FRONT switch, as shown in Fig. 4.17. Configure the PM2530 as indicated below, before connecting the input signal. The input is protected by a 2.5 A-F (Fast Blow) fuse. Note that a blown fuse interrupts the measured current! For replacement see Section 6.4. Preferably use shielded test leads, which should not run parallel to heavy current cables.

GUARDING

Select EXTERNAL guarding (GUARD INT/EXT switch in EXT position) for sensitive and accurate measurements. Connect the GUARD terminal to the LO of the input signal source. If you select INTERNAL guarding (GUARD INT/EXT switch in the INT position), the GUARD terminal is connected internally to the COM (A) input terminal. Refer to Section 4.3 for optimal GUARD connections.

NEVER LEAVE THE GUARD TERMINAL DISCONNECTED WHEN USING EXTERNAL GUARDING!

DEFAULT SET-UP

Press  if you want to set the PM2530 to a simple configuration (autoranging, INTERNAL triggering, READING RATE SPD2, all FILTERS and auxiliary functions off). For further details refer to Section 4.17.

RANGING

Select AUTO or MAN ranging. For ranging details see Section 4.5.1. Available ranges are 2 μ A, 20 μ A, 200 μ A, 2 mA, 20 mA, 200 mA, 2 A.

NOTE 1: No automatic down ranging is provided to the 2 μ A range at SPD1.

NOTE 2: The scale length is reduced by 1 digit in the 2 μ A range at SPD1.

TRIGGERING

 triggers are internally supplied.

 you must supply a trigger manually by pressing .

 other trigger inputs can be selected via the TIMING menu (invoked after pressing .

For further details refer to Section 4.5.2.

INPUT OFFSET COMPENSATION (NULLING)

You can compensate for input offsets (e.g. input leakage current) during low level measurements using the NULLING function, as follows:

open the input, press **ACQUIS**, **2** (NULLING menu), **1** (NULLING ON), **3** (to NULL the current range) or **4** (to NULL all ranges), **SINGLE** (to establish null), **0** (to RETURN), **0** (to EXIT).

Refer to Section 4.8. for a more detailed description.

READING RATE - RESOLUTION

Press READING RATE **UP** **DOWN**, to select the required READING RATE, indicated by SPD.. in the display. SPD1 indicates the lowest and SPD5 the highest reading rate. A higher reading rate results in a lower resolution, see Table 4.5. Select a high reading rate if measurement speed is critical, and a low reading rate if a high resolution is required. Refer to Section 4.5.3 for further details.

NOTE: The measurement time for SPD1 is approximately 5 seconds. However, a provisional result is displayed after 0.5 seconds, when SPD1 is selected or on a step input change.

Table 4.5. Scale length, resolution, and reading rate for Current DC.

SPEED MODE	READINGS PER SECOND ¹⁾		RANGE	FULL SCALE	RESOLUTION
	50 Hz	60 Hz			
1	0.2	0.24	2 μ A 20 μ A - 2 A	2000000 20000000	1 pA 1 pA - 100 nA
2	2	2.4	2 μ A - 2 A	2000000	1 pA - 1 μ A
3	20	24	2 μ A - 2 A	200000	10 pA - 10 μ A
4	200	200	2 μ A - 2 A	20000	100 pA - 100 μ A
5	1000	100	2 μ A - 2A	20000	100 pA - 100 μ A

¹⁾ In internal trigger mode with default trigger interval.

RELATIVE REFERENCE MEASUREMENTS

Press **REL** to select or cancel the relative reference function. If the function is not active (no REL in the display), the present reading is taken as the new reference value and the function becomes active. If it was in use, it is cancelled when you press **REL**. For further details refer to Section 4.7.

FILTERS

You can select the following filters:

- WEIGHTING FILTER (exponential smoothing filter), e.g. to reduce variations in the reading, due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.2.

4.4.7 CURRENT AC MEASUREMENTS

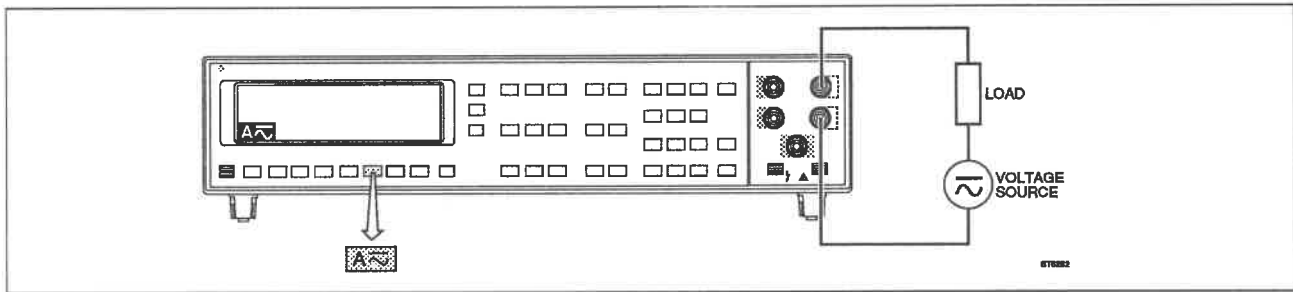


Fig. 4.18 Current AC measurements.

MEASUREMENT FUNCTION

Press to select the Current AC measurement function. In this function, the RMS value of the AC + DC component of the current is measured (DC coupled input).

INPUT CONNECTIONS

Connect the test leads to the REAR or FRONT input terminals, as selected with the INPUTS REAR/FRONT switch, as shown in Fig. 4.18. Configure the PM2530 as indicated below, before connecting the input signal. The input is protected by a 2.5 A-F (Fast Blow) fuse. Note that a blown fuse interrupts the measured current! For replacement see Section 6.4. Preferably use shielded test leads, which do not run parallel to heavy current cables.

GUARDING

Select EXTERNAL guarding (GUARD INT/EXT switch in EXT position) for sensitive and accurate measurements. Connect the GUARD terminal to the LO of the input signal source. If you select INTERNAL guarding (GUARD INT/EXT switch in the INT position), the GUARD terminal is connected internally to the COM (A) input terminal. Refer to Section 4.3 for optimal GUARD connections.

NEVER LEAVE THE GUARD TERMINAL DISCONNECTED WHEN USING EXTERNAL GUARDING!

DEFAULT SET-UP

Press if you want to set the PM2530 to a simple configuration: autoranging, INTERNAL triggering, READING RATE SPD2, RMS-filter >40 Hz (normal response), WEIGHTING and AVERAGING FILTER off, and auxiliary functions off. Refer to section 4.17 for further details.

RANGING

Select AUTO or MAN ranging. For ranging details see Section 4.5.1. Available ranges are 200 μ A, 2 mA, 20 mA, 200 mA, 2 A.

TRIGGERING

triggers are internally supplied.

you must supply triggers manually by pressing .

other trigger inputs can be selected via the TIMING menu (invoked after pressing .

For further details refer to Section 4.5.2

FILTERS

You can select the:

- RMS FILTER frequency, for an optimal response depending on the input signal frequency. The filter setting (>10Hz, >40 Hz, >400Hz) indicates the lowest frequency that can be measured with maximum accuracy and reading stability. e.g. for a 60 Hz input signal you should select >40Hz; if you select >400Hz, you will have a faster response, but a larger measurement error (instable reading).
- WEIGHTING FILTER (exponential smoothing filter), e.g. to reduce variations in the reading, due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.

READING RATE - RESOLUTION

Press READING RATE  , to select the required READING RATE, indicated by SPD.. in the display. SPD1 indicates the lowest and SPD3 the highest reading rate. A higher reading rate results in a lower resolution (reduced scale length), see Table 4.6. Select a high reading rate if measurement speed is critical, and a low reading rate if a high resolution is required. Refer to Section 4.5.3 for further details.

NOTE: The measurement time for SPD1 is approximately 5 seconds. However, a provisional result is displayed after 0.5 seconds, when SPD1 is selected or on a step input change.

Table 4.6. Scale length, resolution, and reading rate for Current AC + DC.

SPEED MODE	READINGS PER SECOND ¹⁾	RANGE	FULL SCALE	RESOLUTION
SPD1	0.2	200 μ A - 2 A	2000000	100 pA - 1 μ A
SPD2	2	200 μ A - 2 A	200000	1 nA - 10 μ A
SPD3	20	200 μ A - 2 A	20000	10 nA - 100 μ A

¹⁾ In internal trigger mode with default trigger interval.

RELATIVE REFERENCE MEASUREMENTS

Press  to select or cancel the relative reference function. If the function is not active (no REL in the display), the present reading is taken as new reference value and the function becomes active. If it is in use, it is cancelled when you press . For further details refer to Section 4.7.

CREST FACTOR

For AC current measurements, the PM2530 can handle an input peak value (positive or negative) of **3.3 x RANGE** maximum. If a signal outside this range is connected to the instrument, the input circuit clamps the signal. This causes an erroneous measurement result. In this situation the crest factor limit of the instrument is exceeded. This is indicated on the display, by "4" next to the (erroneous) reading, and in the remote control Status Subsystem Questionable Condition register bit 10 refer to section 5.3.14.

The crest factor limit of an instrument is the maximum ratio between the positive or negative peak value of the input signal and its rms value for a correct measurement result. The crest factor limit increases as the measurement result goes down scale, according to:

crest factor limit = $3.3 \times \frac{\text{full scale}}{\text{reading}}$, with a maximum of 10.

At full scale, the PM2530 crest factor limit is 3.3, at mid-scale it is 6.6, etc.

4.4.8 FREQUENCY / PERIOD MEASUREMENTS

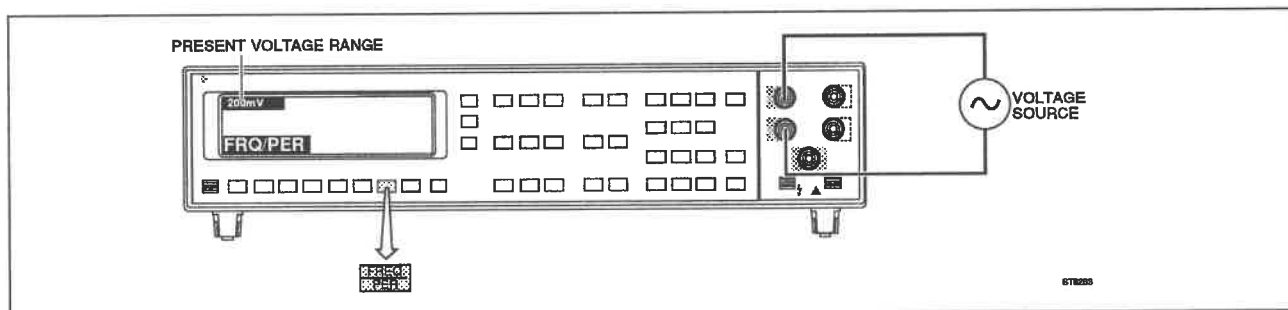


Fig. 4.19 Frequency / Period measurements.

MEASUREMENT FUNCTION

Press to toggle between the frequency and period measurement functions. From the AC coupled input, the frequency and period of the input signal are derived by detecting the zero crossings. The frequency /period range that can be measured is dependent on the reading rate (see READING RATE below). The required minimum input voltage is 5% of the selected voltage range.

Table 4.7 Frequency/ period to be measured.

READING RATE	FREQUENCY RANGE	PERIOD RANGE
SPD1	2 Hz - 2 MHz	500 ms - 1 μ s
SPD2	20 Hz - 2 MHz	50 ms - 1 μ s
SPD3	200 Hz - 2 MHz	5 ms - 1 μ s

If the frequency/period is outside the stated values, or if the input voltage is too low, the display shows dashes (-----).

INPUT CONNECTIONS

Connect the test leads to the REAR or FRONT input terminals, as selected with the INPUTS REAR/FRONT switch, as shown in Fig. 4.19. Preferably use shielded test leads, which do not run parallel to heavy current cables. Configure the PM2530 as indicated below before connecting the input signal. The maximum input voltage is 700 V RMS, 1000 V peak.

GUARDING

Select EXTERNAL guarding (GUARD INT/EXT switch in EXT position) for sensitive and accurate measurements. Connect the GUARD terminal to the LO of the input signal source. If you select INTERNAL guarding (GUARD INT/EXT switch in the INT position), the GUARD terminal is connected internally to the COM ($V\Omega$) input terminal. Refer to Section 4.3 for optimal GUARD connections.

NEVER LEAVE THE GUARD TERMINAL DISCONNECTED WHEN USING EXTERNAL GUARDING!

DEFAULT SET-UP

Press **RESET** if you want to set the PM2530 to a simple configuration (autoranging, INTERNAL triggering, READING RATE SPD2, all FILTERS and auxiliary functions off). Refer to section 4.17 for further details.

RANGING

Frequency and Period ranges are selected automatically. Select AUTO or MAN ranging for the voltage range in which the measurements must be performed. The voltage range is indicated in the upper left part of the display. For ranging details see Section 4.5.1. Available voltage ranges are 200 mV, 2 V, 20 V, 200 V, 700 V.

TRIGGERING

INT triggers are internally supplied.

SINGLE you must supply a trigger manually by pressing **SINGLE**.

ACQUIS other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

For further details refer to Section 4.5.2.

FILTERS

You can use the following filters:

- WEIGHTING FILTER (exponential smoothing filter), to reduce variations in the reading due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.

READING RATE RESOLUTION

Press READING RATE **UP** **DOWN**, to select the required READING RATE, indicated by SPD.. in the display. SPD1 indicates the lowest and SPD3 the highest available reading rate. A higher reading rate results in a lower resolution, see Table 4.8 and 4.9.

Table 4.8 Scale length, resolution, measurement timing for Frequency measurements.

SPEED MODE	READINGS PER SECOND ^{1) 2)}	MEASUREMENT TIME ²⁾	FULL SCALE (DIGITS)	RESOLUTION
SPD1	2 - 1	0.5 - 1 s	7	1 μ Hz - 1 Hz
SPD2	20 - 10	50 - 100 ms	6	100 μ Hz - 10 Hz
SPD3	200 - 100	5 - 10 ms	5	10 mHz - 100 Hz

¹⁾ In internal trigger mode with default trigger interval.

²⁾ The reading rate and measurement time also depend on the input frequency. See below "Measuring principle".

Table 4.9 Scale length, resolution, measurement timing for Period measurements.

SPEED MODE	READINGS PER SECOND ^{1) 2)}	MEASUREMENT TIME ²⁾	FULL SCALE (DIGITS)	RESOLUTION
SPD1	2 - 1	0.5 - 1 s	7	100 ns - 0.1 ps
SPD2	20 - 10	50 - 100 ms	6	100 ns - 1 ps
SPD3	200 - 100	5 - 10 ms	5	100 ns - 10 ps

¹⁾ In internal trigger mode with default trigger interval.

²⁾ The reading rate and measurement time also depend on the input frequency. See below "Measuring principle".

RELATIVE REFERENCE MEASUREMENTS

Press **REL** to select or cancel the relative reference function. If the function is not active (no REL in the display), the present reading is taken as new reference value and the function becomes active. If it was in use, it is cancelled when you press **REL**. For further details refer to Section 4.7.

MEASURING PRINCIPLE (Fig. 4.20)

The AC coupled input rejects any DC level in the input signal. The signal is measured as follows:

1. After a trigger, the signal is monitored during a fixed gate time for a negative to positive zero crossing.
2. The first zero crossing of the signal (in positive direction) during the gate time will start a counter running at a fixed frequency.
3. The first zero crossing in the same direction after the end of the gate time will stop the counter.
4. The counter result is used to calculate the frequency and period of the input signal.

At long period times (low frequencies), the measurement time greatly depends on the input signal period time.

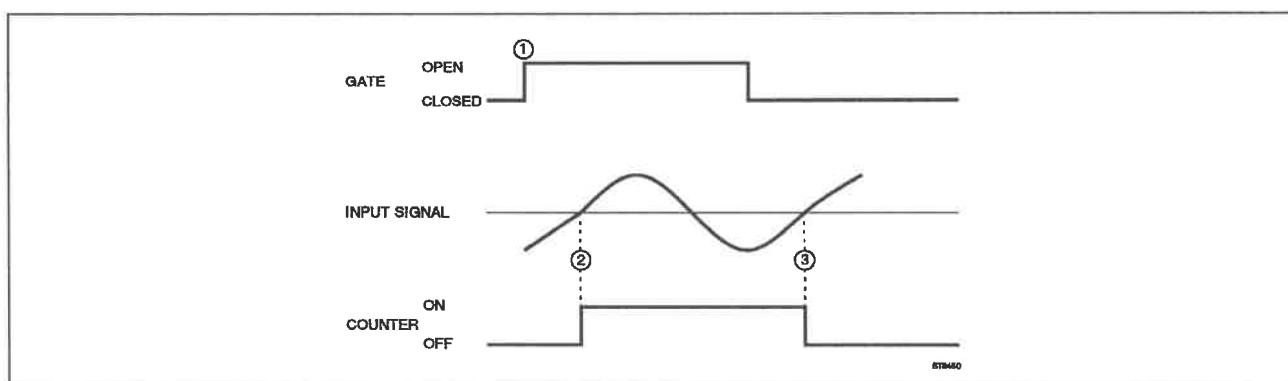


Fig. 4.20 Frequency/Period measurements.

4.4.9 TEMPERATURE MEASUREMENTS

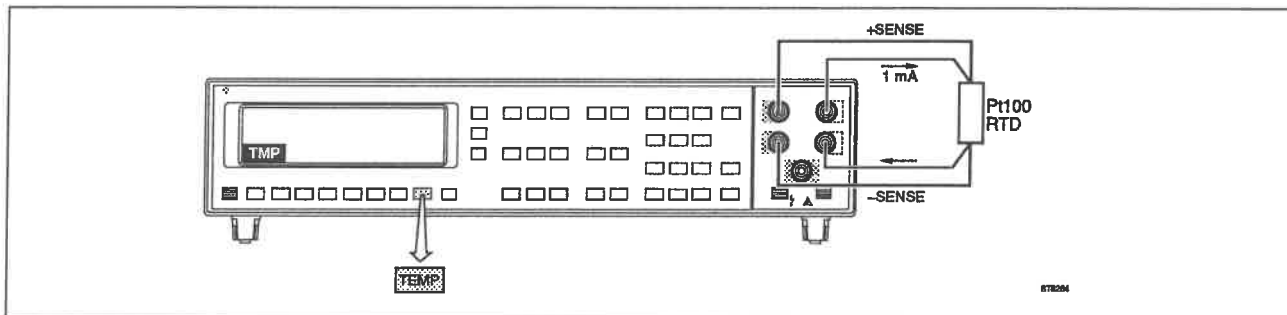


Fig. 4.21 Temperature measurements.

MEASUREMENT FUNCTION

Press **TEMP** to select the temperature measurement function. This function provides temperature measurements with Pt100 RTD's, e.g. the FLUKE/PHILIPS PM9252. Linearization according to three different standards, and the temperature unit can be selected from the displayed menu.

LINEARIZATION AND UNIT SELECTION

Select the required probe type and temperature unit, as follows (see Fig. 4.22):

- Press **1**, **2** or **3** to select Pt100 probe type 385 Pt (DIN43760 table), 390 Pt or 392 Pt respectively.
- Press **4** to invoke the UNITS menu, and press **1**, **2**, or **3** to select °C, °F or K respectively.
- Press **0** or **CE** to return to the TEMP menu or to exit the TEMP menu.

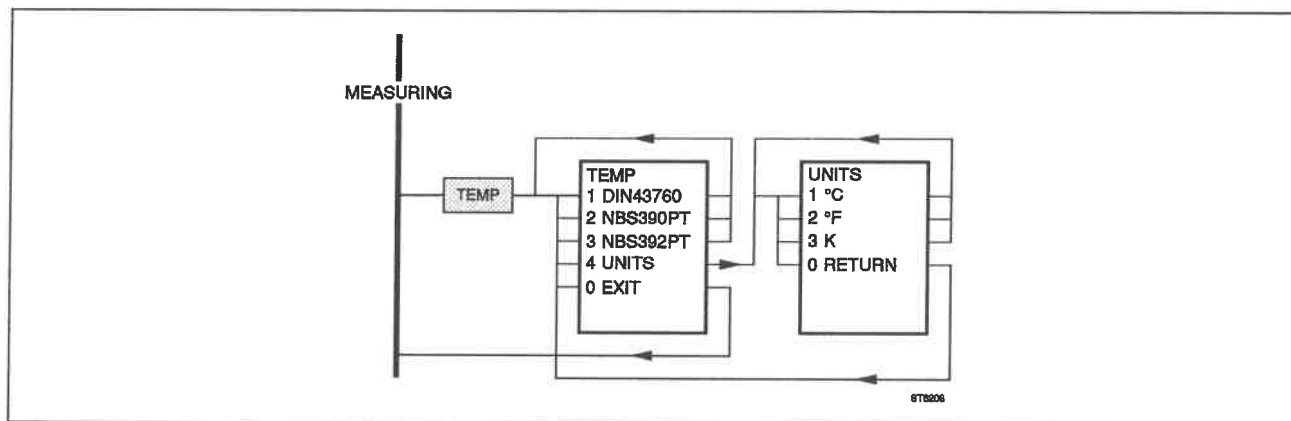


Fig. 4.22 Temperature probe and unit selection.

INPUT CONNECTIONS

Select the REAR or FRONT input terminals with the INPUTS REAR/FRONT switch. Connect the test leads to the input terminals as shown in Fig. 4.21. If the used Pt100 RTD is non-floating (at ground or some potential), you must connect the non-floating side to the upper (A-V Ω) terminals.

GUARDING

Select INTERNAL guarding (GUARD INT/EXT switch in INT position), and connect the GUARD terminal to the cable shield (if present). Refer to Section 4.3 for optimal GUARD connections.

DEFAULT SET-UP

Press **RESET** if you want to set the PM2530 to a simple configuration (autoranging, INTERNAL triggering, READING RATE SPD2, all FILTERS and auxiliary functions off). Refer to section 4.17 for further details.

RANGING

TEMPERATURE measurement uses a single range from -200 °C up to +850 °C (-328 °F up to +1562 °F, 73 K up to 1132 K). The measuring current is 1 mA.

TRIGGERING

INT triggers are internally supplied.
SINGLE you must supply a trigger manually by pressing **SINGLE**.
ACQUIS other trigger inputs can be selected via the TIMING menu (invoked after pressing **ACQUIS**).

For further details refer to Section 4.5.2.

READING RATE RESOLUTION

Press READING RATE **UP** **DOWN**, to select the required READING RATE, indicated by SPD.. on the display. SPD2 indicates the lowest and SPD3 the highest reading rate. A higher reading rate results in a lower resolution (reduced scale length), see Table 4.10. Select SPD3 if measurement speed is critical, and SPD2 if a high resolution is required. Refer to Section 4.5.3 for further details.

Table 4.10 Temperature measurements, measurement timing and resolution.

SPEED MODE	READINGS PER SECOND ¹⁾	MEASUREMENT TIME	FULL SCALE	RESOLUTION
SPD2	2	0.5 s	6 digits	0.01 °C
SPD3	20	50 ms	5 digits	0.1 °C

¹⁾ At internal triggering with default trigger interval.

RELATIVE REFERENCE MEASUREMENTS

Press **REL** to select or cancel the relative reference function. If the function is not active (no REL in the display), the present reading is taken as the new reference value and the function becomes active. If it is active, it is cancelled when you press **REL**.

For further details refer to Section 4.7.

FILTERS

You can select the following filter:

- ANALOG FILTER (low pass, 5 Hz, 20 dB/decade), to reduce the influence of external noise.
- WEIGHTING FILTER (exponential smoothing filter), e.g. to reduce variations in the reading, due to unstable input signals.
- AVERAGING FILTER (takes e.g. 10 samples and shows the average value).

For selection and details refer to Section 4.6.

4.5 MEASUREMENT CONTROL FUNCTIONS

4.5.1 RANGING (, ,)

- Press  to toggle between autorange and manual ranging
- Press  or  to select a higher or lower range. Pressing  or  automatically selects manual ranging if the instrument is in autorange.
- The decimal point and unit annunciators indicate what range the instrument is in. In manual ranging the display shows MAN. Autoranging is not indicated.

AUTORANGE

In autorange, the PM2530 automatically selects the next higher range when a reading is greater than full scale. If no higher range is available, "OL" (overload) is displayed. The meter automatically selects a lower range when a reading is less than approximately 9% of full scale. Notice that autoranging influences the response time.

The PM2530 does not autorange:

- in Volts DC down to the 20 mV range.
- in Volts DC down to the 200 mV range in speed mode SPD1.
- in Volts AC down to the 20 mV range in speed mode SPD1.
- in Current DC down to the 2 μ A range in speed mode SPD1.
- in Ω 2W/ Ω 4W down to the 20 Ω range in speed mode SPD1

These ranges can only be selected manually.

MANUAL RANGING

In manual ranging, the PM2530 remains in the selected range regardless of the size of the input signal, until you select another range or switch to autoranging.

NOTE: In the measurement function FREQ/PER (frequency/period), the RANGE buttons ,  and  determine the voltage range in which the measurement is made. The frequency/period range is selected automatically. Refer to Section 4.4.8 for further details.

4.5.2 TRIGGERING

A trigger signal causes the PM2530 to sample the input, and to calculate and to display the result. When a measurement is in progress, the trigger annunciator (Δ) blinks. The trigger annunciator inverts each time when a measurement value is displayed.

A reading complete signal is provided via the rear panel RDY1 connector, see "READING COMPLETE OUTPUT RDY1" below.

If no measurement result is available (e.g. the PM2530 is waiting for a trigger), the display will show dashes (- - - - -).

Two trigger modes are provided, the INTERNAL and the SINGLE trigger mode.

SELECTING THE INTERNAL TRIGGER MODE

- Press **INT**
or
- Select 1. INTERNAL from the TIMING menu, see "SELECTING TRIGGER SOURCES" below.

In the internal trigger mode readings are triggered by a continuous stream of (equidistant) pulses, from an internal source. The interval between each trigger pulse can be set. When the default interval is selected, a trigger delay can be inserted. Refer to "SETTING THE TRIGGER INTERVAL" below.

NOTE: pressing **RESET** selects the INTERNAL trigger mode, DEFAULT trigger interval, and trigger DELAY OFF.

SELECTING THE SINGLE TRIGGER MODE

In the single trigger mode each trigger must be supplied by the user, via a selectable trigger source. A trigger delay can be inserted. A measurement is started after supplying a trigger, when the delay is expired. See "SETTING THE TRIGGER DELAY" below.

- press **SINGLE** to select MANUAL triggering. A trigger is supplied by pressing **SINGLE** again. The display shows SGL.
- MANUAL triggering can also be selected via the **ACQUIS** TIMING menu, see "SELECTING TRIGGER SOURCES".

NOTE: If your application requires accurate timing it is recommended to use the auto correction ONCE option of the AUTO COR menu to prevent auto corrections to occur while measuring. See AUTO CORRECTIONS section 4.8.

SELECTING TRIGGER SOURCES

- press **ACQUIS** to invoke the ACQUIS menu (Fig. 4.23).

- select

1. TIMING

to select the TIMING menu.

3. SMPL SRC

to select the (sample) trigger sources:

1. INTERNAL

to select the internal (timed) trigger mode.

2. EXT1

to trigger measurements via the rear panel EXT1 connector, see "EXTERNAL TRIGGER INPUTS EXT1 AND EXT2" below.

3. EXT2

to trigger measurements via the rear panel EXT2 connector, see "EXTERNAL TRIGGER INPUTS EXT1 AND EXT2" below.

4. MANUAL

to trigger a measurement manually. A measurement is started after pressing **SINGLE**, when the set trigger delay is expired.

5. INTERFACE

to trigger measurements via the GPIB interface, see Command Reference Manual TRIGGER SUBSYSTEM.

0. RETURN

to return to the TIMING menu.

0. RETURN

to return to the ACQUIS menu.

0. EXIT

to exit the ACQUIS menu.

Press **CE** to exit the ACQUIS menu at from any level.

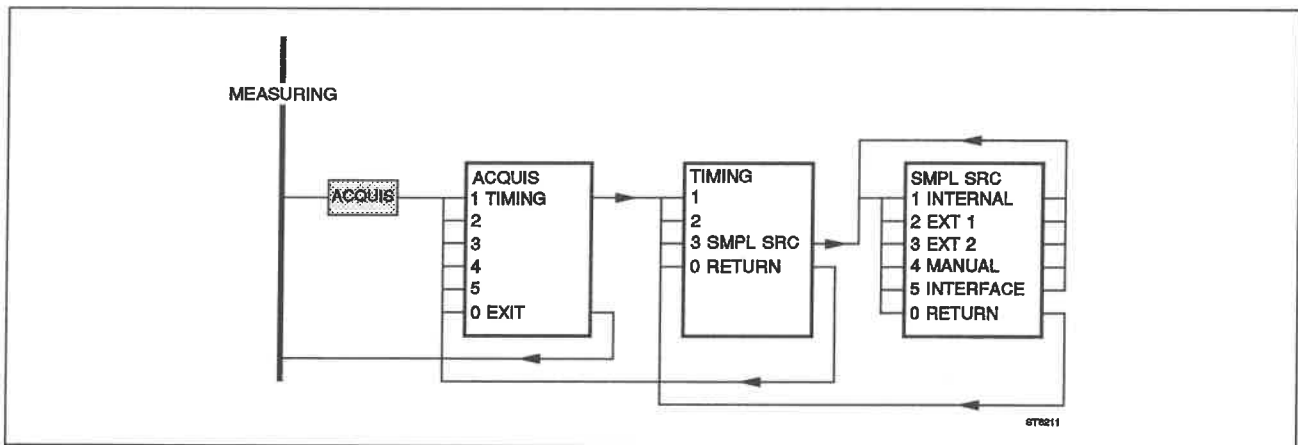


Fig. 4.23 Trigger input selection menu.

NOTE: If none of the SAMPLE INPUT menu options is indicated, the IMMEDIATE trigger mode is selected.

EXTERNAL TRIGGER INPUTS EXT1 AND EXT2

If you select one of the, rear panel, trigger inputs EXT1 or EXT2, a measurement starts after the falling edge of a pulse on the respective connector, when the trigger delay is expired. Since EXT1 and the EXT2 inputs are pulled high internally, they can also be controlled by a normally open switch. A measurement starts when the switch is closed.

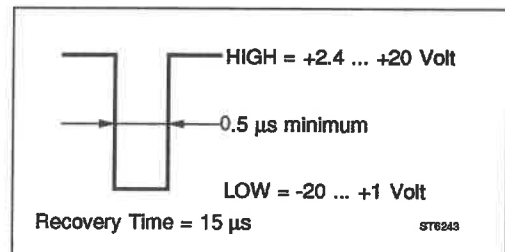


Fig. 4.24 Trigger pulse for the EXT1 or EXT2 trigger input.

SETTING THE TRIGGER INTERVAL FOR INTERNAL (TIMED) TRIGGERING

- press **ACQUIS** to invoke the ACQUIS menu (Fig. 4.25).

- select

1. TIMING

1. INTERVAL

1. USER VAL

2. DEFAULT

0. RETURN

0. RETURN

0. EXIT

to select the TIMING menu.

to set the trigger interval time. If it is not default, the display shows INTERVAL. to enter an arbitrary interval time up to 24 hours, with a minimum resolution of 1 μ s (0-1 s). The minimum value depends on the required measurement time (See Fig. 4.26). Entering a value below the required measurement time sets the interval to the measurement time, and INT >> is displayed. The trigger delay is always discarded.

to select the default (is minimum) measurement interval time, for maximum reading rate. The interval time is equal to the measurement time. The trigger delay can be inserted. See "SETTING THE TRIGGER DELAY" below. If a delay is required, it must be set when the internal trigger mode has already been selected. Entering the internal trigger mode when DEFAULT is active will automatically set the delay OFF. If a delay is inserted, the trigger interval is the sum of the measurement time and the delay.

to return to the INTERVAL menu.

to return to the TIMING menu.

to exit the ACQUIS menu.

Press **CE** to exit the menus from any level.

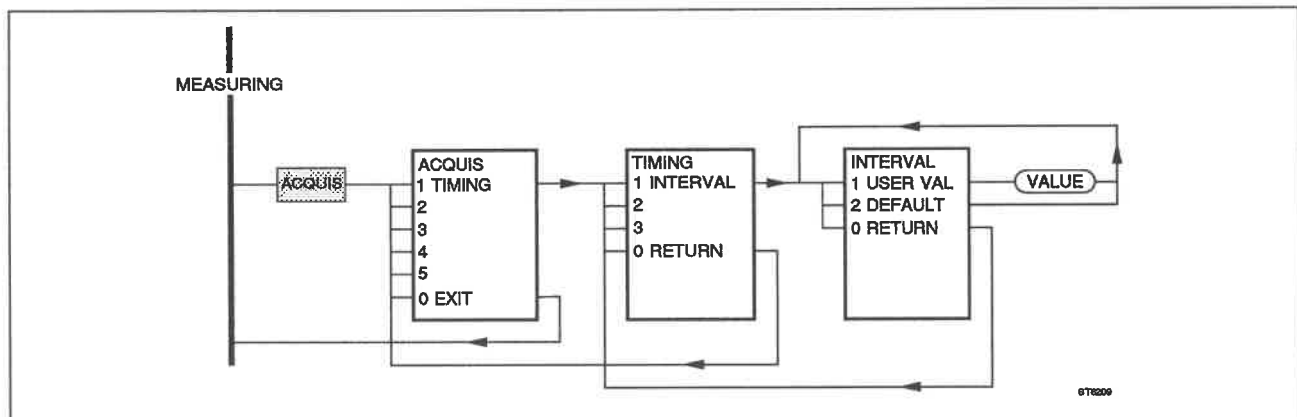


Fig. 4.25 Trigger interval menu.

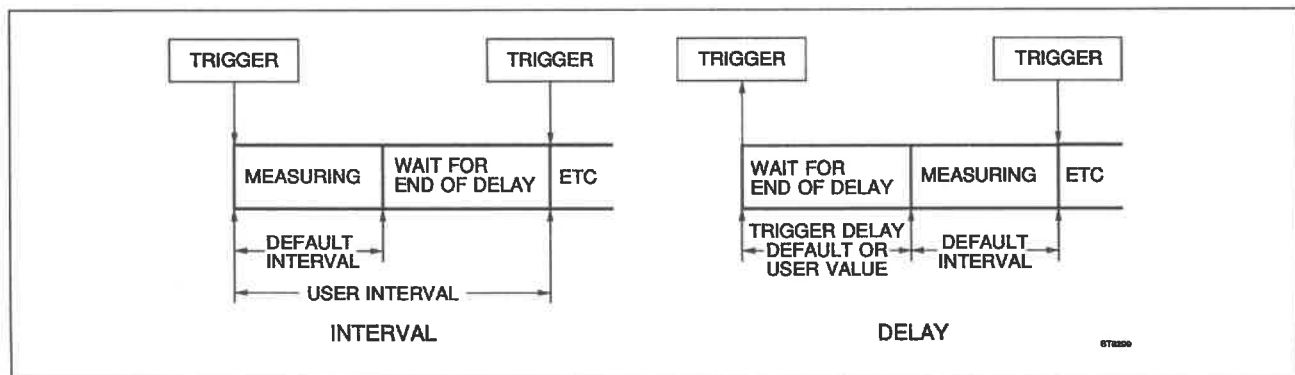


Fig. 4.26 Trigger interval and delay.

SETTING THE TRIGGER DELAY

- press **ACQUIS** to invoke the ACQUIS menu (Fig. 4.27).
- select

1. TIMING

2. DELAY

1. USER VAL

2. DEFAULT

3. OFF

0. RETURN

0. RETURN

0. EXIT

to select the TIMING menu.

to set the trigger delay (see "TRIGGER DELAY" below). If it is not default, the display shows DELAY.

to enter a value between 0 and 10 seconds (resolution 1 μ s).

to set the trigger delay to default.

to cancel the trigger delay. This state is automatically selected when selecting the INTERNAL trigger source (SMPL SRC) with DEFAULT trigger INTERVAL.

to return to the TIMING menu.

to return to the ACQUIS menu.

to exit the ACQUIS menu.

Press **CE** to exit the TIMING menus from any level.

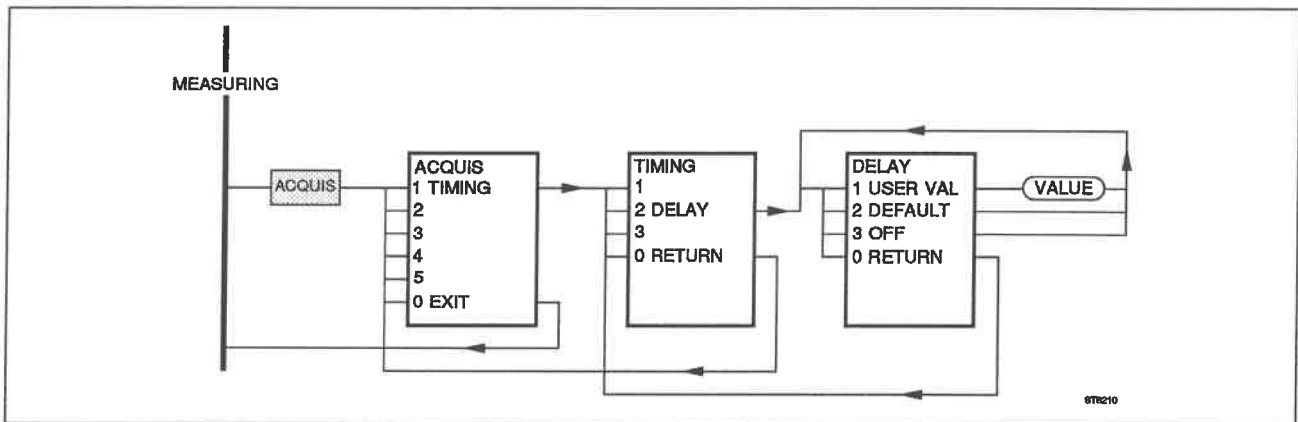


Fig. 4.27 Trigger delay menu.

TRIGGER DELAY (Fig. 4.26)

The trigger delay is the time between a trigger and the start of a measurement. If the default delay is selected, the delay is just long enough so that the reading will be correct (within a specified number of counts of the final value) even if the trigger event occurs as the input makes a step change between zero and full scale. In this situation, the delay covers the PM2530 response time.

You can set the trigger delay below the default value, to accommodate special timing considerations. If you set the delay to zero or OFF, the measurement is started immediately after receiving a trigger. This can cause a reading error when an input step signal arises simultaneously with the trigger signal. See Appendix A "Specifications" for Response times.

READING COMPLETE OUTPUT RDY1

A low pulse (2 μ s, TTL) on the rear panel RDY1 output (see Fig. 4.28) indicates when a reading is complete.

NOTE: in remote control, the RDY1 or RDY2 connector can be assigned as reading complete, input complete or burst complete output. This setting is maintained when returning to local control. Press **RESET** to set RDY1 to "reading complete" again. In this case RDY2 is set to OFF.

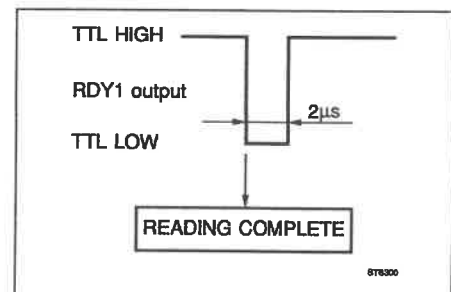


Fig. 4.28 Trigger delay.

4.5.3 READING RATE, RESOLUTION, MEASUREMENT TIME (READING RATE ,)

The reading rate in internal trigger mode is inversely proportional to the trigger interval. The minimum interval depends on the selected measurement speed. Reading rates in the internal trigger mode with default trigger interval are specified in Appendix A. Reading rates in the single trigger mode depend on the trigger rate, refer to Section 4.5.2 for detailed information.

Pressing READING RATE , , selects another measurement speed mode. In each speed mode a different measurement time is established. Measurement times are specified in Appendix A.

- Press  to select a higher speed mode; this results in a shorter measurement time, and a lower resolution.
- Press  to select a lower speed mode; this results in a longer measurement time, and a higher resolution.

Depending on the selected measurement function, the PM2530 has up to 6 speed modes. The actual speed mode is indicated in the display as "SPD1... SPD6". SPD1 is the lowest, SPD6 the highest speed mode.

Select a low speed mode:

- if high resolution is required.
- for maximum line related noise suppression (normal mode rejection), for the functions Volts DC and Current DC. In these measurement functions, speed modes SPD1... SPD3 provide increased normal mode rejection for line frequency related disturbances.

Select a high speed mode if a high measurement speed is required.

NOTE 1: You can select the number of displayed digits from the OUTPUT function "DISPLAY". Note that this function only changes the display length, it does not affect the measurement speed or the measurement resolution.

NOTE 2: In speed mode SPD1, a provisional reading is available after 0.5 seconds:

- when the first measurement in SPD1 is performed.
 - when a step change of the input signal occurs.
- The last digit of the provisional reading is replaced with a dash (-).

Table 4.11 Measurement timing.¹⁾

SPEED MODE	SAMPLING TIME		PER MEASUREMENT		READING RATE		MEASUREMENT TIME	
	50 Hz	60 Hz	50 Hz	60 Hz	50 Hz	60 Hz		
SPD1	100 x 20 ms	100 x 16 2/3 ms	0.2	0.24	5 s	4.17 s		
SPD2	10 x 20 ms	10 x 16 2/3 ms	2	2.4	0.5 s	0.417 s		
SPD3	1 x 20 ms	1 x 16 2/3 ms	20	24	50 ms	41.7 ms		
SPD4	1 x 2 ms		200		5 ms			
SPD5	1 x 200 μs		1000		1 ms			
SPD5 2)	1 x 200 μs		5000		200 μs			
SPD6	1 x 20 μs		1000		1 ms			
SPD6 2)	1 x 20 μs		50000		50 μs			

- 1) Low frequency voltage, Frequency and Period readings are derived from voltage measurements. For timing of these functions refer to Section 4.4.4 (VLF) and 4.4.8 (FREQ/PER).
- 2) In BURST mode.

4.6 FILTERS

4.6.1 INTRODUCTION

If you press **FILTER**, an operating menu shows the filter types available for the selected measurement function, see Fig. 4.29.

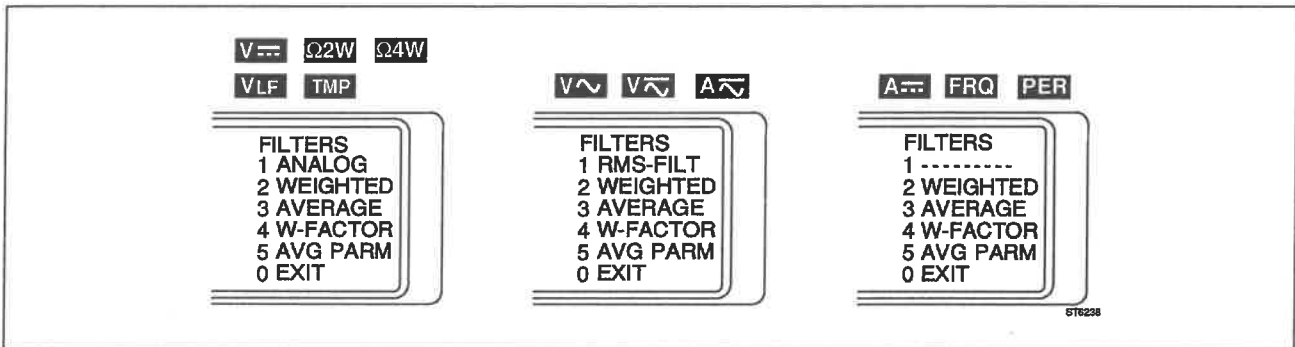


Fig. 4.29 Filter menus.

The filter settings for each measurement function are stored separately. If you alter the filter settings, only those of the selected measurement function are affected.

4.6.2 ANALOG LOW PASS FILTER (**V**, **VLF**, **Ω2W**, **Ω4W**, **TEMP**)

SELECTION OF THE ANALOG LOW PASS FILTER

- press **FILTER** to invoke the filter menu (Fig. 4.30).
- select:
 - 1. ANALOG** to set the analog low pass filter on/off (toggle). If on, the display shows A- FILT.
 - 0. EXIT** to exit the FILTERS menu.

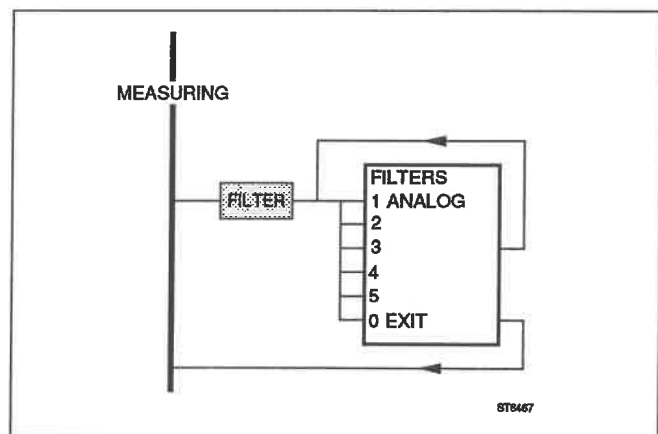


Fig. 4.30 Analog Filter menu.

DESCRIPTION OF THE ANALOG LOW PASS FILTER

The analog filter is located in the input circuit of the PM2530. If selected, it is active for direct as well as BURST measurements. It is a RC filter (roll off is 20 dB/decade) with a 5 Hz corner frequency. Since the filter introduces a delay in response to a change in the input signal, it influences the measurement time/step response. Refer to the Specifications, Appendix A.

Use analog filtering when making measurements in the presence of unwanted environmental noise.

4.6.3 RMS FILTER (,)

SELECTION OF THE RMS FILTER RESPONSE

- press  to invoke the filter menu (Fig. 4.31).
- select:
 - 1. RMS-FILT to set the RMS filter frequency. Select a frequency that is as close as possible below the signal frequency. A lower selected frequency results in a long response time, a higher selected frequency causes an instable reading. The display shows the selected frequency (>10 Hz, >40 Hz, >400 Hz).
 - 1. > 10 Hz to select a 10 Hz RMS filter frequency.
 - 2. > 40 Hz to select a 40 Hz RMS filter frequency.
 - 3. >400 Hz to select a 400 Hz RMS filter frequency.
 - 0. RETURN to return to the FILTER menu.
 - 0. EXIT to exit the FILTERS menu.

Press  to exit the FILTERS menu from any level.

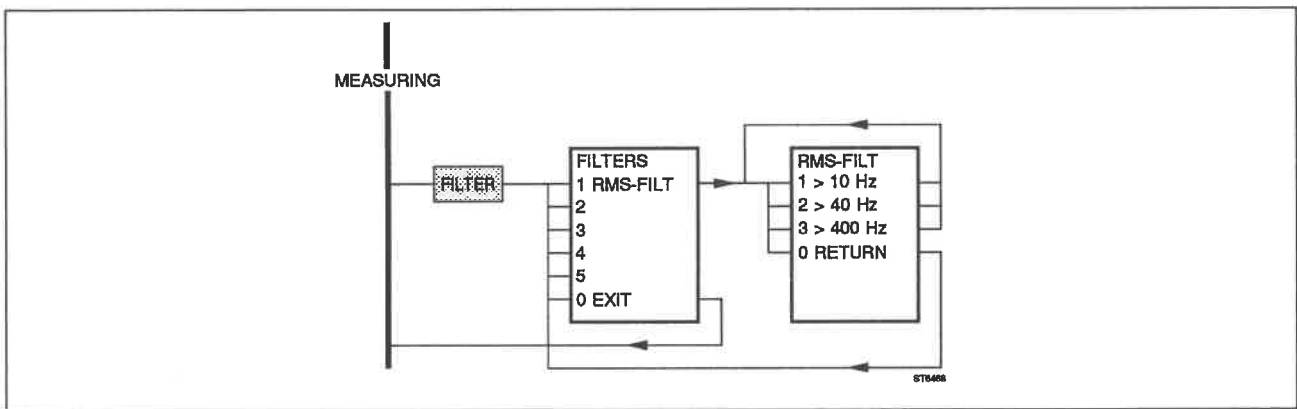


Fig. 4.31 RMS Filter menu.

DESCRIPTION OF THE RMS-FILTER

The RMS filter is the output filter of the PM2530 RMS converter. It rejects the output ripple of the RMS converter to obtain a stable reading. The filter is active for direct as well for BURST measurements. You can set the filter response to slow (>10 Hz), normal(>40 Hz), and fast (>400 Hz). Select the frequency response that is as close as possible below the input signal frequency. This results the optimum balance between ripple rejection and response. Selecting a filter frequency above the input signal frequency will give a fast response, but results in an unstable reading. If the filter frequency is set too far below the input frequency, the response will be slower than necessary. Since the filter introduces a delay in response to a change in the input signal, it influences the measurement time/step response. Refer to the specifications, Appendix A.

Table 4.12 Example for RMS filter setting (sine wave input).

Input frequency	filter setting	response	error	optimal setting
1 kHz	400 Hz	fast	minimal	yes
	10 Hz	slow	minimal	no
50 Hz	40Hz	normal	minimal	yes
	400 Hz	fast	ripple	no
10 Hz	10 Hz	slow	minimal	yes
	400 Hz	fast	ripple	no

4.6.4 WEIGHTING FILTER

SELECTION OF THE WEIGHTING FILTER AND PARAMETERS

- press **FILTER** to invoke the filter menu (Fig. 4.32).
- select:
 - 2. WEIGHTED** to set the WEIGHTED filter on/off (toggle). If on, the display shows WEIGHT. Set the weighting factor, see below 4 W-FACTOR.
 - 4. W-FACTOR** to set the weighting factor of the WEIGHTED filter (1 ... 64).
 - 0. EXIT** to exit the FILTERS menu.

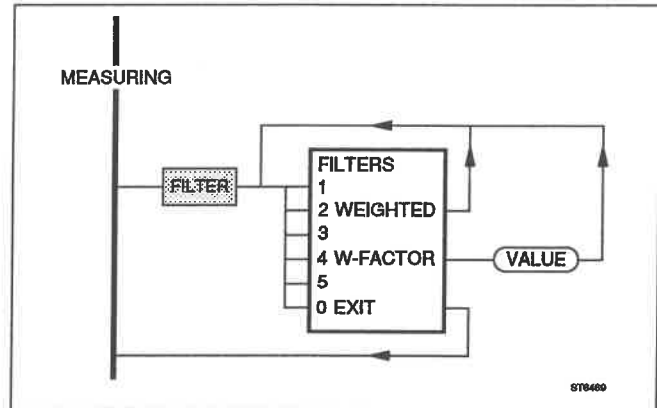


Fig. 4.32 Weighting Filter menu.

DESCRIPTION OF THE WEIGHTING FILTER

The weighting filter is an exponential smoothing filter, with an adjustable weighting factor W . It can be used to reduce variations in the readings due to unstable input signals. The weight factor can be set in the range 1...64. The default factor is 5. Practical W values are from 2 up to 7; variations in the readings will be reduced by a factor 1.5 up to 3 respectively. A higher W factor results in a slower step response.

If the input step signal is more than 0.5 times the range, the filter will be reset, so that the first reading after this step is the actual input value.

In BURST mode, the WEIGHTING filter does NOT process the results until they are read.

THEORY

If the filter is on, each new reading (X_n) is calculated from the previous reading (X_{n-1}) and the actual measured value (X_a), using weight factor W according to the formula:

$$X_n = \frac{W-1}{W} \times X_{n-1} + \frac{1}{W} \times X_a$$

The following example will illustrate the effect of digital filtering on a step input at a weighting factor $W = 2$ (Fig. 4.33):

$$X_n = 0.5 \times X_{n-1} + 0.5 \times X_a$$

The new reading is 0.5 * previous reading + 0.5 * actual measurement result

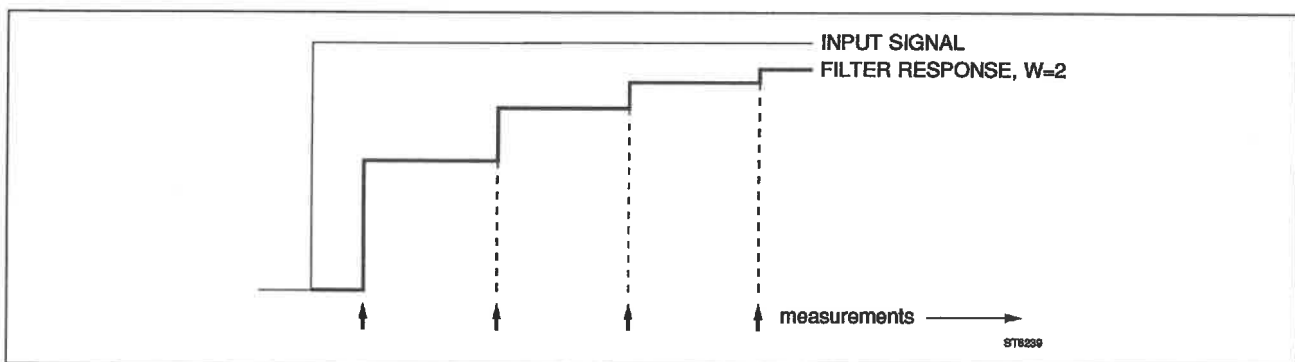


Fig. 4.33 WEIGHTING FILTER response on a step input signal ($W=2$).

4.6.5 AVERAGE FILTER

SELECTION OF THE AVERAGE FILTER AND PARAMETERS

- press **FILTER** to invoke the filter menu (Fig. 4.34).
- select:
 - 3. AVERAGE** to set the AVERAGE filter on/off (toggle). If on, the display shows AVG. Set the AVERAGE filter parameters, see below AVG PARM.
 - 5. AVG PARM** to set the parameters of the AVERAGE filter.
 - 1. #SAMPLES** to set the number of measurements n (1 ... 64) of which the average value must be displayed.
 - 2. EXTREMES** the reading is the average of all n measurement results.
 - 3. SKIP EXTR** the minimum and the maximum measurement result are not included in the average.
 - 0. RETURN** to return to the FILTERS menu.
 - 0. EXIT** to exit the FILTERS menu.

Press **CE** to exit the FILTER menu from any level.

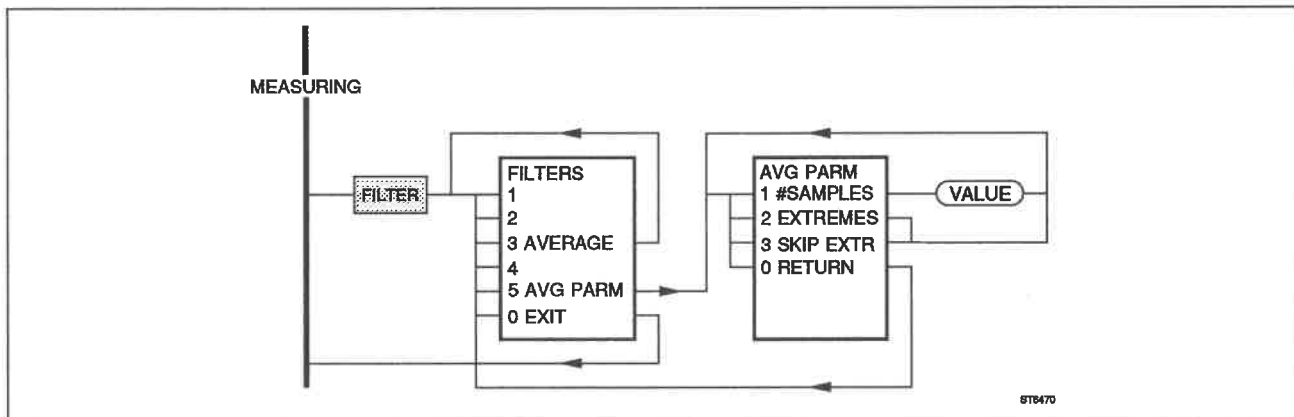


Fig. 4.34 Average Filter menu.

DESCRIPTION OF THE AVERAGE FILTER

If the filter is on, each reading is the average of either:

- the last n measurement results (if EXTREMES included), or
- the last n measurement results, excluding the highest and lowest result (if EXTREMES skipped), in which n is the selected number of samples.
- when EXTREMES are included, the minimum for n=1.
- when EXTREMES are excluded, the minimum for n=3.

If selected, the filter is active for direct as well as BURST measurements.

4.7 RELATIVE REFERENCE MEASUREMENTS

If the relative reference function is in use, readings are the difference between the input values and a reference value. The input value is shown in the upper left display field. For a detailed explanation see below.

WARNING WHEN THE RELATIVE REFERENCE FEATURE IS IN USE, DISPLAYED READINGS ARE RELATIVE AND MAY NOT INDICATE THE PRESENCE OF DANGEROUS POTENTIALS AT THE INPUT CONNECTORS OR TEST LEADS. USE CAUTION TO AVOID ELECTRIC SHOCK OR INSTRUMENT DAMAGE.

You can select the relative reference mode with the **REL** button or via the CALCULATION function.

Fig. 4.35 shows the display at normal measurements (REL REF OFF), at relative measurements (REL REF ON), and during setting the reference value via the CALCULATION function.

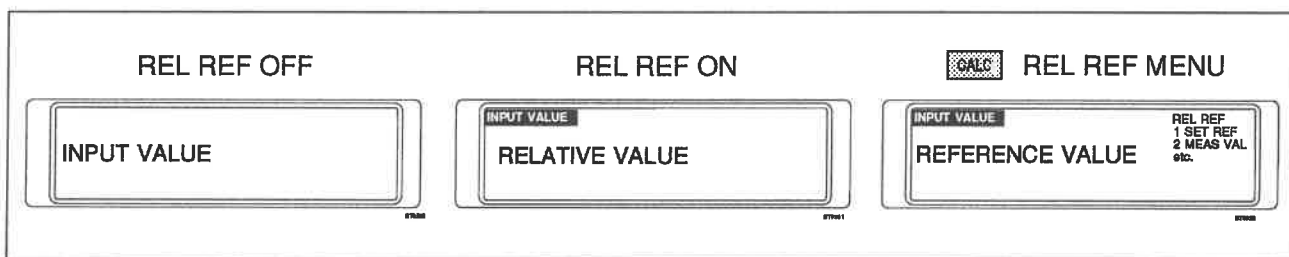


Fig. 4.35 Relative reference function displays.

SELECTING THE RELATIVE REFERENCE MODE WITH THE **REL** BUTTON

- Press **REL** to select or cancel the relative reference mode:

If the relative reference feature is not active (no REL on the display), the present reading is stored as the new relative reference value. Subsequent readings are displayed as the difference between the measured value and the stored reference value. You can enter an arbitrary value as a reference via the CALCULATION menu (see below).

If the relative reference feature is active (REL in the display), it is cancelled and the REL indication is cleared from the display.

SELECTION OF THE RELATIVE REFERENCE MODE VIA THE CALCULATION MENU (FIG. 4.36)

- press **CALC** to invoke the calculation menu.
- select
 - 5. REL REF** to invoke the REL REF menu; the display shows the current reference value, and the actual input value (see Fig. 4.35).
 - 1. SET REF** to select the relative reference mode and to set a new reference value: key in a value and press **ENTER**.
 - 2. MEAS VAL** to select the relative reference mode. The actual input value is taken as a new reference.
 - 3. ON** to select the relative reference function.
 - 4. OFF** to cancel the relative reference function.
 - 0. RETURN** to return to the CALCULATION menu. The reading is the input reference.
 - 0. EXIT** to leave the CALCULATION menu.

Press **CE** to exit the CALCULATION menu from any level.

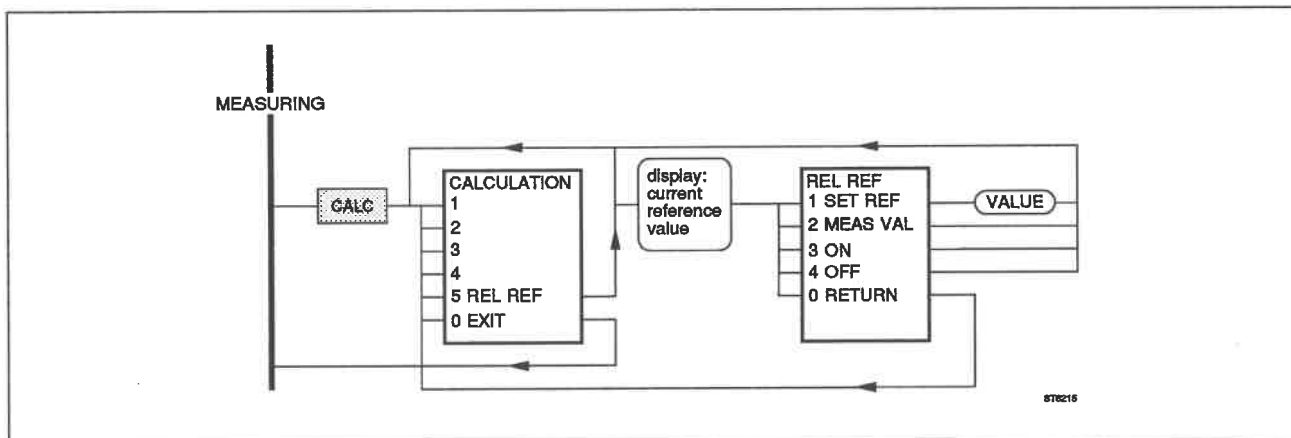


Fig. 4.36 Relative reference menu.

The relative reference mode can be used in combination with the calculation functions; SCALING, DELTA% and dBm (see Section 4.10). As selecting one of these functions cancels the relative reference mode, you must select the relative reference mode after selecting SCALING, DELTA% or dBm. The relative reference value and the reading are both the calculated (scaled, delta% or dBm) values.

DESCRIPTION OF THE RELATIVE REFERENCE FUNCTION

The relative reference feature may be used for all measuring functions. Separate reference values can be stored for each function. When changing functions while relative reference is active the stored value for the new function is automatically selected. When you return to the original function, its relative reference is restored (and the REL indication reappears).

Since the display represents a numeric difference, it can be positive (no sign) or negative (- sign), even in the resistance and AC functions.

If a reading is overrange or unavailable when the REL button is pressed, the PM2530 indicates OL and does not store the relative reference.

Note that input overload limits are not changed by the use of the relative reference feature.

While a relative reference is enabled, the PM2530 indicates an overrange condition if the input signal is overrange. For example, if the instrument is in the 20 V range, Volts DC function and you store a relative reference of +15 V. The maximum positive voltage relative reading that can be displayed, without overranging, is +5 V (approximately), which is actually a +20 V input signal. The maximum negative voltage reading that can be displayed without overranging is -35 V, which is actually a -20 V input signal.

When in autorange, the PM2530 selects the range appropriate for the input signal, regardless of any relative reference. If, for example, a +10 V relative reference is stored, and a +1 V input is applied, the PM2530 will autorange to the 2 V range.

Applications of the relative reference feature include correcting for test lead resistance in 2-wire resistance measurements, measuring voltage deviations, and matching resistors.

4.8 CORRECTING OFFSETS

4.8.1 NULLING: COMPENSATING INPUT OFFSETS (V=, A=, Ω=)

The NULLING function compensates for unwanted input offset in the Volts DC, Current DC and resistance measurement functions. Input offsets can be internal offsets and external offsets, such as thermal e.m.f.'s on the input terminals (Voltage DC and Resistance) or input leakage current (Current DC). As the influence of an input offset is range dependent, each range has a different compensation value (NULL VALUE). The lower the range, the greater the effect of NULLING will be.

Proceed as follows:

- press **ACQUIS** to invoke the ACQUIS menu (Fig. 4.37)
- select
 - 2. NULLING** the current NULL VALUE (the value that has to be considered as zero) in the current range, is displayed.
to select the NULLING function. If ON, readings are measured values minus the NULL VALUE. The display will indicate "NULL". Select 3. or 4. to set new NULL VALUES.
 - 1. ON** to cancel the NULLING function.
 - 2. OFF** to set a new NULL value in the current range:
 - prepare the input (see below) and press **SINGLE**.The new NULL VALUE is the actual input value. Select another range (press **RANGE UP** or **DOWN**) and repeat if more ranges must be nulled.
 - 3. NEW NULL**
 - 4. NULL ALL** to set new NULL values in all ranges:
 - prepare the input (see below) and press **SINGLE**; wait until the NULLING menu appears again.In each range the new NULL VALUE will become the actual input value.
 - 0. RETURN** to return to the ACQUIS menu.
 - 0. EXIT** to leave the ACQUIS menu.

Press **CE** to exit the NULLING menu from any level.

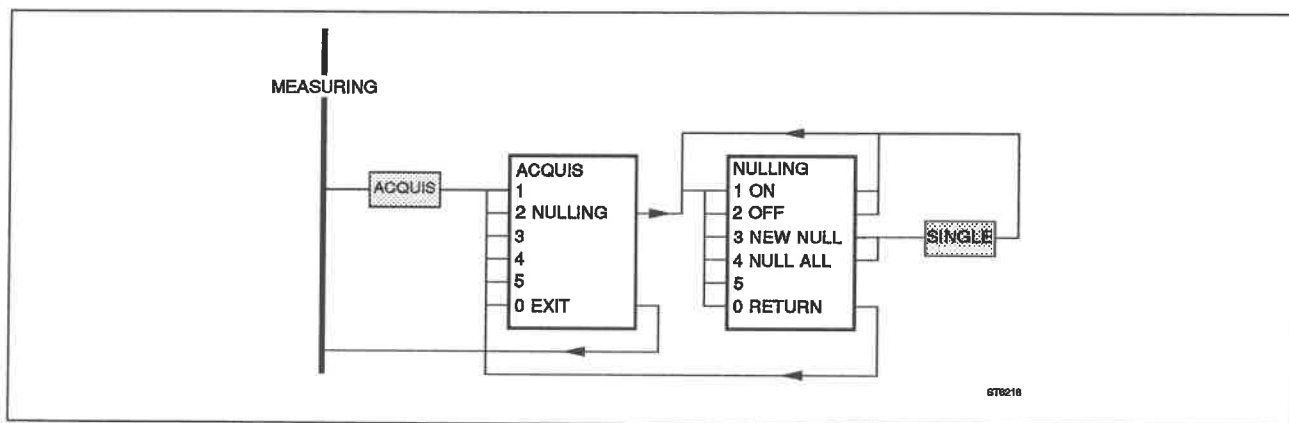


Fig. 4.37 Input offset nulling menu.

PREPARING THE INPUT FOR NULLING

- | | |
|------------|---|
| Volts DC | - Short the input with the test leads. The nulling function compensates for the remaining input voltage. |
| Resistance | - Connect the unknown resistance via the test leads (2-wire or 4-wire). The nulling function compensates for the voltage on the sense (VΩ-COM) input with no current applied. |
| Current DC | - Remove the input test leads (open input). The nulling function compensates for the leakage current. |

4.8.2 AUTO CORRECTION

To obtain long term accuracy the PM2530 is equipped with AUTO CORRECTION functions. The AUTO CORRECTION function is always selected when **RESET** is pushed to maintain the instrument specifications according to Appendix A. While measuring internal measurements are taken to update the correction values applied. If your application requires accurate timing conditions (not interrupted by correction measurements) it is however possible to disable the AUTO CORRECTION function during a time critical measurement sequence.

Proceed as follows:

- press **ACQUIS** to invoke the ACQUIS menu (Fig 4.38)
- select
 - 2. NULLING** to select the NULLING menu. If the nulling function is not available the AUTO CORRECTION menu will appear directly in the ACQUIS menu.
 - 5. AUTO COR** to set the auto correction conditions.
 - 1. ON** to enable AUTO CORRECTION, while measuring internal measurements are taken to update the correction values applied.
 - 2. OFF** to disable AUTO CORRECTION, while measuring the correction values applied are not updated, but the last correction values remain active.
 - 3. ONCE** to update the correction values once using an internal measurement and then sets AUTO CORRECTION OFF. This option allows accurate measurements without measurement time disturbance from the auto correction function.
 - 0. RETURN** to return to the NULLING menu. If the nulling function is not available you will return to the ACQUIS menu directly.
 - 0. RETURN** to return to the ACQUIS menu.
 - 0. EXIT** to exit the ACQUIS menu.

Press **CE** to exit the AUTO COR menus from any level.

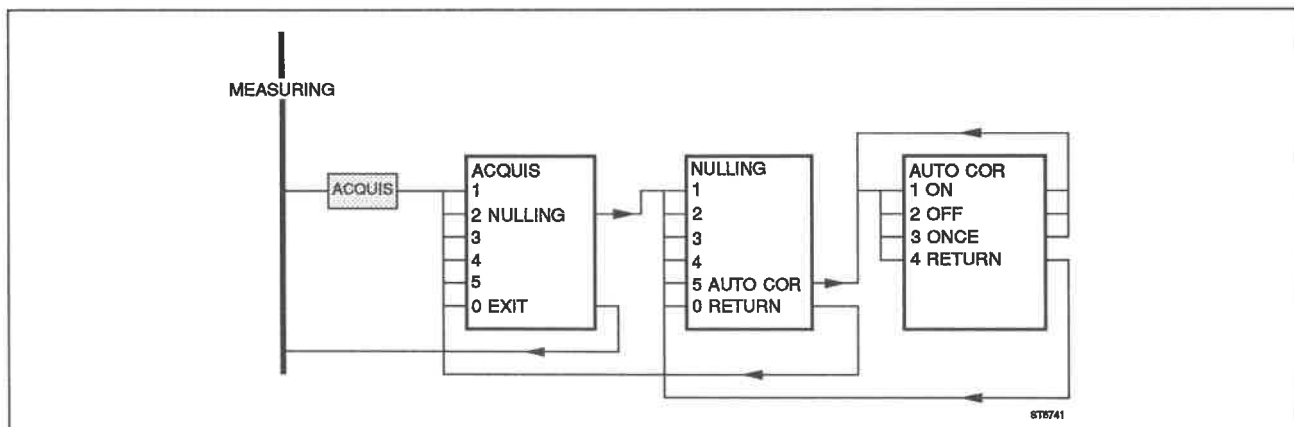


Fig 4.38 Auto correction menu.

DESCRIPTION OF THE AUTO CORRECTION FUNCTION

The auto correction function is only active at the function, speed combinations indicated at table 4.13. Using remote control it is possible to make a more precise selection of the auto correction functions. See Command Reference Manual CALIBRATION SUBSYSTEM.

SPD FUNCTION	1	2	3	4	5	6
	ON	ON	ON	ON	ON	ON
	ON	ON	ON	OFF	OFF	OFF
	ON	ON	ON	ON	ON	OFF
	ON	ON	ON	ON	ON	OFF
	ON	ON	ON	ON	ON	OFF
	ON	ON	ON	OFF	OFF	OFF
	OFF	OFF	OFF	OFF	OFF	OFF
	OFF	ON	ON	ON	ON	OFF

Table 4.13 Auto Correction selection overview.

To obtain accurate measurement results according to the specifications in Appendix A it is recommended to use AUTO CORRECTION ON as selected after pressing . If your application needs accurate timing conditions it is possible to disable AUTO CORRECTION. It is advised to use the AUTO CORRECTION ONCE option in those cases you also need accurate measurements. AUTO CORRECTION ONCE performs a single AUTO CORRECTION and then turns the AUTO CORRECTION OFF so your measurements are accurate and there is no measurement time increase due to AUTO CORRECTIONS being performed while measuring.

4.9 MEASUREMENTS IN THE BURST MODE

4.9.1 INTRODUCTION

The BURST mode allows you to continuously store the most recent measurement results (up to 15000) in the BURST memory. After a trigger and a defined trigger delay, N more results are stored, and then the storage of results stops. N is the selectable post trigger value. The BURST measurement mode enables you to perform high speed measurements (up to 50000 measurements per second in Volts DC), and to examine the results by reading the BURST memory. The right display part shows BURST if the BURST mode is active. During a BURST cycle, results are not displayed in SPD4, SPD5, and SPD6.

The BURST memory can be read via the OUTPUT function "RD BURST", see Section 4.9.3. The Burst memory length depends on the speed mode in which the measurements are performed, see table 4.13.

Table 4.14 BURST memory length.

SPEED MODE	SPD1	SPD2	SPD3	SPD4	SPD5	SPD6
MEMORY LENGTH	6000	7500	7500	10000	10000	15000

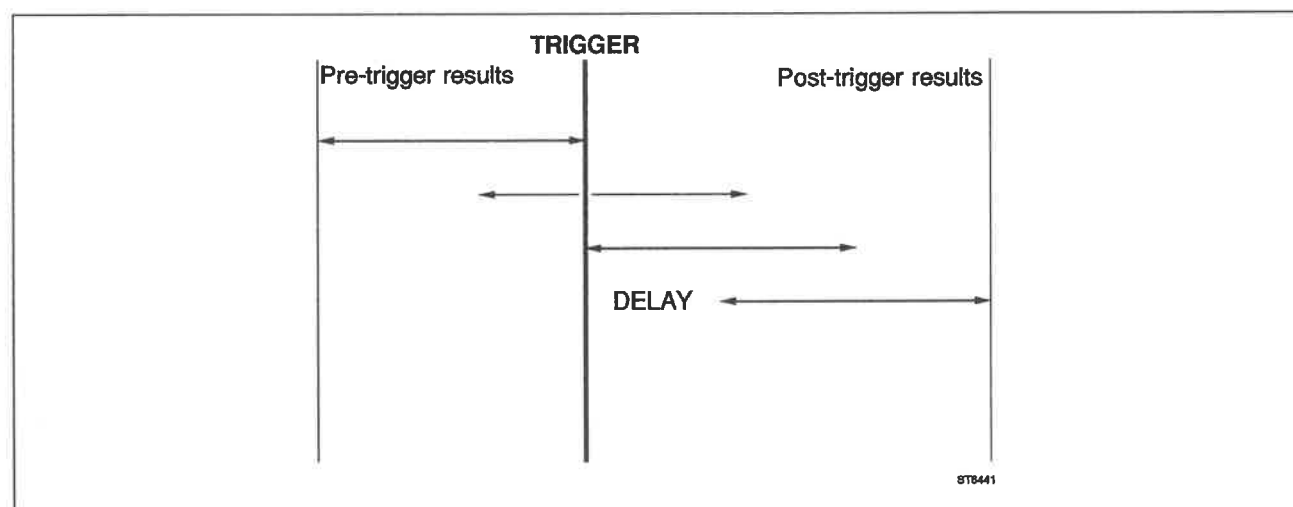


Fig. 4.39 BURST mode.

The WEIGHTING FILTER, RELATIVE REFERENCE mode, CALCULATION functions (SCALING, DELTA%, dBm), MIN/MAX function, and LIMITS function are not operative during a BURST measurement.

When reading a result from BURST memory, it is processed by these functions, if active. Refer to the measurement set up flow diagram Fig. 4.6 in Section 4.2.

START BURST (Fig. 4.40)

A BURST measurement cycle is started by an ARM EVENT, e.g. a falling edge on the rear panel EXT1 connector, when the BURST mode is set ON. Each SAMPLE EVENT (e.g. a falling edge on the rear panel EXT2 connector) after an ARM EVENT, will cause the instrument to take a sample (perform a measurement) and to store the result in the BURST MEMORY. All possible ARM EVENTS and SAMPLE EVENTS are described in Section 4.9.2.

When the BURST MEMORY is full, the first stored result will be erased when a new result is stored.

Example: See Fig. 4.40. The BURST MEMORY size is 7500. When result number 7504 is stored, results numbers 5 to 7504 remain stored in the BURST MEMORY, results numbers 1 to 4 are lost.

POST TRIGGER, STOP BURST (Fig. 4.40)

After a TRIGGER EVENT and the end of the TRIGGER DELAY, N more results will be stored; then the BURST measurement stops. N is the set POST TRIGGER VALUE (see Section 4.9.2).

Example: Speed mode SPD2 is selected (BURST memory length is 7500). The post trigger value N is set to 1000. When the BURST measurement is terminated, the BURST MEMORY contains:

- in memory location 0 ("LOC 0"), the first result (1) after the end of the TRIGGER DELAY (first post-trigger result).
- in memory locations 1 ("LOC 1") up to 999, the other results after the trigger delay (post-trigger results).
- in the remaining memory locations ("LOC -1" to "LOC -6500") the 6500 results stored before the end of the TRIGGER DELAY (pre-trigger results).

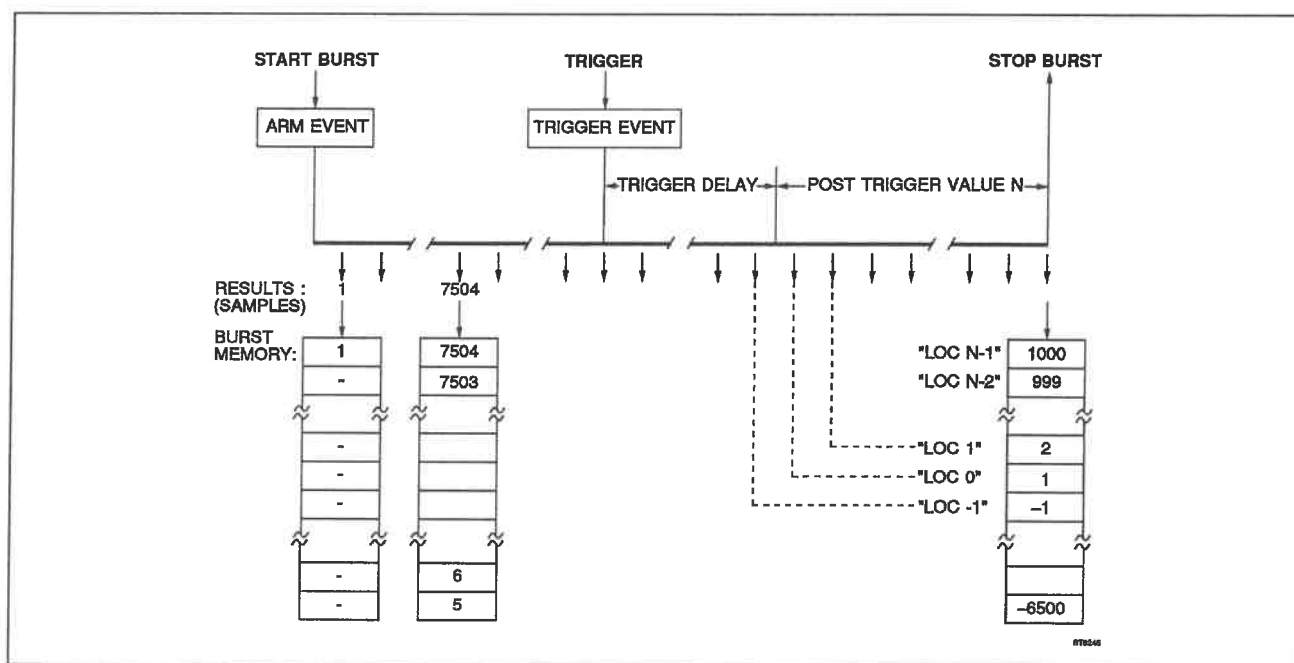


Fig. 4.40 The BURST MODE.

4.9.2 MAKING MEASUREMENTS IN THE BURST MODE

NOTE: *BURST measurements can not be made in the Volts Low Frequency (Vlf) and Frequency/ Period (FREQ/PER) measurement function.*

To make measurements in the BURST mode, proceed as follows:

- switch the BURST mode on (see below "Switching the BURST mode ON/OFF").
- supply the ARM, SAMPLE and TRIGGER events as specified at setting the BURST conditions.
The rear panel RDY1 and RDY2 outputs provide INPUT COMPLETE, READING COMPLETE, and BURST COMPLETE signals for timing synchronization. Refer to "BURST and INPUT complete outputs RDY1, RDY2" below.

The following BURST conditions can be set:

- post trigger value (see below "Setting the POST TRIGGER value").
- arm event input (see below "Selecting the ARM EVENT INPUT").
- sample interval, delay, input (see below "Selecting the SAMPLE INTERVAL, DELAY, INPUT").
- trigger delay, trigger event input (see below "Select the TRIGGER DELAY and TRIGGER EVENT INPUT").

NOTE: *the trigger delay, trigger interval and trigger input in the normal (direct) measurement mode (Section 4.5.2) have the same function and value as the sample delay, sample interval and sample input respectively in the BURST mode. So if you change the sample delay for the BURST mode, the trigger delay for the direct measurement mode is set to the same value.*

SWITCHING THE BURST MODE ON/OFF (FIG. 4.41)

- press **ACQUIS** to invoke the ACQUISITION menu
- select:
 - 5. BURST** to invoke the BURST menu.
 - 1. ON** to select the BURST mode. The PM2530 waits for an arm event and sample events.
 - 2. OFF** to cancel the BURST mode.
 - 0. RETURN** to return to the ACQUIS menu.
 - 0. EXIT** to exit the ACQUIS menu.

Press **CE** to exit the ACQUIS menu from any level.

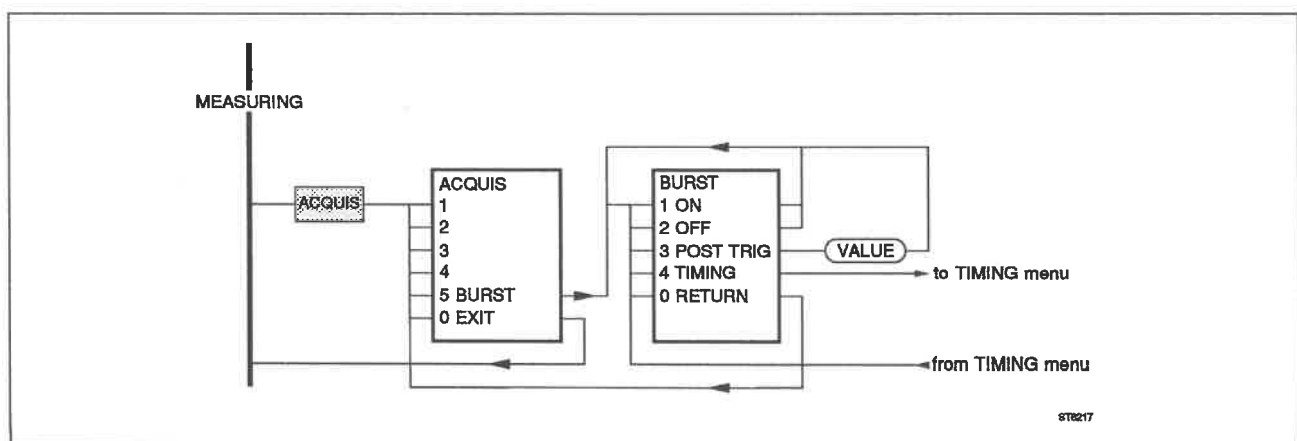


Fig. 4.41 BURST mode selection menu.

SETTING THE POST TRIGGER VALUE (FIG. 4.41)

- press **ACQUIS** to invoke the ACQUISITION menu.
- select:

5. BURST

4. POST TRIG

to invoke the BURST menu.

to enter the number of results to be stored after a trigger; key in a value and press **ENTER**. The maximum value depends on the speed mode:

SPD1	SPD2	SPD3	SPD4	SPD5	SPD6
6000	7500	7500	10000	10000	10000

If a higher value than maximum is set automatically no more than the maximum number of samples is taken. The minimum value is 2. In remote control up to 15000 measurements can be stored. See the Command Reference manual TRIGGER SUBSYSTEM.

to return to the ACQUIS menu.

to exit the ACQUIS menu.

0. RETURN

0. EXIT

Press **CE** to exit the ACQUIS menu from any level.

SELECTING THE ARM EVENT INPUT (FIG. 4.42)

Each BURST measurement cycle must be started by an ARM EVENT. An ARM EVENT also clears the BURST MEMORY.

- press **SINGLE** to invoke the ACQUISITION menu.
- select:

5. BURST

4. TIMING

1. ARM SRC

1. EXT1

2. EXT2

3. MANUAL

4. IMMEDIATE

5. INTERFACE

to invoke the BURST menu.

to invoke the TIMING menu.

the following ARM EVENTS will start a BURST cycle:

a falling edge on the rear panel EXT1 input (see Fig. 4.48).

a falling edge on the rear panel EXT2 input (see Fig. 4.48).

pressing **SINGLE**.

a new sequence is armed immediately after the preceding sequence is finished. The BURST memory is always cleared (Default Setting).

a trigger via the GPIB interface if the instrument is in remote control. Refer to the Command Reference Manual TRIGGER SUBSYSTEM.

- select 0. RETURN to return to previous menus, select 0. EXIT to exit the ACQUIS menu.

Press **CE** to exit the menus from any level.

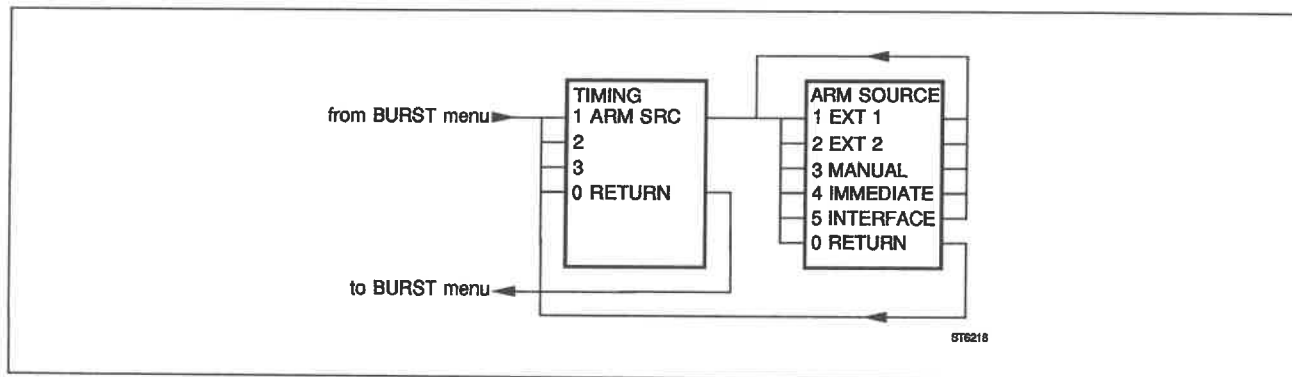


Fig. 4 42 ARM EVENT INPUT menu.

SELECTING THE SAMPLE INTERVAL, -DELAY, AND SOURCE (FIG. 4.42)

Each individual measurement (sample) must be started by a sample event.

- press  to invoke the ACQUISITION menu.
- select:
 - 5. BURST to invoke the BURST menu.
 - 4. TIMING to invoke the TIMING menu.
 - 2. SAMPLING to select the conditions for taking individual measurements.
 - 1. INTERVAL to set the time between successive individual measurements if INTERNAL SAMPLING (see 3. SAMPLE INP) is selected. See Fig. 4.44.
 - 1. USER VAL to enter an arbitrary sample interval time up to 24 hours, minimum resolution 1 μ s (0-1 s) . The minimum value depends on the required measurement time. Entering a value below the required measurement time sets the interval to the measurement time, and INT>> is displayed. The sample delay is discarded.
 - 2. DEFAULT to select the default (is minimum) sample interval time for maximum measurement speed. The sample delay can be made operative. However, when changing over to sample source INTERNAL, the sample delay will be switched OFF if the default sample INTERVAL was operative.
 - 2. DELAY to set the delay between the SAMPLE EVENT and the start of a sample. See "Sample delay" below.
 - 1. USER VAL enter a value up to 10 seconds (1 μ s resolution).
 - 2. DEFAULT to set the default delay; it covers the PM2530 response time.
 - 3. OFF to cancel the sample delay. OFF is automatically selected when changing over to sample source INTERNAL, when the default sample INTERVAL is operative.
 - 3. SAMPLE SRC to select one of the following EVENTS on which a new sample is taken:
 - 1. INTERNAL a new sample is taken when the defined sampling interval time elapsed (Fig. 4.44). When sampling interval DEFAULT is operative, the sampling delay can be made operative. At sampling interval USER VAL the delay is always discarded.
 - 2. EXT 1 a new sample is taken after a falling edge on the rear panel EXT 1 input, when the DELAY is expired (Fig. 4.45). See below "EXT1, EXT2 input".
 - 3. EXT 2 a new sample is taken after a falling edge on the rear panel EXT 2 input, when the DELAY is expired (Fig. 4.45). See below "EXT1, EXT2 input".
 - 4. IMMEDIATE a new sample is taken immediately after termination of the preceding sample, when the delay is expired (see Fig. 4.46).
 - 5. INTERFACE a new sample is taken after a sample event via the GPIB bus (remote control only). Refer to the Command Reference Manual TRIGGER SUBSYSTEM.
- select 0.RETURN to return to previous menus, select 0. EXIT to exit the ACQUIS menu.

Press  to exit the menus from any level.

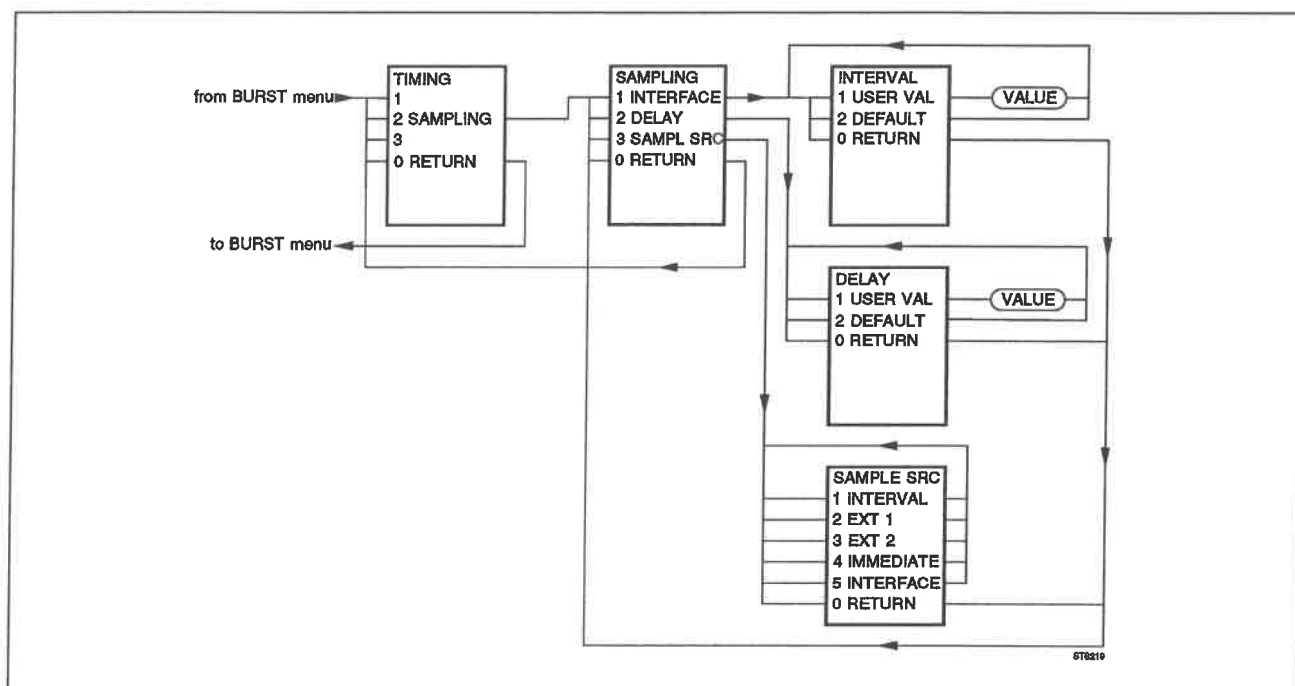


Fig. 4.43 Sampling conditions menu.

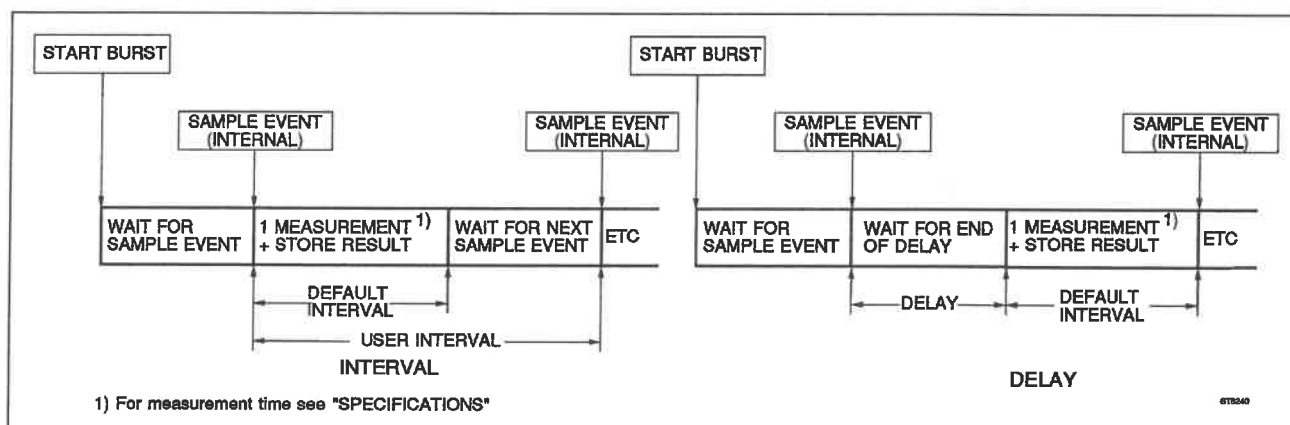


Fig. 4.44 Sample timing at sample input "INTERNAL".

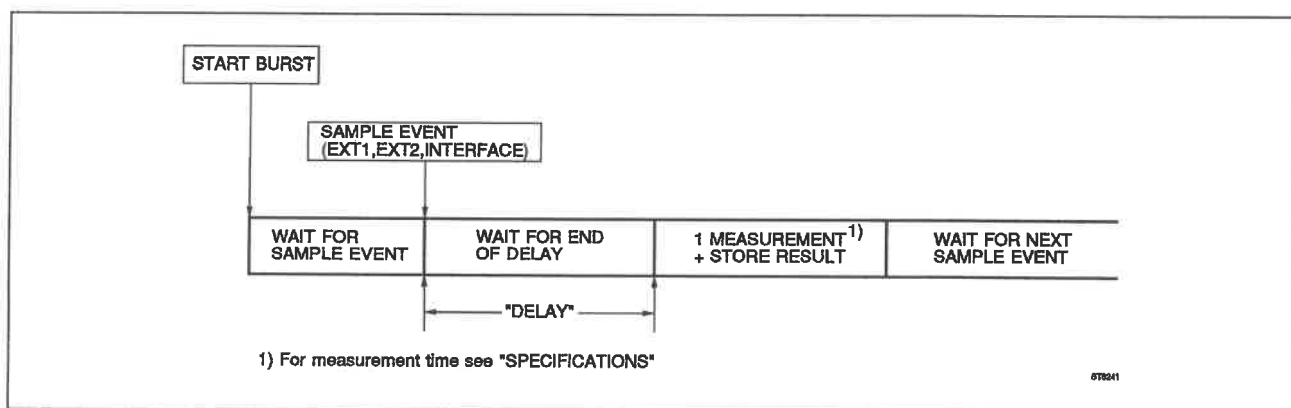


Fig. 4.45 Sample timing at sample input EXT1, EXT2, or INTERFACE.

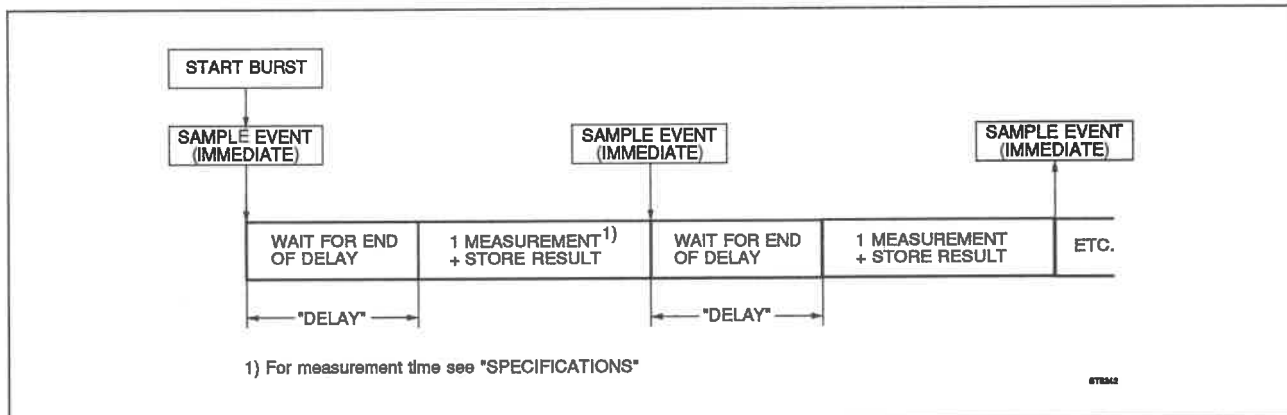


Fig. 4.46 Sample timing at sample input IMMEDIATE.

SELECTING THE TRIGGER DELAY AND THE TRIGGER EVENT INPUT (FIG. 4.47)

After a trigger event and the defined delay, a specified number of samples (post trigger value) are taken. The BURST measurement cycle then stops.

- press **ACQUIS** to invoke the ACQUISITION menu
- select:
 - 5. **BURST** to invoke the BURST menu.
 - 4. **TIMING** to invoke the TIMING menu.
 - 3. **TRIGGER** to select the trigger DELAY and EVENT. After the TRIGGER EVENT and DELAY, N more measurements will be taken, and then the BURST cycle stops. N is the POST TRIGGER VALUE.
 - 1. **DELAY** enter a DELAY up to 65 s, with a 1 ms resolution.
 - 2. **TRIG SRC** to select one of the following TRIGGER EVENTS:
 - 1. **EXT 1** the trigger event is a falling edge on the rear panel EXT 1 input. See below "EXT1, EXT2 input".
 - 2. **EXT 2** the trigger event is a falling edge on the rear panel EXT 2 input. See below "EXT1, EXT2 input".
 - 3. **MANUAL** a trigger event is generated by pressing **SINGLE**.
 - 4. **IMMEDIATE** the TRIGGER EVENT is internally generated immediately after a BURST measurement is started.
 - 5. **INTERFACE** a trigger event must be supplied via the GPIB bus if the instrument is in remote control. Refer to the Command Reference Manual TRIGGER SUBSYSTEM.
- select 0. **RETURN** to return to previous menus.

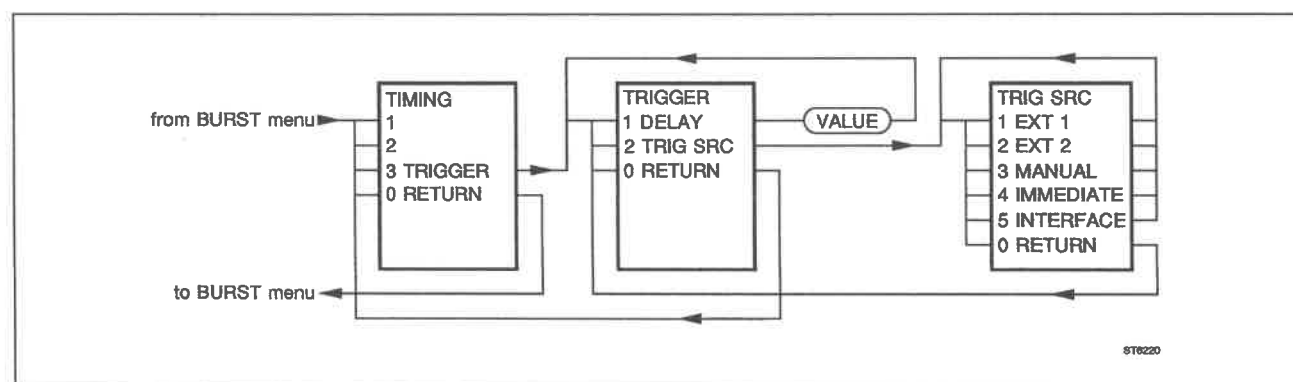


Fig. 4.47 Trigger conditions menu.

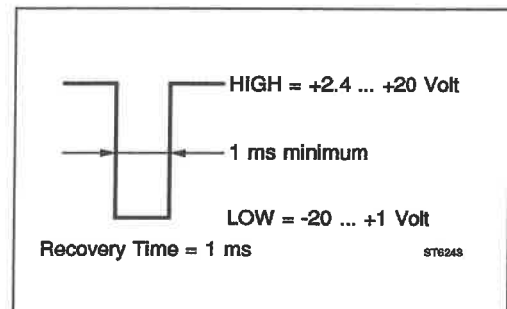
SAMPLE DELAY

The sample delay is the time between the sample event and the start of an individual measurement. The DEFAULT sample delay is just long enough so that the reading will be correct (within a specified number of counts of the final value) even if the sample event occurs as the input makes a step change between zero and full scale. In this situation the delay covers the PM2530 response time. For response times refer to Appendix A "Specifications". You can set the sample delay below the default value to accommodate special timing considerations. If you set the delay to zero (or OFF), a measurement is started immediately after receiving a sample event, regardless of the PM2530 response time. This can cause a reading error when an input step signal arises simultaneously with the sample event.

EXT1 AND EXT2 INPUTS

If you select ARM, SAMPLE, or TRIGGER input EXT1 or EXT2, the event is a falling edge on the rear panel EXT1 or EXT2 connectors. The EXT1 and EXT2 inputs are pulled high internally, as such they can be controlled by a normally open switch. An event occurs when the switch is closed. If new sample events (falling edges) occur before a sample is completed, the first one is latched. Any further events are lost. For timing synchronization see below "The BURST and SAMPLE complete outputs RDY1 and RDY2".

Fig. 4.48 Input pulse for the EXT1 or EXT2 input.



THE READING COMPLETE OUTPUT RDY1 (FIG. 4.49)

A low output pulse (2 μ s, TTL) from the rear panel RDY1 output, indicates when a READING is complete.

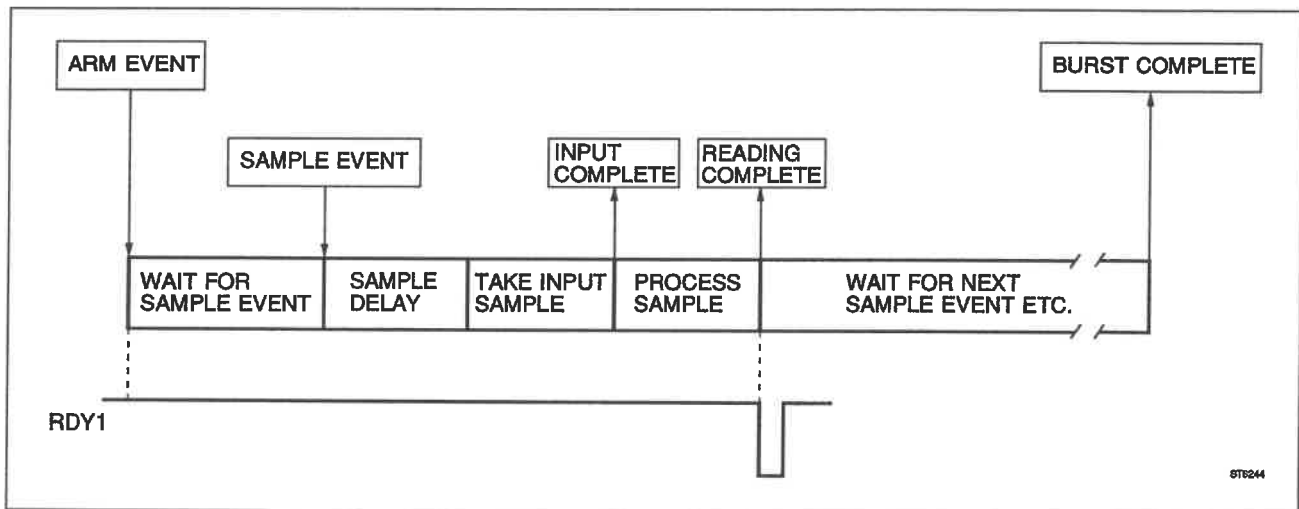


Fig. 4.49 The RDY1 and RDY2 output.

NOTE: In remote control, the RDY1 and RDY2 connector can be assigned as READING COMPLETE, INPUT (sample) COMPLETE, or BURST COMPLETE (Section 5. OUTPUT SUBSYSTEM). The configuration is maintained when returning to local operation. Pressing  will set RDY1 to READING COMPLETE and RDY2 to OFF again.

WHAT HAPPENS IF THE SAME INPUT IS USED FOR THE ARM EVENT, TRIGGER EVENT AND SAMPLE EVENT

The same event input can be selected for the ARM event, TRIGGER event and SAMPLE event. The order in which the successive events on the same input are interpreted is: 1. ARM event, 2. TRIGGER event, 3. SAMPLE event.

Example:

- | | | |
|--------------------------|---|---|
| Post trigger value 100 | : | after a trigger event 100 more samples will be taken, then the BURST stops. |
| ARM event input EXT1 | : | first TTL high to low transition on EXT1 is ARM event. |
| TRIGGER event input EXT1 | : | second TTL high to low transition on EXT1 connector is TRIGGER event. |
| SAMPLE event input EXT1 | : | successive 100 TTL high to low transitions on the EXT1 connector are SAMPLE events. The next pulse on EXT1 is again an ARM event etc. |

4.9.3 READING THE BURST MEASUREMENT RESULTS

In the BURST mode, measurement results are stored in the BURST memory. To read these results, select the measurement function in which the measurements were taken. You can:

- read the number of stored results.
- read the measured highest value, lowest value and highest to lowest value.
- read the results by cycling through the memory locations.
- dump the results at a selectable rate to the display.

Refer to Fig. 4.50 for the OUTPUT BURST menu.

READING THE NUMBER OF STORED RESULTS

- press  and select:
 - 3. BURST to invoke the BURST menu.
 - 1. LENGTH to display the number of stored measurement results.
 - 0. RETURN to return to the OUTPUT menu.

READING THE HIGHEST, LOWEST, AND PEAK TO PEAK VALUE

- press  and select:
 - 3. BURST to invoke the BURST menu.
 - 2. EXTREMES to invoke the EXTREMES menu.
 - 1. HIGHEST to display the highest stored measurement result, including its memory location number.
 - 2. LOWEST to display the lowest stored measurement result, including its memory location number.
 - 3. PK to PK to display the difference between the highest and lowest measured value.
 - 0. RETURN to return to the RD BURST menu.

CYCLING THROUGH THE MEMORY LOCATIONS

- press  and select:
 - 3. BURST to invoke the BURST menu.
 - 3. SELECT to display the stored results and location numbers sequentially. Press  or  to select higher or lower location numbers.
- press  to leave the SELECT mode and to return to the RD BURST menu.

DUMPING RESULTS TO THE DISPLAY

- press **OUTPUT** and select:

3. BURST

to invoke the BURST menu.

4. READ

to display selectable memory contents at a selectable rate.

1. START LOC

to select the first location to be displayed. After entering the location number, the cursor is at 2. LENGTH.

2. LENGTH

to define the length of the memory part to be examined (last minus first location number). After entering the number, the cursor is at 3. STEPSIZE.

3. STEPSIZE

to define the interval between memory locations to be displayed. After entering the number, the cursor is at 4. STEPRATE.

4. STEPRATE

to define the time before a next result is displayed. It can be set up to 10 seconds, in 0.1 second steps. After entering the time, the cursor is at 5. DO DUMP.

5. DO DUMP

the contents of the selected memory locations will be displayed at the selected rate. The displaying can be stopped by pressing any key.

0. RETURN

to return to the RD BURST menu.

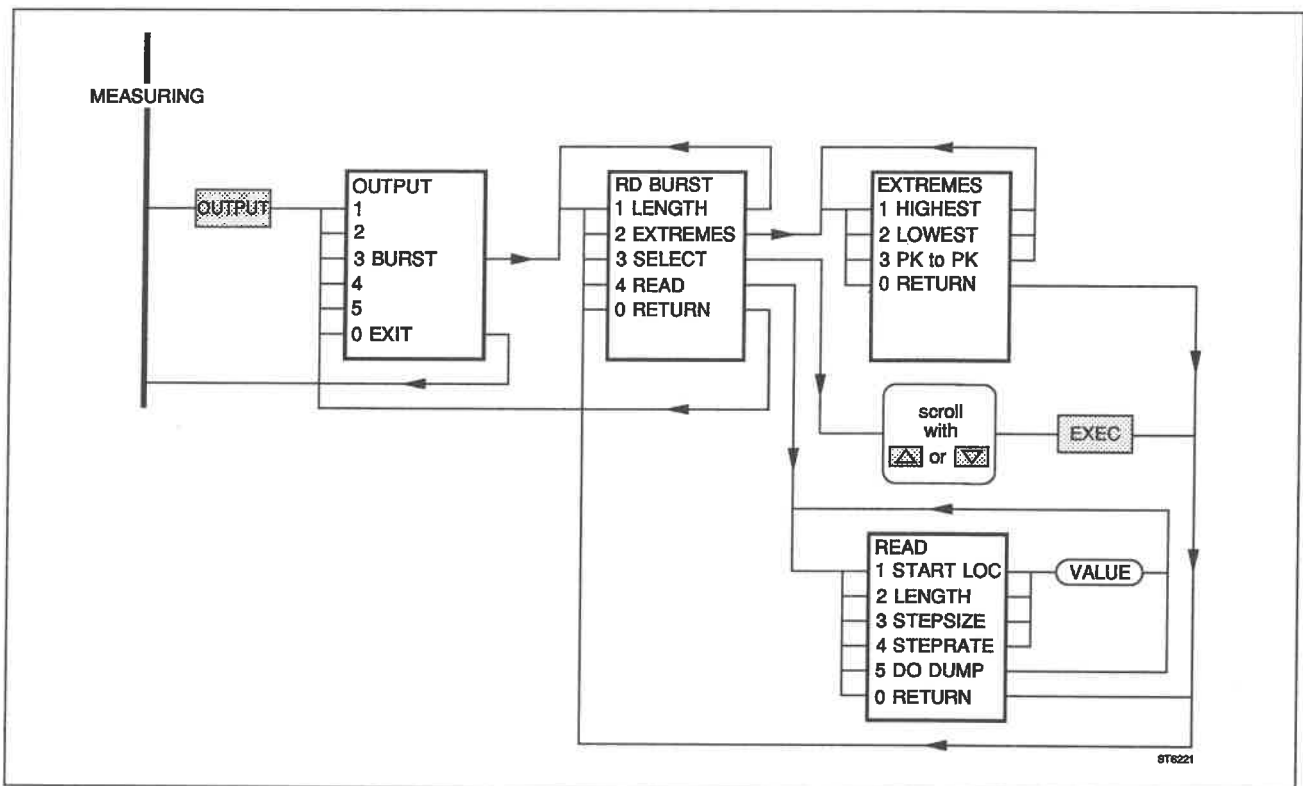


Fig. 4.50 The OUTPUT BURST menu.

4.10 CALCULATION FUNCTIONS

4.10.1 INTRODUCTION

- press **CALC** to invoke the CALCULATIONS menu. The following calculation functions are provided:
 1. **SCALE** the reading Y is the input value X, scaled using the equation $Y=(AX + B)/Z$. Factors A, B and Z can be set. The result is displayed in Engineering notation.
 2. **DELTA%** the reading is the deviation of the measured value from a reference value in terms of percentage.
 3. **dBm** the reading is the measurement result expressed in dBm.
 4. **OFF** cancel calculation functions SCALE, DELTA% and dBm.
 5. **REL REF** the reading is the difference between the measured value and a reference value. The result is displayed in Engineering notation except for DELTA% and dBm reference. In this case the reading is in % or dBm. The REL REF function is described in Section 4.7.
- 0. **EXIT** exit the CALCULATIONS menu.

The calculated result is displayed in the main display field. The upper left display field shows the raw (unprocessed) result. The active calculation function is indicated in the right display part, see Fig. 4.51. The functions 1. SCALE, 2. DELTA%, and 3. dBm cannot be combined. They can be used in combination with the REL REF function.

Selecting one of the calculation functions cancels any previously selected calculation function (including REL REF, even if REL REF is selected from the front panel **REL** key).

To use the REL REF function in combination with one of the other calculation functions, you must first select the calculation function and then REL REF.

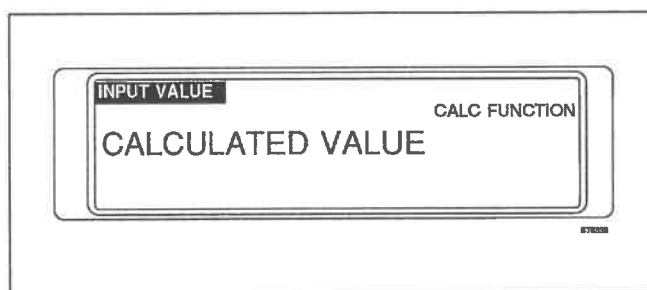


Fig. 4.51 Display at calculated results.

MINIMUM/MAXIMUM READINGS AND PARAMETER VALUES

The minimum/maximum values to be entered via the front panel, and the minimum/maximum readings are limited by the display length. Limits are ± 0.0001 to $99999E\pm 9$ or 10^{-21} to 10^{+21} in Engineering notation. In remote control (GPIB), values $\pm 9.9E\pm 37$ can be used. If a result can not be displayed, the display shows dashes (-----). If a calculated result is out of range ($\pm 9.9E\pm 37$), the OFL (overflow) message is displayed.

4.10.2 SCALING (AX + B)/Z

In the Scaling function, the measurement result X is scaled using the formula $Y=(AX+B)/Z$, where Y is the reading, not provided with a unit. Parameters A, B, and Z can be selected. Refer to "Description of the scaling function" below for detailed information.

SELECTION OF THE SCALING FUNCTION

- press **CALC** to invoke the CALCULATIONS menu.
- select
 - 1. SCALE** to switch on the SCALING function, and to invoke the SCALE menu.
 - 1. SET A** to set A (gain); key in a value and press **ENTER**. For a 25 mA offset, you must enter 25E-3 (or 0.025), etcetera.
 - 2. SET B** to set B (offset); key in a value and press **ENTER**. For a 50 mA normalize value, you must enter 50E-3 (or 0.050), etcetera.
 - 3. SET Z** to set Z (norm); key in a value and press **ENTER**.
 - 4. NORMALIZ** to set Z to AX+B for the current input value X. X is shown in the upper left display part.
 - 0. RETURN** to return to the CALCULATION menu; readings are scaled now.
 - 0. EXIT** to exit the CALCULATION menu.

Press **CE** to exit the CALCULATION menu from any level.

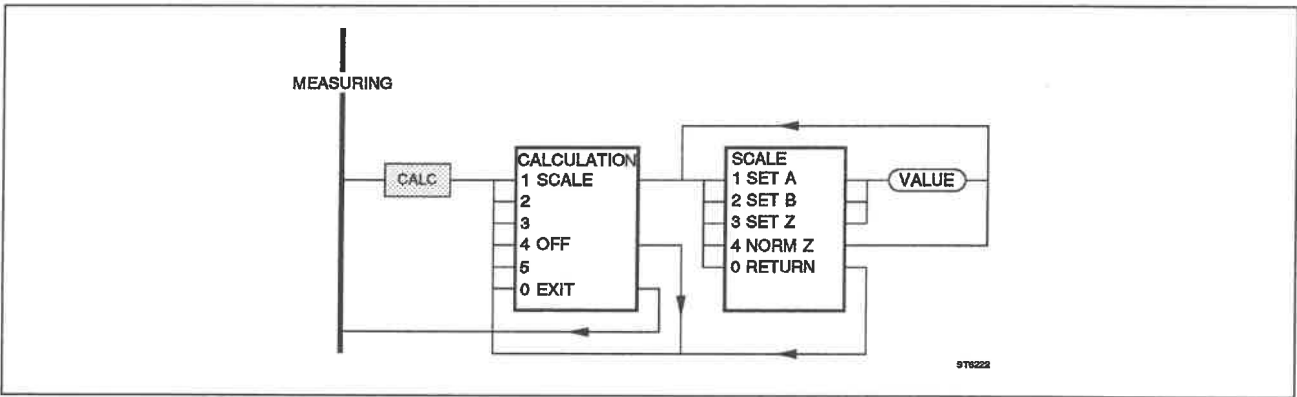


Fig. 4.52 Scaling menu.

DESCRIPTION OF THE SCALING FUNCTION

The effect of scaling the measurement results is explained below by means of an example.

The measured value X is a current between 4 mA and 20 mA. The reading is Y.

Y=X (unscaled)	4	8	12	16	20	mA
scaled $Y=AX$; A=6.25 (B=0, Z=1)	25	50	75	100	125	m
scaled $Y=AX+B$; B=-0.025 (Z=1)	0	25	50	75	100	m
scaled $Y=(AX+B)/Z$; Z=0.05	0	0.5	1.0	1.5	2.0	

If from the SCALE menu, option 4. NORM Z is selected, the Z value will be set to AX+B for the actual input value X, resulting in a reading 1. In this example, selecting 4. NORM Z at an input value of 12 mA results in a reading 1, and a Z value 0.05.

4.10.3 DELTA%

In the DELTA % function, the reading is the deviation of the input value from a reference value in terms of percentage. The reference value can be entered via the keyboard. The reference value can also be set to the current input value.

SELECTION OF THE DELTA% FUNCTION

- press **CALC** to invoke the CALCULATIONS menu.
- select
 - 2. DELTA%** to switch on the DELTA% function and to invoke the DELTA% menu; the current reference value C is shown in the main display field, the current input value in the upper left display field.
 - 1. SET C** to set the reference value C. Key in a value and press **ENTER**. For a 10 mV reference value you must key-in 10E-3 (or 0.01), etcetera.
 - 2. NEW REF** the reference value C is set to the current input value.
 - 0. RETURN** to return to the CALCULATIONS menu; readings are percentual deviations from the reference value now.
 - 0. EXIT** to exit the CALCULATION menu.

Press **CE** to exit the CALCULATION menu from any level.

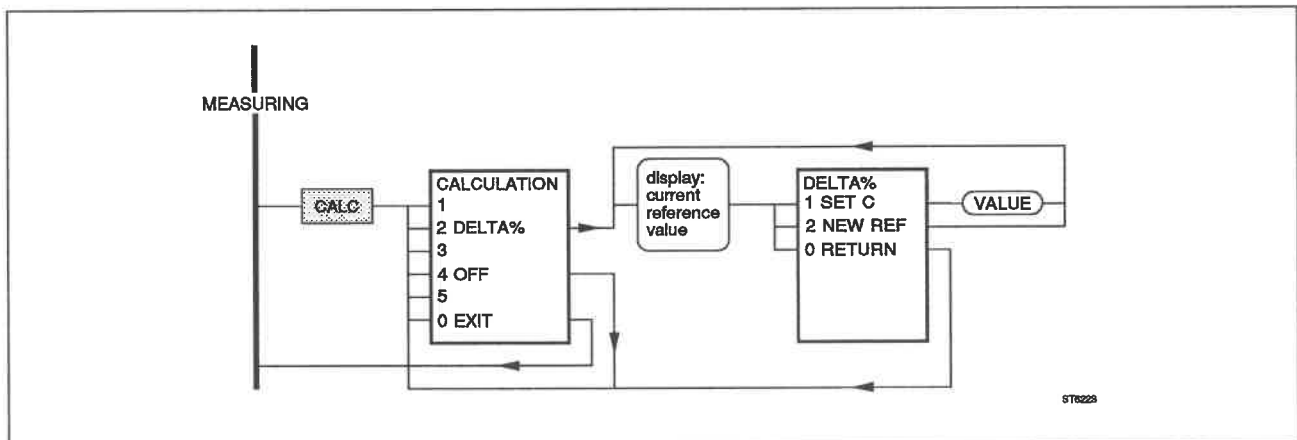


Fig. 4.53 DELTA% menu.

4.10.4 dBm CALCULATION

In the dBm calculation function, the PM2530 takes a voltage or current measurement, converts it to dBm (measure of decibels relative to 1 mW), and displays the result. The reference impedance can be set. The dBm function can be selected only in the measurement functions Volts DC, Volts AC, Volts AC+DC, Current DC, or Current AC+DC.

SELECTION OF THE dBm FUNCTION

- press **CALC** to invoke the calculation menu.
- select
 - 3. dBm** to switch the dBm function on: The current reference impedance value is shown.
 - Key in a value and press **ENTER** to change the reference impedance value.
 - 0. EXIT** to exit the calculation menu.

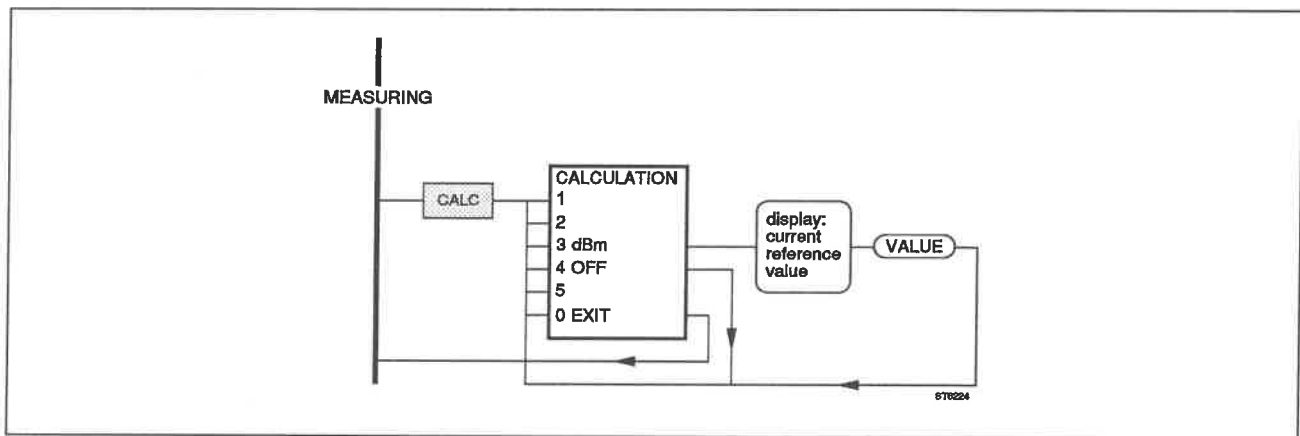


Fig. 4.54 dBm menu.

Decibels are always displayed in a single, fixed range. A voltage measurement is converted to dBm using the formula: $\text{dBm} = 10 \cdot \log_{10}(1000 \cdot \text{value}^2 / \text{reference impedance})$, where "value" is the input value. A current measurement is converted to dBm using the formula: $\text{dBm} = 10 \cdot \log_{10}(1000 \cdot \text{value}^2 \cdot \text{reference impedance})$, where value is the input value. An input value 0 (zero) in the dBm function results in the message UL (underload).

4.11 STORING AND SIGNALLING MINIMUM/MAXIMUM READINGS

If the ACQUISITION function MIN/MAX is active, the minimum and maximum measurement readings are stored, including the time of occurrence. When a new minimum or maximum reading occurs, this can be signalled on the display or by the beeper. Readings are processed values if a CALCULATION function is active. Separate MIN and MAX memories are available for each measurement function. They can be read using the OUTPUT function RD MIN/MAX, see below.

SELECTION OF THE MIN/MAX FUNCTION. (FIG. 4.55)

- press **ACQUIS** to invoke the ACQUISITION menu
- select
 - 3. MIN/MAX** to select the MIN/MAX menu.
 - 1. ON** to switch the MIN/MAX function on.
 - 2. OFF** to switch the MIN/MAX function off.
 - 3. CLEAR** to clear the MIN and MAX memory of the current measurement function. If you don't clear the memories before starting new measurements, you may read "old" minimum and maximum values when examining the MIN and MAX memory.
 - 4. SIGNAL TO** to select the "new MIN MAX value" signalling.
 - 1. DISPLAY** the display shows ▼ or ▲, when a new MIN or MAX value is measured (toggle ON/OFF). Overruled by display settings at OUTPUT DISPLAY, Section 4.13.
 - 2. BEEPER** the beeper sounds when a new MIN or MAX value is measured (toggle ON/OFF).
 - 0. RETURN** to return to the MIN/MAX menu.
 - 0. RETURN** to return to the ACQUIS menu.
 - 0. EXIT** to exit the ACQUIS menu.

Press **CE** to exit the ACQUIS menu from any level.

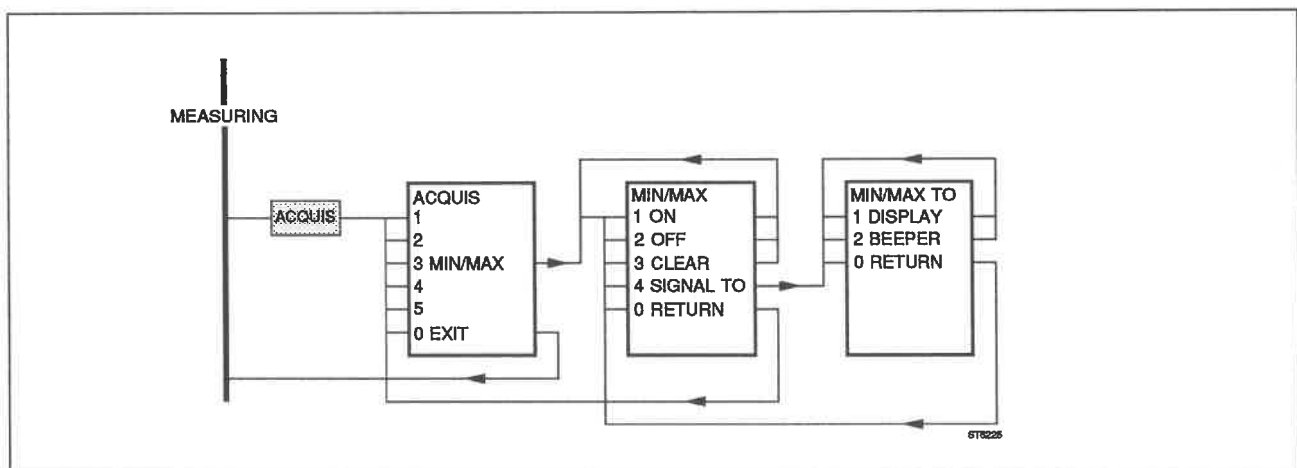


Fig. 4.55 MIN/MAX function selection menu.

READING THE MIN AND MAX MEMORY (FIG. 4.56)

- press **OUTPUT** to invoke the output menu.
- select
 1. **MIN/MAX** to invoke the MIN/MAX menu.
 1. **MIN** the lowest measured value, and its time of occurrence are displayed.
 2. **MAX** the highest measured value, and its time of occurrence are displayed.
 3. **CLEAR** to clear the MIN and the MAX memory of the current measurement function.
 0. **RETURN** to return to the OUTPUT menu.

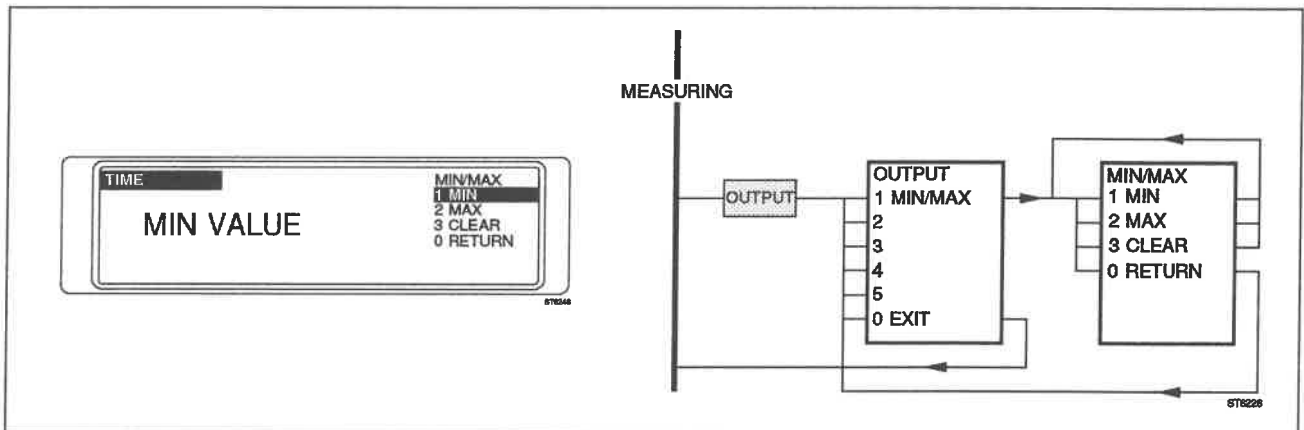


Fig. 4.56 Display at reading the MIN or MAX memory.

Fig. 4.57 Read MIN/MAX memory menu.

EXAMPLE OF THE MIN/MAX FUNCTION OPERATION

The input signal varies between +2 and -2. Before starting the measurements, the MIN and MAX memories are cleared. If a reading is higher than the previously stored maximum, or lower than the previously stored minimum value, the MAX or MIN memory, respectively, is updated. The occurrence of a new MAX/MIN value is signaled.

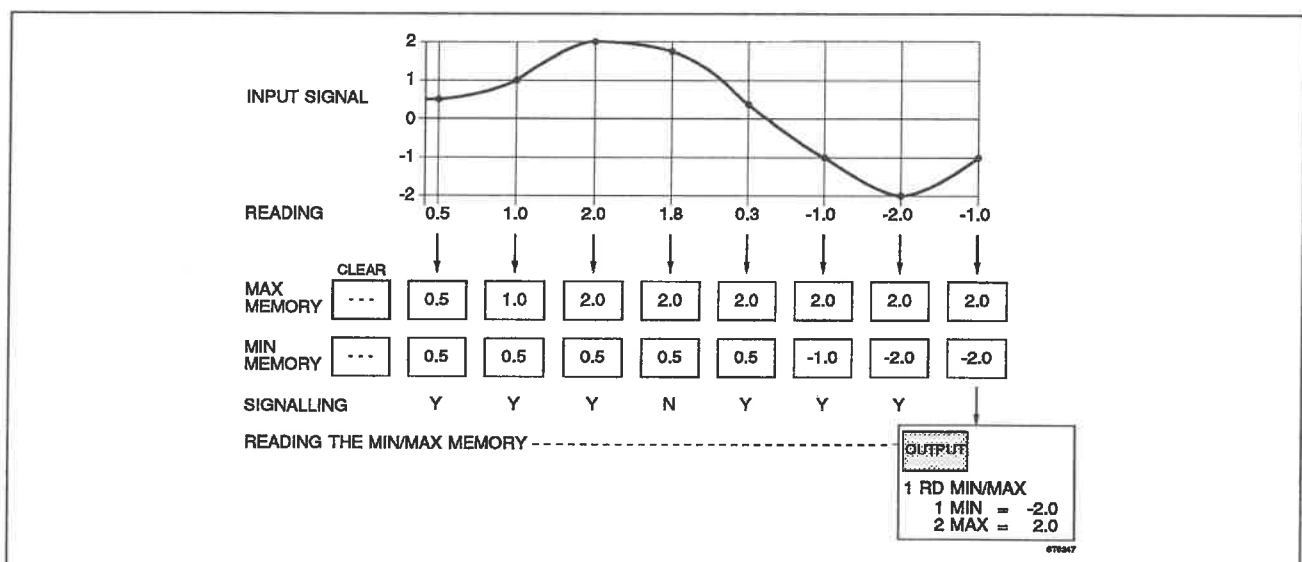


Fig. 4.58 MIN/MAX function example.

4.12 THE LIMITS FUNCTION

The LIMITS function determines if a reading falls within a designated range of values. UPPER and LOWER LIMITS of this range can be defined for each measurement function. A reading that is outside these limits is counted by the LIMIT COUNTER. It can also be signalled on the display ($\triangle \nabla \diamond$), by the beeper and the rear panel LIMIT connector. LIMIT COUNTER memories are provided, to store:

- the number of readings above the UPPER LIMIT, including the time of the first occurrence.
- the number of readings below the LOWER LIMIT, including the time of the first occurrence.
- the total number of passed readings. A reading is PASSED when it is within the UPPER and LOWER limit if the UPPER limit is above the LOWER limit. If the UPPER limit is below the LOWER limit, a reading is PASSED if it is not within the UPPER and LOWER limit. See also fig. 4.59 and 4.60.
- the total number of readings involved.

Readings are processed values if one of the CALCULATION functions is active. Separate LIMIT COUNTER memories are available for each measurement function. They can be read using the OUTPUT function LIM-COUNT, see below. Fig. 4.59 and Fig. 4.60 below, show the output levels at the LIMITS connector.

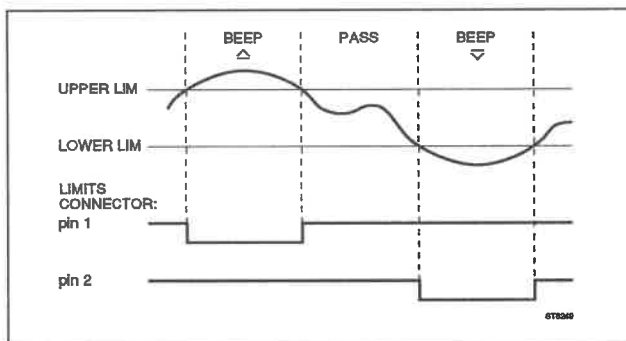


Fig. 4.59 UPPER LIMIT > LOWER LIMIT.

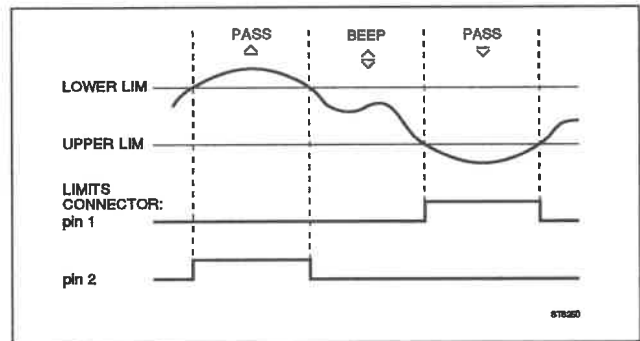


Fig. 4.60 LOWER LIMIT > UPPER LIMIT.

SETTING THE LIMITS FUNCTION (FIG. 4.60)

- press **ACQUIS** to invoke the ACQUISITION menu
- select

4. LIMITS

1. ON
2. OFF
3. CLEAR

to invoke the LIMITS menu.

to select the LIMITS function.

to cancel the LIMIT function.

to clear the LIMIT COUNTER memories of the current measurement function. The display will be temporary inverted. If you don't clear the memories before starting new measurements, you may read "old" values when examining the memories.

4. SET LIMIT

1. UPPER
2. LOWER
0. RETURN

to invoke the SET LIMIT menu.

to set the UPPER limit: key in a value and press **ENTER**.

to set the LOWER limit: key in a value and press **ENTER**.

to return to the LIMITS menu.

5. SIGNAL TO

1. DISPLAY

to invoke the SIGNAL TO menu. From this menu you can select the signalling destination (toggle ON/OFF).

the display shows $\triangle \nabla \diamond$ when a reading is above the upper, below the lower or both above the upper and below the lower limit. The latter if the upper limit is below the lower limit. Overruled by display settings at OUTPUT DISPLAY, Section 4.13.

2. BEEPER

the beeper sounds when a reading is not passed.

3. LIM OUT

a low output (TTL level) is generated on the rear panel LIMITS connector, when a reading is above the UPPER (pin 1) or below the LOWER (pin 2) LIMIT. See below "The rear panel LIMITS connector".

0. RETURN

to return to the LIMITS menu.

0. RETURN

to return to the ACQUIS menu.

0. EXIT

to exit the ACQUIS menu.

Press **OE** to exit the ACQUIS menu from any level.

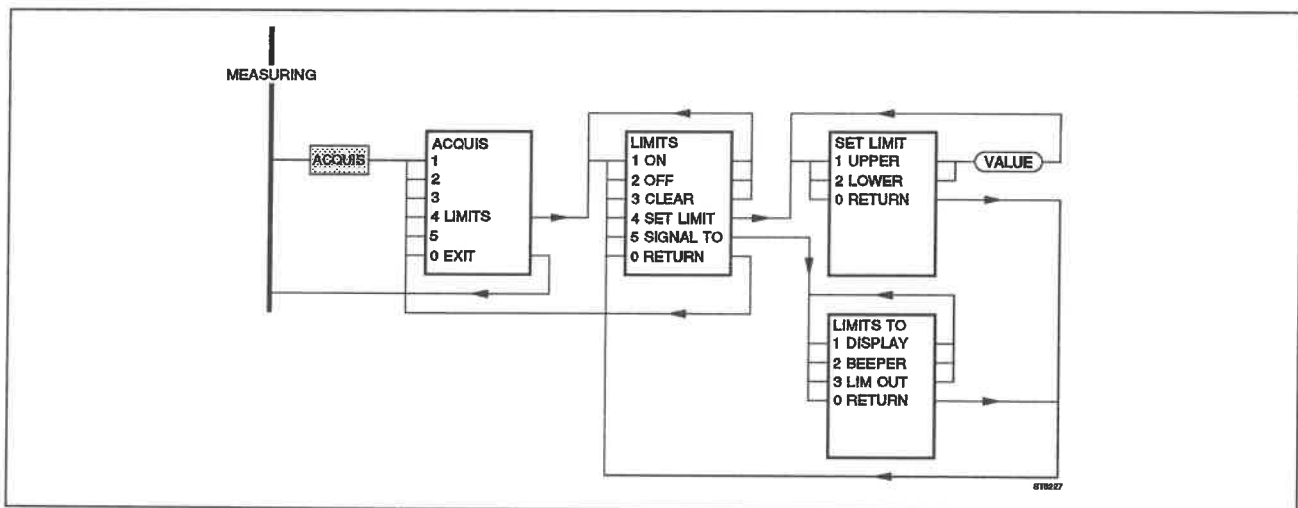


Fig. 4.61 The LIMIT setting menu.

THE REAR PANEL LIMIT CONNECTOR

5. +5 V supply voltage (10 mA), 50 Ω output resistance.
2. "LO LIM", LOWER LIMIT out; normally open, TTL low if reading < LOWER LIMIT.
1. "HI-LIM", UPPER LIMIT out; normally open, TTL low if reading > UPPER LIMIT.
- 3,4 GROUND
- 6-9. GROUND

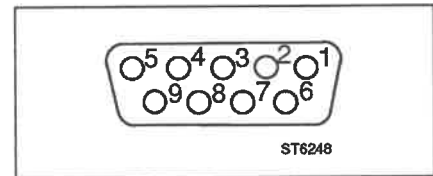


Fig. 4.62 LIMITS connector.

READING THE LIMIT COUNTER MEMORIES (FIG. 4.64)

- select the measurement function in which the measurements are taken.
- press **OUTPUT** to invoke the output menu.
- select
 2. **LIM-COUNT** to invoke the LIM-COUNT menu.
 1. **> UPPER** the number of readings that was over the set UPPER limit is displayed, including the time of the first passing.
 2. **< LOWER** the number of readings that was below the set LOWER limit is displayed, including the time of the first passing.
 3. **PASS** the number of PASSED readings is displayed.
 4. **TOTAL** the total number of measurements is displayed.
 5. **CLEAR** the LIMIT COUNTER memories of the present measurement function are cleared.
 0. **RETURN** to return to the OUTPUT menu.
 0. **EXIT** to exit the OUTPUT menu.

Press **GE** to exit the OUTPUT menu from any level.

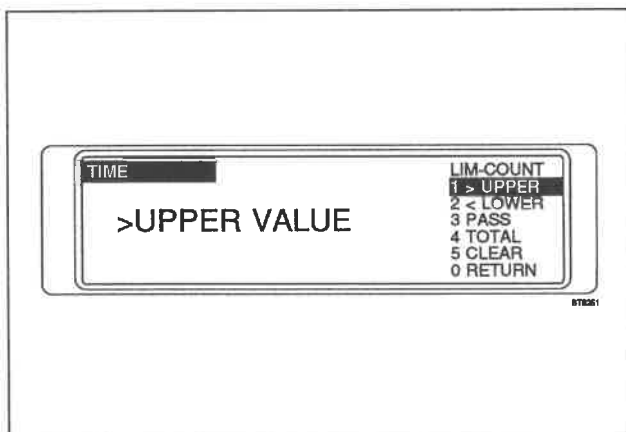


Fig. 4.63 Reading the UPPER LIMIT exceedings.

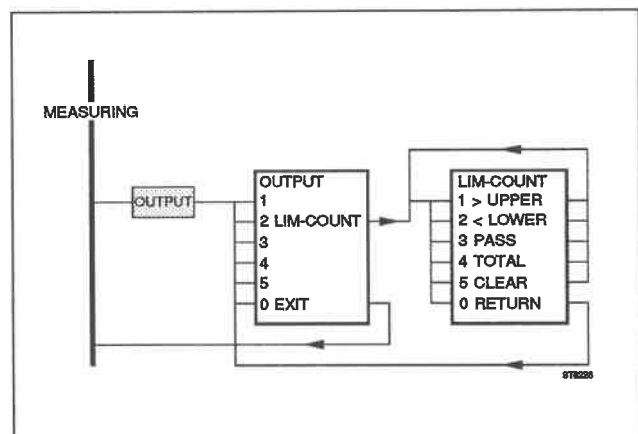


Fig. 4.64 Menu for reading the LIMIT COUNTERS.

4.13 SETTING THE DISPLAY LENGTH, ANNUNCIATORS AND BACKLIGHT

Using the OUTPUT function DISPLAY, you can set the display length, switch the annunciators on/off and control the display backlighting and contrast. The settings are valid for all measurement functions.

- press **OUTPUT** to invoke the OUTPUT menu (see Fig. 4.65).

- select:

4. DISPLAY

1. DIGITS

to invoke the DISPLAY menu.

to set the number of displayed digits. If set to 0 (zero), the main display is switched off. Notice that the display maximum display length depends on the measurement resolution. The latter depends on the selected reading rate.

2. ANNUNC

to set a number of annunciators OFF (toggle ON/OFF). The following annunciators will not be set OFF:

measurement result, the actual measurement function, and the OL (overload) indication in the highest Voltage and Current ranges.

3. BACKLIGHT

to set the display backlight ON or OFF (toggle function).

4. CONTRAST

to control the display contrast. Press **▲** or **▼** to increase or respectively decrease the contrast. Press **EXEC** to return to the DISPLAY menu.

0. RETURN

to return to the OUTPUT menu.

0. EXIT

to exit the OUTPUT menu.

Press **CE** to exit the OUTPUT menu from any level.

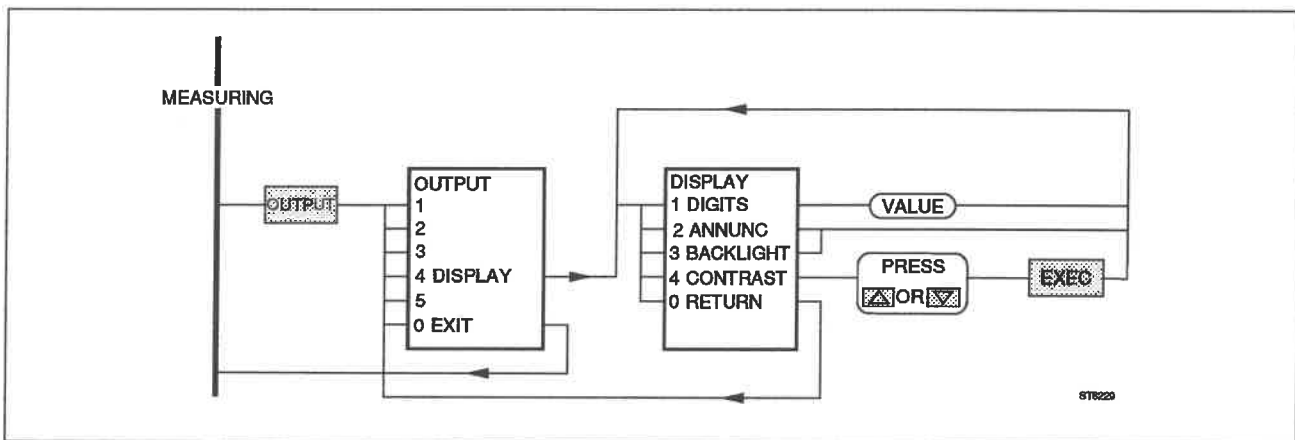


Fig. 4.65 DISPLAY control menu.

4.14 SETTING THE BEEPER CONDITIONS

Using the OUTPUT function BEEPER, you can set the beeper conditions. These conditions are valid in all measurement ranges.

NOTE: A beeper signal is always generated at overload in the highest range of the Voltage and Current measurement functions.

- press **OUTPUT** to invoke the OUTPUT menu.
- select

- | | |
|--------------------|--|
| 5. BEEPER | to invoke the BEEPER menu. |
| 1. ON | the beeper sounds on the following conditions;
* new MIN or MAX value, see Section 4.11.
* LIMIT exceeded, see Section 4.12.
* one of the conditions selected at 3. SELECT (below). |
| 2. OFF | to set the beeper to OFF. If you set the beeper ON at MIN/MAX (4.11) or LIMITS (4.12), this will cancel the OFF state. See the Note below. |
| 3. SELECT | to invoke the BEEP ON menu. |
| 1. OVERLOAD | the beeper is activated on overload. |
| 2. OVERFLOW | the beeper is activated on an arithmetic overflow. |
| 3. CREST | the beeper is activated when the crest factor is exceeded. |
| 4. CLIP | the beeper is activated if the PM2530 input circuit clips. |
| 0. RETURN | to return to the BEEPER menu. |
| 0. RETURN | to return to the OUTPUT menu. |
| 0. EXIT | to exit the OUTPUT menu. |

Press **CE** to exit the OUTPUT menu from any level.

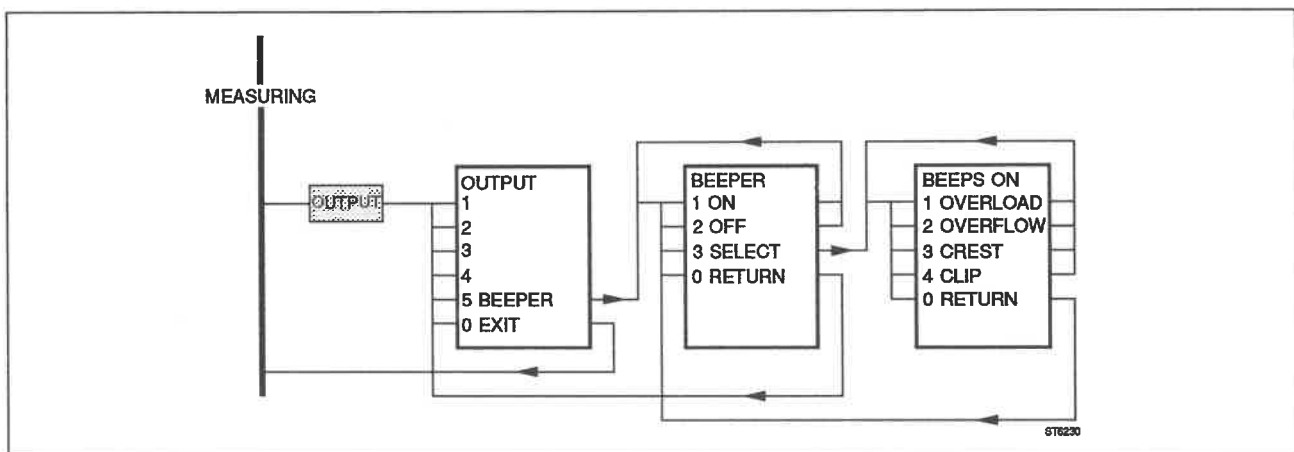


Fig. 4.66 BEEPER control menu.

4.15 SAVE AND RECALL INSTRUMENT CONFIGURATIONS

The SAVE function enables you to assign a number to the current instrument configuration and to store this numbered configuration. You can restore one of the saved instrument configurations using the RECALL function. Up to 24 instrument configurations can be stored. Configuration 0 and 1 are stored in non-volatile memory.

Proceed as follows:

- press **MISC** to invoke the MISC menu.
- select
 1. **RECALL** to restore a SAVED configuration:
 - key in the number (0 ... 24) of the required configuration.
 - press **ENTER**.
 2. **SAVE** to store the current instrument configuration:
 - key in the number (0 ... 24) to be assigned to this configuration.
 - press **ENTER**.
- 0. **EXIT** For details see below "STORED CONFIGURATION".
to leave the MISC menu.

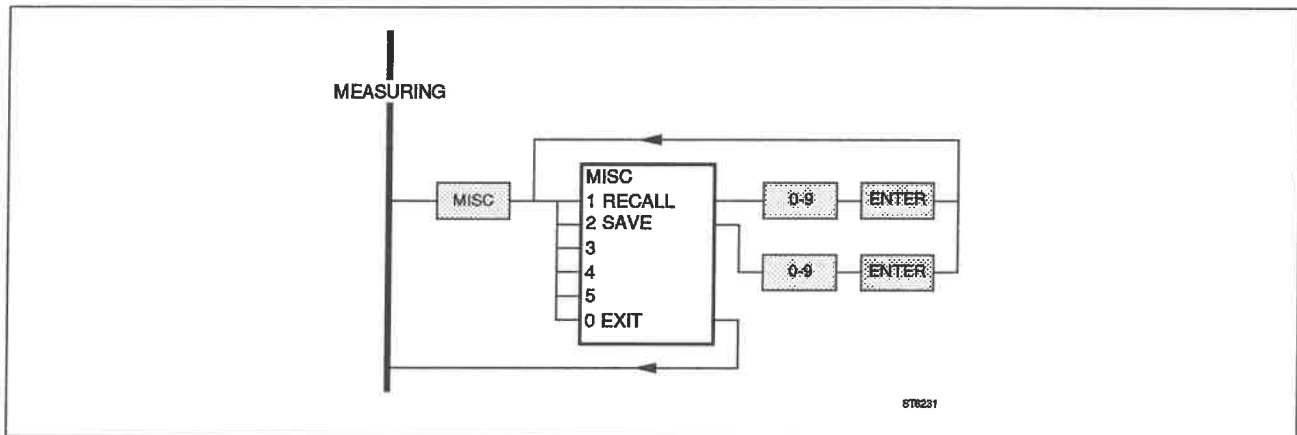


Fig. 4.67 SAVE/RECALL menu.

STORED CONFIGURATION

The stored configuration includes all settings and parameters, except:

- * parameters for reading the BURST memory (start location, length, step size, step rate).
- * display backlight and contrast setting.
- * GPIB address.
- * time and date.

Refer also to Appendix A, Section A.2.15.

NOTE: At power-up, the instrument configuration saved at "SAVE 0" is recalled. This enables you to assign any configuration as power-up configuration. Refer to Section 4.16.

4.16 CHANGING THE POWER-UP CONFIGURATION

On power-up (switching the meter on or after power loss), the settings stored in the SAVE/RECALL function in memory location 0 (SAVE 0, refer to Section 4.15), are recalled. This means that the meter assumes the saved measurement function including all its settings. Settings of other measurement functions (not included in the SAVE/RECALL function) are set to their default values (see Appendix A.2.15, settings after **RESET**).

You can change the power-up configuration to one that more closely meets your needs and preferences. Any combination of settings can become the power-up configuration.

Proceed as follows to save the present configuration as the new power-up configuration:

- Press **MISC**, **2**, **0**, **ENTER**. The current configuration is now the power-up configuration.

4.17 SELECTING DEFAULT INSTRUMENT SETTINGS

When **RESET** is pressed the instrument resumes the default settings for the present measurement function. The main default settings are shown in the table below. Settings of other measurement functions are not affected.

Table 4.15 Default instrument settings.

MEASUREMENT FUNCTION	UNCHANGED	BURST MODE	OFF
RANGING	AUTO	CALC. FUNCTIONS	OFF
TRIGGER MODE	INTERNAL	MIN MAX FUNCTION	OFF
READING RATE/SPEED MODE	SPD 2	LIMITS FUNCTION	OFF
FILTERS	OFF	BEEPER	OFF
RMS FILTER FREQUENCY	>40 Hz	DISPLAY LENGTH	MAX
REL. REF. MEASUREMENTS	OFF	ANNUNCIATORS	ON
INPUT OFFSET COMPENSATION	UNCHANGED		

Refer to Appendix A, Section A.2.15 for a detailed list of default settings.

4.18 SETTING THE INSTRUMENT DATE AND TIME

The instrument features a built-in, battery backed clock and calendar. Verify or change the settings as follows:

- press **MISC** to invoke the MISC menu.
- select

4. DATE/TIME

the current date is displayed. If required, key in a new date, in the format YY MM DD (Year Month Day). Press **ENTER** to complete the calendar setting. The current time is now displayed. If required, key in a new time, in the format HH : MM : SS (Hour : Minute : Second). Press **ENTER** to complete the TIME setting.

0. EXIT

to leave the MISC menu.

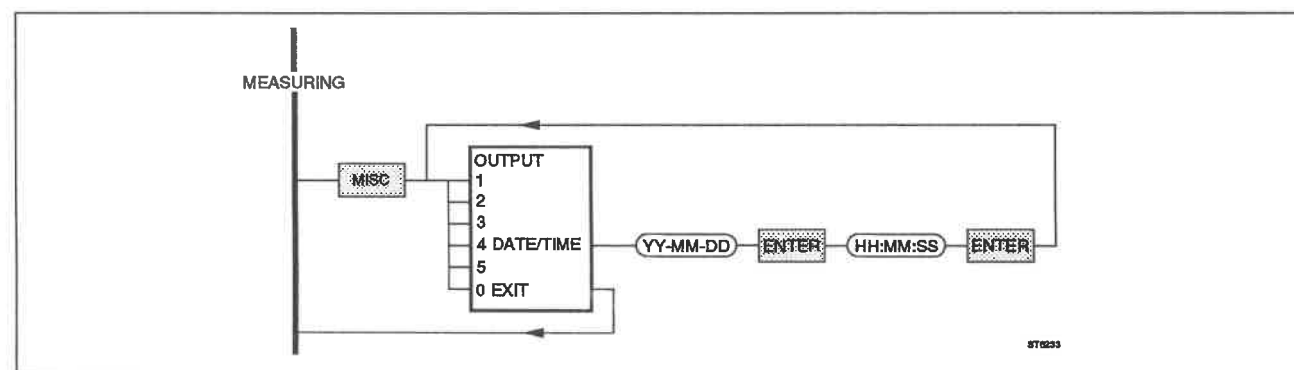


Fig. 4.69 DATE/TIME selection menu.

4.19 SETTING THE GPIB ADDRESS

If the PM2530 is to be operated under remote control, via its GPIB interface, it must be assigned an address ("0" to "30", inclusive). If you want to change this address, proceed as follows:

- press **MISC** to invoke the MISC menu.
- select:
 - 3. IEEE ADDR** the current GPIB address is displayed. If you want to change the address, key in a new value (0 ... 30) and press **ENTER**. The new address is now valid.
 - 0. EXIT** to leave the MISC menu.

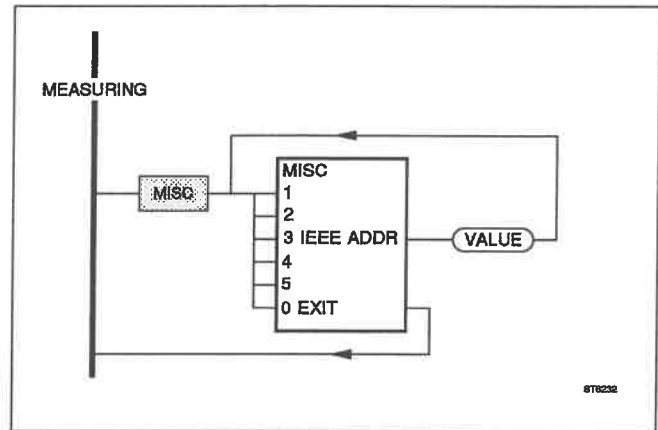


Fig. 4.68 GPIB address setting menu.

Note: The GPIB address cannot be changed when the PM2530 is in REMOTE.

4.20 RETURNING FROM REMOTE TO LOCAL CONTROL

If the PM2530 is in the remote state without front panel lock out (Local Lock Out), pressing **LOCAL** returns it to local (i.e., front panel) control.

4.21 REQUESTING SERVICE VIA THE SRQ BUTTON

If the PM2530 is in the remote state without front panel lock out (Local Lock Out), a service request can be sent from the front panel by pressing **SRQ**. If **SRQ** is pressed, the URQ (User Request) bit in the Standard Event Status Register will be set (i.e., 1). Refer to the Command Reference Manual *ESR.

5 REMOTE CONTROL

With the GPIB (=IEEE-488.2) interface the PM 2530 is a fully programmable instrument for use in an automated instrumentation system. This section assumes you are familiar with GPIB interface and Standard Commands for Programmable Instruments (SCPI). For an introduction to the GPIB interface and SCPI refer to the book "SCPI Standard Commands for Programmable Instruments" which is supplied with the PM2530.

Both the short and long forms of the SCPI commands may be used. The short form is shown in upper case in a syntax specification. The remaining part of the long form is shown in lower case. Note that this does not specify the use of upper and lower case characters in commands entered. This notation provides a distinction between the long and short forms of the SCPI commands.

The GPIB interface provides remote control of all front panel controls except for the POWER, INPUTS FRONT/REAR, GUARD INT/EXT switches, and setting of the DISPLAY CONTRAST and the GPIB bus address.

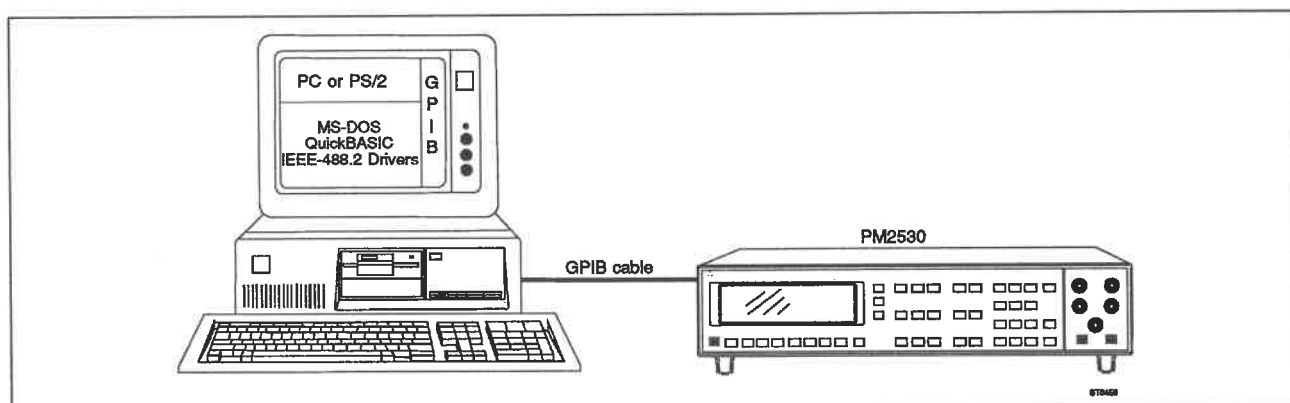
The PM2530 supports SCPI, this results in it being very flexible in accepting a wide range of command and parameter formats. This makes programming the PM2530 very straight forward.

5.1 GETTING STARTED

5.1.1 PROGRAMMING THE PM2530

To communicate with the PM2530 through the GPIB interface, you need a programming language which runs on your controller. In order to perform the program examples in this manual you need a Personal Computer + GPIB interface card + software GPIB drivers + a programming language. The Philips/Fluke IEEE-488.2 drivers (PM2201 V3.0 / PM2202 V2.0) provide the hardware interface and software drivers (PC2A.2 or MC-GPIB.2 from National Instruments may also be used). QuickBASIC 4.5 has been chosen as programming language environment.

Tutorial set-up:



Before starting this tutorial, the following actions and checks must be carried out:

- 1) An GPIB interface card must be installed in the controller, e.g.:
 - the PM2201 V3.0 interface board for IBM PC/XT/AT/PS/2 model 30 or compatible controllers
 - the PM2202 V2.0 interface board for PS/2 model 50 (and higher) controllers
- 2) Connect the PM2530 to the PC or PS/2 (MS-DOS) via an GPIB interface cable.
- 3) QuickBASIC and the IEEE-488.2 drivers must have been installed.

- 4) Make sure that the quick library QBIB.QLB is produced. This can be done by typing the command:
LINK /Q QBIB.OBJ,,,BQLB45.LIB (see section 1.3.1 in the PM2201-PM2202 IEEE-4882 Software Drivers "Reference Manual").
- 5) Start the QuickBASIC interactive environment with the command: qb /1 QBIB.QLB
This will load the QBIB.QLB Quick Library when the program is started.
- 6) Now type the following GPIB program.

```

REM $INCLUDE:'qbdecl.bas'
REM '***** PROGRAM NAME: INTERACT.BAS *****'
DIM response AS STRING * 5000                                'Size of response string
board = 0 'Board number
CALL SendIFC(board)                                           'Clear the GPIB interface
CALL IBTMO(board, 13)                                         'Timeout 10 seconds
CLS
INPUT "GPIB - address = "; ADDRESS                            'Enter the instruments address
PRINT
DO
  LINE INPUT "Command ? "; command.string$                  'Enter the command/query string
  PRINT
  CALL Send(board, ADDRESS, command.string$, NLen)           'Send the command
  IF INSTR(command.string$, "?") <> 0 THEN                     'Test if a query was send
    RESPONSE $ = CHR$(10)                                     'Clear response string
    CALL Receive(board, ADDRESS, response$, NLen)            'Read the response
    PRINT "Response = ";
    PRINT LEFT$(response$, INSTR(response$, CHR$(10)))       'Print the response left of NLen
  END IF
  DO
    CALL Send(board, ADDRESS, ":SYSTem:ERRor?", NLen)        'Send the query to read error queue
    CALL Receive(board, ADDRESS, response$, NLen)            'Read the response
    IF INSTR(response$, "No error") = 0 THEN                  'Test if error occurred
      PRINT "Error = ";
      PRINT LEFT$(response$, INSTR(response$, CHR$(10)))     'Print error if occurred
    END IF
  LOOP UNTIL INSTR(response$, "No error") <> 0                'Read Error queue until "No error"
LOOP UNTIL command.string$ = ""                                'Test for empty string
CALL SendIFC(board)                                           'Clear the GPIB interface
END                                                            'End of program

```

- 7) Select Save As...from the FILE menu to save the program as INTERACT.BAS.
- 8) Select Start from the Run menu to start the program.

This program allows you to send Commands and Queries to the PM2530 and to receive the responses from Queries, on an interactive base. You have to enter the GPIB address of the instrument (for example 22 is the address on delivery, see Section 4.18 on setting the GPIB interface address). Now you may enter command or query strings. When a query is sent, the program will detect the "?" in the string and ask the response from the instrument. After the response is read, the program will read the Error Queue and display its contents on the screen if any error was made. If you just press the Enter key, you enter an empty string terminating the program. You may restart the program to send commands and queries to other instruments with other addresses. This program may be used to run the following tutorial examples, so that we can concentrate on the instrument program data rather GPIB programming matters.

5.1.2 LEARNING TO PROGRAM THE PM2530

When learning to program an SCPI compatible instrument, such as the PM2530 multimeter, it is essential to check that the commands have been entered correctly. Incorrect commands are not executed by the instrument. When an error occurs during programming, an error message is placed in the instrument's Error Queue. This error message remains in the queue until it is read and automatically cleared. The next error is added to the queue, this continues until the queue is full. When you read the Error Queue by sending the :SYSTem:ERRor? query to the instrument, it responds with the first (oldest) error in the Queue. The next query gives the second error and so forth, until all errors have been read and the Error Queue is empty. The program described above will automatically read and display the contents of the error queue after each command or query. When you start writing your own programs, it is best to read the instrument's Error Queue each time a command is sent to the instrument. Especially in the beginning, when you are not yet familiar with the instrument's command set.

Example using the program described above:

Command	? :ABC	Undefined command.
Error	= -113, "Undefined header"	The error is reported.
Command	? *IDN?	This is the Identity Query.
Response	= PHILIPS,PM2530,<serial.nr>,Vxx.yy	This is the PM2530 identity data, Vxx.yy is the firmware revision level.
Command	? :MEASure:VOLTage:DC?	Perform a DC voltage measurement.
Response	= -1.741716E+00	Reading of the instrument.
Command	? :CONFigure:VOLTage:DC;:INPut:FILTer ON;:READ?	Setup the instrument for a DC voltage measurement enable the analog low pass filter and start the DC voltage measurement .
Response	= -1.650000E+00	Reading of the instrument.
Command	? :SYSTem:TIME?;:MEASure:VOLTage:AC;	Read the system time of the instrument and perform a AC voltage measurement
Response	= 9,59,15.11	No measurement result of the instrument due to an ERROR, ? is missing.
Error	= -110, "Command header error"	The ? is missing at the end of the command string.
Command	? :SYSTem:TIME?;:MEASure:VOLTage:AC?	The error is corrected.
Response	= 10,2,13.67;-0.01102E-02	System time and measurement reading.
Command	? :CONFigure:ARRay:VOLTage:DC (10),DEFault	Setup BURST measurement mode.
Command	? :SAMPLE:SOURCE TIMER;TIMER 1	Take a sample every second.
Command	? :INITiate:IMMediate	Start the measurement.
Command	? :TRACE:DATA? POSTtrigger,2,1	Retrieve the BURST results 2 samples starting at posttrigger sample 1.
Response	= 1.008331E+01,1.008335E+01	Result at sample 1 and 2

In this way it is easy to become familiar with SCPI, and the programming benefits it offers in programming your PM2530.

5.2 SETTING UP REMOTE CONTROLLED MEASUREMENTS

This section provides a number of examples of how to set up the PM2530 to make measurements under the control of the GPIB bus. This section assumes that you are familiar with Section 4, and the basic principles of setting up a measurement. You can use the program from Section 5.1.1 to set up the PM2530 with the SCPI commands according to the following descriptions. The commands shown are in short form notation and omit default nodes.

Note: *Commands related to the instruments measurement configuration can only be accepted in the IDLE state of the trigger model (see section 5.3.18). If the instrument is not in the IDLE state ERROR -221 "Settings conflict" is generated.*

After the Power is switched ON or when a measurement configuration change is made via the front panel the instrument will not be in the IDLE state, but it is continuously measuring for "BENCH" use.

The instrument can be put in the idle state with the command;

```
:INITiate:CONTinuous OFF;:ABORt
```

The :MEASure and :CONFigure commands will automatically turn the instrument in the idle state. The command:

```
:INITiate:CONTinuous ON
```

will enable continuous measuring again for "BENCH" use.

5.2.1 STRAIGHTFORWARD MEASUREMENTS

The measure commands provide a powerful tool for performing measurements. While use of these commands is very simple, they do not offer the possibility of using the auxiliary measurement functions. With the measure commands you may specify the expected measurement value and the desired resolution. The instrument then selects the most suitable range and measuring speed for that value. If you omit these parameters, autoranging and speed mode SPD2 are automatically selected. The measure commands always select immediate sampling with no delay. See the Command Reference Manual for more details on the :MEASure commands. See Table in appendix A.2.15 (SET BY :MEAS) for details of the settings affected by the measure command.

Volts DC measurement

MEAS:VOLT?

MEAS:VOLT? 15,0.001

Autoranging, speed mode SPD2.

Range 20V (15V is expected), speed mode SPD5 (resolution of 0.001V is specified).

Volts AC measurement

MEAS:VOLT:AC?

MEAS:VOLT:AC? 15,0.001

Autoranging, speed mode SPD2.

Range 20V (15V is expected), speed mode SPD3 (resolution of 0.001V is specified).

Volts AC+DC measurement

MEAS:VOLT:ACDC?

MEAS:VOLT:ACDC? 15,0.001

Autoranging, speed mode SPD2.

Range 20V (15V is expected), speed mode SPD3 (resolution of 0.001V is specified).

Low frequency volts measurement

MEAS:VOLT:LFAC?

MEAS:VOLT:LFAC? 15,0.001

Autoranging, speed mode SPD2.

Range 20V (15V is expected), speed mode SPD2 selected automatically.

Frequency measurement

MEAS:VOLT:FREQ?

MEAS:VOLT:FREQ? (15,DEF),DEF,MAX

Autoranging, speed mode SPD2.

The volt range in which the measurement is made is set to 20V (15V is expected), the voltage speed mode can not be set, the speed is mode is determined by the frequency mode. The frequency range is always selected automatically, the frequency speed is set to speed mode SPD3 (Maximum).

Period measurement

MEAS:VOLT:PER?

MEAS:VOLT:PER? 15,DEF)DEF,MAX

Autoranging, speed mode SPD2.

The volt range in which the measurement is made is set to 20V (15V is expected) the voltage speed mode can not be set, the speed is determined by the period mode. The period range is always selected automatically, the speed mode SPD3 (Maximum).

Current DC measurement

MEAS:CURRE?

MEAS:CURRE? 15E-3,1E-6

Autoranging, speed mode SPD2.

Range 20mA (15mA is expected), speed mode SPD5 (resolution of 1μA is specified).

Current AC+DC measurement

MEAS:CURRE:ACDC?

MEAS:CURRE:ACDC? 1.5,0.0001

Autoranging, speed mode SPD2.

Range 2A (1.5A is expected), speed mode SPD3 (resolution of 0.0001A is specified).

2 Wire resistance measurement

MEAS:RES?

MEAS:RES? 1E3,1

Autoranging, speed mode SPD2.

Range 2kΩ (1kΩ is expected), speed mode SPD2 (resolution of 1Ω is specified).

4 Wire resistance measurement

```
MEAS:FRES?  
MEAS:FRES? 1E3,1
```

Autoranging, speed mode SPD2.

Range $2k\Omega$ ($1k\Omega$ is expected), speed mode SPD2 (resolution of 1Ω is specified).

Temperature measurement

```
UNIT:TEMP CEL  
MEAS:TEMP?  
  
MEAS:TEMP? FRTD,390,25,0.1
```

Select temperature unit degree Celsius.

Autoranging, speed mode SPD2, DIN 43760 linearisation characteristic.

Transducer type FRTD, NBS 390 PT linearisation characteristic, Range -200°C .. 850°C only one range possible, speed mode SPD3 (resolution of 0.1°C is specified).

Quick Basic Example for (PM2201-PM2202) IEEE-488.2 Drivers

```
REM $INCLUDE: 'qbdecl.bas'  
***** VOLTAGE AC, DC, AC+DC MEASUREMENT EXAMPLE *****  
DIM response AS STRING * 80  
board = 0  
ADDRESS = 22  
CALL SendIFC(board)  
CALL IBTMO(board, 13)  
CALL Send(board, ADDRESS, ":MEAS:VOLT:DC? 15,0.001", NLen)  
CALL Receive(board, ADDRESS, response$, NLen)  
PRINT "DC value = ";  
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))  
CALL Send(board, ADDRESS, ":MEAS:VOLT:AC? 15,0.001", NLen)  
CALL Receive(board, ADDRESS, response$, NLen)  
PRINT "AC value = ";  
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))  
CALL Send(board, ADDRESS, ":MEAS:VOLT:ACDC? 15,0.001", NLen)  
CALL Receive(board, ADDRESS, response$, NLen)  
PRINT "RMS value = ";  
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))  
CALL SendIFC(board)  
END
```

'Size of response string
'Board number = 0
'Address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Measure Volt DC range 20V SPD5
'Read the response

'Print the response left of NLen
'Measure Volt AC range 20V, SPD3
'Read the response

'Print the response left of NLen
'Measure Volt AC+DC range 20V, SPD3
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2 ADVANCED MEASUREMENTS

This section gives an overview of how a measurement can be set up using the PM2530 auxiliary measurement functions. The examples in this section use the configure subsystem commands to select the measurement function. This ensures the instrument is configured to a known state. See Table in appendix A.2.15 (SET BY :CONF) for settings affected by the configure commands.

If the measurement function is required to have different settings from the :CONFigure command, the :SENSe:FUNCTion command can be used. This selects the measurement function and restores the settings that were last used with that function. See Table in appendix A.2.15 (STORED PER MEAS.FUNC) for details of which settings are recalled when the measurement function is reselected.

All examples regarding "Reading rate" are for 50 Hz line voltage. For 60 Hz line voltage settings refer to sections 4.5.3 and section 14.2 of the Command Reference Manual.

5.2.2.1 Volts DC Measurements

Measurement function

:CONF:VOLT

The PM2530 is configured for a single volt DC measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on :CONFigure commands.

Ranging

:VOLT:RANG 20E-3

Select the 20 mV range. The PM2530 provides no automatic down ranging to the 20 mV range, and in speed mode SPD1 to 200 mV range. If this range is to be used it must be set explicit.

Reading rate

:VOLT:APER 2E-2

Set the aperture of the Volt DC measurement function. This will select speed mode SPD 3. See also the :SENSe:VOLTage:DC:RESolution command and section 4.5.3 for details.

Input offset compensation

:CORR:STAT ON

:CORR:COLL STAN

Enable input offset compensation.

A measurement is taken and the measurement data is used as nulling data for the actual range of the volt DC measurement function. Refer to section 4.8 for a detailed description.

Relative reference measurements

:CALC:REF:ACQ

:CALC:REF:STAT ON

The last available measurement data is used as reference value for the relative reference measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

Filters

:INP:FILT ON

:CALC:AVER:EXP:WEIG 2

:CALC:AVER:EXP ON

:CALC:AVER:COUN 50

:CALC:AVER ON

Enable the analog low pass, 5 Hz, 20 dB/decade filter to eliminate external noise.

Set the weight factor to be used for the exponential average calculation.

Enable the digital weighting filter (exponential averaging).

Set the number of measurements to be used for averaging.

Enable the digital averaging filter.

Extremes function

```
:SYST:BEEP:STAT ON
:CALC:EXTR:SIGN:BEEP ON
:CALC:EXTR:STAT ON
```

Enable the beeper

Enable the beeper to report the measurement of an extreme value. Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

Triggering

```
:SAMP:TIM 30
:SAMP:SOUR TIM
```

Select the timer as sample source to trigger the measurements. Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.

Start the measurement

```
:INIT
```

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The :READ? command contains the :ABORT;:INIT;:FETC? commands and may also be used here to start the measurement and read the data.

Read the measurement data

```
:FETC?
```

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

```
:CALC:LOG:STAT ON
```

Enable the logarithmic calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

```
:READ?
```

A new measurement is started and the measurement data is retrieved, the reading is now a dBm value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** VOLTAGE DC MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:VOLT", NLen)
CALL Send(board, ADDRESS, ":VOLT:RANG 20E-3", NLen)
CALL Send(board, ADDRESS, ":INP:FILT ON", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Analog filtered Volt DC measurement = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Volts DC Autoranging SPD2
'Select 20 mV range
'Enable the analog filter
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.2 Volts AC Measurements

Measurement function

:CONF:VOLT:AC

The PM2530 is configured for a single volt AC measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on :CONF:VOLT:AC commands.

Ranging

:VOLT:AC:RANG 20E-3

Select the 20 mV range. The PM2530 provides no automatic down ranging to the 20 mV range in speed mode SPD1. If this range is to be used it must be set explicit. For ranging details see section 4.5.1.

Reading rate

:VOLT:AC:APER 2E-2

Set the speed of the Volt DC measurement function to speed mode SPD3. See also the :SENSe:VOLTage:AC:RESolution commands and section 4.5.3 for details.

Relative reference measurements

:CALC:REF:ACQ

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 8 of the Command Reference Manual for details. Enable the relative reference function.

:CALC:REF:STAT ON

Filters

:INP:FILT:RMS:FREQ 400

Set the RMS filter frequency. For optimal result a frequency that is as close as possible below the signal frequency must be selected. (Possible selections are 10Hz, 40Hz, 400Hz.) Selecting a lower frequency will result in a longer response time and selecting a higher frequency will result in a instable reading. Set the weight factor to be used for the exponential average calculation.

:CALC:AVER:EXP:WEIG 2

Enable the digital weighting filter (exponential averaging). Set the number of measurements to be used for averaging. Enable the digital averaging filter.

:CALC:AVER:EXP ON

:CALC:AVER:COUN 50

:CALC:AVER:STAT ON

Extremes function

:SYST:BEEP:STAT ON

Enable the beeper

:CALC:EXTR:SIGN:BEEP ON

Enable the beeper to report the measurement of an extreme value. Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

:CALC:EXTR:STAT ON

Triggering

:SAMP:SOUR TIM

Select the timer as sample source to trigger the measurements. Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.

:SAMP:TIM 30

Start the measurement

:INIT

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The :READ? command contains the :ABORT::INIT::FETC? commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:LOG:STAT ON

Enable the logarithmic calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

READ?

A new measurement is started and the measurement data is retrieved, the reading is now a dBm value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** VOLTAGE AC MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:VOLT:AC", NLen)
CALL Send(board, ADDRESS, ":VOLT:AC:RANG 20E-3", NLen)
CALL Send(board, ADDRESS, ":CALC:LOG:STAT ON", NLen)

CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "dBm reading Volt AC measurement = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Volts AC Autoranging SPD2
'Select 20 mV range
'Enable the dBm "logarithmic" calculation
'function
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.3 Volts AC+DC Measurements (ACDC)

Measurement function

`:CONF:VOLT:ACDC`

The PM2530 is configured for a single volt AC+DC measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONF` commands.

Ranging

`:VOLT:ACDC:RANG 2`

Select the 2 V range. For ranging details see section 4.5.1.

Reading rate

`:VOLT:ACDC:APER 2e-2`

Set the speed of the volt AC+DC measurement function to speed mode SPD 3. See also the `:SENSe:VOLTage:ACDC:RESolution` commands and section 4.5.3 for details.

Relative reference measurements

`:CALC:REF:ACQ`

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

`:CALC:REF:STAT ON`

Filters

`:INP:FILT:RMS:FREQ 400`

Set the RMS filter frequency. For optimal result a frequency that is as close as possible below the signal frequency must be selected. (Possible selections are 10Hz, 40Hz, 400Hz.) Selecting a lower frequency will result in a longer response time and selecting a higher frequency will result in a instable reading. Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:WEIG 2`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:EXP:STAT ON`

Set the number of measurements to be used for averaging.

`:CALC:AVER:COUN 50`

Enable the digital averaging filter.

`:CALC:AVER:STAT ON`

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities. Select the timer as sample source to trigger the measurements

`:SAMP:SOUR TIM`

Start the measurement

`:INIT`

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The `:READ?` command contains the `:ABORT;:INIT;:FETC?` commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:LOG:STAT ON

Enable the logarithmic calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now a dBm value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** VOLTAGE ACDC MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:VOLT:ACDC", NLen)
CALL Send(board, ADDRESS, ":VOLT:ACDC:RANG 20E-3", NLen)
CALL Send(board, ADDRESS, ":CALC:LOG:STAT ON", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "dBm reading Volt AC+DC measurement = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Volts AC+DC
'Autoranging SPD2
'Select 20 mV range
'Enable the dBm "logarithmic"
'calculation function
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.4 Low Frequency Voltage Measurements (VLF)

Measurement function

`:CONF:VOLT:LFAC`

The PM2530 is configured for a single low frequency volt AC+DC measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONF` commands.

Ranging

`:VOLT:LFAC:RANG 2`

Select the 2 V range. For ranging details see section 4.5.1.

Reading rate

`:VOLT:LFAC:APER 20`

Set the speed at the VLF measurement function to speed mode SPD 3. See also the `:SENSe:VOLTage:LFAC RESolution` commands and section 4.5.3 for details

Relative reference measurements

`:CALC:REF:ACQ`

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details
Enable the relative reference function.

`:CALC:REF:STAT ON`

Filters

`:INP:FILT:LPAS:STAT ON`

Enable the analog low pass, 5 Hz, 20 dB/decade filter to eliminate external noise.

`:CALC:AVER:EXP:WEIG 2`

Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:STAT ON`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:COUN 50`

Set the number of measurements to be used for averaging.

`:CALC:AVER:STAT ON`

Enable the digital averaging filter.

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:SOUR TIM`

Select the timer as sample source to trigger the measurements

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 20 of the Command Reference Manual for more information on the various trigger possibilities.

Start the measurement

`:INIT`

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The `:READ?` command contains the `:ABORT`; `:INIT`; `:FETC?` commands and may also be used here to start the measurement and read the data.

Read the measurement data

`:FETC?`

The measurement data is retrieved when the measurement is finished. The `FETCH?` query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

CALC:LOG:STAT ON

Enable the logarithmic calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:FETC?

The same measurement data is retrieved again, but the reading is now a dBm value.

NOTE: Burst measurements are not possible in the Low Frequency Voltage measurement function.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** LOW FREQUENCY VOLTAGE MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:VOLT:LFAC", NLen)
CALL Send(board, ADDRESS, ":VOLT:LFAC:RANG 20", NLen)
CALL Send(board, ADDRESS, ":CALC:LOG:STAT ON", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Volt Low Frequency measurement = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Volts Low Frequency
'Autoranging SPD2
'Select 20 V range
'Enable the dBm "logarithmic" calculation
'function
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.5 Frequency Measurements

Measurement function

`:CONF:VOLT:FREQ`

The PM2530 is configured for a single frequency measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONF` commands.

Ranging

`:VOLT:FREQ:RANG 2`

Select the 2 V range. For ranging details see section 4.5.1.

Reading rate

`:FREQ:APER 2E-2`

Set the speed of the frequency measurement function to speed mode SPD3. See also the `:SENSe:FREQuency:RESolution` commands and section 4.5.3 for details.

Relative reference measurements

`:CALC:REF:ACQ`

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

`:CALC:REF:STAT ON`

Filters

`:CALC:AVER:EXP:WEIG 2`

Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:STAT ON`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:COUN 50`

Set the number of measurements to be used for averaging.

`:CALC:AVER:STAT ON`

Enable the digital averaging filter.

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.

`:SAMP:SOUR TIM`

Select the timer as sample source to trigger the measurements

Start the measurement

`:INIT`

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The `:READ?` command contains the `:ABORT`; `:INIT`; `:FETC?` commands and may also be used here to start the measurement and read the data.

Read the measurement data

`:FETC?`

The measurement data is retrieved when the measurement is finished. The `FETCH?` query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:PERC:REF:ACQ

The last available measurement data is used as reference value for the percentage calculation.

:CALC:PERC:STAT ON

Enable the percentage calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now the deviation in terms of percentage of the reference value.

NOTE: Burst measurements are not possible in the frequency measurement function.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** FREQUENCY MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:VOLT:FREQ", NLen)

CALL Send(board, ADDRESS, ":VOLT:FREQ:RANG 20", NLen)

CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Frequency measurement = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Frequency Measurement
'Autoranging SPD2
'Select 20 V range to be used for
'frequency measurements
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.6 Period Measurements

Measurement function

`:CONF:VOLT:PER`

The PM2530 is configured for a single period time measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONFigure` commands.

Ranging

`:VOLT:PER:RANG 2`

Select the 2 V range. For ranging details see section 4.5.1.

Reading rate

`:PER:APER 2E-2`

Set the speed of the period measurement function to speed mode SPD3. See also the `:SENSe:PERiod:RESolution` commands and section 4.5.3 for details.

Relative reference measurements

`:CALC:REF:ACQ`

A measurement is taken and the measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details

`:CALC:REF:STAT ON`

Enable the relative reference function.

Filters

`:CALC:AVER:EXP:WEIG 2`

Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:STAT ON`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:COUN 50`

Set the number of measurements to be used for averaging.

`:CALC:AVER:STAT ON`

Enable the digital averaging filter.

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.

`:SAMP:SOUR TIM`

Select the sample as trigger source to trigger the measurements

Start the measurement

`:INIT`

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The `:READ?` command contains the `:ABORT`; `:INIT`; `:FETC?` commands and may also be used here to start the measurement and read the data.

Read the measurement data

`:FETC?`

The measurement data is retrieved when the measurement is finished. The `FETCH?` query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:PERC:REF:ACQ

The last available measurement data is used as reference value for the percentage calculation.

:CALC:PERC:STAT ON

Enable the percentage calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now the deviation in terms of percentage of the reference value.

NOTE: Burst measurements are not possible in the period measurement function.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** PERIOD MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:VOLT:PER", NLEnd)
CALL Send(board, ADDRESS, ":VOLT:PER:RANG 20", NLEnd)
CALL Send(board, ADDRESS, ":INIT", NLEnd)
CALL Send(board, ADDRESS, ":FETC?", NLEnd)
CALL Receive(board, ADDRESS, response$, NLEnd)
PRINT "Period measurement = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Period Measurement
'Autoranging SPD2
'Select 20 V range to be used
'for period measurements
'Start the measurement
'Retrieve the result
'Read the response
'Print the response left of NLEnd
'Clear the GPIB interface
'End of program
```

5.2.2.7 Current DC Measurements

Measurement function

:CONF:CURR

The PM2530 is configured for a single current DC measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on :CONFigure commands.

Ranging

:CURR:RANG 2E-6

Select the 2 uA range. The PM2530 provides no automatic down ranging to the 2 uA range in speed mode SPD1. If this range is to be used it must be set explicit. For ranging details see section 4.5.1.

Reading rate

:CURR:DC:APER 2E-2

Set the speed of the current DC measurement function to speed mode SPD3. See also the :SENSe:CURRent:DC:RESolution commands and section 4.5.3 for details.

Relative reference measurements

:CALC:REF:ACQ

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

:CALC:REF:STAT ON

Filters

:CALC:AVER:EXP:WEIG 2

Set the weight factor to be used for the exponential average calculation.

:CALC:AVER:EXP:STAT ON

Enable the digital weighting filter (exponential averaging).

:CALC:AVER:COUN 50

Set the number of measurements to be used for averaging.

:CALC:AVER:STAT ON

Enable the digital averaging filter.

Extremes function

:CALC:EXTR:STAT ON

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

:CALC:EXTR:SIGN:BEEP ON

Enable the beeper to report the measurement of an extreme value.

Triggering

:SAMP:TIM 30

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities. Select the timer as sample source to trigger the measurements

:SAMP:SOUR TIM

Start the measurement

:INIT

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The :READ? command contains the :ABORT;:INIT;:FETC? commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:LOG:STAT ON

Enable the logarithmic calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now a dBm value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM ' ***** CURRENT DC MEASUREMENT EXAMPLE ***** '
REM $INCLUDE: 'qbdecl.bas'                                'Program declarations
DIM response AS STRING * 80                               'Size of response string
board = 0                                                  'Board number = 0
ADDRESS = 22                                               'PM2530 address = 22 at delivery
CALL SendIFC(board)                                       'Clear the GPIB interface
CALL IBTMO(board, 13)                                     'Timeout 10 seconds
CALL Send(board, ADDRESS, ":CONF:CURR", NLEnd)            'Configure for Current DC Autoranging SPD2
CALL Send(board, ADDRESS, ":CURR:RANG 20E-3", NLEnd)      'Select 20 mA range
CALL Send(board, ADDRESS, ":CALC:AVER:COUN 10", NLEnd)    '10 Measurements used for averaging
CALL Send(board, ADDRESS, ":CALC:AVER:STAT on", NLEnd)    'Enable averaging filter
CALL Send(board, ADDRESS, ":INIT", NLEnd)                'Start the measurement
CALL Send(board, ADDRESS, ":FETC?", NLEnd)               'Retrieve the result
CALL Receive(board, ADDRESS, response$, NLEnd)           'Read the response
PRINT "Average of 10 Current DC measurements = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))      'Print the response left of NLEnd
CALL SendIFC(board)                                       'Clear the GPIB interface
END                                                       'End of program
```

5.2.2.8 Current AC+DC Measurements

Measurement function

`:CONF:CURR:ACDC`

The PM2530 is configured for a single current AC+DC measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONFigure` commands.

Ranging

`:CURR:ACDC:RANG 200E-3`

Select the 200 mA range. For ranging details see section 4.5.1.

Reading rate

`:CURR:ACDC:APER 2E-2`

Set the speed of the current AC+DC measurement function to speed mode SPD3. See also the `:SENSe:CURRent:ACDC:RESolution` commands and section 4.5.3 for details.

Relative reference measurements

`:CALC:REF:ACQ`

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

`:CALC:REF:STAT ON`

Filters

`:INP:FILT:RMS:FREQ 400`

Set the RMS filter frequency. For optimal result a frequency that is as close as possible below the signal frequency must be selected. (Possible selections are 10Hz, 40Hz, 400Hz.) Selecting a lower frequency will result in a longer response time and selecting a higher frequency will result in a instable reading. Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:WEIG 2`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:EXP:STAT ON`

`:CALC:AVER:COUN 50`

Set the number of measurements to be used for averaging.

`:CALC:AVER:STAT ON`

Enable the digital averaging filter.

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 5.3.2 for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities. Select the timer as sample source to trigger the measurements

`:SAMP:SOUR TIM`

Start the measurement

`:INIT`

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The `:READ?` command contains the `:ABORT::INIT::FETC?` commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:LOG:STAT ON

Enable the logarithmic calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now a dBm value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** CURRENT ACDC MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:CURR:ACDC", NLen)

CALL Send(board, ADDRESS, ":CURR:RANG 20E-3", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:COUN 10", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:STAT on", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Average of 10 Current AC + DC measurements = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Current ACDC
'Autoranging SPD2
'Select 20 mA range
'10 Measurements used for averaging
'Enable averaging filter
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.9 Resistance Measurements 2 Wire

Measurement function

`:CONF:RES`

The PM2530 is configured for a single 2 wire resistance measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONF` commands.

Ranging

`:RES:RANG 20`

Select the 20 Ω range. For ranging details see section 4.5.1.

Reading rate

`:RES:APER 2E-2`

Set the speed of the resistance measurement function to speed mode SPD3. See also the `:SENSe:RESistance:RESolution` commands and section 4.5.3 for details.

Relative reference measurements

`:CALC:REF:ACQ`

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

`:CALC:REF:STAT ON`

Filters

`:INP:FILT:LPAS:STAT ON`

Enable the analog low pass, 5 Hz, 20 dB/decade filter to eliminate external noise.

`:CALC:AVER:EXP:WEIG 2`

Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:STAT ON`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:COUN 50`

Set the number of measurements to be used for averaging.

`:CALC:AVER:STAT ON`

Enable the digital averaging filter.

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.

`:SAMP:SOUR TIM`

Select the timer as sample source to trigger the measurements

Start the measurement

`:INIT`

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The `:READ?` command contains the `:ABORT::INIT::FETC?` commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:PERC:REF:ACQ

The last available measurement data is used as reference value for the percentage calculation.

:CALC:PERC:STAT ON

Enable the percentage calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now the deviation in terms of percentage of the reference value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** 2 WIRE RESISTANCE MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:RES", NLen)

CALL Send(board, ADDRESS, ":RES:RANG 20E3", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:COUN 10", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:STAT on", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Average of 10 Two Wire Resistance measurements = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for 2 Wire Resistance
'Autoranging SPD2
'Select 20 K Ohm range
'10 Measurements used for averaging
'Enable averaging filter
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.10 Resistance Measurements 4 Wire

Measurement function

:CONF:FRES

The PM2530 is configured for a single 4 wire resistance measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on :CONFigure commands.

Ranging

:FRES:RANG 20

Select the 20 Ω range. For ranging details see section 4.5.1.

Reading rate

:FRES:APER 2E-2

Set the speed of the resistance measurement function to speed mode SPD3. See also the :SENSe:FRESistance:RESolution commands and section 4.5.3 for details.

Input offset compensation

:CORR:COLL ALL

:CORR:STAT ON

A measurement is taken and the measurement data is used as nulling data for all ranges of the 4 wire resistance measurement function. Refer to section 4.10 for a detailed description. Enable input offset compensation.

Relative reference measurements

:CALC:REF:ACQ

:CALC:REF:STAT ON

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details. Enable the relative reference function.

Filters

:INP:FILT:LPAS:STAT ON

:CALC:AVER:EXP:WEIG 2

:CALC:AVER:EXP:STAT ON

:CALC:AVER:COUN 50

:CALC:AVER:STAT ON

Enable the analog low pass, 5 Hz, 20 dB/decade filter to eliminate external noise.

Set the weight factor to be used for the exponential average calculation.

Enable the digital weighting filter (exponential averaging).

Set the number of measurements to be used for averaging.

Enable the digital averaging filter.

Extremes function

:CALC:EXTR:STAT ON

:CALC:EXTR:SIGN:BEEP ON

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

Enable the beeper to report the measurement of an extreme value.

Triggering

:SAMP:TIM 30

:SAMP:SOUR TIM

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities. Select the timer as sample source to trigger the measurements

Start the measurement

:INIT

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The :READ? command contains the :ABORT::INIT::FETC? commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:PERC:REF:ACQ

The last available measurement data is used as reference value for percentage calculation.

:CALC:PERC:STAT ON

Enable the percentage calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now the deviation in terms of percentage of the reference value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** 4 WIRE RESISTANCE MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:FRES", NLen)

CALL Send(board, ADDRESS, ":FRES:RANG 20E3", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:COUN 10", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:STAT on", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Average of 10 Four Wire Resistance measurements = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for 4 Wire Resistance
'Autoranging SPD2
'Select 20 K Ohm range
'10 Measurements used for averaging
'Enable averaging filter
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.2.11 Temperature Measurements

Measurement function

`:CONF:TEMP`

The PM2530 is configured for a single temperature measurement, with autoranging enabled and speed mode SPD2 selected. See section 5 of the Command Reference Manual for more details on `:CONFigure` commands.

Linearization

`:TEMP:TYPE 390`

Select Pt100 probe type 390 Pt linearization table to be used.

Unit

`:UNIT:TEMP K`

Select temperature unit Kelvin.

Ranging

The temperature measurements provide one range 73 K up to 1132 K (-328 °F up to +1562 °F, -200 °C up to 850 °C)

Reading rate

`:TEMP:APER 2E-2`

Set the speed of the temperature measurement function to speed mode SPD3. See also the `:SENSe:TEMPerature:RESolution` commands and section 4.5.3 for details.

Relative reference measurements

`:CALC:REF:ACQ`

The last available measurement data is used as reference value for the relative measurement function. See section 4.7 and section 3.7 of the Command Reference Manual for details
Enable the relative reference function.

`:CALC:REF:STAT ON`

Filters

`:INP:FILT:LPAS:STAT ON`

Enable the analog low pass, 5 Hz, 20 dB/decade filter to eliminate external noise.

`:CALC:AVER:EXP:WEIG 2`

Set the weight factor to be used for the exponential average calculation.

`:CALC:AVER:EXP:STAT ON`

Enable the digital weighting filter (exponential averaging).

`:CALC:AVER:COUN 50`

Set the number of measurements to be used for averaging.

`:CALC:AVER:STAT ON`

Enable the digital averaging filter.

Extremes function

`:CALC:EXTR:STAT ON`

Enable monitoring of the extreme measured values. See section 3 of the Command Reference Manual for more information on the extremes and/or limit functions.

`:CALC:EXTR:SIGN:BEEP ON`

Enable the beeper to report the measurement of an extreme value.

Triggering

`:SAMP:TIM 30`

Set the timer value to 30 seconds. See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.
Select the timer as sample source to trigger the measurements

`:SAMP:SOUR TIM`

Start the measurement

:INIT

Initiate the measurement sequence. The PM2530 will now wait for the sample event to be detected. (In this example we selected the timer with 30 seconds as sample event source). The :READ? command contains the :ABORT;:INIT;:FETC? commands and may also be used here to start the measurement and read the data.

Read the measurement data

:FETC?

The measurement data is retrieved when the measurement is finished. The FETCH? query may be given before the measurement is completed, after completion the data is returned.

Calculate functions

:CALC:PERC:REF:ACQ

The last available measurement data is used as reference value for percentage calculation.

:CALC:PERC:STAT ON

Enable the percentage calculation. See section 3 of the Command Reference Manual for details on the calculation functions.

Read the measurement data

:READ?

A new measurement is started and the measurement data is retrieved, the reading is now a dBm value.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```
REM '***** TEMPERATURE MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 80
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 13)
CALL Send(board, ADDRESS, ":CONF:TEMP", NLen)

CALL Send(board, ADDRESS, ":UNIT:TEMP CEL", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:COUN 10", NLen)
CALL Send(board, ADDRESS, ":CALC:AVER:STAT on", NLen)
CALL Send(board, ADDRESS, ":INIT", NLen)
CALL Send(board, ADDRESS, ":FETC?", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Average of 10 Temperature measurements = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END
```

'Program declarations
'Size of response string
'Board number = 0
'PM2530 address = 22 at delivery
'Clear the GPIB interface
'Timeout 10 seconds
'Configure for Temperature
'Autoranging SPD2
'Select Temperature unit Celcius
'10 Measurements used for averaging
'Enable averaging filter
'Start the measurement
'Retrieve the result
'Read the response

'Print the response left of NLen
'Clear the GPIB interface
'End of program

5.2.3 BURST MEASUREMENTS

The burst measurement mode (see section 4.9) can be selected with all the previously described measurement functions (except for the low frequency voltage, period, and frequency measurement modes). One method of selecting burst measurements is using :CONFigure:ARRay command. This sets the instrument up to a known state (See Table in appendix A.2.15 SET BY :CONF), it also specifies the number of posttrigger measurements to be taken. You can also use the :CONFigure command to select burst measuring mode, but the number of posttrigger measurements must then be specified using the :SAMPle:COUNT command. The :SENSe:FUNCTion command may also be used to select the measurement function, with the settings last used with this function (see Table in appendix A.2.15 (STORED PER MEAS.FUNC.) for details of which settings are recalled when the measurement function is reselected). The following example shows how the main burst measurement features can be set, after the instrument is configured using the :CONFigure:ARRay command. This example can be used in conjunction with the previously described measurement set up examples. In this example settings not concerning burst mode are left out to clarify the burst operation.

Measurement function

:CONF:ARR:VOLT (100),DEF

The PM2530 is configured to perform 100 posttrigger volt DC measurements, with autoranging and speed mode SPD2. The parentheses are used to separate the parameters for the :ARRay node and the :VOLTage node. Because the :ARRay node parameter is obligatory and parameter groups may only be omitted from the left, the :VOLTage node must also be specified. The :CONFigure commands automatically select 'immediate' as the arm, trigger, and sample source. As an example volt DC is selected, but any other measurement function may be selected here (except VLF, Frequency and Period measurement function). See Section 5 of the Command Reference Manual for more details of :CONFigure:ARRay commands. When you don't need auxiliary measurement functions or any triggering other than 'immediate', you can use the :MEAS:ARR:VOLT? (100), DEF command, which immediately starts the specified number of measurements and returns the reading.

Arming

:ARM:SOUR EXT1

Select the rear panel input EXTERNAL1 as arm event source to arm the burst measurements. When this event is detected the actual sampling will start.

Triggering

:TRIG:SOUR BUS

Select the rear GPIB interface as trigger event source to start the posttrigger measurements. When a trigger event is detected on the bus the specified 100 posttrigger samples are taken. All previous taken samples are considered as pretrigger samples. Set the trigger delay to 50 seconds. This start the 100 posttrigger measurements 50 seconds after the trigger event is detected.

:TRIG:DEL 50

Sampling

:SAMP:SOUR EXT2

:SAMP:DEL:AUTO OFF

:SAMP:DEL 20e-3

Select the rear panel input EXTERNAL2 as sample event source to "trigger" each measurement.

Disable the auto delay mode. The auto delay mode automatically selects the minimum delay between the detection of the sample event and the actual data acquisition to guarantee a correct measurement result for the current selected range and speed. See the appendix for internal settling time specification of each measurement function.

Set the sample delay to 20 mSec. The sample is taken 20 mSec after the sample event is detected.

See section 4.5.2 and the trigger subsystem in section 19 of the Command Reference Manual for more information on the various trigger possibilities.

Start the measurement

:INIT

Initiate the measurement sequence. The PM2530 will now wait for the arm event to be detected. (In this example we selected the EXTERNAL1 input as arm source). When the arm event is detected, the sampling starts taking a sample when the sample event is detected on the EXTERNAL2 input. These samples are stored as pretrigger samples, until the trigger event is detected on the BUS and the 50 sec. trigger delay is passed (The BUS trigger event may be an GET Group Execute Trigger or the *TRG command). After the trigger event detection and the delay the 100 posttrigger samples taken and stored.

Start posttrigger sampling

*TRG

Send trigger via GPIB interface to start the posttrigger measurements.

Read the measurement data

:TRAC? PRET, 100, MAX

The last measured 100 pretrigger burst data elements are retrieved.

:TRAC? POST, MAX, MIN

The 100 posttrigger burst data elements (MAX) are retrieved.

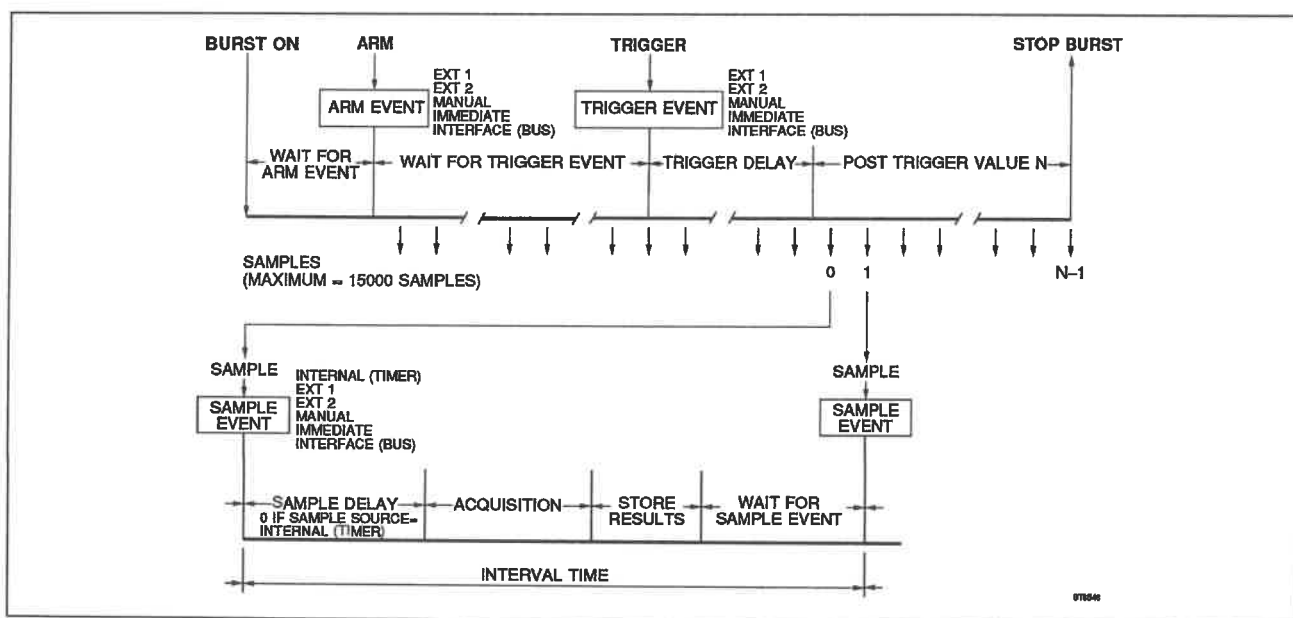


Fig. 5.2.1 Burst Timing Diagram.

Quick Basic Example Program using PM2201-PM2202 IEEE-488.2 Drivers

```

REM '***** VDC BURST MEASUREMENT EXAMPLE *****'
REM $INCLUDE: 'qbdecl.bas'
DIM response AS STRING * 200
board = 0
ADDRESS = 22
CALL SendIFC(board)
CALL IBTMO(board, 14)
CALL send(board, ADDRESS, ":CONF:ARR:VOLT (10),20,0.01", NLen)

CALL send(board, ADDRESS, ":TRIG:SOUR BUS", NLen)
CALL send(board, ADDRESS, ":SAMP:TIM 0.5", NLen)
CALL send(board, ADDRESS, ":SAMP:SOUR TIM", NLen)
CALL send(board, ADDRESS, ":INIT", NLen)
PRINT "Taking PRE-TRIGGER samples"
delay = TIMER + 5
WHILE delay > TIMER: WEND
PRINT "Send TRIGGER"
CALL Trigger(board, ADDRESS)
PRINT "Taking POST-TRIGGER samples"
CALL send(board, ADDRESS, "**WAI", NLen)
completed
CALL send(board, ADDRESS, ":TRAC? PRET,MAX,MIN", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Pre-trigger Burst = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL send(board, ADDRESS, ":TRAC? POST,MAX,MIN", NLen)
CALL Receive(board, ADDRESS, response$, NLen)
PRINT "Post-trigger Burst = ";
PRINT LEFT$(response$, INSTR(response$, CHR$(10)))
CALL SendIFC(board)
END

```

'Program declarations
 'Size of response string
 'Board number = 0
 'PM2530 address = 22 at delivery
 'Clear the GPIB interface
 'Timeout 30 seconds
 'Configure for VDC Burst,
 '10 posttrigger samples,
 '20 V range, SPD6
 'Select triggering via the bus
 'Set timer to 0.5 seconds
 'Select timed sampling
 'Start the measurement
 'Delay before sending trigger
 'Send trigger
 'Wait until previous commands are
 'Retrieve the pretrigger result
 'Read the response
 'Print the response left of NLen
 'Retrieve the posttrigger result
 'Read the response
 'Print the response left of NLen
 'Clear the GPIB interface
 'End of program

5.2.4 GENERAL SET UP REMOTE CONTROLLED MEASUREMENTS

This section provides an overview of the steps required to set up the PM2530 for making measurements under remote control. The set up is described in the order in which the signal is processed.

Details of each step are described in Section 4 and details of the SCPI commands used to program the PM2530 are described in the Command Reference Manual.

5.2.4.1 Basic Measurement Setup

You can use the :MEASure commands for making basic measurements. These commands select the configuration according to Figure 5.2.2. You can select a measurement function, range and Reading RATE. The measurement is taken immediately and the result can be read out via the GPIB interface controller.

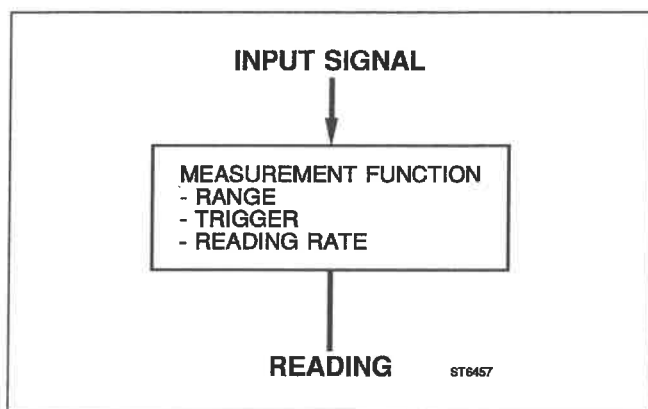


Fig. 5.2.2 Basic Measurement Setup.

Configuration used when a :MEASure command is given.

- All filters off
- All auxiliary functions off
- Measurement function as specified with the command
- Range according to the specified expected value (if no value is specified, autoranging will be selected)
- The measurement is triggered immediate
- Speed mode according to the specified desired resolution. (if no resolution is specified, Speed mode SPD2 is selected)

5.2.4.2 Enhanced Measurement Setup

Fig 5.2.3 gives an overview of the additional functions of the PM2530, it shows the order in which the input signal is influenced by those functions.

1 INPUT TERMINALS

The selections made by the INPUTS REAR/FRONT and GUARD INT/EXT switches cannot be altered via remote control. You can however check the position of these switches. This is done using the :ROUTe subsystem for the INPUTS REAR/FRONT switch and using the :INPut subsystem for the GUARD INT/EXT switch. It is also possible to detect a change of position of either of these switches, via the :STATus subsystem. The position of these switches is reported in the Operation Condition Register (INPUTS switch: FRONT position bit 10 is set "1", REAR position bit 10 is cleared "0". GUARD switch: EXT position bit 9 is set to "1", INT position bit 9 is cleared "0"). The :STATus subsystem can be set up in such a way that a service request (SRQ) is generated when the position of either of these switches changes.

2 MEASUREMENT FUNCTION

The measurement function can be selected with the :CONFigure or :SENSe:FUNCTION commands.

:CONFigure provides a quick way to set the instrument to a known state. See Table in appendix A.2.15 (SET BY :CONF) and Section 5 of the Command Reference Manual for a detailed description of the settings affected by :CONFigure.

:SENSe:FUNCTION restores the settings used the last time this function was selected. Each time a new function is selected, the settings of the old function are stored. See Table in appendix A.2.15 (STORED PER MEAS.FUNC.) for details of which settings are stored.

3 RANGE

The :SENSe:RANGe subsystem can be used to control the ranging. The range can be explicitly set to a certain range, or autoranging can be selected.

4 TRIGGER

The triggering of each measurement can be controlled via the :SAMPle subsystem. You can select various sources to trigger each sample (EXTErnal1, EXTErnal2, BUS, TIMer, IMMEDIATE). You can also set a delay between the detection of the sample trigger event and the sample being taken.

5 READING RATE

The measurement speed mode (reading rate) can be controlled via the :SENSe:APERture or :SENSe:RESolution subsystems. With the :APERture commands you can specify the measurement time, see section 14.2 of the Command Reference Manual for a specification of the measurement time of each measurement function and the speed mode in relation to the line frequency. With the :RESolution commands you can specify the resolution of the measurement, the instrument then automatically selects the reading rate appropriate to this resolution. Selecting a high resolution results in a long measurement time and selecting a short measurement time results in a low resolution.

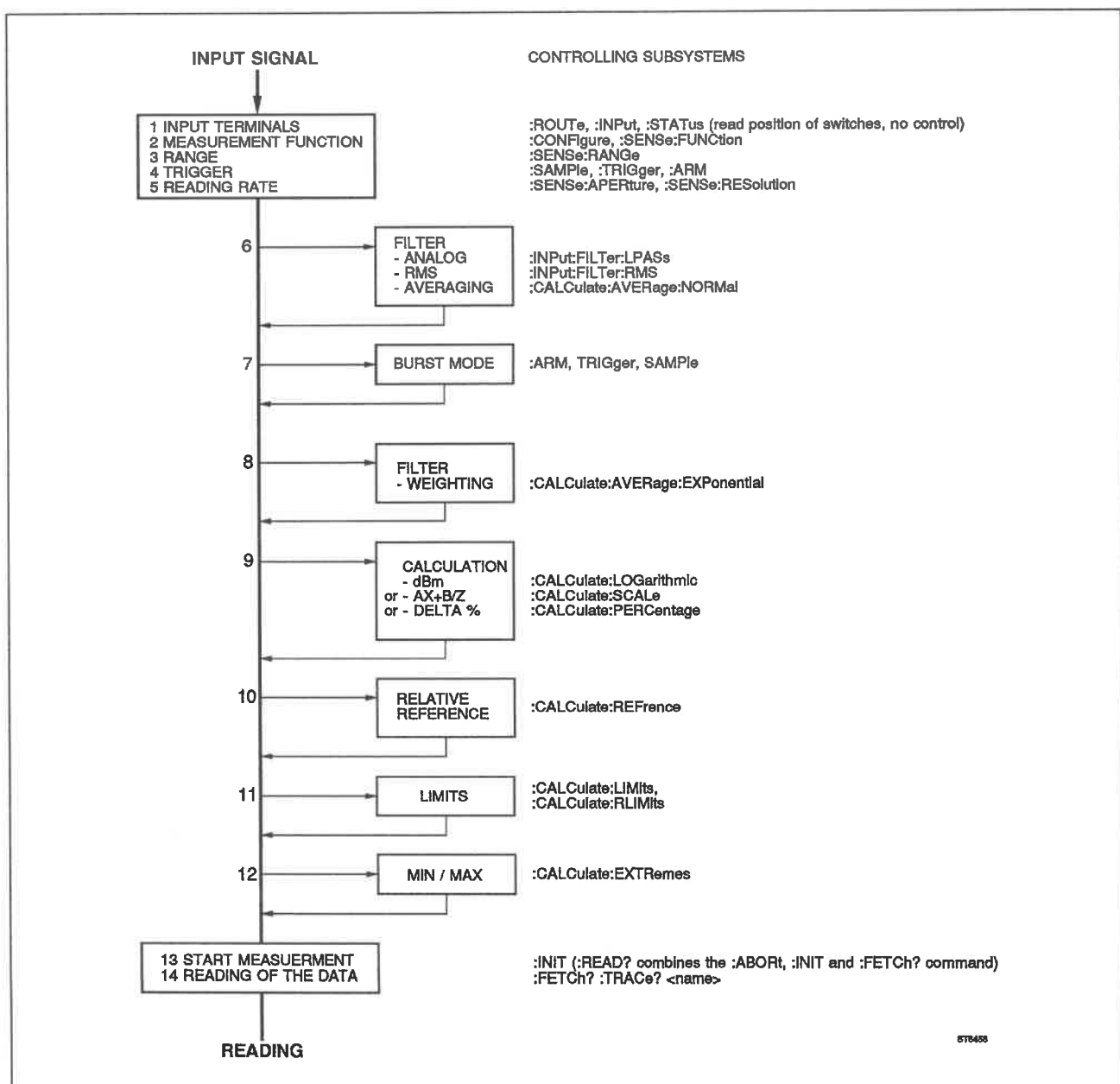


Fig 5.2.3 Remote Measurement Setup Flow diagram.

6 FILTERS

At this level in the signal flow the following filters may be active:

- The analog low pass filter may be selected for the following measurement functions: volt DC, low frequency voltage, resistance 2 wire, resistance 4 wire, and temperature. This analog filter is controlled via the :INPut:FILTER:LPASS subsystem, see section 9 of the Command Reference Manual.
- The RMS filter setting may be changed to get an optimal response for the following functions: volt AC, volt AC+DC, current AC. The RMS filter settings are controlled via the :INPut:FILTER:RMS subsystem.
- The averaging filter controlled via :CALCulate:AVERage:NORMAL may be selected for all measurement functions.

7 BURST MEASUREMENT MODE

The burst mode is used to perform measurements and store up to 15000 of the most recent results, continuously in burst memory. To make full advantage of the burst function the storage operation may be independently triggered, to capture a group of successive results, that may occur before, after, or before and after the trigger event. The :ARM subsystem is used to select the event source on which the measure sequence is started.

The :TRIGger subsystem is used to select the trigger event source on which the burst measurement is triggered, The :TRIGger subsystem is also used to specify the trigger delay.

The :SAMple subsystem is used to select the sample event source on which each sample is taken, and to specify the sample delay. The :SAMple subsystem is also used to specify the number of measurements to be taken after the trigger event is detected. (the number of posttrigger measurements may also be specified using the :MEASure:ARRay or :CONFigure:ARRay command see section 5 and section 10 of the Command Reference Manual).

8 FILTER (WEIGHTING)

The digital weighting filter is active in the signal flow at this level. During burst measurements the results are not processed by the digital weighting filter. The results are processed by the digital weighting filter, if selected, when the burst memory is read.

The :CALCulate:AVERage:EXPonential subsystem is used to control the digital weighting filter.

9 CALCULATION FUNCTIONS (dBm, scaling, delta %)

One of the calculation functions, dBm (calculation of the dBm value), scaling $((AX+B)/Z)$, or delta% (computing the deviation from a preset value in terms of percentage) may be selected to present the measurement data. Only one of these functions can be active at a time. It is not possible to select another calculation function after reading the data, and then reread the measured data using the new calculation function.

The :CALCulate:LOGarithmic, :CALCulate:SCALE, and :CALCulate:PERCentage subsystems are used to control the calculation functions.

10 RELATIVE REFERENCE

The relative reference function presents the data (including all previous processing) relative to a set reference point.

It can be controlled via the :CALCulate:REFerence subsystem.

11 LIMIT EXCEEDINGS

The limit function is used to count the number of times a specified limit is exceeded. This can either be a set of absolute limits, which you may specify, or relative limits (the limits are related to a specified reference value). A measurement result exceeding the limit can be signalled by the display, beeper, and/or rear limits connector, see Section 3 of the Command Reference Manual. The :CALCulate:LIMit subsystem is used for absolute limit checks, and the :CALCulate:RLIMit subsystem is used for checking on limits related to a reference.

12 MIN / MAX READINGS

The MIN / MAX function (extremes) is used to store the minimum and maximum measured values. This function is also capable of signalling on the display and/or beeper when a new extreme value is measured. The :CALCulate:EXTRemes subsystem is used to control the MIN / MAX function.

13 START THE MEASUREMENT

The measurement can be started using the :INIT command. If you want to start the measurement and immediately retrieve the data you may use the :READ? command which combines the :ABORT, :INIT and FETCH? commands.

14 RETRIEVE THE MEASUREMENT DATA

The :FETCH? subsystem is used to read data from a single measurement and the :TRACE? subsystem is used to read data from a burst measurement. If the last measurement was a burst measurement, :FETCh? will return the contents of the posttrigger buffer. E.g. :MEAS:ARR:VOLT? (10),DEF::FETC? will return the same data twice.

5.3 PROGRAMMING EXAMPLES

This section shows some examples on programming the PM2530. These programmes can be run in the same way as described in section 5.1.1

```

REM $INCLUDE: 'qbdecl.bas'
REM '***** PM2530 QUICK BASIC 4.0/4.5 PROGRAM EXAMPLE 1 *****'
REM '***** USES - PM2201 V3.0/PM2202 V2.0 IEEE-488.2 DRIVERS *****'
REM '***** SIMPLE VOLTAGE DC MEASUREMENT SETUP *****'
DIM response AS STRING * 20
Board = 0: Address = 22

CALL SendIFC(Board)
CALL IBTMO(Board, 12)
CLS
CALL Send(Board, Address, "*RST", NLen)

CALL Send(Board, Address, ":MEASure:SCALar:VOLTagE:DC?", NLen)

CALL Receive(Board, Address, response$, NLen)
PRINT "Measured value = "; VAL(response$); " Volt DC "
END

REM $INCLUDE: 'qbdecl.bas'
REM '***** PM2530 QUICK BASIC 4.0/4.5 PROGRAM EXAMPLE 2 *****'
REM '***** USES - PM2201 V3.0/PM2202 V2.0 IEEE-488.2 DRIVERS *****'
REM '***** CURRENT DC MEASUREMENT SETUP AVERAGING 50 MEASUREMENTS *****'
REM '***** SCALING THE SIGNAL 4 - 20 mA = 0 - 100 % *****'
DIM response AS STRING * 20
Board = 0
Address = 22
CALL SendIFC(Board)
CALL IBTMO(Board, 12)
CLS
response = SPACE$(20)
CALL Send(Board, Address, "*RST", NLen)

CALL Send(Board, Address, ":CONFIgure:SCALar:CURRent:DC", NLen)

CALL Send(Board, Address, ":SENSe:CURRent:DC:RANGe:UPPer 20E-3", NLen)
CALL Send(Board, Address, ":SENSe:CURRent:DC:RESolution 20E-6", NLen)
CALL Send(Board, Address, ":CALCulate:AVERAge:NORMAL:COUNT 50", NLen)

CALL Send(Board, Address, ":CALCulate:AVERAge:STATe ON", NLen)
CALL Send(Board, Address, ":CALCulate:SCALE:GAIN 6.250", NLen)

CALL Send(Board, Address, ":CALCulate:SCALE:NORMALize:VALue 0.001", NLen)
CALL Send(Board, Address, ":CALCulate:SCALE:OFFSet -0.025", NLen)

CALL Send(Board, Address, ":CALCulate:SCALE:STATe ON", NLen)

CALL Send(Board, Address, ":INITiate:IMMediate", NLen)
CALL Send(Board, Address, ":FETCh?", NLen)
CALL Receive(Board, Address, response$, NLen)
PRINT "Measured value ="; VAL(response$); "% (4 - 20 mA)"
END

```

'Size of response string
'Board number,
'22 = address
'at delivery
'Clear the GPIB interface
'Timeout 3 seconds
'Clear screen
'Set the instrument
'to a known state
'Perform a single
'Voltage DC measurement
'Read the response
'Print the response"
'End of program

'Size of response string
'Board number
'22 = Address at delivery
'Clear the GPIB interface
'Timeout 3 seconds
'Clear screen
'Initial string with spaces
'Set the instrument
'to a known state
'Set up for a
'Current DC measurement
'Select 20 mA range
'Select 20 uA resolution
'Set averaging count
'to 50
'Enable averaging
'Set scale gain factor
'to 6.240
'Set scale normalize
'value to 0.001
'Set scale offset
'to -0.025
'Enable scalling
'4..20 mA to 0..100%
'Start the measurement
'Retrieve the result
'Read the response
'Print the response"
'End of program

```

REM $INCLUDE: 'qbdecl.bas'
REM '***** PM2530 QUICK BASIC 4.0/4.5 PROGRAM EXAMPLE 3 *****'
REM '***** USES - PM2201 V3.0/PM2202 V2.0 IEEE-488.2 DRIVERS *****'
REM '***** 10 VOLTAGE DC MEASUREMENTS SETUP USING RELATIVE LIMITS*****'
REM '***** SIGNAL VIA BEEPER AND DISPLAY IF SIGNAL IS OUT OF LIMIT *****'
DIM response AS STRING * 20
DIM commstr AS STRING * 20

Board = 0
Address = 22

CALL SendIFC(Board)

CALL IBTMO(Board, 12)
CLS
response = SPACE$(20)

CALL Send(Board, Address, "*RST", NLen)

CALL Send(Board, Address, ":CONFigure:SCALar:VOLTagE:DC", NLen)

CALL Send(Board, Address, ":CALCulate:RLIMit:LOWer:DATA 0.95", NLen)

CALL Send(Board, Address, ":CALCulate:RLIMit:UPPer:DATA 1.05", NLen)

CALL Send(Board, Address, ":CALCulate:RLIMit:STATe ON", NLen)

CALL Send(Board, Address, ":CALCulate:LIMit:SIGNAL:BEEPer:STATe ON", NLen)

CALL Send(Board, Address, ":CALCulate:LIMit:SIGNAL:DISPlay:STATe ON", NLen)

FOR i = 1 TO 10
    INPUT "Reference value for the input signal = "; reference
    commstr$ = "READ?" + STR$(reference)
    CALL Send(Board, Address, commstr$, NLen)
    CALL Receive(Board, Address, response$, NLen)
    PRINT "Measured value = "; VAL(response$); "Volt"
NEXT i
END

```

'Size of response string
'Size of the command
'string
'Board number
'22 = Address at
'delivery
'Clear the GPIB
'interface
'Timeout 3 seconds
'Clear screen
'Initial string with
'spaces
'Set the instrument
'to a known state
'Set up for a
'Voltage DC measurement
'Set the lower relative
'limit to 5%
'Set the upper relative
'limit to 5%
'Enable relative
'limit test
'Enable reporting via
'the beeper
'Enable reporting via
'the display
'Loop 10 times
'Enter reference value
'Create command string
'Start the measurement
'Read the response
'Print the response

'End of program

```

REM $INCLUDE: 'qbdecl.bas'
REM '***** PM2530 QUICK BASIC 4.0/4.5 PROGRAM EXAMPLE 4 *****'
REM '***** USES - PM2201 V3.0/PM2202 V2.0 IEEE-488.2 DRIVERS *****'
REM '***** VOLTAGE ACDC BURST MEASUREMENT SETUP 100 POSTTRIGGER SAMPLES *****'
REM '***** TRIGGER SOURCE MANUAL, SAMPLE SOURCE TIMER *****'
DIM Response AS STRING * 20

DIM ESRResponse AS STRING * 20

DIM PointResponse AS STRING * 20

DIM commstr AS STRING * 40

board = 0
address = 22

CALL SendIFC(board)

CALL IBTMO(board, 12)
CLS
Response = SPACE$(20)

ESRResponse = SPACE$(20)

PointResponse = SPACE$(20)

CALL Send(board, address, "*RST;*CLS", NLEnd)

CALL Send(board, address, ":CONFigure:ARRay:VOLTage:ACDC (100),DEFault", NLEnd)

CALL Send(board, address, ":TRIGger1:SEQuence1:SOURce MANual", NLEnd)

CALL Send(board, address, ":SAMPle:SOURce TIMer", NLEnd)

CALL Send(board, address, ":SAMPle:TIMer 1", NLEnd)
CALL Send(board, address, ":INITiate:IMMediate;*OPC", NLEnd)
'Start the measurement

PRINT "Press SINGLE on front panel to start 100 post-trigger measurements"
KeyPressed% = 32
DO UNTIL KeyPressed% = 0

    pause = TIMER + 1
    WHILE pause > TIMER: WEND

    LOCATE 12, 25
    PRINT "Press SINGLE on front panel"
    pause = TIMER + 1
    WHILE pause > TIMER: WEND

    LOCATE 12, 25
    PRINT " "
    CALL Send(board, address, ":STATus:OPERation:CONDition?", NLEnd)

    CALL Receive(board, address, Response$, NLEnd)
    KeyPressed% = VAL(Response$) AND 32

```

```

'Size of response
'string
'Size of ESRresponse
'string
'Size of
'Pointresponse
'string
'Size of command
'string
'Board number
'22 = Address at
'delivery
'Clear the GPIB
'interface
'Timeout 3 seconds
'Clear screen
'Clear response
'string
'Clear ESRresponse
'string
'Clear Pointresponse
'string
'Set the instrument
'to a known state
'Set up for
'Burst Voltage ACDC
'Set trigger source
'to manual
'Set sample source
'to timer
'Set the timer value

'Standard Event Bit 1
'OPC is set
'if measurement is
'finished
'Print message

'Wait until SINGLE
'is pressed
'Delay value
'Wait for end of
'delay
'Position cursor
'Print message
'Delay value
'Wait for end of
'delay
'Position cursor
'Erase message
'Request Operation
'Condition register
'Read response
'Test Wait for
'trigger Bit

```

LOOP	
LOCATE 2, 1	'Position cursor
PRINT "Taking 100 posttrigger samples now"	'Print message
LOCATE 24, 10: COLOR 0, 7	'Progress indicator
	'color
PRINT " ";	'Progress indicator
	'bar
LOCATE 24, 10: COLOR 7, 0	'Normal color
OPC% = 0	
DO UNTIL OPC% = 1	'Test if OPC is 1
	'measurement
	'complete
pause = TIMER + 2	'Delay value
WHILE pause > TIMER: WEND	'Wait for end of
	'delay
PRINT " ";	'Progress indication
CALL Send(board, address, "*ESR?", NLen)	'Request standard
	'event status
CALL Receive(board, address, ESRResponse\$, NLen)	'Retrieve response
OPC% = VAL(ESRResponse\$) AND 1	'Test OPC bit
LOOP	
LOCATE 3, 1	'Position cursor
CALL Send(board, address, "TRACe:POINTs? PRETrigger", NLen)	'Request number of
	'posttrigger
	'measurements taken
CALL Receive(board, address, PointResponse\$, NLen)	'Retrieve response
FOR n = 1 TO FIX(VAL(PointResponse\$))	'Loop for pretrigger
	'samples
commstr\$ = ":TRACe:DATA? PRETrigger,1," + STR\$(n)	'Create command
	'string
CALL Send(board, address, commstr\$, NLen)	'Request for
	'pretrigger sample
CALL Receive(board, address, Response\$, NLen)	'Read the response
PRINT "sample "; -(FIX(VAL(PointResponse\$)) - n + 1); TAB(13); " = "; TAB(16);	
PRINT USING "###.#####"; VAL(Response\$); : PRINT " Volt"	'Print the response
NEXT n	
FOR n = 0 TO 99 STEP 1	'Loop for posttrigger
	'samples
commstr\$ = ":TRACe:DATA? POSTtrigger,1," + STR\$(n)	'Create command
	'string
CALL Send(board, address, commstr\$, NLen)	'Request for
	'posttrigger sample
CALL Receive(board, address, Response\$, NLen)	'Read the response
PRINT "sample "; n; TAB(13); " = "; TAB(16);	
PRINT USING "###.#####"; VAL(Response\$); : PRINT " Volt"	'Print the response
NEXT n	
END	'End of program

6 MAINTENANCE

6.1 INTRODUCTION

This section describes basic maintenance procedures that the user can perform. Do not attempt any maintenance not described in this section. For additional maintenance, service, and calibration procedures, qualified personnel can refer to the PM2530 Service Manual.

6.2 CLEANING

WARNING: To avoid electrical shock or damage to the instrument, never get water inside the case. To avoid damage to the instrument housing do not use solvents.

For cleaning, wipe the instrument with a cloth that is lightly dampened with water or mild detergent. Do not use aromatic hydrocarbons, chlorinated solvents, or methanol-based fluids.

6.3 LINE FUSE

The instrument uses a 250 V Slow Blow line fuse in series with the power supply. For line fuse ratings and replacement refer to Section 3.1.3.

6.4 CURRENT INPUT PROTECTION FUSE

The current input is protected by a user-replaceable fuse rated at F 2.5 A, 250 V (Fast Blow), high breaking capacity, IEC-127 Sheet I. The Parts Number is 4822 070 12502.

REPLACING THE CURRENT INPUT FUSE

CAUTION: for protection against fire, replace a blown fuse only with one of identical rating.

The fuse holder is located on the rear panel, next to the AMPS input terminal. It is marked "FUSE 2.5A-F/250V".

Proceed as follows:

- unplug the line cord and disconnect the instrument from all other voltage sources
- remove the fuse holder with the fuse by turning it counter clock wise , using a screw driver.
- replace a blown fuse with one of identical rating and reinsert the fuse holder. Secure the fuse holder by pressing in and turning it clock wise.

The instrument is shipped with a replacement fuse.

6.5 PERFORMANCE TESTS

When received, the PM2530 is calibrated and in operating condition. The following Performance Verification Procedures are provided for acceptance testing upon initial receipt or to verify correct operation at any time. The tests compare the performance of the PM2530 with the specifications given in Appendix A.

6.5.1 SELF-TEST DIAGNOSTICS AND ERROR CODES

Self-test diagnostics are performed each time the instrument is powered up. See table 6.1 for an overview of tests. If an error is detected, an error-code E xxxx is displayed at the bottom of the "menu display field", where xxxx represents the error code number. If a serious error is detected, and the instrument is not capable of displaying an error code, a beep is generated and the instrument can not be operated.

Any error indicates that a malfunction has occurred and maintenance is required. If you encounter an error, note the error code, and consult Table 6.1. See if the instrument repeats the error. If the problem persists, contact the nearest FLUKE/PHILIPS Service Centre. A list of Service Centers is provided on the last pages of this manual.

Besides automatic Self Test Diagnostics after turning the power on, the PM2530 enables you to perform Self Test Diagnostics at any time.

SELFTESTS IN LOCAL CONTROL (front panel operation)

Any errors encountered during these tests are displayed momentarily. In addition to the self-test, a display test and a keyboard test can be carried out.

- press  to invoke the MISC (MISCellaneous functions) menu (Fig. 6.1).
- select
 - 5. CHECK to invoke the CHECK menu
 - 1. TEST to invoke the TEST menu
 - 1. PATTERN 1 to start display test 1. The pattern shown below is displayed.
Press any key except  or  to exit the test.



- 2. PATTERN 2 to start display test 2. The inverse of PATTERN 1 is shown.
Press any key except  or  to exit the test.

From PATTERN 1 and PATTERN 2 you can see if any display dot is missing.

- 3. KEYBOARD to check the front panel push button operation. Press the button asked for in the display. If the switch functions, the next button to be pressed will be shown. The message "FAILED", "PRESS ANY KEY TO LEAVE KEY TEST" will be shown if:
 - the pressed switch fails
 - the button asked for is not pressed within 10 seconds
 The KEYBOARD test is terminated on pressing any button after this message is shown, or after all buttons have been tested.

- 4. SELFTEST the instrument performs the self-test diagnostics routine, consisting of 13 tests. If a test fails, an error code will be displayed at the bottom of the "menu display field". For error descriptions see Table 6.1. The instrument tries to complete the self-test routine and begin normal operation. If no test failed, the display will show 0.

- 0. RETURN to return to the CHECK menu
- 0. RETURN to return to the MISC menu
- 0. EXIT to leave the MISC menu

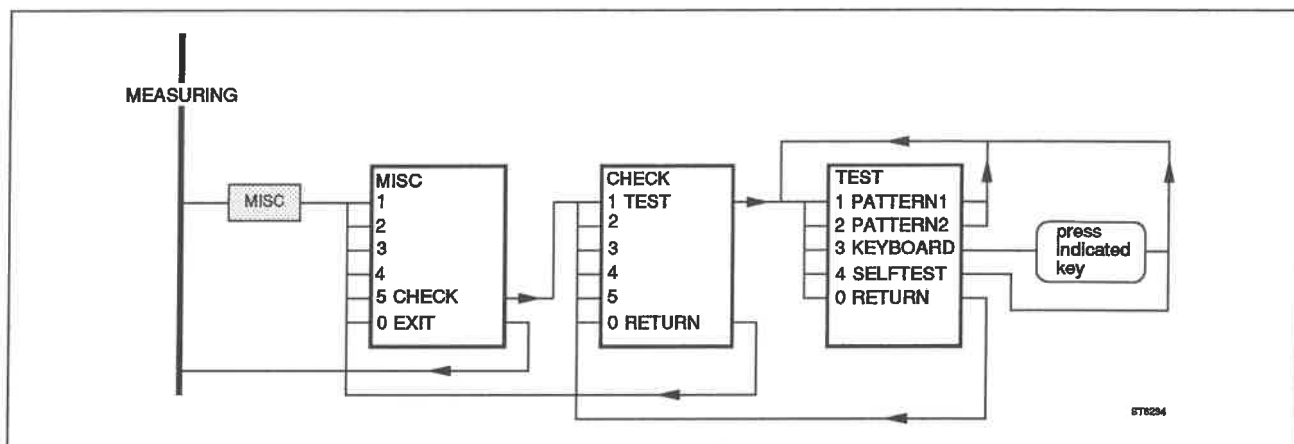


Fig. 6.1 Self Test menu structure.

Table 6.1 Diagnostic Power On and Self Test error codes

ERROR CODE ¹⁾	During Power On	During Self Test	Description
0	NO	YES	No error detected
1	YES	YES	ROM 1 checksum test
2	YES	YES	RAM 1 test
4	YES	YES	Power supply 1 selftest
8	YES	YES	Line Synchronisation test
16	YES	YES	Dual slope Analog to Digital Converter test
32	YES	YES	Successive Approximation Analog to Digital Converter test
64	YES	YES	RMS Convertor test
256	YES	YES	ROM 2 checksum test
512	YES	YES	RAM 2 test
1024	NO	YES	Power supply 2 selftest
2048	NO	YES	I ² C RAM test
4096	YES	YES	Communication test
8192	NO	YES	EEPROM test

The error code displayed after the instrument is Power up, or when a selftest is performed, may consist of an addition of several error codes, indicating the errors detected.

SELFTEST IN REMOTE CONTROL (via the GPIB interface)

Selftests will be performed when the common command *TST? is send to the PM2530. The command :TEST:SElect can be used to specify a particular test to be performed on *TST?. The PM2530 response is 0 (zero) if no error is encountered. Any other response indicates an error. Refer to Section 17 of the Command Reference Manual.

6.5.2 ACCURACY VERIFICATION TEST

The following procedures may be used to verify the accuracy (90 days spec.) of the various measurement functions. All tests must be carried out in the order given below. Allow a recovery time of several minutes for your calibration and meter if you applied a high voltage or current (700V, 1000V or 2A) before you continue with the next accurate verification test.

The PM2530 enables you to store and recall the date of verification. Refer to "Store/Recall the verification date", at the end of this section.

TOLERANCE LIMITS

The accuracy and traceability of a user's standards affects the manner in which the performance of the PM2530 can be verified. You will need to include the tolerances of your own equipment, in order to calculate the overall tolerance limit. The Tables 6.2. up to 6.15 indicate:

- **RELATIVE TOLERANCE** the relative tolerance limits, derived from the PM2530 specifications listed in Appendix A (relative to calibration standards).
- **FACTORY UNCERTAINTY** the factory calibration standards uncertainty. After re- calibration, re- calibration uncertainties must be used here.
- **USER'S TOLERANCE** the user's measurement tolerance limits, to be filled in by the user. Notice that these limits include test equipment and transport (e.g. thermal e.m.f. of cable connections) errors.

To calculate the allowable readings, you must add these three for each verification point, see the example below.

Table 6.2 Example for tolerance calculation

STEP	RANGE	INPUT (Vdc)	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE ¹⁾	
			MIN	MAX			MIN	MAX
1	20 V	+20 V	+19.99960	+20.00040	00.000420	00.000200	+19.998980	+20.001020

- ¹⁾ OVERALL TOLERANCE MIN. = RELATIVE TOLERANCE MIN. - FACTORY UNCERT'Y - USER'S TOLERANCE
OVERALL TOLERANCE MAX. = RELATIVE TOLERANCE MAX. + FACTORY UNCERT'Y + USER'S TOLERANCE

RECOMMENDED TEST EQUIPMENT

The test equipment requirements are listed in Table 6.3.

Table 6.3 Test equipment requirements

PM2530 measurement function	Equipment required ¹⁾
Volts, Current, Ohms	Fluke 5700A calibrator. Shorting bar.
Temperature	Pt100 simulator.
Frequency/Period	Function generator, Philips PM5191 or equivalent.

- ¹⁾ depending on the desired traceability, the 5700A calibrator may require characterization (comparison to external standards).

LOCAL OR REMOTE CONTROLLED VERIFICATION

The accuracy verification tests are described for front panel operation (LOCAL control). To establish the required steps via the GPIB (REMOTE control), you can use the following SCPI commands:

:CONF:"function"	selects measurement function and default settings (all filters and auxiliary functions off, SPD2).
:SENS:"function":RANG<value>	selects the range; <value> is the range in main units, e.g. for 200 k Ω <value> is 200000 (or 2E5, 20E4 etc.).
:SENS:"function":APER<value>	selects the speed mode (aperture); for SPD2 <value> is 2E-,1 for 50 Hz line frequency and 167E-3, for 60 Hz line frequency, for SPD4 <value> is 2E-3 (for 50 and 60 Hz line frequency).
:READ?	takes a reading and stores the result in the PM2530 output buffer.
"function"	indicates the required measurement function: VOLT:DC (volts DC), VOLT:AC (volts AC), VOLT:ACDC (volts AC + DC), FRES (resistance 4-wire), RES (resistance 2-wire), CURR:DC (current DC), CURR:ACDC (current AC + DC), TEMP (temperature Pt100), LFAC (volts low frequency), VOLT:FREQ (frequency), VOLT:PER (period).

VOLTS DC ACCURACY TEST

The following procedure may be used to verify the accuracy of the Volts DC function.

- Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C. The maximum relative humidity is 80%.
- Press  to select the Volts DC measurement function.
- Press  to switch all filters and auxiliary functions off, and to select SPD2.
- Press RANGE  to select manual ranging.
- Connect the short or DC calibrator to provide a voltage input to the VOLTS INPUT terminals.
- For each step in Table 6.4, select the indicated range (RANGE  ), supply the specified input, and verify that the displayed reading is within the limits shown.
- Select speed mode SPD4 (READING RATE ).
- For each step in table 6.5, select the indicated range (RANGE  ), supply the specified input, and verify that the display reading is within the limits shown.
- Set the calibrator for zero volts and disconnect it from the PM2530.

Table 6.4 Volts DC test, SPD2 (90 days spec)

STEP	RANGE	INPUT (Vdc)	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE	
			MIN	MAX			MIN	MAX
1	20 mV	0 V (short)	- 00.0004_	00.0004_	00.00000			
2	200 mV	0 V (short)	- 000.0008	000.0008	000.0000			
3	2 V	0 V (short)	- 0.000004	0.000008	0.000000			
4	20 V	0 V (short)	- 00.00004	00.00008	00.00000			
5	200 V	0 V (short)	- 000.0004	000.0004	000.0000			
6	2000 V	0 V (short)	- 0000.004	0000.004	0000.000			
7	20 mV	20 mV	19.9986_	20.0014_	00.00066			
8		-20 mV	- 20.0014_	- 19.9986_	00.00066			
9	200 mV	200 mV	199.9942	200.0058	000.0014			
10		-200 mV	- 200.0058	- 199.9942	000.0014			
11	2 V	700 mV	0.699983	0.700017	0.000011			
12		-700 mV	- 0.700017	- 0.699983	0.000011			
13		1.4 V	1.399971	1.400029	0.000011			
14		-1.4 V	- 1.400029	- 1.399971	0.000011			
15		2 V	1.999960	2.000040	0.000011			
16		-2 V	- 2.000040	- 1.999960	0.000011			
17	20 V	20 V	19.99960	20.00040	00.00008			
18		-20 V	- 20.00040	- 19.99960	00.00008			
19	200 V	200 V	199.9946	200.0054	000.0008			
20		-200 V	- 200.0054	- 199.9946	000.0008			
21	2000 V	1000 V	0999.971	1000.029	0000.008			
22		-1000 V	- 1000.029	- 0999.971	0000.004			

Table 6.5 Volts DC test, SPD4 (90 days spec)

STEP	RANGE	INPUT (Vdc)	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE	
			MIN	MAX			MIN	MAX
1	20 mV	0 V (short)	- 00.01_	00.01_	00.00000			
2	200 mV	0 V (short)	- 000.02	000.02	000.0000			
3	2 V	0 V (short)	- 0.0002	0.0002	0.000000			
4	20 V	0 V (short)	- 00.002	00.002	00.00000			
5	200 V	0 V (short)	- 000.02	000.02	000.0000			
6	2000 V	0 V (short)	- 0000.2	0000.2	0000.000			
7	20 mV	20 mV	19.98_	20.02_	00.00066			
8		-20 mV	- 20.02_	- 19.98_	00.00066			
9	200 mV	200 mV	199.95	200.05	000.0014			
10		-200 mV	- 200.05	- 199.95	000.0014			
11	2 V	0.7 V	0.6997	0.7003	0.000011			
12		-0.7 V	- 0.7003	- 0.6997	0.000011			
13		1.4 V	1.3997	1.4003	0.000011			
14		-1.4 V	- 1.4003	- 1.3997	0.000011			
15		2 V	1.9996	2.0004	0.000011			
16		-2 V	- 2.0004	- 1.9996	0.000011			
17	20 V	20 V	19.996	20.004	00.00008			
18		-20 V	- 20.004	- 19.996	00.00008			
19	200 V	200 V	199.96	200.04	000.0008			
20		-200 V	- 200.04	- 199.96	000.0008			
21	2000 V	1000 V	0999.7	1000.3	0000.008			
22		-1000 V	- 1000.3	- 0999.7	0000.004			

VOLTS AC ACCURACY TEST

The following procedure may be used to verify the accuracy of the Volts AC function. The procedure consists of an AC test, to be carried out for the Volts AC function, and an additional DC test, to be carried out for the Volts AC+DC function only.

- A. Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C.
- B. Press  to select the Volts AC measurement function.
- C. Press  to select SPD2, to switch all auxiliary functions off, and to set the RMS filter to >40 Hz.
- D. Press RANGE  to select manual ranging.
- E. Connect the calibrator to provide a voltage input to the VOLTS INPUT terminals.
- F. For each step in Table 6.6, select the indicated range (RANGE  ), supply the specified input (voltage, frequency), and verify that the displayed reading is within the limits shown.
- G. Press  to select the AC+DC Volts measurement function. Proceed with the AC+DC Volts test by carrying out the steps of Table 6.7.
- H. Set the calibrator for zero volts and disconnect it from the PM2530.

Table 6.6 Volts AC test, SPD2 (90 days spec)

STEP	RANGE	INPUT (Vac)	FREQ	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE	
				MIN	MAX			MIN	MAX
1	20 mV	20 mV	50 Hz	19.9780	20.0220	00.0022			
2	200 mV	200 mV	50 Hz	199.820	200.180	000.030			
3	2 V	0.2 V	50 Hz	0.19946	0.20054	0.00016			
4		1 V	50 Hz	0.99890	1.00110	0.00016			
5		2 V	50 Hz	1.99820	2.00180	0.00016			
6		2 V	20 kHz	1.99820	2.00180	0.00016			
7		2 V	100 kHz	1.99100	2.00900	0.00058			
8	20 V	20 V	50 Hz	19.9820	20.0180	00.0016			
9		20 V	20 kHz	19.9820	20.0180	00.0016			
10		20 V	100 kHz	19.9100	20.0900	00.0054			
11	200 V	200 V	50 Hz	199.820	200.180	000.017			
12	2000 V	700 V	50 Hz	0699.11	0700.89	0000.06			

Table 6.7 Volts AC+DC test (in addition to the Volts AC test), SPD2 (90 days spec)

STEP	RANGE	INPUT (Vdc)	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE	
			MIN	MAX			MIN	MAX
1	20 mV	20 mV	19.9660	20.0340	00.00066			
2	200 mV	20 mV	019.916	020.084	000.0014			
3		100 mV	099.820	100.180	000.0014			
4		200 mV	199.700	200.300	000.0014			
5	2 V	2 V	1.99700	2.00300	0.000011			
6	20 V	20 V	19.9700	20.0300	00.00008			
7	200 V	200 V	199.700	200.300	000.0008			
8	2000 V	700 V	0698.56	0701.44	0000.004			

RESISTANCE ACCURACY TEST

The following procedure may be used to verify the accuracy of the Ohms 4-wire as well as the Ohms 2-wire function.

- Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C.
- Press **RANGE** **Ω4W** to select the Ohms 4-wire (Ω4W) function.
- Press **RESET** to select SPD2, and to switch all auxiliary functions off.
- Press **RANGE** **MAN** to select manual ranging.
- Connect the calibrator to the PM2530 for 4-wire Ohms. To obtain correct readings connect the Fluke 5700A calibrator as follows:

Fluke 5700A Calibrator	PM2530
SENSE HI	SENSE COM
SENSE LO	OHMS SENSE
OUTPUT HI	SOURCE COM
OUTPUT LO	OHMS SOURCE
GUARD	GUARD (guard switch in Ext. position)

- For each step in Table 6.8, select the indicated range (RANGE **UP** **DOWN**), supply the specified input, and verify that the displayed reading is within the limits shown.
- Select speed mode SPD4 (READING RATE **UP**)
- For each step in table 6.9 select the indicated range (RANGE **UP** **DOWN**), supply the specified input, and verify that the displayed reading is within the limits shown.
- Disconnect the calibrator from the PM2530.

Table 6.8 Resistance test, SPD2 (90 days spec)

STEP	RANGE	INPUT	Rd 1)	RELATIVE TOLERANCE	FACTORY UNCERTY	USER'S TOLERANCE	OVERALL TOLERANCE MIN	OVERALL TOLERANCE MAX
1	20 Ω	0 Ω (short)		00.00020	00.00000			
2	200 Ω	0 Ω (short)		000.00020	000.00000			
3	2 kΩ	0 Ω (short)		0.000020	0.000000			
4	20 kΩ	0 Ω (short)		00.00020	00.00000			
5	200 kΩ	0 Ω (short)		000.00020	000.00000			
6	2 MΩ	0 Ω (short)		0.000020	0.000000			
7	20 MΩ	0 Ω (short)		00.0020_	00.00000			
8	200 MΩ	0 Ω (short)		000.02_	000.00000			
9	20 Ω	19 Ω	00.00089	1)	00.00008			
10	200 Ω	190 Ω	000.0089	1)	000.0010			
11	2 kΩ	1.9 kΩ	0.000089	1)	0.000008			
12	20 kΩ	19 kΩ	00.00089	1)	00.00007			
13	200 kΩ	190 kΩ	000.0089	1)	000.0018			
14	2 MΩ	1.9 MΩ	0.000248	1)	0.000014			
15	20 MΩ	1.9 MΩ	00.0039_	1)	00.00042			
16		19 MΩ	00.0210_	1)	00.00042			
17	200 MΩ	100 MΩ	000.12_	1)	000.0115			

Table 6.9 Resistance test, SPD 4 (90 days spec)

STEP	RANGE	INPUT	Rd 1)	RELATIVE TOLERANCE	FACTORY UNCERTY	USER'S TOLERANCE	OVERALL TOLERANCE	
							MIN	MAX
1	20 Ω	0 Ω (short)		00.002	00.00000			
2	200 Ω	0 Ω (short)		000.02	000.0000			
3	2 k Ω	0 Ω (short)		0.0002	0.000000			
4	20 k Ω	0 Ω (short)		00.002	00.00000			
5	200 k Ω	0 Ω (short)		000.02	000.0000			
6	2 M Ω	0 Ω (short)		0.0020	0.000000			
7	20 Ω	19 Ω	00.005	1)	00.00008			
8	200 Ω	190 Ω	000.05	1)	000.0010			
9	2 k Ω	1.9 k Ω	0.0005	1)	0.000008			
10	20 k Ω	19 k Ω	00.005	1)	00.00007			
11	200 k Ω	190 k Ω	00.005	1)	000.0018			
12	2 M Ω	1.9 M Ω	000.39	1)	0.000014			

- 1) The resistance values supplied by the 5700A calibrator (Rref) are not exactly the values indicated in table 6.8. The relative tolerance of the reading depends on the value supplied by the 5700A. This value (Rref) is indicated on the 5700A display. To calculate the relative tolerance, proceed as follows:

$$\left. \begin{array}{l} \text{RELATIVE TOLERANCE MIN.} = R_{\text{ref}} - R_d \\ \text{RELATIVE TOLERANCE MAX.} = R_{\text{ref}} + R_d \end{array} \right\} \text{ for } R_d \text{ see Table 6.8}$$

To calculate the overall tolerance, proceed as follows:

$$\begin{array}{l} \text{OVERALL TOLERANCE MIN.} = \text{RELATIVE TOLERANCE MIN.} - \text{FACTORY UNCERTY} - \text{USER'S TOLERANCE} \\ \text{OVERALL TOLERANCE MAX.} = \text{RELATIVE TOLERANCE MAX.} + \text{FACTORY UNCERTY} + \text{USER'S TOLERANCE} \end{array}$$

Example: Suppose that at step 9 in table 6.8, the 5700A calibrator supplies 19.10000 Ω . Rd is 0.00089. The RELATIVE TOLERANCE MIN is $(19.10000 - 0.00089) = 19.09911$; the RELATIVE TOLERANCE MAX is $(19.10000 + 0.00089) = 19.10089$.

CURRENT DC ACCURACY TEST

The following procedure may be used to verify the accuracy of the Current DC function.

- A. Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C.
- B. Press **MODE** to select the Current DC measurement function
- C. Press **RESET** to select SPD2, and to switch all auxiliary functions off.
- D. Press **RANGE** **AUTO** to select manual ranging.
- E. Connect the calibrator to provide a current input to the AMPS INPUT terminals.
- F. For each step in Table 6.10, select the indicated range (RANGE **UP** **DOWN**), supply the specified input, and verify that the displayed reading is within the limits shown.
- G. Select speed mode SPD4 (READING RATE **UP**).
- H. For each step in table 6.11 select the indicated range (RANGE **UP** **DOWN**), supply the specified input and verify that the displayed reading is within the limits.
- I. Set the calibrator for zero mA and disconnect it from the PM2530.

Table 6.10 Current DC test, SPD2 (90 days spec)

STEP	RANGE	INPUT (Idc)	RELATIVE TOLERANCE		FACTORY UNCERTY	USER'S TOLERANCE	OVERALL TOLERANCE	
			MIN	MAX			MIN	MAX
1	2 μ A	0 μ A (open)	- 0.000020	0.000020	0.000000			
2	20 μ A	0 μ A (open)	- 00.00020	00.00020	00.00000			
3	200 μ A	0 μ A (open)	- 000.0020	000.0020	000.0000			
4	2 mA	0 mA (open)	- 0.000020	0.000020	0.000000			
5	20 mA	0 mA (open)	- 00.00060	00.00060	00.00000			
6	200 mA	0 mA (open)	- 000.0060	000.0060	000.0000			
7	2 A	0 A (open)	- 0.000060	0.000060	0.000000			
8	2 μ A	2 μ A	1.999780	2.000220	0.000980			
9		-2 μ A	- 2.000220	- 1.999780	0.000980			
10	20 μ A	20 μ A	19.99780	20.00220	00.00095			
11		-20 μ A	- 20.00220	- 19.99780	00.00095			
12	200 μ A	200 μ A	199.9780	200.0220	000.0036			
13		-200 μ A	- 200.0220	- 199.9780	000.0036			
14	2 mA	2 mA	1.999780	2.000220	0.000012			
15		-2 mA	- 2.000220	- 1.999780	0.000012			
16	20 mA	20 mA	19.99640	20.00360	00.00012			
17		-20 mA	- 20.00360	- 19.99640	00.00012			
18	200 mA	20 mA	019.9910	020.0090	000.0018			
19		-20 mA	- 020.0090	- 019.9910	000.0018			
20		100 mA	099.9820	100.0360	000.0018			
21		-100 mA	- 100.0360	- 099.9820	000.0018			
22		200 mA	199.9640	200.0360	000.0018			
23		-200 mA	- 200.0360	- 199.9640	000.0018			
24	2 A	2 A	1.999640	2.000360	0.000030			
25		-2 A	- 2.000360	- 1.999640	0.000030			

Table 6.11 Current DC test, SPD4 (90 days spec)

STEP	RANGE	INPUT (Idc)	RELATIVE TOLERANCE		FACTORY UNCERTY	USER'S TOLERANCE	OVERALL TOLERANCE	
			MIN	MAX			MIN	MAX
1	2 μ A	0 μ A (open)	- 0.0001	0.0001	0.000000			
2	20 μ A	0 μ A (open)	- 00.001	00.001	00.00000			
3	200 μ A	0 μ A (open)	- 000.01	000.01	000.0000			
4	2 mA	0 mA (open)	- 0.0001	0.0001	0.000000			
5	20 mA	0 mA (open)	- 00.002	00.002	00.00000			
6	200 mA	0 mA (open)	- 000.02	000.02	000.0000			
7	2 A	0 mA (open)	- 0.0002	0.0002	0.000000			
8	2 μ A	2 μ A	1.9995	2.0005	0.000980			
9		-2 μ A	- 2.0005	- 1.9995	0.000980			
10	20 μ A	20 μ A	19.995	20.005	00.00095			
11		-20 μ A	- 20.005	- 19.995	00.00095			
12	200 μ A	200 μ A	199.95	200.05	000.0036			
13		-200 μ A	- 200.05	- 199.95	000.0036			
14	2 mA	2 mA	1.9995	2.0005	0.000012			
15		-2 mA	- 2.0005	- 1.9995	0.000012			
16	20 mA	20 mA	19.993	20.007	00.00012			
17		-20 mA	- 20.007	- 19.993	00.00012			
18	200 mA	20 mA	019.98	020.02	000.0018			
19		-20 mA	- 020.02	- 019.98	000.0018			
20		100 mA	099.96	100.04	000.0018			
21		-100 mA	- 100.04	- 099.96	000.0018			
22		200 mA	199.93	200.07	000.0018			
23		-200 mA	- 200.07	- 199.93	000.0018			
24	2 A	2 A	1.9993	2.0007	0.000020			
25		-2 A	- 2.0007	- 1.9993	0.000020			

CURRENT AC ACCURACY TEST

The following procedure may be used to verify the accuracy of the Current AC function.

- Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C.
- Press **ACV** to select the Current AC +DC measurement function.
- Press **RESET** to select SPD2, to switch all auxiliary functions off, and to set the RMS filter to >40 Hz.
- Press **RANGE** **RANGE** to select manual ranging.
- Connect the calibrator to provide a current input to the AMPS INPUT terminals.
- For each step in Table 6.12, select the indicated range (RANGE **UP** **DOWN**), supply the specified input (current, frequency), and verify that the displayed reading is within the limits shown.
- Set the calibrator for zero mA and disconnect it from the PM2530.

Table 6.12 Current AC test, SPD2 (90 days spec)

STEP	RANGE	INPUT (Iac)	FREQ	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE	
				MIN	MAX			MIN	MAX
1	200 μ A	200 μ A	1 kHz	199.700	200.300	000.048			
2	2 mA	2 mA	1 kHz	1.99700	2.00300	0.00032			
3	20 mA	20 mA	50 Hz	19.9700	20.0300	00.0032			
4		20 mA	1 kHz	19.9700	20.0300	00.0032			
5		20 mA	5 kHz	19.9700	20.0300	00.0170			
5		20 mA	10 kHz	19.8900	20.1100	00.0420			
6	200 mA	20 mA	1 kHz	019.880	020.120	000.032			
7		100 mA	1 kHz	099.800	100.200	000.034			
8		200 mA	1 kHz	199.700	200.300	000.034			
9	2 A	2 A	1 kHz	1.99700	2.00300	0.00134			
10	(Idc) 200 mA	200 mA	DC	199.700	200.300	000.0018			

PT100 TEMPERATURE ACCURACY TEST (USING Pt100 SIMULATOR)

The following procedure may be used to verify the accuracy of the Temperature function.

- Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C.
- Press **TEMP** to select the Temperature function.
- Press **1** (DIN385PT), **4** (UNITS), **1** (°C), **0** (RETURN), **0** (EXIT)
- Press **RESET** to select SPD2, and to switch all auxiliary functions off.
- Connect the Pt100 simulator to the PM2530 for Pt100 temperature (see Section 4.4.9).
- For each step in Table 6.13 supply the specified input, and verify that the displayed reading is within the limits shown.
- Disconnect the Pt100 simulator from the PM2530.

Table 6.13 Pt100 Temperature test (using Pt100 simulator), SPD2 (90 days spec)

STEP	INPUT (°C)	RELATIVE TOLERANCE		FACTORY UNCERT'Y	USER'S TOLERANCE	OVERALL TOLERANCE	
		MIN	MAX			MIN	MAX
1	-200	- 0199.92	- 0200.08	0000.01			
2	000	- 0000.02	0000.02	0000.01			
3	100	0099.95	0100.05	0000.01			
4	200	0199.92	0200.08	0000.01			
5	300	0299.89	0300.11	0000.01			

VOLTS LOW FREQUENCY ACCURACY TEST

The following procedure may be used to verify the accuracy of the Volts Low Frequency function. In addition the VOLTS DC test must be done to test the complete Volts Low Frequency function.

- Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C. The VOLTS DC accuracy test must have been performed previously.
- Press **VLF** to select the Volts Low Frequency measurement function.
- Press **RESET** to select SPD2, and to switch all auxiliary functions off.
- Press RANGE **AUTO** to select manual ranging.
- Connect the calibrator to provide a voltage input to the VOLTS INPUT terminals.
- Select the 2 volt range (RANGE **UP** **DOWN**) and, supply the specified input (voltage, frequency) according to table 6.14, and verify that the displayed reading is within the limits shown.
- Set the calibrator for zero volts and disconnect it from the PM2530.

Table 6.14 Volts Low Frequency test, SPD2 (90 days spec)

STEP	RANGE	INPUT (Vac)	FREQ	RELATIVE TOLERANCE		FACTORY UNCERTY	USER'S TOLERANCE	OVERALL TOLERANCE	
				MIN	MAX			MIN	MAX
1	2 V	2 V	80 Hz	1.9974	2.0026	0.0002			

FREQUENCY/PERIOD TEST

The following procedure may be used to verify the accuracy of the FREQUENCY and PERIOD measurement function. In addition the VOLTS AC test, Table 6.6 only, must be done to test the complete FREQUENCY/PERIOD function.

- Ensure the PM2530 has warmed up for at least 1 hour, and that the ambient temperature is between +18 and +28 °C.
- Press **FREQ** to select the Frequency measurement function.
- Press **RESET** to select SPD2, and to switch all auxiliary functions off.
- Press RANGE **AUTO** to select manual voltage ranging. Select voltage range 2 V displayed in the left upper display corner.
- Connect the frequency signal source to provide a voltage input to the VOLTS INPUT terminals.
- For each step in Table 6.15, supply the specified input (voltage, frequency), and verify that the displayed reading is within the limits shown.
- Set the calibrator for zero volts and disconnect it from the PM2530.

Table 6.15 Frequency/Period test, SPD2 (90 days spec)

STEP	INPUT (Vac)	FREQ	RELATIVE TOLERANCE		FACTORY UNCERTY	USER'S TOLERANCE	OVERALL TOLERANCE	
			MIN	MAX			MIN	MAX
1	200 mV	2 Hz	1.999979	2.000021	0.000001			
2	2 V	2 Hz	1.999979	2.000021	0.000001			
3	2 V	2 kHz	1.999979	2.000021	0.000001			
4	2 V	2 MHz	1.999979	2.000021	0.000001			

STORE/RECALL THE VERIFICATION DATE

The PM2530 enables you to store and to recall the verification date either by front panel operation (LOCAL) or via the GPIB (REMOTE).

LOCAL:

- press **MISC** to invoke the MISC (MISCellaneous functions) menu (Fig. 6.3).
- select
 - 5. CHECK to invoke the CHECK menu
 - 2. VER. DATE to invoke the VER.DATE menu. The last set verification date is displayed.
 - 1. NEW DATE to set the verification date to the actual date. Key in the password, and press **ENTER** (the initial password is 2530).
 - 2. NEW PASSW to define a new password: key in the old password, press **ENTER**, key in the new password, press **ENTER**. The password is a whole decimal number in the range 0 up to 65536. The initial password is 2530.

NOTE: if a wrong password is entered, the display shows the message "ERROR PUSH ANY KEY TO CONTINUE". In this case, Press a key and select 2. NEW PASSW again to enter the correct password.

- 0. RETURN to return to the CHECK menu
- 0. RETURN to return to MISC the menu
- 0. EXIT to exit the MISC menu

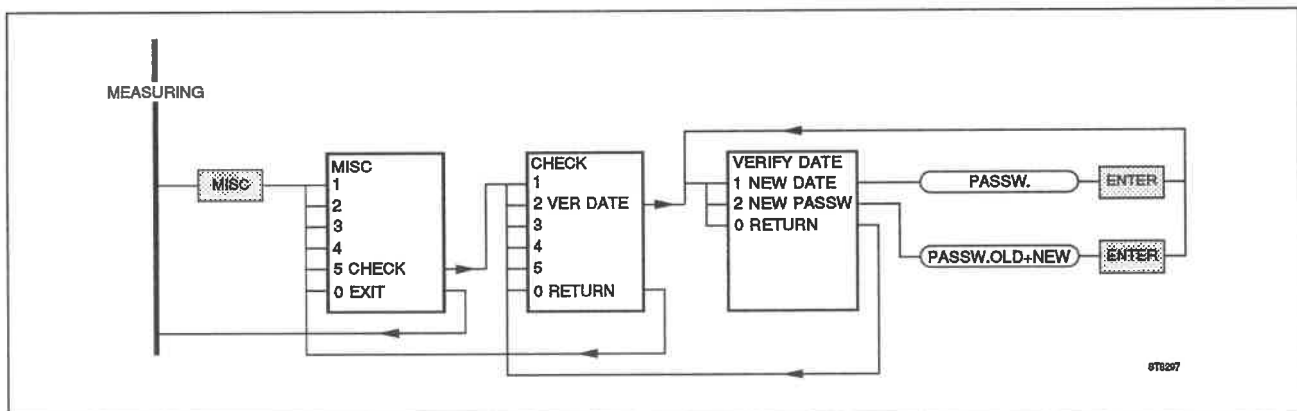


Fig. 6.3 Store/Recall verification date menu.

REMOTE:

The following command can be used to read the date of last verification:

:CAL:DATE? VER the PM2530 returns <year>,<month>,<day>

The following commands can be used to change the verification date:

:SYST:UNPR:STAT ON to select the PM2530 unprotect state.

:SYST:UNPR:SEL <passw>, VER to select the **calibration verification mode**; <passw> is the PM2530 verification password, at delivery set to 2530.

:SYST:UNPR:STAT OFF to leave the unprotected state. The verification date is now set to the actual system date.

Refer to Section 4 of the Command Reference Manual for more detailed informaton.

6.6 CALIBRATION

Refer to the PM2530 Service Manual, for calibration procedures.

NOTE: Do not press the pencil operated CAL switch at the front panel, unless you have a copy of the Service Manual and intend to calibrate the instrument.

The instrument features closed-case calibration controlled over the front panel or over the GPIB interface. Using known reference sources, closed-case calibration has many advantages. There are no parts to disassemble, no mechanical adjustments to make, and the instrument can be calibrated by an automated instrumentation system.

The instrument should normally be calibrated on a regular cycle, typically every 90 days or once a year. The chosen calibration cycle depends on the accuracy specification you wish to maintain. The instrument should also be calibrated if it fails the accuracy verification tests (see Section 6.5.2) or has been repaired. The calibration date is stored in the instrument memory. The PM2530 enables you to read the calibration date either via the (GPIB REMOTE), or by front panel operation (LOCAL).

READING THE CALIBRATION DATE (LOCAL)

- press **MISC** to invoke the MISC (MISCellaneous functions) menu (Fig. 6.4).
- select
 - 5. CHECK to invoke the CHECK menu
 - 3. CAL DATE the display shows the calibration date
 - 0. RETURN to return to the CHECK menu
 - 0. RETURN to return to the MISC menu
 - 0. EXIT to exit the MISC menu

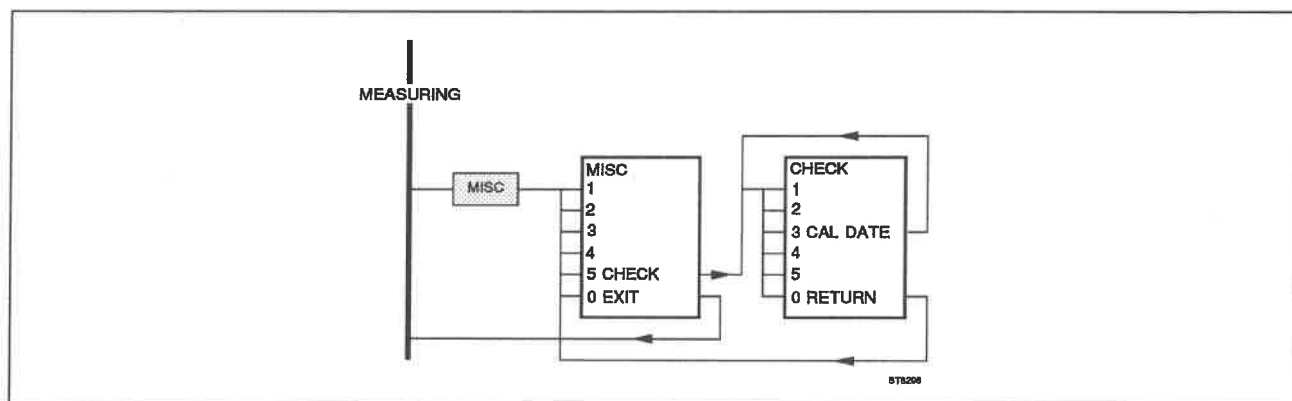


Fig. 6.4 Read calibration date menu.

READING THE CALIBRATION DATE (REMOTE)

The following command can be used to read the date of last calibration:

`:CAL:DATE? CAL` The 2530 returns the calibration date in the format <year>,<month>,<day>.

Refer to Section 4 of the Command Reference Manual for more detailed information.

6.7 SERVICE

If the instrument fails, check that operating instructions presented earlier in this manual are being followed. If the problem cannot be solved, forward the instrument to the nearest FLUKE/PHILIPS Service Centre. A list of Service Centers is provided on the last pages of this manual. Be sure to pack the instrument securely; use the original container if available. Include a description of the problem.

6.8 SERIAL NUMBER AND FIRMWARE VERSION

The serial number of your PM2530 is given on the type plate at the rear. It is also stored in non-volatile memory. Beside the serial number, the firmware version is stored. To recall both, proceed as follows:

READING THE SERIAL NUMBER AND FIRMWARE VERSION (LOCAL)

- press **MISC** to invoke the MISC (MISCellaneous functions) menu (Fig. 6.5).
- select
 - 5. CHECK to invoke the CHECK menu
 - 5. SERIAL NO the display shows the serial number and Firmware Version
 - 0. RETURN to return to the CHECK menu
 - 0. RETURN to return to the MISC menu
 - 0. EXIT to exit the MISC menu

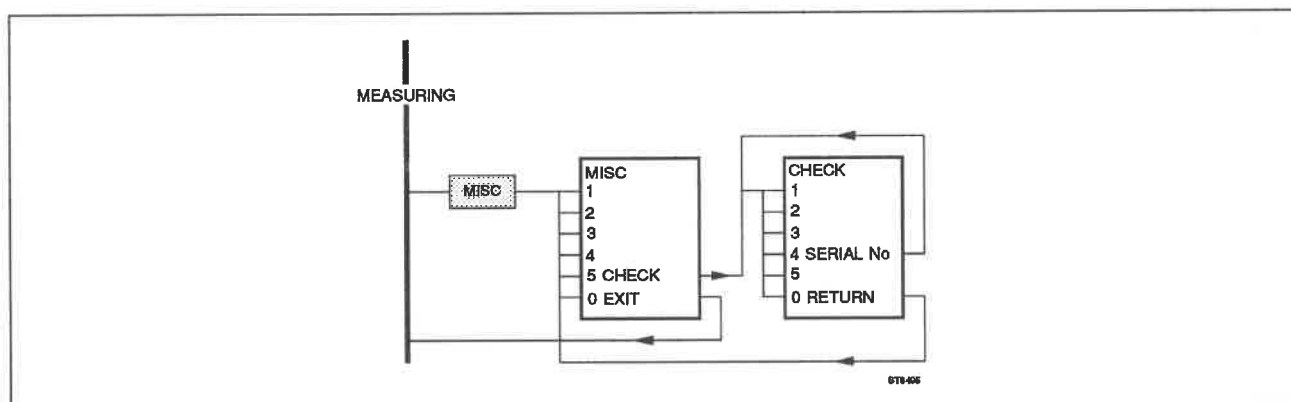


Fig. 6.5 Serial number and firmware version menu.



Fig. 6.6 Displayed serial number and firmware version.

READING THE SERIAL NUMBER AND FIRMWARE VERSION (REMOTE)

The following command can be used to read the serial number and firmware version.

***IDN?** The 2530 returns "Philips, PM2539, aaaaaaaa, Vxx.xx" aaaaaaaa is the serial number and xx.xx is the firmware version.

Refer to section 2 of the Command Reference Manual for more detailed information.

7 OPERATING FEATURE REFERENCE

7.1 FRONT PANEL FEATURES

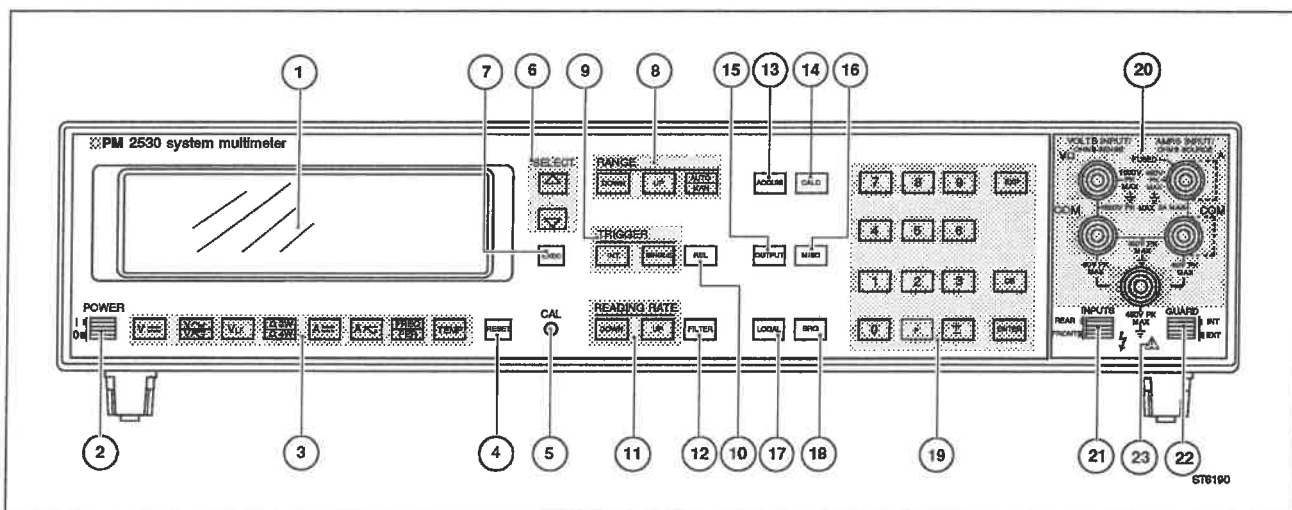
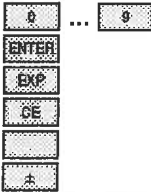


Fig. 7.1. PM2530 front panel.

Item	Name	Description	Section
1	Display	Shows results, operating menus, annunciators, etc.	7.2
2		POWER switch.	4.1.1
3	 	MEASUREMENT FUNCTION selection buttons. Selects Volts DC. Toggles between Volts AC and AC+DC. Selects Volts low frequency. Toggles between 2-wire and 4-wire resistance. Selects Current DC. Selects current AC+DC. Toggles between Frequency and Period. Selects Pt100 temperature measurements.	4.4 4.4.2 4.4.3 4.4.4 4.4.5 4.4.6 4.4.7 4.4.8 4.4.9
4		Selects default settings for the current measurement function.	4.17
5	o CAL	Pencil operated CALIBRATION switch; for calibration adjustment only. Refer to the PM2530 Service Manual, P/N 4822 872 35315	
6	SELECT and	Moves the cursor in the operating menus.	7.4
7		Selects the function indicated by the cursor in the operating menu.	7.4

Item	Name	Description	Section
8	RANGE    	Ranging push buttons. Toggles between auto ranging and manual ranging mode. Selects a lower range and manual ranging. Selects a higher range and manual ranging.	4.5.1
9	TRIGGER  	Trigger push buttons. Selects internal (continuous) measurement triggers. Selects manual trigger mode with first use, and initiates a measurement with each additional use.	4.5.2
10		Selects the relative reference function and stores the present reading as a relative reference value if the function is off; switches off the relative reference function if it is on (toggle function).	4.7
11	READING RATE  	Reading rate push buttons. Selects a higher reading rate (results in a lower measurement resolution). Selects a lower reading rate (results in a higher measurement resolution).	4.5.3
12		Invokes a menu from which a filter type can be selected/deselected.	4.6
13		Invokes a menu from which the following ACQUISITION functions can be selected: - TIMING, to set trigger timing parameters (interval and delay). - NULLING, to eliminate input offsets. - MIN/MAX, to store highest and lowest readings. - LIMITS, to count and signal limit exceeding. - BURST, to select the BURST measurement mode.	4.5.2 4.8 4.11 4.12 4.9
14		Invokes a menu from which the following CALCULATION functions can be selected: - (AX + B):Z, to scale measurements. - DELTA%, to calculate the percentual deviation from a selected reference value. - dBm, to calculate the dBm value. - REL REF, to select/deselect the relative reference function and to set a reference value.	4.10.1 4.10.2 4.10.13 4.7
15		Invokes a menu from which the following OUTPUT functions can be selected: - RD MIN/MAX, to read the MIN/MAX memories. - RD LIMCTR, to read the LIMIT memories. - RD BURST, to read the BURST memory. - DISPLAY, to set the display length, backlight, number of annunciators and contrast. - BEEPER, to switch the beeper on/off, and to set the BEEPER conditions	4.11 4.12 4.9 4.13

Item	Name	Description	Section
16		Invokes a menu from which MISCELLANEOUS functions can be selected: - SAVE, to assign a number to the current instrument settings and save them. - RECALL, to restore saved instrument settings. - DATE/TIME, to set the internal clock/calendar. - IEEE ADDR, to set the interface settings. - CHECK, to enable TEST and verification facilities.	4.15 4.15 4.19 4.18 6
17		Sets the instrument to the LOCAL state when it is in the REMOTE state (without Local Lock Out).	4.20
18		Generates a service request from the front panel.	4.21
19	NUMERIC key pad 	To enter numerics, exponents and polarity signs. Numbers. Enters keyed in numbers, selects pointed menu item. Establishes exponential values (power of 10) Clears the displayed parameter value or menu. Decimal point. Changes the polarity (toggles between + and -).	7.4
20	INPUT TERMINALS	Measurement connection terminals, connected if the INPUTS REAR/FRONT switch is in the FRONT position.	4.4
21	 REAR/FRONT	Selects either front or rear input terminals.	4.4
22	 INT/EXT	Connects the instrument guard internally (INT), or to connect the instrument guard to the GUARD terminal (EXT).	4.3
23		Refers to the maximum input values.	A.2.14.

7.2 DISPLAY FEATURES

Figure 7.2 shows the PM2530 display fields, and an example of displayed items. All possible display messages are listed below.

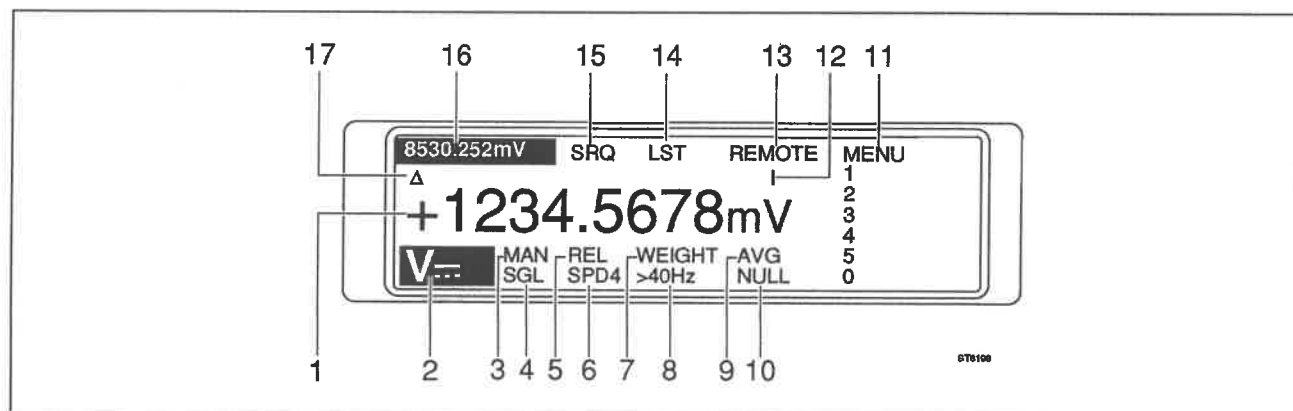


Fig. 7.2 The PM2530 display.

Field	Indicates	Annunciator	Description	Section
1	Reading	"±0...9 E"	Reading, up to 7 ¹ / ₂ digit, including sign (±) and unit (mV, V, µA, mA, A, Ω, kΩ, MΩ, Hz, kHz, MHz, ms, s, °C, °F, K, dBm).	7.4
	Parameter	"±0...9 E"	The value of current or new parameters. The min/max value to be entered by front panel operation is +/-99999E+/-9.	
		-----	- no result available (e.g. waiting for a trigger). - result of a calculation can not be displayed as it is out of the range +/- (0.0001 ... 99999)E+/-9.	
	Errors	"OL"	Input overload (select higher range or supply lower input value).	7.5
		"UL"	Underload; the input value in the dBm function is 0 (zero).	4.10.4
		"OFL"	Overflow; the result of a calculation (dBm, SCALE, DELTA%, REL REF) is out of range +/-9.9E+/-37.	4.10.1
2	Measurement Function	"HH:MM:SS"	PM2530 system time (hours:minutes:seconds).	4.19
		"YY-MM-DD"	PM2530 system date (year-month-day).	4.19
		"V=="	Volts DC.	4.4.2
		"V~"	Volts AC.	4.4.3
		"V~"	Volts AC + DC.	4.4.3
		"Vlf"	Volts low frequency.	4.4.4
		"Ω2W"	Resistance 2-wire.	4.4.5
		"Ω4W"	Resistance 4-wire.	4.4.5
		"A=="	Current DC.	4.4.6
		"A~"	Current AC + DC.	4.4.7
		"FRQ"	Frequency.	4.4.8
		"PER"	Period.	4.4.8
		"TMP"	Temperature.	4.4.9
3	Range mode	"MAN" BLANK	Manual ranging. Auto ranging.	4.5.1
4	Trigger mode	"SGL" BLANK	Single (manual/external) triggering. Internal (continuous) triggering.	4.5.2
5	Relative ref.	"REL" BLANK	Relative reference mode is on. Relative reference mode is off.	4.7
6	Reading rate	"SPD1..6"	Lowest ... highest measurement speed, depending on the selected reading rate.	4.5.3
7	Weighting filter state	"WEIGHT" BLANK	The weighting filter is on. The weighting filter is off.	4.6
8	Analog filter state	"> ...Hz"	Lower frequency of the RMS filter (>10 Hz, >40 Hz, >400 Hz in ).	4.6.3
		"A-FILT" BLANK	Analog filter on ( ,  ,  , ). Analog filter or RMS filter off.	4.6.2

Field	Indicates	Annunciator	Description	Section
9	Averaging filter state	"AVG-FILT" BLANK	The averaging filter is on. The averaging filter is off.	4.6
10	Nulling	"NULL" BLANK	The input offset nulling function is on. Only in  ,  , and  . The input offset nulling function is off.	4.8
11	Operating menu Auxiliary functions	"BURST" "SCALE" "DELTA%" "dBm" "INTERVAL" "DELAY" "MIN/MAX" "LIMITS"	A function dependant text when an operating menu is used. The burst mode is on. The SCALE function is on. The DELTA% function is on. The dBm function is on. Triggering with interval selected. Triggering with trigger delay selected. The MIN/MAX function is on. The LIMITS function is on.	7.4 4.9 4.10.1 4.10.2 4.10.3 4.5 4.5 4.11 4.12
12	Symbols	"  " "  " "↑"	The reading is out of limits. New maximum/minimum reading. Input circuit clipping, crest factor exceeded.	4.12 4.11 7.5
13	Remote control	"REMOTE" BLANK	The PM2530 is in remote control. The PM2530 can be operated from the front panel (local).	5 *) 4
14	Remote state	"LSTN" "TALK"	The PM2530 is in the LISTENER state. The PM2530 is in the TALKER state.	5 *) 5 *)
15	Service requested	"SRQ" BLANK	The PM2530 asks for service; "SRQ" disappears after polling. No PM2530 Service Request pending.	5 *)
16	Raw result Input voltage Burst memory location Time	"±0...9 E" "±0...9 E" "LOC .." "HH:MM:SS"	The unprocessed measurement value at Relative Reference, SCALED, DELTA%, and dBm measurements. The input voltage at Frequency Period measurements. The memory location number at reading the burst memory. The time of updating of the MIN/MAX or LIMIT memory, at reading these memories.	4.7 4.10 4.8 4.9 4.11 4.12
17	Trigger	Δ	Indicates that a trigger is received, and that a measurement is in progress.	4.5.2

*) See also the Command Reference Manual

7.3 REAR PANEL FEATURES

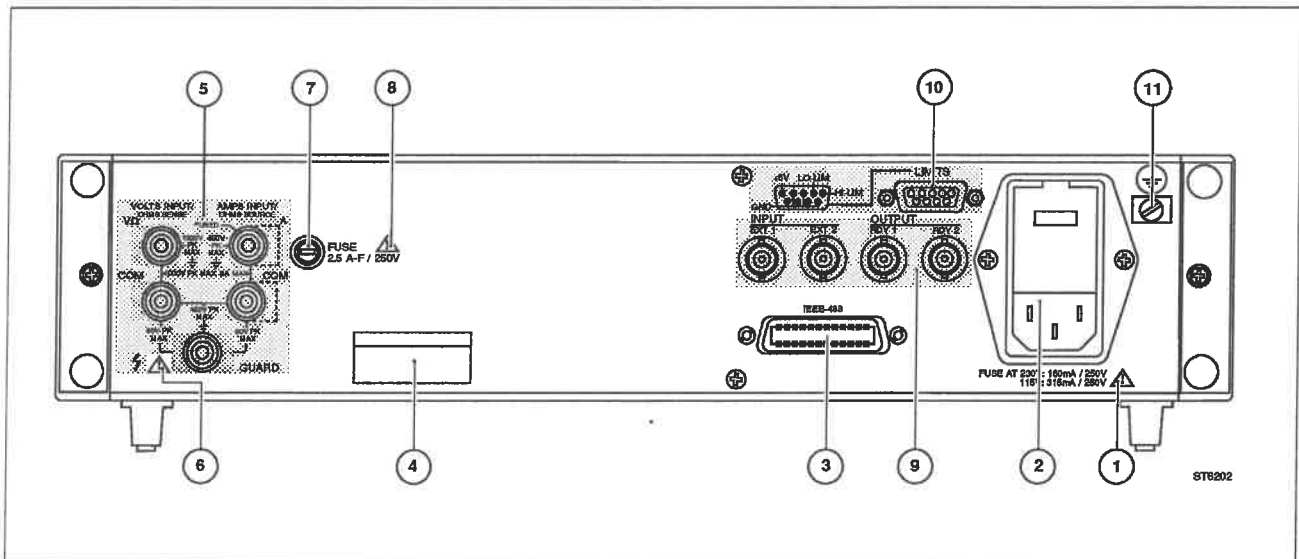


Fig. 7.3 The PM2530 rear panel.

Item	Name	Description	Section
1		Refers to the required line voltage and fuse ratings.	3.1
2	Power connector	Three prong connector with fuse holder, accepting line cord with ground wire.	3.1
3	GPIB connector	Connector in accordance with IEEE-488.2 1987, to connect the PM2530 to the GPIB.	5
4	Type plate	Indicates the PM2530 type number "PM2530/...", 9447 325 30...), serial number "No DM", required line voltage and frequency, and power consumption.	I
5	Input terminals	Measurement connection terminals, connected if front switch INPUTS REAR/FRONT is in the REAR position.	4.4
6		Refers to the maximum input values.	A.2.14
7	Fuse holder	Provided with 2.5 A fast blow, 250 V fuse in series with the A (current) input terminal.	4.4.6 4.4.7
8		Refers to the correct current input fuse rating.	6.4.

Item	Name	Description	Section
9	Trigger, ready connectors RDY1	External (single) trigger inputs and measurement complete outputs. LOCAL: reading complete output (BURST). REMOTE: see SCPI "Output Subsystem".	4.9 4.5.2 5.3.10
	RDY2	LOCAL: BURST complete output. REMOTE: See section ? "Output subsystem".	4.9 11 *)
	EXT1, EXT2	LOCAL: sample, arm, or trigger input (BURST), trigger input (normal). REMOTE: See "Trigger Subsystem".	4.9 4.5.2 5.3.19
10	LIMITS	Output to signal limit exceeding.	4.12
11	Safety earth terminal	Can be used as additional safety earth connection.	3.1.1

*) Command Reference Manual

7.4 USING THE OPERATING MENUS

Pressing any of the following push buttons causes an operating menu to be displayed:

TEMP, FILTER, ACQUIS, CALC, OUTPUT, MISC.

You can select the required option from the menu. The selected setting is shown in inverse text. The cursor is also shown in inverse text, it highlights the item number for the option to be selected. In the example below, ITEM A is the selected setting and 4 is the cursor position.

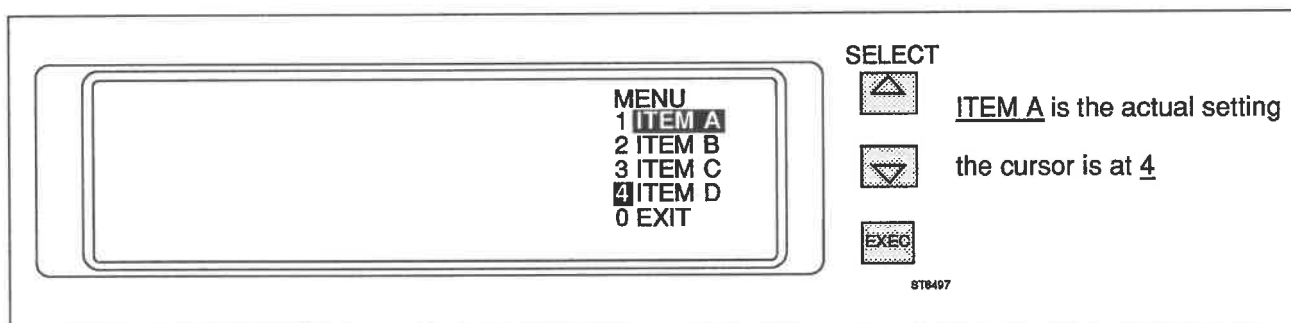


Fig. 7.4 The PM2530 operating menu.

SELECTING AN ITEM (SEE FIG. 7.4)

- move the cursor with SELECT or to the required figure and press or
- OR
- key in the corresponding figure (no or required).

When an item is selected, either:

- The same or a new menu is displayed. You can select further items as described above.
- OR
- the PARAMETER VALUE of the selected setting is displayed (see Fig. 7.4). It can be changed as detailed below.

LEAVING A MENU

- press **CE** to leave the menus
OR
- press **0** to return to a previous menu (0. RETURN) and to leave the first menu (0. EXIT).

CHANGING A NUMERIC PARAMETER VALUE.

key in a new value and press **ENTER**.

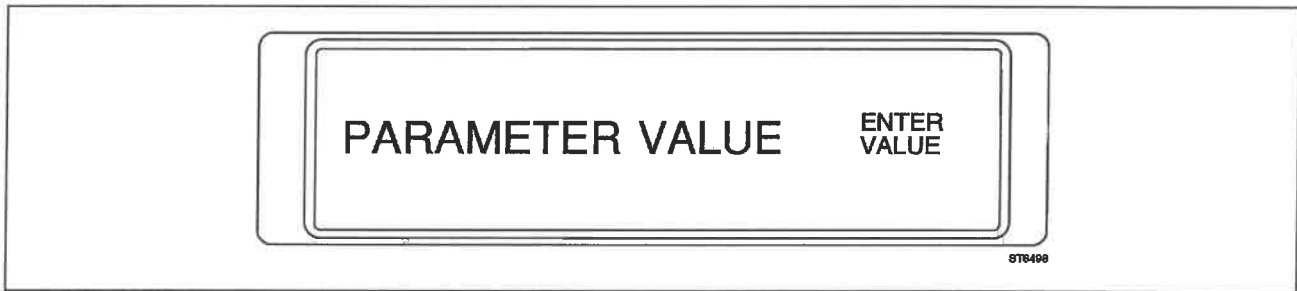


Fig. 7.5 Setting a parameter value.

Pressing **ENTER** without keying in a new value leaving the current value unchanged. Pressing **CLEAR**, only clears the display; the stored parameter value remains unchanged.

To set exponential values (powers of 10), press **EXP**. e.g. to set 6E3 (6000), press **6**, **EXP**, **3**. The value to be entered via the keyboard must be in the range +/- (0.0001 ... 99999)E+/-9 (limited by the display length).

Menu operation is represented in diagrams throughout this manual. The example below shows all aspects of using the operating menus.

EXAMPLE OF MENU ASSISTED OPERATION (SEE ALSO FIG. 7.6)

This example shows how to set a 10 seconds trigger delay in the single trigger mode, for the volts DC measuring function.

- press **V_{DC}** to select the Volts DC measurement function.
- press **RESET** to select the default PM2530 set-up.
- press **TRIGGER SINGLE** to select the single trigger mode.
- press **ACQUIS**; the ACQUIS menu will be displayed.
- press **1**, **2**, **1**, to select successively:
 - 1. TIMING the TIMING menu will be displayed.
 - 2. DISPLAY the DISPLAY menu will be displayed.
 - 1. USER VAL the current trigger delay time will be displayed.
- press **1**, **0**, **ENTER** the trigger delay is set to 10 seconds. The DELAY menu will be displayed again.
- press **CE** to leave the menus.
- press **TRIGGER SINGLE** to start a measurement.

Notice that the PM2530 displays a new result after approximately 10 seconds.

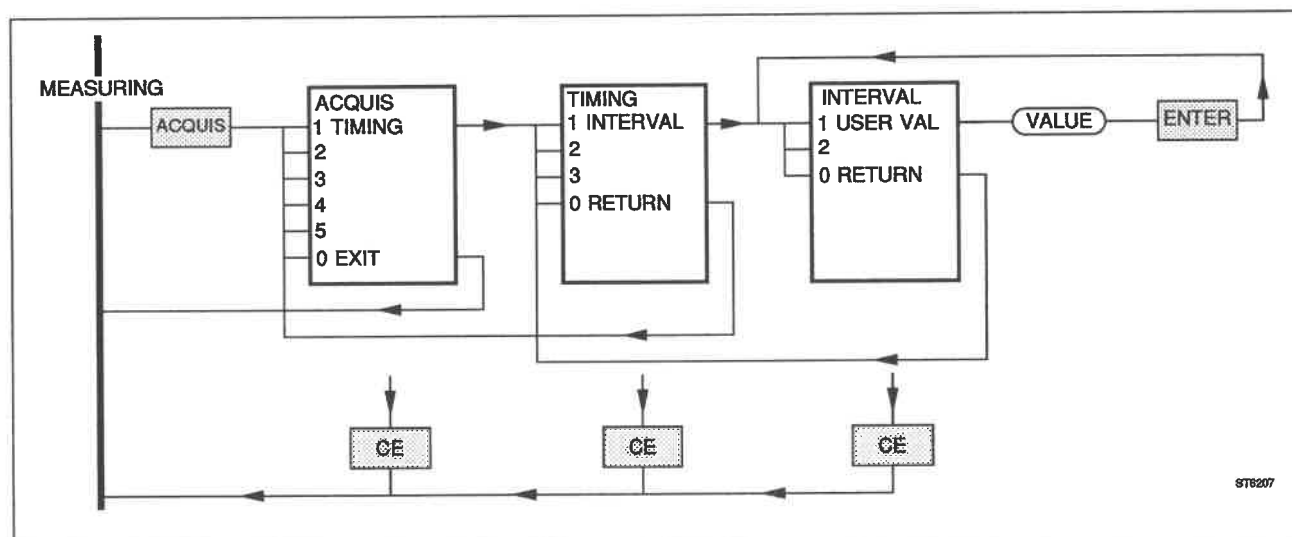


Fig. 7.6 Example operating menu structure.

7.5 ERROR MESSAGES

Instrument, operating, and measurement errors are reported by means of a displayed message, by an audible beeper signal, and via the GPIB if the PM2530 is remote controlled.

DISPLAYED ERROR MESSAGES

Message	Error	Description
OL	overload	The input signal is too large. Reduce the input signal amplitude, or select a higher range. The OL message can be disabled, refer to Section 4.13.
UL	underload	An input value 0 (zero) is measured in the dBm function, refer to section 4.10.4.
OFL	numeric overflow	The result of a calculation (dBm, SCALE, DELTA%, REL REF) is out of range $\pm 9.9E\pm 37$.
↑	input clipped	The reading is not correct. The input signal causes the PM2530 input amplifier to be clamped. This can be caused by spikes on the input signal. In the Volts AC and Current AC functions, the message indicates that the PM2530 crest factor limit is exceeded. Refer to Sections 4.4.3 (Volts AC) or 4.4.7 (Current AC).
E xxxx	instrument error	An instrument error is detected after power up, or when performing the self test xxxx is the error code (Section 6.5.1). See if the instrument repeats the error. If the error persists, contact the nearest FLUKE/PHILIPS Service Centre. A list of Service Centers is provided on the last pages of this manual.
INT >>	interval time too short	When a function, range and speed combination is selected, which would result in a too short interval time INT >> is shown in the display and the interval time is adapted to the minimum possible.
CAL LOST	calibration data invalid	If this error occurs during normal operation contact the nearest FLUKE/PHILIPS service centre.

BEEPER ERROR MESSAGES

The following errors will cause the beeper to sound (refer to display messages above):

- overload (see OL)
- overflow (see OFL)
- crest factor limit exceeding (see ↑)
- input circuit clipping (see ↑)

Beeper error messages, except for OL in the highest Voltage and Current ranges, can be disabled. Refer to Section 4.14.

ERROR REPORTING IN REMOTE CONTROL

For error reporting in remote control refer to the command **:SYSTem:ERRor?** in Section 17 of the Command Reference Manual SYSTEM Subsystem, and also to the book "SCPI Standard Commands for Programmable Instruments" supplied with your PM2530 for an overview of SCPI Error Messages.

APPENDIX A

SPECIFICATIONS

A.1 SAFETY CHARACTERISTICS

This apparatus has been designed and tested in accordance with Safety Class I requirements of IEC Publication 348, Safety Requirements for Electronic Measuring Apparatus, and CSA C22.2 No 231, and has been supplied in a safe condition. It meets the requirements of UL1244 and VDE 0411.2

- This instrument is approved by the Canadian Standards Association (CSA) according to CSA 22.2 Nr 231.

This manual contains information and warnings which must be followed to ensure safe operation and to retain the instrument in a safe condition.

A.2 TECHNICAL SPECIFICATIONS

A.2.1 General

- Properties expressed in numerical values with stated tolerance are guaranteed by PHILIPS. Specified non-tolerance numerical values indicate those that could be nominally expected from the mean of a range of identical instruments.
- This specification is valid after the instrument has warmed up for 60 minutes to reach the 90 days accuracy. The warm up time to reach the 24 hours specification is 2 hours. The reference temperature is $23^{\circ}\text{C} \pm 1^{\circ}\text{C}$.
- This specification is valid within the rated range of use unless otherwise stated.

A.2.2 Volts DC

Ranges	20 mV, 200 mV, 2 V, 20 V, 200 V, 1000 V. Range 20 mV not in speed modes SPD5 and SPD6.
Ranging	Manual or automatic. No automatic down ranging to the 20 mV range, and to the 200 mV range in speed mode SPD1.
Input nulling	Input offset voltages (e.g. thermal contact potentials) can be compensated via the "NULL" function.

Accuracy, $\pm$ (ppm of reading + ppm of range). ¹⁾

SPEED MODE	RANGE	24 HOUR, Tcal $\pm$ 1 °C		90 DAY, Tcal $\pm$ 5 °C		1 YEAR, Tcal $\pm$ 5 °C	
SPD1,2	20 mV	36	10	50	20	60	20
	200 mV	18	3	25	4	36	4
	2 V	15	2	18	2	24	2
	20 V	15	2	18	2	24	2
	200 V	18	2	25	2	36	2
	1000 V	18	2	25	2	36	2
SPD3	20 mV	40	10	50	20	60	20
	200 mV	20	15	25	10	35	10
	2 V	15	5	20	5	25	5
	20 V	15	5	20	5	25	5
	200 V	20	5	25	5	35	5
	1000 V	20	5	25	5	35	5
SPD4	20 mV	500	500	500	500	500	500
	200 mV	100	100	150	100	200	100
	2 V	50	100	100	100	150	100
	20 V	50	100	100	100	150	100
	200 V	50	100	100	100	150	100
	1000 V	50	100	100	100	150	100
SPD5	200 mV	150	100	200	100	250	100
	2 V	100	100	150	100	200	100
	20 V	100	100	150	100	200	100
	200 V	100	100	150	100	200	100
	1000 V	100	100	150	100	200	100
SPD6	200 mV	500	500	1000	1000	1000	1000
	2 V	500	500	1000	1000	1000	1000
	20 V	500	500	1000	1000	1000	1000
	200 V	500	500	1000	1000	1000	1000
	1000 V	500	500	1000	1000	1000	1000

- ¹⁾ - Relative to calibration values and within 24 hours after nulling.
 - Zero point drift included in accuracy and temperature coefficient.
 - At 1000 V range, ppm of range is based on 2000 V full scale.
 - For 20 mV range (SPD1,2,3) and 200 mV range (SPD1,2) the stated accuracies are with analog filter on.

Transfer accuracy (SPD1, SPD2) $\pm$ (ppm of reading + ppm of range).

Within 10 minutes, and Tref $\pm$ 0.5 °C.

RANGE	ACCURACY	
20 mV	20 +	5
200 mV	10 +	2
2 V	5 +	2
20 V	5 +	2
200 V	8 +	2
1000 V	8 +	2

Temperature coefficient in range 0-18 °C and 28-50 °C: $\pm(\text{ppm of reading} + \text{count})/\text{K}$ (SPD2).

RANGE	TEMPERATURE	COEFFICIENT
20 mV	12	1.0
200 mV	8	1.0
2 V	5	0.3
20 V	5	0.3
200 V	8	0.3
1000 V	8	0.3

Measurement timing (for sampling times see A.2.11).

SPEED MODE	READINGS PER SECOND ¹⁾		MEASUREMENT TIME	
	50 Hz	60 Hz	50 Hz	60 Hz
SPD1 ²⁾	0.2	0.24	5 s	4.17 s
SPD2	2	2.4	0.5 s	0.417 s
SPD3	20	24	50 ms	41.7 ms
SPD4	200	50	2 ms	2 ms
SPD5 (Direct)	1000 ³⁾		250 μs	
SPD5 ⁴⁾ (Burst)	4000		250 μs	
SPD6 (Direct)	1000		20 μs	
SPD6 ⁴⁾ (Burst)	50000		20 μs	

¹⁾ In internal trigger mode with default trigger interval.

²⁾ Display update after 0.5 seconds; correct value after stated measurement time.

³⁾ Reduced to >500 measurements per second if triggered and read via the IEEE488 interface bus.

⁴⁾ In BURST mode; no real time output to display or interface.

Response time. ¹⁾

SPEED MODE	WITHOUT RANGING		WITH RANGING ²⁾		NUMBER OF COUNTS FROM FINAL VALUE ³⁾
	INTERNAL TRIGGER	SINGLE TRIGGER	INTERNAL TRIGGER	SINGLE TRIGGER	
1	5.5 s	5.1 s	6 s	5.5	50
2	1 s	0.52 s	1.5 s	1 s	10
3	110 ms	60 ms	600 ms	550 ms	10
4	15 ms	10 ms	500 ms	500 ms	5
5	3 ms	2 ms	500 ms	500 ms	5
6	1 ms	1 ms	500 ms	500 ms	5

¹⁾ Analog filter off; add 200 ms to the response times for analog filter on.

²⁾ Over maximum number of ranges.

³⁾ Difference between first reading and final value for a step input change. At single trigger for a step coincident with trigger.

Scale length and resolution.

SPEED MODE	RANGE	FULL SCALE	RESOLUTION
SPD1	20 mV 200 mV 2 V up to 1000 V	200000 2000000 20000000	0.1 μ V 0.1 μ V 0.1 μ V up to 0.1 mV
SPD2	20 mV 200 mV up to 1000 V	200000 2000000	0.1 μ V 0.1 μ V up to 1 mV
SPD3	20 mV 200 mV up to 1000 V	20000 200000	1 μ V 1 μ V up to 10 mV
SPD4	20 mV 200 mV up to 1000V	2000 20000	10 μ V 10 μ V up to 100 mV
SPD5	200 mV up to 1000 V	20000	10 μ V up to 100 mV
SPD6	200 mV up to 1000 V	2000	100 μ V up to 1 V

Input impedance

ranges 20 mV...20 V: 1 G Ω // 150 pF.
 ranges 200V, 1000 V: 10 M Ω // 150 pF.
 at overload : 100 k Ω // 150 pF.

Input current

< 30 pA; temperature coefficient 1 pA/K

Normal mode rejection.

SPEED MODE	NORMAL MODE REJECTION RATIO ¹⁾		
	50 or 60 Hz $\pm$ 0.1%	50 or 60 Hz $\pm$ 1%	400 Hz $\pm$ 1%
SPD1	> 80 dB	> 60 dB	> 50 dB
SPD2	> 70 dB	> 50 dB	> 40 dB
SPD3	> 60 dB	> 40 dB	> 30 dB
SPD4	0	0	0
SPD5	0	0	0

¹⁾ Valid for a maximum normal mode voltage of 1.3 x the range.

Common mode rejection.

SPEED MODE	COMMON MODE REJECTION RATIO ¹⁾			
	50 or 60 Hz $\pm$ 0.1%	50 or 60 Hz $\pm$ 1%	400 Hz $\pm$ 1%	DC
SPD1	> 160 dB	> 140 dB	> 110 dB	> 140 dB
SPD2	> 150 dB	> 130 dB	> 100 dB	> 140 dB
SPD3	> 140 dB	> 120 dB	> 90 dB	> 140 dB
SPD4	> 80 dB	> 80 dB	> 60 dB	> 140 dB
SPD5	> 80 dB	> 80 dB	> 60 dB	> 140 dB
SPD6	> 80 dB	> 80 dB	> 60 dB	> 140 dB

¹⁾ Measured with 1 k Ω in series with either lead, GUARD connected to CM voltage, analog filter off.

Maximum Common Mode Voltage

between COM-V Ω and GUARD: 60 Vac or dc; 85 Vpeak.
between GROUND and GUARD : 450 Vac or dc; 630 Vpeak.
slew rate: 10 kV/ μ s maximum.

Input voltage characteristics:

- maximum input voltage
- maximum VHz product
- maximum slew rate

1000 V continuously.
1400 V peak for <10 ms.
10⁶.
5 kV/ μ s.

Overload recovery time

< 300 ms up to 1000 V.

A.2.3 Volts AC

Measurement quantities	Volts AC+DC, RMS value; dc coupled input. Volts AC, RMS value; ac coupled input.
Ranges	20 mV, 200 mV, 2 V, 20 V, 200 V, 700 V.
Ranging	Manual or automatic. No automatic down ranging to the 20 mV range in speed mode SPD1.
Frequency range Volts AC	Depending on the selected RMS filter: - fast response 400 Hz to 1 MHz. - normal response 40 Hz to 1 MHz. - slow response 10 Hz to 1 MHz.

Accuracy Volts AC, $\pm(\% \text{ of reading} + \% \text{ of range}).1)$.

RESPONSE (FILTER)	FREQUENCY RANGE	24 HOUR, Tcal $\pm 1^\circ\text{C}$		90 DAY, Tcal $\pm 5^\circ\text{C}$		1 YEAR, Tcal $\pm 5^\circ\text{C}$	
normal	40 Hz - 20 kHz	0.05	0.01	0.07	0.02	0.1	0.03
fast	400 Hz - 20 kHz	0.05	0.01	0.07	0.02	0.1	0.03
slow	10 Hz - 40 Hz	0.07	0.01	0.1	0.02	0.15	0.03
	40 Hz - 20 kHz	0.05	0.01	0.07	0.02	0.1	0.03
all	20 kHz - 50 kHz	0.125	0.025	0.175	0.05	0.25	0.075
	50 kHz - 100 kHz	0.25	0.05	0.35	0.1	0.5	0.15
	100 kHz - 300 kHz	0.75	0.15	1.05	0.3	1.5	0.45
	300 kHz - 500 kHz	1.25	0.25	1.75	0.5	2.5	0.75
	500 kHz - 1 MHz	2.50	1.00	3.50	2.0	5.0	3.0

- 1) - Relative to calibration values.
 - Valid over 1-100% of range for sinusoidal input signals.
 - Add 0.01% of reading + 0.01% of range in the 20 mV range.
 - For Volts AC+DC:
 add $\pm (0.05\% \text{ of reading} + 0.01\% \text{ of range})$. The maximum dc component for the stated accuracy is 50 x range or 400 V (whichever is less).
 - Temperature coefficient $\pm (0.005\% \text{ of reading} + 0.005\% \text{ of range}) / \text{K}$, in range 0-18 °C and 28- 50 °C (frequency range up to 20 kHz).

Measurement timing and resolution.

SPEED MODE	READINGS PER SECOND ¹⁾	MEASUREMENT TIME	RANGE	FULL SCALE	RESOLUTION
1	0.2	5 s	20 mV 200 mV - 700 V	200000 2000000	0.1 μV 0.1 μV - 1 mV
2	2	0.5 s	20 mV - 700 V	200000	0.1 μV - 10 mV
3	20	50 ms	20 mV - 700 V	20000	1 μV - 100 mV

- 1) In internal trigger mode with default trigger interval.

Response time.

SPEED MODE	RESPONSE	WITHOUT RANGING		WITH RANGING ¹⁾		NUMBER OF COUNTS FROM FINAL VALUE ²⁾
		INTERNAL TRIGGER	SINGLE TRIGGER	INTERNAL TRIGGER	SINGLE TRIGGER	
1	normal	15s	15s	17s	17s	50
	fast	10s	10s	12s	12s	50
	slow	15s	15s	17s	17s	50
2	normal	2s	1.5s	4s	3.5s	10
	fast	1s	0.7s	3s	2.7s	10
	slow	3s	2.5s	6s	5.5s	10
3	normal	0.7s	0.6s	2.7s	2.6s	10
	fast	0.2s	0.2s	1.5s	1.5s	10
	slow	1.2s	1.1s	4.2s	4.2s	10

1) Over maximum number of ranges.

2) Difference between first reading and final value for a step input change (coincident with trigger at single trigger).

Input impedance

1 M Ω // 150 pF ($\pm 15\%$)

at overload: 20 V - 1000 V
20 mV - 2 V

1 M Ω // 150 pF ($\pm 15\%$)
min. 11 k Ω

Common mode rejection

120 dB for DC signals.

80 dB for ac signals up to 60 Hz, decreasing with 20 dB/decade.

GUARD connected to COM, 1 k Ω in series with either lead.

Maximum common mode voltage

between COM and GUARD 60 Vac or dc, 85 Vpeak.

between GUARD and GROUND 450 Vac or dc, 630 Vpeak.

Maximum slew rate: 10 kV/ μ s.

Input voltage characteristics:

- Maximum input voltage
- Maximum slew rate
- Maximum VHz product
- Crest factor limit

700 Vac or dc, 1000 Vpeak on all ranges.

5 kV/ μ s.

2×10^7 .

4 at full scale, increasing down scale via

4 x (full scale : reading), to 10 maximum.

Overload recovery time

< 1 s up to 700 V.

A.2.4 Resistance Ω 4W, Ω 2W

Measurement functions

2 wire via "V-Ohm" "Vcom" terminals
4 wire via "V-Ohm" "Vcom" and "A"-"Acom" terminals

Ranging

Manual or automatic. No automatic down ranging to the 20 Ω range, and to the 200 Ω range in speed mode SPD1.

Input nulling (TRUE OHM)

Input offset can be compensated via the "NULL" function (see Section 4.10).

Ranges, measurement current.

RANGE	20 Ω	200 Ω	2 k Ω	20 k Ω	200 k Ω	2 M Ω	20 M Ω	200 M Ω
CURRENT	10 mA	1 mA	1 mA	100 μ A	10 μ A	1 μ A	100 nA	10 nA

Accuracy, $\pm$ (% of reading + % of range). ¹⁾

SPEED MODE	RANGE	24 HOUR, Tcal $\pm$ 1 $^{\circ}$ C		90 DAY, Tcal $\pm$ 5 $^{\circ}$ C		1 YEAR, Tcal $\pm$ 5 $^{\circ}$ C	
SPD1,2	20 Ω - 200 k Ω	0.0024	0.0005	0.0036	0.001	0.0085	0.001
	2 M Ω	0.0072	0.0005	0.012	0.001	0.024	0.001
	20 M Ω - 200 M Ω	0.036	0.01	0.1	0.01	0.1	0.01
SPD3	20 Ω - 200 k Ω	0.0030	0.0005	0.0040	0.001	0.0095	0.001
	2 M Ω	0.0080	0.0005	0.013	0.001	0.025	0.001
	20 M Ω - 200 M Ω	0.040	0.01	0.07	0.01	0.1	0.01
SPD4	20 Ω - 200 k Ω	0.006	0.005	0.012	0.01	0.020	0.01
	2 M Ω	0.05	0.05	0.10	0.10	0.10	0.10
SPD5	20 Ω - 200 k Ω	0.01	0.01	0.015	0.01	0.025	0.01
	2 M Ω	0.05	0.05	0.10	0.10	0.10	0.10

¹⁾ Relative to calibration values.

Temperature coefficient in range 0-18 $^{\circ}$ C and 28-50 $^{\circ}$ C, $\pm$ % of reading/K.

RANGE ¹⁾	20 Ω	200 Ω	2 k Ω	20 k Ω	200 k Ω	2 M Ω	20 M Ω	200 M Ω
TEMP. COEFFICIENT	0.002	0.001	0.001	0.001	0.001	0.003	0.003	0.01

Measurement timing, resolution.

SPEED MODE	READINGS PER SECOND ¹⁾		MEASUREMENT TIME		RANGE	FULL SCALE	RESOLUTION
	50 Hz	60 Hz	50 Hz	60 Hz			
1 ²⁾	0.2	0.24	5 s	4.17s	20 Ω - 200 Ω 2 k Ω - 2 M Ω 20 M Ω 200 M Ω	2000000 20000000 2000000 200000	10 $\mu\Omega$ - 100 $\mu\Omega$ 100 $\mu\Omega$ - 100 m Ω 10 Ω 1 k Ω
2	2	2.4	0.5 s	0.417s	20 Ω - 2 M Ω 20 M Ω 200 M Ω	2000000 200000 20000	10 $\mu\Omega$ - 1 Ω 100 Ω 10 k Ω
3	20	24	50 ms	41.7ms	20 Ω - 2 M Ω 20 M Ω - 200 M Ω	200000 20000	100 $\mu\Omega$ - 10 Ω 1 k Ω - 10 k Ω
4	200		5 ms		20 Ω - 200 k Ω 2 M Ω	20000 2000	1 m Ω - 10 Ω 1 k Ω
5	1000		1 ms		20 Ω - 200 k Ω 2 M Ω	20000 2000	1 m Ω - 10 Ω 1 k Ω
5 ³⁾	4000		0.25 ms		20 Ω - 200 k Ω 2 M Ω	20000 2000	1 m Ω - 10 Ω 1 k Ω

1) In internal trigger mode with default trigger interval.

2) Display update after 0.5 seconds; correct value after stated measurement time.

3) in BURST mode only; no direct output to display and interface.

Response time for ranges up to 200 k Ω (analog filter off, add 200 ms if filter on).

SPEED MODE	WITHOUT RANGING		WITH RANGING ¹⁾		NUMBER OF COUNTS FROM FINAL VALUE ²⁾
	INTERNAL TRIGGER	SINGLE TRIGGER	INTERNAL TRIGGER	SINGLE TRIGGER	
SPD1	5.5 s	5 s	6 s	5.5 s	50
SPD2	1.0 s	0.5 s	1.5 s	1.0 s	10
SPD3	110 ms	60 ms	600 ms	550 ms	10
SPD4	15 ms	10 ms	500 ms	500 ms	5
SPD5	3 ms	2 ms	500 ms	500 ms	5

1) Over maximum number of ranges, for ranges up to 200 k Ω .

2) Difference between first reading and final value for a step input change (coincident with trigger in single trigger mode).

Maximum lead resistance	Ohms 4-Wire:100 Ω (10 Ω in the 20 Ω range). Ohms 2-Wire: lead resistance included in reading.
Voltage at open input	10 V max.
Input protection	The input circuitry is protected against voltages up to 500 V ac or dc and 750 V _{peak} on the 2 wire and 4 wire terminals.
Recovery time from input voltage overload	< 300 ms for voltages up to 30 V. < 15 s for voltages up to 250 V.
Testing semiconductor junctions	In forward (upto $V_f=2$ V on 2 k Ω range) and reverse direction

Common mode voltage influence. ¹⁾

SPEED MODE	ADDITIONAL ERROR FOR COMMON MODE VOLTAGE ($\pm$ % RANGE/VOLT)		
	DC	50 or 60 Hz $\pm$ 0.1%	50 or 60 Hz $\pm$ 1%
SPD1,2,3	0.00002	0.00002	0.0002
SPD4,5	0.002	0.01	0.01

1) "GUARD"and "COM" connected to the CM voltage.

Maximum Common Mode voltage	between COM and GUARD 60 V ac or dc, 85 V _{peak} . between GUARD and GROUND 450 Vac or dc, 630 V _{peak} . Max. slew rate: 10 kV/ μ s
-----------------------------	--

A.2.5 Current DC

Ranges	2 μ A, 20 μ A, 200 μ A, 2 mA, 20 mA, 200 mA, 2 A.
Ranging	Manual or automatic. No automatic ranging to the 2 μ A range in speed mode SPD1.

Accuracy, $\pm$ (% of reading + % of range). ¹⁾

SPEED MODE	RANGE	24 HOUR, T _{cal} $\pm$ 1 $^{\circ}$ C		90 DAY, T _{cal} $\pm$ 5 $^{\circ}$ C		1 YEAR, T _{cal} $\pm$ 5 $^{\circ}$ C	
SPD1,2,3	2 μ A - 2 mA	0.002	0.001	0.010	0.001	0.015	0.001
	20 mA - 2 A	0.003	0.003	0.015	0.003	0.025	0.003
SPD4,5	2 μ A - 2 mA	0.010	0.005	0.020	0.005	0.025	0.005
	20 mA - 2 A	0.011	0.007	0.025	0.007	0.035	0.007

- 1) - Relative to calibration values.
- Temperature coefficient $\pm$ (0.003% of reading + 0.0005% of range) / K, in range 0 -18 $^{\circ}$ C and 28-50 $^{\circ}$ C.
- Add 50 I²ppm of reading, because of selfheating of the shunt.

Measurement timing, resolution.

SPEED MODE	READINGS PER SECOND 1)		MEASUREMENT TIME		RANGE	FULL SCALE	RESOLUTION
	50 Hz	60 Hz	50 Hz	60 Hz			
1 2)	0.2	0.24	5 s	4.17s	2 μ A 20 μ A - 2 A	2000000 20000000	1 pA 1 pA - 100 nA
2	2	2.4	0.5 s	0.417s	2 μ A - 2 A	2000000	1 pA - 1 μ A
3	20	24	50 ms	41.7ms	2 μ A - 2 A	200000	10 pA - 10 μ A
4	200		5 ms		2 μ A - 2 A	20000	100 pA - 100 μ A
5	1000		0.5 ms		2 μ A - 2 A	20000	100 pA - 100 μ A

1) In internal trigger mode with default trigger interval.

2) Display update after 0.5 seconds; correct value after stated measurement time.

Response time.

SPEED MODE	WITHOUT RANGING		WITH RANGING 1)		NUMBER OF COUNTS FROM FINAL VALUE 2)
	INTERNAL TRIGGER	SINGLE TRIGGER	INTERNAL TRIGGER	SINGLE TRIGGER	
SPD1	5.5 s	5.0 s	6.0 s	5.5 s	50
SPD2	1.0 s	0.5 s	1.5 s	1.0 s	10
SPD3	110 ms	60 ms	600 ms	550 ms	10
SPD4	15 ms	10 ms	500 ms	500 ms	10
SPD5	9 ms	8 ms	500 ms	500 ms	10

1) Over maximum number of ranges.

2) Difference between first reading and final value for a step change coincident with trigger.

3) For the 2 μ A range the response time is > 400 ms

Voltage drop (at end of range).

RANGE	2 μ A	20 μ A	200 μ A	2 mA	20 mA	200 mA	2 A
VOLTAGE DROP	0.25 mV	2.5 mV	25 mV	250 mV	250 mV	350 mV	950 mV

Protection of the input

current limited by fuse 2.5 AF, high breaking capacity;
protection up to 250 Vac or dc, 350 Vpeak.

Series mode rejection.

SPEED MODE	50 or 60Hz $\pm 0.1\%$	50 or 60 Hz $\pm 1\%$	400 Hz $\pm 1\%$
SPD1	> 80 dB	> 60 dB	< 50 dB
SPD2	> 70 dB	> 50 dB	< 40 dB
SPD3	> 60 dB	> 40 dB	< 30 dB
SPD4	0	0	0
SPD5	0	0	0

Maximum series mode signal

Peak value: 130% of range.

Common mode voltage influence, $\pm$ % of range/V.

SPEED MODE	DC	50 or 60 Hz $\pm 0.1\%$	50 or 60 Hz $\pm 1\%$
SPD1,2,3	0.00002	0.00002	0.0002
SPD4	0.002	0.01	0.01

Maximum common mode voltage

between COM and GUARD 60 V ac or dc, 85 V_{peak}.
between GUARD and GROUND 450 V ac or dc, 630 V_{peak}.Maximum slew rate: 10 kV/ μ s

Overload recovery time

< 300 ms up to 2 A.

A.2.6 Current AC+DC

Measurement quantities
(measurement function)

Current AC+DC, RMS value; dc coupled input.

Ranges

200 μ A, 2 mA, 20 mA, 200 mA, 2 A.

Ranging

Manual or automatic

Frequency range

Depending on the selected RMS filter:

- fast response 400 Hz to 10 kHz (>400 Hz).
- normal response 40 Hz to 10 kHz (>40 Hz).
- slow response 10 Hz to 10 kHz (>10 Hz).

Accuracy, $\pm$ (% of reading + % of range).

RESPONSE (FILTER)	FREQUENCY RANGE	24 HOUR Tcal $\pm$ 1 °C	90 DAY Tcal $\pm$ 5 °C	1 YEAR Tcal $\pm$ 5 °C
NORMAL	40 Hz - 1 kHz	0.07 + 0.05	0.1 + 0.05	0.15 + 0.05
FAST	400 Hz - 1 kHz	0.07 + 0.05	0.1 + 0.05	0.15 + 0.05
SLOW	10 Hz - 40 Hz	0.1 + 0.05	0.15 + 0.05	0.2 + 0.05
	40 Hz - 1 kHz	0.07 + 0.05	0.1 + 0.05	0.15 + 0.05
ALL	1 kHz - 5 kHz	0.1 + 0.05	0.1 + 0.05	0.1 + 0.1
	5 kHz - 10 kHz	0.45 + 0.05	0.5 + 0.05	0.5 + 0.2

- 1) - Valid over 1-100% of range, for sinusoidal inputs. Relative to calibration values.
 - Temperature coefficient $\pm$ 0.01% of reading/K, in range 0 -18 °C and 28 50 °C.
 - Add 50 l²ppm of reading, because of selfheating of the shunt.

Measurement timing, resolution.

SPEED MODE	READINGS PER SECOND ¹⁾	MEASUREMENT TIME	RANGE	FULL SCALE	RESOLUTION
SPD1	0.2	5 s	200 μ A - 2 A	2000000	100 pA - 1 μ A
SPD2	2	0.5 s	200 μ A - 2 A	200000	1 nA - 10 μ A
SPD3	20	50 ms	200 μ A - 2 A	20000	10 nA - 100 μ A

- 1) In internal trigger mode with default trigger interval.

Response time.

SPEED MODE	RESPONSE	WITHOUT RANGING		WITH RANGING ¹⁾		NUMBER OF COUNTS FROM FINAL VALUE ²⁾
		INTERNAL TRIGGER	SINGLE TRIGGER	INTERNAL TRIGGER	SINGLE TRIGGER	
1	normal	15s	15s	17s	17s	50
	fast	10s	10s	12s	12s	50
	slow	15s	15s	17s	17s	50
2	normal	2s	1.5s	4s	3.5s	10
	fast	1s	0.7s	3s	2.7s	10
	slow	3s	2.5s	6s	5.5s	10
3	normal	0.7s	0.6s	2.7s	2.6s	10
	fast	0.2s	0.2s	1.5s	1.5s	10
	slow	1.2s	1.1s	4.2s	4.2s	10

1) Over maximum number of ranges.

2) Difference between first reading and final value for a step change coincident with trigger.

Voltage drop (at end of range).

RANGE	200 μ A	2 mA	20 mA	200 mA	2 A
VOLTAGE DROP	25 mV	250 mV	250 mV	350 mV	950 mV

Protection of the input

current limited by fuse 2.5 AF, high breaking capacity.
protection up to 250 Vac or dc, 350 Vpeak.

Common mode voltage influence

$\pm 0.0001\%$ of range / V for dc.
 $\pm 0.0001\%$ of range / V for ac up to 100 Hz.

Maximum common mode voltage

between COM and GUARD 60 V ac or dc, 85 Vpeak.
between GUARD and GROUND 450 V ac or dc, 630 Vpeak.
slew rate 10 kV/ μ s maximum.

Crest factor

3.3 at full scale.
crest factor = $3.3 * (\text{full scale} : \text{reading})$, with a maximum of 10.

Overload recovery time

< 1 s up to 2 A

A.2.7 Volts Low frequency

Measurement quantities The RMS value of the input signal are calculated from the input samples. The input is dc coupled.

Ranges 200 mV, 2 V, 20 V, 200 V, 1000 V.

Ranging Manual or automatic.

Measurement timing, frequency range.

READING RATE	ACQUISITION TIME ¹⁾	RESPONSE TIME	FREQUENCY RANGE
1	2000 s	< 2006 s	1 mHz - 500 Hz
2	200 s	< 206 s	10 mHz - 500 Hz
3	20 s	< 26 s	100 mHz - 500 Hz
4	2 s	8 s	1 Hz - 500 Hz
5	0.2 s	8 s	10 Hz - 500 Hz

1) The acquisition time is automatically selected depending on the input signal frequency.

Full scale 20000 counts.

Accuracy, $\pm$ (% of reading + % of range). 1)

24 HOUR, Tcal $\pm$ 1 °C	90 DAY, Tcal $\pm$ 5 °C	1 YEAR, Tcal $\pm$ 5°C
0.08 0.02	0.1 0.03	0.15 0.05

Temperature coefficient $\pm$ 0.005% of reading /K in range 0-18 °C and 28-50 °C.

Zero point drift included in accuracy and temperature coefficient.

Input impedance ranges 200 mV-20 V : > 1 G Ω // 150 pF.
 ranges 200 V, 1000 V : > 10 M Ω // 150 pF.
 at overload : 100 k Ω // 150 pF.

Input current < 30 pA; temp. coefficient 1 pA/K.

Maximum Common Mode Voltage between COM and GUARD: 60 Vac or dc, 85 Vpeak.
 between GROUND and GUARD : 450 Vac or dc, 630 Vpeak.
 Maximum slew rate: 10 kV/ μ s.

Input voltage characteristics:

- maximum input voltage 1000 V continuously.
 1400 V peak for <10 ms.
 - maximum VHz product 10^6 .
 - maximum slew rate 5 kV/ μ s.

Overload recovery time <300 ms up to 1000V

A.2.8 Frequency/Period

Measurement quantities

Frequency and period are derived from the zero crossings of an Volts AC measurement (AC coupled).

Ranges

Voltage ranges : 200 mV - 700V
Frequency ranges: see the table below.
Period ranges: see the table below.

Ranging

Voltage ranging: automatic or manual.
Freq/Per ranging: automatic.
RANGE UP DOWN apply only to voltage ranging.

Input voltage sensitivity

5% of full scale.

Accuracy

$\pm 0.001\% \pm 1$ count

Frequency ranges, measurement timing and resolution.

SPEED MODE	READINGS PER SECOND ^{1) 2)}	MEASUREMENT TIME ²⁾	END OF RANGE ³⁾	FULL SCALE	RESOLUTION
SPD1	1 - 2	0.5 - 1.0 s	10 Hz 100 Hz 1 kHz 10 kHz 100 kHz 1 MHz 2 MHz	9.999999 99.99999 999.9999 9.999999 99.99999 999.9999 2.000000	1 μ Hz 10 μ Hz 100 μ Hz 1 mHz 10 mHz 100 mHz 1 Hz
SPD2	10 - 20	50 - 100 ms	100 Hz 1 kHz 10 kHz 100 kHz 1 MHz 2 MHz	99.9999 999.999 9.99999 99.9999 999.999 2.00000	100 μ Hz 1 mHz 10 mHz 100 mHz 1 Hz 10 Hz
SPD3	100 - 200	5 - 10 ms	1 kHz 10 kHz 100 kHz 1 MHz 2 MHz	999.99 9.9999 99.999 999.99 2.0000	10 mHz 100 mHz 1 Hz 10 Hz 100 Hz

1) In internal trigger mode with default trigger interval.

2) The reading rate and measurement time depend on the input frequency. See Section 4.4.8.

3) The minimum input frequency is 2 Hz (SPD1), 20 Hz (SPD2), 200 Hz (SPD3).

Period ranges, measurement timing, and resolution.

SPEED MODE	READINGS PER SECOND ^{1) 2)}	MEASUREMENT TIME ²⁾	RANGE ³⁾	FULL SCALE	RESOLUTION
SPD1	1 - 2	0.5 - 1.0 s	1000 ms	999.9999	100 ns
			100 ms	99.99999	10 ns
			10 ms	9.999999	1 ns
			1000 μ s	999.9999	100 ps
			100 μ s	99.99999	10 ps
			10 μ s	9.999999	1 ps
			1000 ns	999.9999	0.1 ps
SPD2	10 - 20	50 - 100 ms	100 ms	99.9999	100 ps
			10 ms	9.99999	10 ns
			1000 μ s	999.999	1 ns
			100 μ s	99.9999	100 ps
			10 μ s	9.99999	10 ps
			1000 ns	999.999	1 ps
SPD3	100 - 200	5 - 10 ms	10 ms	9.9999	100 ns
			1000 μ s	999.99	10 ns
			100 μ s	99.999	1 ns
			10 μ s	9.9999	100 ps
			1000 ns	999.99	10 ps

1) In internal trigger mode with default trigger interval.

2) The reading rate and measurement time depend on the input frequency. See Section 4.4.8.

3) The maximum period time is 500 ms (SPD1), 50 ms (SPD2), 5 ms (SPD3).

Input impedance:

1 M Ω // 150 pF ($\pm$ 15%)

at overload: 20 V - 1000 V 1 M Ω // 150 pF ($\pm$ 1 5%)
20 mV - 2 V min. 11 k Ω

Input voltage characteristics:

- Maximum input voltage 700 Vac or dc, 1000 V_{peak} on all voltage ranges.
- Maximum VHz product 2×10^7 .
- Maximum slew rate 5 kV/ μ s.

Overload recovery time

< 1 s up to 700 V.

A.2.9 Temperature

Measurement principle

Pt100 RTD measurement in 4-wire configuration.

Probe type

Selectable:

- * 385 Pt (DIN 43760 table), e.g. for PM9252 probe.
- * 390 Pt (ALPHA = 0.0038994, DELTA = 1.494,
A4 = -0.265668×10^{-4} , C4 = 0.205984×10^{-11}). ¹⁾
- * 392 Pt (ALPHA = 0.0039221, DELTA = 1.493,
A4 = -0.38668×10^{-5} , C4 = $+0.192912 \times 10^{-13}$). ¹⁾

¹⁾ See IPTS 68 equations in NBS Monograph 126.

Measurement current

1 mA.

Range

-200 °C up to +850 °C.
-328 °F up to +1562 °F.
73 K up to 1123 K.
Unit can be selected.

Accuracy

± (0.3% of reading + 0.2 °C), excluding probe.

Temperature coefficient

± (0.001% of reading + 0.001% of range) in ambient temperature range 0-18 °C and 28-50 °C.

Measurement speed

SPEED MODE	50 Hz	60 Hz
SPD2	2 meas/s.	2.4 meas/s
SPD3	20 meas/s.	24 meas/s

Resolution

SPD2 : 0.01 °C.

SPD3 : 0.1 °C .

Response time

internal triggering: SPD2 < 750 ms, SPD3 < 75 ms.
single triggering : SPD2 < 550 ms, SPD3 < 65 ms.
Times are excluding probe.

Maximum lead resistance

4-Wire configuration: 100 Ω.

Voltage at open input sockets

10 V max. between A and COM terminal.

Input protection

The input circuitry is protected against voltages up to 500 V ac or dc and 750 Vpeak on the 2 wire and 4 wire terminals.

Recovery time from input voltage overload

< 300 ms for voltages up to 30 V.
< 15 s for voltages up to 250 V.

Maximum voltage at probe tip

depending on used probe.

Maximum Common Mode voltage

between COM and GUARD 60 V ac or dc, 85 Vpeak.
between GUARD and GROUND 450 Vac or dc, 630 Vpeak.

A.2.10 Operational data

Warming up time

To avoid condensation the instrument shall be temperature preconditioned after bringing it from a cold into a warm environment.

- operational within 5 seconds
- 60 minutes to meet 90 days accuracy
- 2 hours to meet 24 h specification
- 2 hours before calibration

Calibration

On delivery the instrument is calibrated with equipment traceable to national standards.

Calibration interval: 90 days or 1 year.

Closed case calibration, controlled by front panel operation or via IEEE-488 interface.

Backup battery lifetime

> 5 years

A.2.11 Measurement control

Input sampling time.

SPEED MODE	SAMPLES PER MEASUREMENT	APERTURE PER SAMPLE	
		50 Hz, 400 Hz	60 Hz
1	100	20 ms	16 ² / ₃ ms
2	10	20 ms	16 ² / ₃ ms
3	1	20 ms	16 ² / ₃ ms
4	1	2 ms	2 ms
5		200 µs	200 µs
6		20 µs	20 µs

Ranging

Manual or automatic. DOWN ranging at approx. 9% of full scale, UP ranging at 100% of full scale.

Triggering

INTERNAL (continuous) or SINGLE (external or manual) trigger mode.

A.2.12 Remote/external control

IEEE-488 interface

interface according to IEEE 488 (1 and 2) is standard built-in.

Supported interface function subsets:

SH1, AH1, T6, L4, SR1, RL1, DC1, DT1, PP0.

The PM2530 is SCPI 1990.0 compatible.

External triggering

via rear panel BNC connectors EXT1 and EXT2.

Ready indication

via rear panel BNC connectors RDY1 and RDY2.

Limits indication

LIMIT passings are signalled via the rear panel LIMITS connector.

A.2.13 Cabinet, dimensions and weight

Cabinet

Metal case.

Dimensions

see Fig. 2.1.

Weight

5 kg

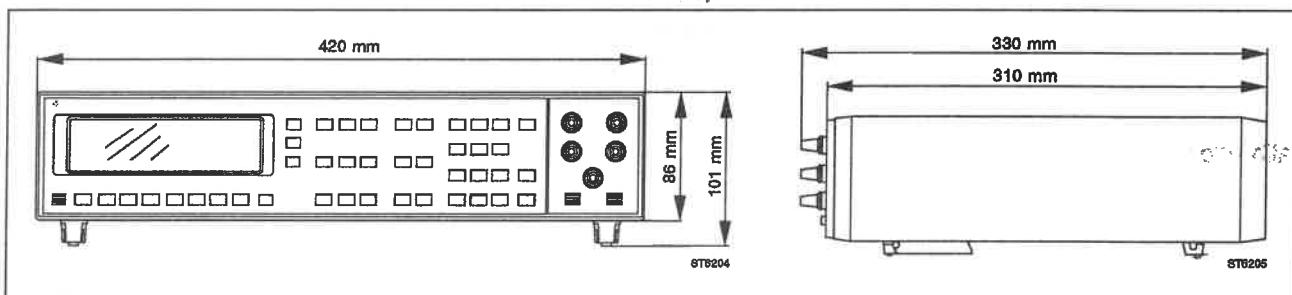


Fig. 2.1. PM2530 external dimensions.

A.2.14 Input terminals, input limits

Number of sockets

5x safe 4 mm terminals on the front and rear.

Input configuration

asymmetrical, floating, guarded.

Guarding

internal or external guarding.

Maximum voltage, input impedance

INPUT TERMINALS	FUNCTION	MAXIMUM VALUE	IMPEDANCE
GUARD - GROUND	ALL	450 V ac or dc, 630 V peak	10 G Ω // 500 pF
GUARD - COM(V Ω)	ALL	60 V ac or dc, 85 V peak	10 G Ω // 500 pF
COM(V Ω) - GROUND	ALL	450 V ac or dc, 630 V peak	20 G Ω // 300 pF
V Ω - GROUND	ALL	1000 V ac or dc, 1400 V peak	
V Ω - COM(V Ω) A - COM(A)	See the relevant measurement function		

A.2.15 Instrument configurations.

The table below shows:

FUNCTION	the instrument functions, the SCPI command (short form) is shown between brackets.
PON	the instrument settings after power up.
	the instrument settings after pressing the  button (DEFAULT settings).
*RST	the instrument settings after the common SCPI command *RST.
MEAS:FUNC	if a setting is stored per measurement function apart.
SAV/*SAV	if a setting is included when an instrument configuration is stored in the SAVE/RECALL function.
:SET	if a setting is included when the SCPI command SYSTem:SET or SYSTem:SET? is sent.
:CONF/MEAS	if a setting is set by the SCPI :MEASure or :CONFIgure command. It can be set to a specified, or to a pre-defined state.

Used abbreviations:

R0	the settings saved with SAV/*SAV (SAVE/RECALL function, Section 4.17, Section 5.3.15) in memory location 0 are recalled.
CLR	cleared.
-	not affected
INT	internal
DEF	default
IMM	immediate
MIN	minimum
MAX	maximum
INF	Infinitive positive
NINF	Infinitive negative

The following settings are only influenced if explicitly set:

TIME, DATE, BUS ADDRESS, DISPLAY CONTRAST, DISPLAY BACKLIGHT ON/OFF, TEMPERATURE UNIT.

Instrument configurations

FUNCTION		SETTING AFTER			STORED PER MEAS. FUNC.	STORED AT SAV/ *SAV	INCL. IN :SET	SET BY :CONF/ :MEAS
		PON		*RST				
MEASUREMENT FUNCTION	(:SENS:FUNC)	R0	-	V=		Y	Y	Y
TEMP PROBE TYPE	(:TEMP:TYPE)	R0	-	43760	Y	Y	Y	Y
TEMP UNIT	(:UNIT:TEMP)	-	-	-	Y	Y	Y	N
RANGE	(:SENS:RANGE)	R0	AUTO	AUTO	Y	Y	Y	Y
TRIGGERING								
- MODE	(:SAMP:SOUR)	R0	IMM	IMM	N	Y	Y	IMM
- DELAY	(:SAMP:DEL:VAL)	R0	OFF	DEF	N	Y	Y	DEF
- INTERVAL	(:SAMP:TIM)	R0	DEF	MIN	N	Y	Y	N
- SAMPLE TRIGGER INPUT	(:SAMP:SOUR)	R0	IMM	IMM	N	Y	Y	IMM
READING RATE (SPD..)	(SENS:<function>:APER)	R0	SPD2	SPD2	Y	Y	Y	Y

FUNCTION	SETTING AFTER			STORED PER MEAS.	STORED AT SAV	INCL IN SET	SET BY :CONF/ :MEAS
	PON	RESET	*RST				
FILTERS							
- ANALOG ON/OFF (:INP:FILT:LPAS:STAT)	R0	OFF	OFF	Y	Y	Y	OFF
- RMS FILTER (:INP:FILT:RMS:FREQ)	R0	40	40	Y	Y	Y	40
- WEIGHTING ON/OFF (:CALC:AVER:EXP:STAT)	R0	OFF	OFF	Y	Y	Y	OFF
- WEIGHTING PARAM. (:CALC:AVER:EXP:WEIG)	R0	-	5	Y	Y	Y	N
- AVERAGING ON/OFF (:CALC:AVER:NORM:STAT)	R0	OFF	OFF	N	Y	Y	OFF
- AVERAGING PARAM. (:CALC:AVER:NORM:VAL)	R0	-	1	N	Y	Y	N
- AVERAGING EXTREMES (:CALC:AVER:EXTR:STAT)	R0	ON	ON	Y	Y	Y	N
RELATIVE REFERENCE							
- REL REF ON/OFF (:CALC:REF:STAT)	R0	OFF	OFF	Y	Y	Y	OFF
- REFERENCE VALUE (:CALC:REF:VAL)	R0	-	0	Y	Y	Y	N
INPUT OFFSET NULLING							
- NULLING ON/OFF (:SENS:CORR:STAT)	R0	-	OFF	Y	Y	Y	N
- NULL VALUE	-	-	-	-	-	-	-
BURST MODE							
- BURST ON/OFF	OFF	OFF	OFF	N	N	N	N
- POST TRIGGER VALUE (:SAMP:COUN)	R0	1	1	N	Y	Y	Y
- ARM INPUT (:ARM:SOUR)	R0	IMM	IMM	N	Y	Y	IMM
- SAMPLE INTERVAL (:SAMP:TIM)	R0	MIN	MIN	N	Y	Y	N
- SAMPLE DELAY (:SAMP:DEL)	R0	DEF	0	N	Y	Y	DEF
- SAMPLE INPUT (:SAMP:SOUR)	R0	IMM	IMM	N	Y	Y	IMM
- (POST) TRIGGER DELAY (:TRIG:DEL)	R0	0	0	N	Y	Y	0
- (POST) TRIGGER INPUT (:TRIG:SOUR)	R0	IMM	IMM	N	Y	Y	IMM
BURST MEMORY CONTENTS (LENGTH, EXTREMES)	CLR	-	CLR	N	N	N	CLR
- READ START LOCATION	0	-	-	N	N	N	N
- READ LENGTH	1	-	-	N	N	N	N
- READ STEP SIZE	1	-	-	N	N	N	N
- READ STEP RATE	1	-	-	N	N	N	N
CALC FUNCTIONS							
- SCALE ON/OFF (:CALC:SCAL:STAT)	R0	OFF	OFF	Y	Y	Y	OFF
- SCALE GAIN (A) (:CALC:SCAL:GAIN)	R0	-	1	Y	Y	Y	N
- SCALE OFFSET (B) (:CALC:SCAL:OFFS)	R0	-	0	Y	Y	Y	N
- SCALE NORM (Z) (:CALC:SCAL:NORM)	R0	-	1	Y	Y	Y	N
- DELTA% ON/OFF (:CALC:PERC:STAT)	R0	OFF	OFF	Y	Y	Y	OFF
- DELTA % PARAM (:CALC:PERC:REF:VAL)	R0	-	1	Y	Y	Y	N
- dBm ON/OFF (:CALC:LOG:VAL)	R0	OFF	OFF	Y	Y	Y	OFF
- dBm PARAM (:CALC:LOG:REF:VAL)	R0	-	600	Y	Y	Y	N
MIN/MAX FUNCTION							
- ON/OFF (:CALC:EXTR:STAT)	R0	OFF	OFF	Y	Y	Y	N
- SIGNAL TO BEEPER (:CALC:EXTR:SIGN:BEEP:STAT)	R0	-	OFF	N	Y	Y	N
- SIGNAL TO DISPLAY (:CALC:EXTR:SIGN:DISP:STAT)	R0	-	OFF	N	Y	Y	N
- MEMORY CONTENTS (:CALC:EXTR:CLE)	CLR	CLR	CLR	N	N	N	CLR

Instrument configurations, continued

FUNCTION	SETTING AFTER	STORED PER MEAS. FUNC.	STORED AT SAV/*SAV	INCL. IN :SET	SET BY :CONF/:MEAS
	PON	 *RST			
LIMITS FUNCTION					
- ON/OFF (:CALC:LIM:STAT)	R0	OFF	OFF	Y	Y
- UPPER LIMIT (:CALC:LIM:UPP)	R0	-	-INF	Y	Y
- LOWER LIMIT (:CALC:LIM:LOW)	R0	-	+INF	Y	Y
- RELATIVE LIMIT ON/OFF (:CALC:RLIMIT:STAT)	R0	OFF	OFF	Y	Y
- RELATIVE LIMIT REF (:CALC:RLIMIT:REF)	R0	-	0	Y	Y
- RELATIVE UPPER LIMIT (:CALC:RLIMIT:UPP)	R0	-	1	Y	Y
- RELATIVE LOWER LIMIT (:CALC:RLIMIT:LOW)	R0	-	1	Y	Y
- SIGNAL TO DISPLAY (:CALC:LIM:SIGN:DISP:STAT)	R0	-	OFF	N	Y
- SIGNAL TO BEEPER (:CALC:LIM:SIGN:BEEP:STAT)	R0	-	OFF	N	Y
- EXTERNAL LINES (:CALC:LIM:SIGN:EXT:STAT)	R0	-	OFF	N	Y
- MEMORY CONTENTS (:CALC:LIM:CLE)	CLR	CLR	CLR	N	N
OUTPUT					
- DISPLAY LENGTH (:DISP:LENG)	R0	MAX	MAX	N	Y
- DISPLAY ANNUNCIATORS (:DISP:ANN)	R0	ON	ON	N	Y
- BEEPER ON/OFF (:SYST:BEEP:STAT)	R0	OFF ¹⁾	OFF ¹⁾	N	Y
- RDY 1 LINE (:OUTP:EXT1:STAT)	R0	ON	ON	N	Y
- RDY 2 LINE (:OUTP:EXT2:STAT)	R0	OFF	OFF	N	Y
- RDY 1 SOURCE (:OUTP:EXT1:SOUR)	R0	RCOM	RCOM	N	Y
- RDY 2 SOURCE (:OUTP:EXT2:SOUR)	R0	RCOM	RCOM	N	Y
- MEASUREMENT DATA (:FORMAT)	R0	-	ASCII	N	Y
AUTO CORRECTION					
ADC (CAL:AUTO:ADC)	R0	ON	ON	N	Y
RMS (CAL:AUTO:RMS)	R0	ON	ON	N	Y
ZERO (CAL:ZERO:AUTO)	R0	ON	ON	N	Y
BUS					
SRQ MASK (*SRE)	0	-	-	N	N
EVENT REGISTERS	CLR	-	-	N	N
BUS TRIGGER DEFINITIONS (*DDT)	*RST	-	*RST	N	N

- 1) The beeper always signals overload in the highest range for Volts DC, Volts AC and Current measurements.

Note: At delivery the Power ON settings stored in Ro are equal to the settings according to the *RST column.

A.3 ENVIRONMENTAL CONDITIONS

Instrument conditions and limits

The environmental data mentioned in this manual are based on the results of the manufacturer's checking procedures.

Details on these procedures and failure criteria are supplied on request by the FLUKE or PHILIPS organization in your country, or by PHILIPS, INDUSTRIAL ELECTRONICS DIVISION, EINDHOVEN, THE NETHERLANDS.

The instrument meets the environmental requirements of MIL-T-28800 D type III, Class 5, Style E, Colour R.

A.3.1 Climatic conditions.

The Temperature and Humidity specifications are in accordance with MIL-T-28800D part 3.7.2.1 unless otherwise specified below.

TEMPERATURE

Reference value
Operating range
Limit range of storage
and transport

23 °C ± 1 °C

0 °C ... 50 °C

-40 °C ... +71 °C

(comment: LCD storage range -25 °C ... +71 °C)

HUMIDITY

Reference rel. hum.
Rated range of use
No condensation
Limit range of operation
Limit range of storage
and transport
Max. dew point

45 - 75% RH

20 - 80% RH

20 - 80% RH

5 - 95% RH

25 °C

ALTITUDE

Operating
Non-operating

up to 3050 m (10 000 ft)

up to 4570 m (15 000 ft)

A.3.2 Mechanical conditions.

VIBRATION

Operating
frequency
g level

MIL-T-28800D 3.7.4.1

5 Hz - 55 Hz

2 at 55 Hz

Test method

resonance search 15 min; cycling 5,
resonance dwell 10 min.

Bounce, loose cargo

MIL-T-28800D 3.7.4.3

Shock functional
peak acceleration
amount of shocks
shock wave form
duration

MIL-T-28800D 3.7.5.1

20 g

18 (3 in each direction x,y, and z).

half sine wave

6-9 ms

Bench handling

MIL-T-28800D 3.7.5.3

Shock high impact

MIL-T-28800D 3.7.5.4

A.3.3 Power supply conditions

LINE VOLTAGE

Rated range of use

230 Vac ± 12 -15%

Reference value

230 Vac $\pm 1\%$

The instrument can be externally set to a line voltage of 115 V.

LINE FREQUENCY

Reference value

50 or 60 Hz $\pm 1\%$ (no adaption required)

Rated range of use

50 or 60 Hz $\pm 1\%$

50 or 60 Hz $\pm 5\%$ with reduced Series Mode Rejection
400 Hz is supported.

Power consumption

30 VA

Allowable power interruptions

up to 10 ms.

A.3.4 Electromagnetic Compatibility

Emission

The instrument meets the requirements of:
VDE 871-B, CISPR 11.

Susceptibility

IEC801 1-6

A.4 OPTIONAL ACCESSORIES, RECOMMENDED FOR THE PM2530

Besides the accessories included in the shipment, see Initial Inspection on page IV, the following PM2530 accessories are available:

PM9268	4 Wire Ohms cable with Kelvin clips and 4 mm banana connectors
5440A-7002	Low thermal copper EMF plug-in cable (FLUKE).
PM9265	Shielded measurement cable.
PM9252	Pt100 Contact Probe with 4 x 4 mm banana connectors. 1) Probe type 385 Pt, linearization according to DIN 43760 table. Probe temperature measurement range -60 °C ... +200 °C. Max. voltage between probe tip and any connection wire 30 Vac, 42 Vdc. Accuracy $\pm 0.5\%$ of reading ± 0.5 °C (to be added to PM2530 accuracy). Maximum temperature of handgrip and cable 80 °C. Response time for a temperature jump of 55 °C to 90% of end value is 3 seconds (pressed with 1N on polished copper).

- 1) Use in non-aggressive liquids is allowed.